Notes on Advanced Macroeconomics

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Spring semester, 2024

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Preface

Aim of this work of mine is to provide clear macroeconomic knowledges in language as simple as possible. Most of this note is taken from the advanced macroeconomics course of Yanfei Deng I took. This note is still a work in progress and needs tons of ammending. See the newest version at `https://github.com/xolarvill/notes-on-economics` along with my other notes. And pull a request if you figure some part of this note is wrong.

This work was first written in markdown language on Obsidian, which provides a lightweight and highly personalized working experience. But it was later transfered into purely latex language on Sublime Text, due to poor support for rendering of long math equation blocks of Obsidian. For the tranfer I used Pandoc at `https://pandoc.org/`. I wrote some LaTeX snippets of Sublime Text to optimize the writing experience, and uploaded on github at `https://github.com/xolarvill/snippets-for-quick-latex-on-st`.

One can find enormous academical help on advanced macroeconomics from books listed below:

- Introduction to Modern Economic Growth, D. Acemoglu
- Economic Growth, R. Barro and S.i.M. Xavi
- The ABCs of RBCs, An Introduction to Dynamic Macroeconomic Models, G. McCandless
- Macroeconomics, A Comprehensive Textbook for First-Year Ph.D. Courses in Macroeconomics. M. Azzimonti, P. Krusell, A. McKay, and T. Mukoyama
- Adavanced Macroeconomics, D. Romer

Part I
Introduction

A short bridge from intermediate macroeconomics to the decisions, equilibrium concepts, and notation used in these notes.

1 Introduction

These notes assume intermediate macroeconomics, including national accounts, basic growth theory, and aggregate demand and supply. We now ask how aggregate outcomes follow from decisions over time, and how those decisions respond to the environment and policy.

Parts II–IV cover growth, fluctuations, and the open economy. Part V studies money, finance, and policy; Part VI applies these mechanisms to the dollar system and China's growth after the pandemic. Chapters and sections marked *planned* contain outlines for later development.

1.1 Growth, fluctuations, and their interaction

Growth theory asks how productive capacity and living standards evolve. Starting with capital accumulation, we introduce saving decisions, population, technology, and resource allocation to explain differences across economies.

Business-cycle theory asks how an economy responds to disturbances in productivity, demand, or financing. We identify the shock, then explain the response of spending, employment, and prices. A mechanism that amplifies a shock need not be its cause.

The questions overlap. A recession can reduce investment and change future capital. Population and productivity growth can change the conditions for stabilization policy. Growth need not be an unaffected trend, and fluctuations can have lasting effects. We begin with separate models to isolate mechanisms, then study their interaction.

1.2 From aggregate relations to individual decisions

An aggregate consumption function such as $C = C(Y - T)$ summarizes a relationship. It does not yet explain how a household responds to an expected tax change, uncertain income, or a borrowing limit. To study those questions, we specify what the household chooses and what constrains it.

A two-period example. Let a household receive real income y_0 today and y_1 tomorrow, with no initial or terminal assets. Saving a_1 earns a known gross real return $R > 0$; negative a_1 is borrowing at that return. The problem is

$$\begin{aligned} \max_{c_0, c_1, a_1} \; & u(c_0) + \beta u(c_1), \qquad 0 < \beta < 1, \\ \text{s.t. } & c_0 + a_1 = y_0, \\ & c_1 = y_1 + R a_1. \end{aligned} \tag{1}$$

Assume $u' > 0$, $u'' < 0$, positive consumption, and an interior solution. Substituting the constraints leaves one choice:

$$\max_{a_1} \; u(y_0 - a_1) + \beta u(y_1 + R a_1). \tag{2}$$

Differentiate with respect to a_1, then substitute back for consumption:

$$-u'(y_0 - a_1) + \beta R u'(y_1 + R a_1) = 0, \tag{3}$$

$$u'(c_0) = \beta R u'(c_1). \tag{4}$$

Saving one more unit costs $u'(c_0)$ units of utility today and yields $\beta R u'(c_1)$ tomorrow. An interior optimum equates the two.

For marginal utility $u'(c) = c^{-\sigma}$, $\sigma > 0$,

$$c_0^{-\sigma} = \beta R c_1^{-\sigma}, \tag{5}$$

$$\left(\frac{c_1}{c_0}\right)^{\sigma} = \beta R, \tag{6}$$

$$\frac{c_1}{c_0} = (\beta R)^{1/\sigma}. \tag{7}$$

Thus consumption growth depends on patience and the return to saving. Its sensitivity to that return is governed by $1/\sigma$, the elasticity of intertemporal substitution in this example. A binding borrowing constraint would require an inequality condition instead of this interior equality.

Expectations and policy. With uncertainty, today's utility cost is known but tomorrow's gain is not. Replace that gain by its conditional expectation. For an asset with gross real payoff R_{t+1}, the interior condition becomes

$$u'(c_t) = \beta E_t[R_{t+1} u'(c_{t+1})], \qquad E_t[\cdot] \equiv E[\cdot \mid \mathscr{I}_t]. \tag{8}$$

Here $\mathscr{I}_t$ is information available when the household chooses saving. An uncertain return stays inside the expectation; its comovement with marginal utility matters. Rational expectations require forecasts consistent with the model and available information, not knowledge of tomorrow's realization.

The same logic matters for policy. An announced future tax change may alter consumption today. A change in a policy rule can change expectations and the aggregate relationships observed under the old rule. This is the concern in the Lucas critique.[1] Writing down individual decisions makes these responses explicit. It does not, by itself, guarantee that the chosen preferences, constraints, or information assumptions remain appropriate under every policy change.

1.3 Reading a dynamic equilibrium

The household takes income and returns as given. In general equilibrium, they must be consistent with production and other agents' choices. A competitive equilibrium combines individual optimality, budget constraints, market clearing, and policy rules at each date and, under uncertainty, each history.

States, choices, and paths. Consider a deterministic economy with fixed labor normalized to one, output $f(k_t)$, and no government or foreign trade. Capital k_t is available at the start of period t. With depreciation δ, feasibility gives

$$k_{t+1} = f(k_t) + (1 - \delta)k_t - c_t. \tag{9}$$

Capital is inherited; consumption is chosen and affects tomorrow's capital. Here k_t is a state variable and c_t a control variable. The set of states depends on the model's timing and assumptions.

If households can save in capital, competitive factor pricing gives the gross return $f'(k_{t+1}) + 1 - \delta$. Substituting this return into (4) yields the intertemporal condition

$$u'(c_t) = \beta\,[f'(k_{t+1}) + 1 - \delta]\,u'(c_{t+1}). \tag{10}$$

Together, (9) and (10) describe feasible accumulation and optimal consumption growth. Given k_0, different initial choices of c_0 generate different candidate paths. Feasibility and the Euler equation alone do not establish that each path is an equilibrium. The model also needs its boundary conditions, including the

[1] Robert E. Lucas, Jr. (1976), "Econometric Policy Evaluation: A Critique," *Carnegie-Rochester Conference Series on Public Policy*, 1, 19–46. https://doi.org/10.1016/S0167-2231(76)80003-6.

appropriate transversality condition for this infinite-horizon saving problem. We derive these conditions in the Ramsey chapter.

A steady state is a particular path. For a positive stationary allocation, set $k_{t+1} = k_t = k^*$ and $c_{t+1} = c_t = c^*$ in the two equations. Since $u'(c^*) > 0$,

$$c^* = f(k^*) - \delta k^*, \tag{11}$$

$$1 = \beta\left[f'(k^*) + 1 - \delta\right]. \tag{12}$$

These equations locate a steady state. They do not describe the transition from an arbitrary k_0, or prove existence or uniqueness of an equilibrium path. With sustained technological or population growth, aggregate quantities may keep growing on a balanced growth path. A stationary system may then require normalization, for example by effective labor, rather than constant levels.

Equilibrium and welfare answer different questions. A planner chooses a feasible allocation directly; a competitive equilibrium also specifies prices that support individual choices. Under suitable conditions, a planner's allocation can be decentralized. Taxes, market power, externalities, incomplete markets, or nominal rigidities may prevent that equivalence. Market clearing alone establishes neither efficiency nor the desirability of a policy. A welfare comparison must state whose utility is evaluated, which constraints apply, and what alternative is feasible.

1.4 Accounting and notation used in these notes

We need only a few accounting conventions at the outset. In a closed economy, using gross investment I_t,

$$Y_t = C_t + I_t + G_t, \tag{13}$$

$$K_{t+1} = (1 - \delta)K_t + I_t. \tag{14}$$

K_t is the stock available at the beginning of the period; I_t is a flow during the period. Depreciation already enters (14). We do not add it again to (13). A household's purchase of a financial claim need not create physical capital; domestic claims and liabilities must be consolidated when deriving aggregate feasibility.

Accounting becomes a model only after we specify behavior, technology, and equilibrium. Government budgets enter with fiscal policy; international accounts enter in Part IV.

Capital letters generally denote aggregates; lower-case letters denote per-person or normalized quantities. Check each chapter's normalization before comparing equations. Unless stated otherwise, β is a discount factor, ρ a rate of time preference, r a net real interest rate, and $R = 1 + r$ a gross real return. In discrete time, the convention $\beta = 1/(1 + \rho)$ differs from continuous-time discounting $e^{-\rho t}$.

For a nominal one-period return $1 + i_t$, realized inflation $\pi_{t+1} = P_{t+1}/P_t - 1$ gives the gross real return

$$R_{t+1} = \frac{1 + i_t}{1 + \pi_{t+1}}. \tag{15}$$

An interest rate quoted in nominal units is therefore not the real return in (8). Under uncertainty, the realized price change must remain inside the relevant expectation.

Finally, for a positive variable X_t and a positive stationary reference $\bar{X}$, define

$$\hat{x}_t \equiv \log X_t - \log \bar{X} \approx \frac{X_t - \bar{X}}{\bar{X}}. \tag{16}$$

The definition is exact; the approximation is local. A deviation from a steady state, a growth rate, and an output gap relative to a flexible-price allocation are different objects. Each chapter states its reference point before using an approximation.

Part II
Growth

How does an economy's productive capacity evolve? Growth facts lead to Harrod and Domar's demand and investment questions, then to Solow's capital adjustment. We move from a given saving rate to optimal saving, overlapping generations, and endogenous technological change. A closing synthesis connects the mechanisms to development, demographic change, and the diagnosis of a growth slowdown before we turn to fluctuations.

2 Growth Facts and Growth Accounting

Why are some economies richer than others, and why do living standards grow? Before choosing a model, we need to separate the outcomes to be explained. This chapter distinguishes income levels from growth rates, then decomposes output growth into measured inputs and productivity. The decomposition helps locate a question; explaining the underlying decisions requires a growth model.

2.1 Income levels, growth rates, and convergence

2.1.1 Which output, and per whom?

Let Y be real GDP, N the population, E the number of employed people, and L total hours worked. For positive quantities,

$$\frac{Y}{N} = \frac{Y}{L}\frac{L}{E}\frac{E}{N}. \tag{17}$$

Output per person therefore depends on output per hour, hours per employed person, and the employment-to-population ratio. Higher GDP growth need not mean faster growth in GDP per person. Likewise, a rise in GDP per person can come from more work per person without any rise in output per hour.

Use real output to study changes over time. For cross-country level comparisons, purchasing power parities (PPPs) adjust for differences in price levels; conversion at market exchange rates can mix output differences with price and exchange-rate movements.[2] Use a common measure and specify the comparison year. GDP per person measures average production, not its distribution or welfare: leisure, health, and consumption opportunities also matter.

2.1.2 A level difference is not a growth-rate difference

For a positive continuous-time variable $X(t)$, define its instantaneous growth rate as

$$g_X(t) \equiv \frac{\dot{X}(t)}{X(t)} = \frac{d\log X(t)}{dt}. \tag{18}$$

Logarithms turn products into sums. Applying this definition to (17) gives the exact identity

$$g_{Y/N} = g_{Y/L} + g_{L/E} + g_{E/N}. \tag{19}$$

The ratios must be defined consistently; employment growth and population growth do not enter in the same way.

For observations at dates t and $t+T$, the average annual log growth rate is

$$\bar{g}_X = \frac{\log X_{t+T} - \log X_t}{T}, \qquad \frac{X_{t+T}}{X_t} = e^{T\bar{g}_X}. \tag{20}$$

The corresponding compound annual percentage change is $e^{\bar{g}_X} - 1$, approximately $\bar{g}_X$ for small rates. Unless stated otherwise, finite-period accounting below uses log changes, $\Delta \log X_t = \log X_t - \log X_{t-1}$.

[2] World Bank, "ICP 2021: Concepts and Definitions," explains PPP-based comparisons of expenditure volumes. `https://www.worldbank.org/en/brief/2021/04/30/icp2021-concepts-definitions`.

At a constant growth rate $g > 0$, doubling takes $T = \log 2/g$. A 2% annual log growth rate doubles output in about 35 years. A one-off 10% increase in output raises its level; it leaves subsequent growth unchanged if the economy then returns to its previous growth rate. A lasting increase in the growth rate keeps widening the gap relative to the previous path.

Facts that guide the models. Jones (2016) documents three useful patterns.[3]

- Sustained growth is a feature of modern economic history. US GDP per person grew at about 2% per year over 1870–2007, with substantial variation across shorter periods.
- Income levels differ widely across countries. Some economies catch up; others remain far behind or lose ground over particular periods.
- A roughly stable capital-output ratio is a useful long-run benchmark. Factor shares and growth rates can change, so balanced growth is an approximation to be assessed, rather than a universal empirical law.

A theory must distinguish growth at the frontier from movement toward it. Matching a long-run average also does not require every shock to be temporary.

2.1.3 What does convergence mean?

Consider a poorer economy i and a richer economy j, with per-person outputs y_i and y_j measured in comparable units. Their log income gap evolves as

$$D(t) \equiv \log y_j(t) - \log y_i(t), \tag{21}$$

$$\dot{D}(t) = g_{y_j}(t) - g_{y_i}(t), \tag{22}$$

$$D(T) - D(0) = \int_0^T \left[g_{y_j}(t) - g_{y_i}(t) \right] dt. \tag{23}$$

The relative gap narrows when the poorer economy grows faster over the interval. Positive growth in both economies is insufficient. Nor does a narrowing relative gap necessarily reduce the absolute difference $y_j - y_i$.

Unconditional convergence means poorer economies tend to grow faster without conditioning on other characteristics. *Conditional convergence* allows economies to approach different long-run paths: an economy further below its own path tends to grow faster, holding the determinants of that path fixed. Saving, population growth, and productivity may differ across economies. Diminishing returns to capital alone therefore do not imply that every poor country must catch up with every rich country.[4]

A tendency for initially poorer economies to grow faster is called β-convergence. A decline in the cross-country dispersion of log income is called σ-convergence. These are different statements: new shocks can increase dispersion even when growth tends to be faster at lower initial income. The sample and observation period are part of any convergence claim.

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[3] Charles I. Jones (2016), "The Facts of Economic Growth," *Handbook of Macroeconomics*, 2A, 3–69, especially Sections 1, 2, and 4. The empirical statements here describe that historical evidence, not a current data release. https://doi.org/10.1016/bs.hesmac.2016.03.002.

[4] Robert J. Barro, N. Gregory Mankiw, and Xavier Sala-i-Martin (1992), "Capital Mobility in Neoclassical Models of Growth," NBER Working Paper 4206, discusses convergence relative to each economy's steady state. https://doi.org/10.3386/w4206.

2.2 Capital, labor, and productivity

2.2.1 From a production function to growth accounting

Start with a one-good aggregate production function in continuous time:

$$Y(t) = B(t)F\big(K(t), H(t)\big), \qquad H(t) = h(t)L(t). \tag{24}$$

K is productive physical capital, L hours worked, and h average labor quality per hour. Thus H is quality-adjusted labor input. B is Hicks-neutral total factor productivity (TFP). It multiplies the whole production function; its growth need not equal the growth of labor-augmenting technology used in the Solow chapter.

Assume positive inputs, differentiable F, constant returns to scale, and no additional inputs. The total derivative is

$$\dot{Y} = \dot{B}F + BF_K\dot{K} + BF_H\dot{H}. \tag{25}$$

Divide by $Y = BF$ and multiply and divide each input term by its input level:

$$\frac{\dot{Y}}{Y} = \frac{\dot{B}}{B} + \frac{BF_K K}{Y}\frac{\dot{K}}{K} + \frac{BF_H H}{Y}\frac{\dot{H}}{H}, \tag{26}$$

$$g_Y = g_B + \varepsilon_K g_K + \varepsilon_H g_H, \quad \varepsilon_K \equiv \frac{BF_K K}{Y}, \quad \varepsilon_H \equiv \frac{BF_H H}{Y}. \tag{27}$$

The weights are output elasticities. Constant returns implies $F_K K + F_H H = F$, so $\varepsilon_K + \varepsilon_H = 1$.

To replace elasticities by observed income shares, we need further assumptions. With competitive factor markets and no wedges, let q be the real rental price of capital services and w_H the price of one unit of quality-adjusted labor. Profit maximization gives

$$q = BF_K, \qquad w_H = BF_H. \tag{28}$$

Consequently,

$$s_K \equiv \frac{qK}{Y} = \varepsilon_K, \qquad s_H \equiv \frac{w_H H}{Y} = \varepsilon_H = 1 - s_K. \tag{29}$$

Here q is a gross rental price before depreciation, not the net interest rate. Labor compensation is $w_H H$; with common quality h, the wage per hour is $w_H h$. In this benchmark all output is paid to these two inputs. Pure profits, markups, taxes, or omitted factors can break the equality between measured income shares and output elasticities.

Substitute the shares and use $g_H = g_h + g_L$:

$$\boxed{g_Y = g_B + s_K g_K + (1 - s_K)(g_h + g_L).} \tag{30}$$

This is an exact instantaneous decomposition under the stated production and pricing assumptions. It does not yet determine any input's growth rate.

2.2.2 Labor productivity and finite-period contributions

Subtract g_L from both sides of (30):

$$g_{Y/L} = g_B + s_K g_K + (1 - s_K)g_h - s_K g_L, \tag{31}$$

$$= g_B + s_K g_{K/L} + (1 - s_K)g_h. \tag{32}$$

Output per hour can rise through capital deepening, better labor quality, or higher TFP. Labor productivity Y/L and TFP B are therefore different objects. Combining this result with (19) also separates productivity growth from changes in work per person.

For annual data, a common implementation uses average adjacent-year income shares, $\bar{s}_{K,t} = (s_{K,t} + s_{K,t-1})/2$:

$$\Delta \log Y_t \approx \bar{s}_{K,t} \Delta \log K_t + (1 - \bar{s}_{K,t}) \Delta \log H_t + \Delta \log B_t, \tag{33}$$

$$\Delta \log B_t^{\text{res}} \equiv \Delta \log Y_t - \bar{s}_{K,t} \Delta \log K_t - (1 - \bar{s}_{K,t}) \Delta \log H_t. \tag{34}$$

The first line approximates the continuous-time decomposition over an interval. The second defines the measured residual exactly for the chosen series and weights. It need not equal a pure technology change. For Cobb–Douglas production with constant exponents, the corresponding log decomposition is exact over a finite interval.

An accounting example. Suppose annual log changes are 0.04 for output, 0.06 for capital, and 0.01 for quality-adjusted labor, with capital share $1/3$. Then

$$\Delta \log B^{\text{res}} = 0.04 - \frac{1}{3}(0.06) - \frac{2}{3}(0.01) = 0.013\overline{3}.$$

Capital contributes 2 percentage points, labor about 0.67, and the residual about 1.33, when log changes are multiplied by 100. These sum to 4. This is an illustration, not an estimate for a country. The calculation does not imply that raising investment would cause exactly 2 percentage points of growth.

2.2.3 What is inside the residual?

The measured residual depends on what the inputs capture. Capital services can change differently from the value of the capital stock. Hours need not track employment, and labor quality need not be constant. Unmeasured utilization, input quality, reallocation, and measurement error can all enter the residual.[5] For example, if true labor input is hL but the calculation uses only L, then (30) implies

$$g_B^{\text{res}} = g_Y - s_K g_K - (1 - s_K) g_L = g_B + (1 - s_K) g_h. \tag{35}$$

Omitted improvements in labor quality appear as TFP growth in that calculation. A change in the residual therefore calls for a mechanism and measurement checks before it is interpreted as innovation or technological regress.

2.3 From accounting to a growth model

2.3.1 Level accounting and joint determination

For a transparent benchmark, assume

$$Y = BK^{\alpha}(hL)^{1-\alpha}, \qquad 0 < \alpha < 1. \tag{36}$$

[5] OECD (2001), *Measuring Productivity: OECD Manual*, especially Chapters 2, 4, and 5, on interpretation, labor input, and capital services. https://doi.org/10.1787/9789264194519-en.

Divide by L and define $y \equiv Y/L$ and $k \equiv K/L$:

$$y = Bk^{\alpha}h^{1-\alpha}. \tag{37}$$

For two economies using the same α and comparable measurements,

$$\frac{y_i}{y_j} = \frac{B_i}{B_j}\left(\frac{k_i}{k_j}\right)^{\alpha}\left(\frac{h_i}{h_j}\right)^{1-\alpha}, \tag{38}$$

$$\log\frac{y_i}{y_j} = \log\frac{B_i}{B_j} + \alpha\log\frac{k_i}{k_j} + (1-\alpha)\log\frac{h_i}{h_j}. \tag{39}$$

This decomposes a level gap; it is distinct from accounting for growth over time. B_i/B_j is inferred after choosing the production function and measuring the inputs. The decomposition does not establish three independent causes.

Higher productivity can raise the return to investment and induce more capital accumulation. Its effects may then appear in both the productivity and capital terms. Similarly, education, saving, and technology adoption may respond to common policies or institutions. An accounting contribution records where output differences appear; a causal explanation specifies what changes first and how the other variables respond.

2.3.2 Connecting TFP to labor-augmenting technology

Chapter 4 writes technology as $Y = F(K, AL)$. To connect the two conventions, set $h = 1$ in (36) and use Cobb–Douglas production:

$$Y = BK^{\alpha}L^{1-\alpha}, \tag{40}$$

$$Y = K^{\alpha}(AL)^{1-\alpha} = A^{1-\alpha}K^{\alpha}L^{1-\alpha}. \tag{41}$$

The two representations describe the same technology when

$$B = A^{1-\alpha}, \qquad g_B = (1-\alpha)g_A. \tag{42}$$

This equivalence uses the Cobb–Douglas form and constant α; it is not a general conversion for every production function.

Suppose a balanced growth path has constant K/Y, constant $h = 1$, and labor growth n. A constant capital-output ratio implies $g_K = g_Y$. Substitution into (30), where $s_K = \alpha$, gives

$$g_Y = g_B + \alpha g_Y + (1-\alpha)n, \tag{43}$$

$$(1-\alpha)(g_Y - n) = g_B, \tag{44}$$

$$g_{Y/L} = \frac{g_B}{1-\alpha} = g_A. \tag{45}$$

Capital deepening accompanies growth on this path. It does not follow that capital accumulation supplies an independent permanent growth engine. We have only derived what the growth rates must satisfy *if* this path exists.

What a growth model must add. Growth accounting leaves saving, capital accumulation, labor supply, and productivity unexplained. Harrod and Domar ask how investment plans and demand can remain consistent with expanding capacity. Solow begins with a given saving rate and technology path. Ramsey makes saving an intertemporal choice; OLG changes who saves and when; endogenous growth models study the forces sustaining productivity or other reproducible inputs.

For Solow's closed one-good economy without government, feasibility and a constant saving rate s imply

$$Y = C + I, \qquad I = sY, \qquad \dot{K} = sY - \delta K. \tag{46}$$

Here I is gross investment and δ is the continuous-time depreciation rate. Together with production, population growth, and a technology path, these assumptions determine capital dynamics from an initial stock. Chapter 4 uses that system to study convergence, income levels, and sustained growth per person. First, Chapter 3 asks how investment creates demand and capacity, and what is required for a growth path to remain consistent with firms' plans and full employment.

3 Harrod–Domar: Investment and Growth

Investment is spending on current production and a source of future productive capacity. Once capacity changes, a demand level sufficient to employ resources today may be insufficient tomorrow. Harrod and Domar approach this moving problem from different directions. We develop their questions separately before connecting them to the fixed-proportions benchmark that Solow will change.

3.1 From a spending equilibrium to a growth problem

Start with a closed economy without government. At a given date, planned investment supports demand through the multiplier. But a machine purchased today can also produce goods later. Repeating the static spending calculation while keeping productive capacity unchanged would discard one of investment's two effects.

The next question is therefore about a path. How quickly must spending expand to use the capacity being created? And would firms' responses bring the economy toward that path if actual growth differed from what they expected? These are questions about consistency and adjustment, before they are questions about an optimal saving rate.

The exposition below uses continuous time, a constant saving propensity $s \in (0, 1)$, and no depreciation initially. Thus gross and net investment coincide, $\dot{K} = I$. Prices are held fixed for the demand calculation. The notation and the explicit adjustment example are teaching devices, not a reconstruction of either author's private sequence of thought.

01 Common accounting

Output is consumed or invested. Realized saving and investment are equal:

$$Y = C + I, \qquad S = sY = I.$$

02 Domar's question

Investment supports demand and adds capacity. What path keeps the two consistent?

$$Y^d = I/s, \qquad \dot{Q} = \kappa I.$$

03 Harrod's question

Desired investment responds to expected expansion. What growth rate validates firms' plans?

$$I^d = v_r \dot{Y}^e.$$

04 Two distinct tests

Does actual growth return to the warranted rate? Can warranted growth persist at full employment?

$$g_a \text{ vs. } g_w, \qquad g_w \text{ vs. } g_n.$$

The two closures in the map are studied in turn. They are not imposed together as a complete off-equilibrium model. In particular, realized investment I and desired investment I^d need not coincide when expectations are disappointed.

3.2 Domar: demand must keep up with capacity

3.2.1 A flow affects demand; its accumulation affects capacity

Let $Q(t)$ be productive capacity, measured as a flow of potential real output. Let $\kappa > 0$ be the increase in capacity generated by a unit of investment. Domar's central distinction is between the demand and capacity effects of investment.[6] The supply-side relation is

$$\dot{Q}(t) = \kappa I(t). \tag{47}$$

κ measures added productive capacity, not the financial return earned by an investor. With positive investment and constant κ, capacity expands even when investment itself is not growing.

For the demand side, write desired consumption as $C^d = (1 - s)Y^d$ and treat investment as the autonomous expenditure at that date. The multiplier gives

$$Y^d = (1 - s)Y^d + I, \tag{48}$$

$$sY^d = I, \qquad Y^d = \frac{I}{s}, \tag{49}$$

$$\dot{Y}^d = \frac{\dot{I}}{s}. \tag{50}$$

This determines demand-supported output at fixed prices when capacity is sufficient. It does not guarantee that output above Q can be produced.

The distinction between I and $\dot{I}$ is crucial. The *level* of investment adds capacity; an *increase* in investment raises the multiplier-supported level of demand. Keeping investment at a high but constant level can therefore leave demand unchanged while capacity keeps rising.

3.2.2 Deriving the required investment path

Suppose demand initially matches capacity: $Y^d(0) = Q(0)$. To preserve that equality, their changes must also match:

$$\dot{Y}^d = \dot{Q}, \tag{51}$$

$$\frac{\dot{I}}{s} = \kappa I, \tag{52}$$

$$\frac{\dot{I}}{I} = s\kappa \equiv g_D. \tag{53}$$

For positive $I(0)$, this differential equation has the solution

$$I(t) = I(0)e^{g_D t}, \qquad Y^d(t) = \frac{I(0)}{s}e^{g_D t}. \tag{54}$$

Integrating the capacity equation gives

$$Q(t) = Q(0) + \frac{I(0)}{s}\left(e^{g_D t} - 1\right). \tag{55}$$

[6] Evsey D. Domar (1946), "Capital Expansion, Rate of Growth, and Employment," *Econometrica*, 14(2), 137–147, especially pp. 139–141. https://doi.org/10.2307/1905364. We write s for the saving propensity and κ for the capacity effect, rather than using his original symbols.

The initial condition $Q(0) = I(0)/s$ makes $Q(t) = Y^d(t)$ at every date. Matching the derivatives alone would preserve an initial level gap, not remove it.

A numerical example. Measure output and investment flows in units per year. Let $s = 0.2$, $\kappa = 0.25$ per year, and $I(0) = 20$. Initially,

$$Y^d(0) = 100, \qquad \dot{Q}(0) = 0.25 \times 20 = 5, \qquad \dot{I}(0) = 0.2 \times 5 = 1.$$

The derivatives measure changes in those annual flows per year. Investment must initially grow at $1/20 = 5\%$ per year to raise demand at the same rate as capacity. These numbers illustrate the equations; they are not estimates.

If investment instead stays at $\bar{I} > 0$, with $Q(0) = \bar{I}/s$, then

$$Y^d(t) = \frac{\bar{I}}{s}, \qquad Q(t) = \frac{\bar{I}}{s} + \kappa \bar{I} t, \qquad \frac{Y^d(t)}{Q(t)} = \frac{1}{1 + s\kappa t}. \tag{56}$$

Utilization falls even though positive investment continues. A construction expansion can support current spending while creating capacity that needs future demand. Whether a particular investment program faces this problem depends on its capacity effect and the demand it generates; the equations do not establish that every program creates excess capacity.

Conversely, if demand tries to grow faster than capacity from a fully utilized position, the multiplier calculation runs into a supply constraint. Prices, imports, or other adjustments would require mechanisms absent here. We have derived a path that keeps demand and capacity consistent, not a rule ensuring that private investment follows it or a complete theory of unemployment.

3.3 Harrod: which growth rate validates investment plans?

3.3.1 Actual and warranted growth

Suppose firms add capacity when they expect sales to expand. Saving responds to the level of income, while desired investment responds to its expected increase. Harrod's warranted rate describes growth that validates the resulting investment plans; his natural rate addresses the scope for sustained growth at full employment.[7]

First distinguish an accounting relation from a behavioral condition. Define actual growth and, for $\dot{Y} > 0$, an actual incremental capital-output ratio:

$$g_a \equiv \frac{\dot{Y}}{Y}, \qquad v_a \equiv \frac{I}{\dot{Y}}. \tag{57}$$

Because realized saving equals realized investment,

$$g_a v_a = \frac{\dot{Y}}{Y}\frac{I}{\dot{Y}} = \frac{I}{Y} = s. \tag{58}$$

[7] R. F. Harrod (1939), "An Essay in Dynamic Theory," *Economic Journal*, 49(193), 14–33. https://doi.org/10.2307/2225181. We distinguish his actual and required incremental capital coefficients using v_a and v_r, and express the argument in continuous-time notation.

This identity does not determine g_a: v_a changes with the realized outcome. It is also an incremental ratio, not automatically the stock ratio K/Y.

Now let $v_r > 0$ be the addition to capital that firms regard as required per unit of additional output at normal utilization. A simple investment rule is

$$I^d = v_r \dot{Y}^e, \tag{59}$$

where $\dot{Y}^e$ denotes the expected increase in output. On a path that validates expectations, $\dot{Y}^e = \dot{Y}$ and planned investment equals saving:

$$v_r \dot{Y} = sY, \tag{60}$$

$$g_w \equiv \frac{s}{v_r}. \tag{61}$$

g_w is the warranted growth rate. At this rate, firms' capital additions are consistent with the realized expansion in sales. Warranted does not mean socially optimal, forecast with certainty, or necessarily equal to actual growth.

3.3.2 Why a deviation can amplify

Compare positive actual growth near g_w, holding s and v_r fixed. When $g_a > g_w$, (58) gives

$$v_a = \frac{s}{g_a} < \frac{s}{g_w} = v_r. \tag{62}$$

Capital additions have been small relative to the growth of sales. If firms interpret this as a shortage of capacity or inventories, they may increase planned investment. That spending raises demand and can reinforce the expansion. When $g_a < g_w$, the corresponding $v_a > v_r$ can leave firms with excess capacity or unwanted inventories; cutting investment then weakens demand further.

The sign of this response is a behavioral assumption, not a consequence of $S = I$. Realized investment can include unplanned inventory accumulation; accounting equality does not make all investment plans mutually consistent. Imposing $I^d = I = sY = v_r \dot{Y}$ at every date would force $g_a = g_w$ and leave no off-path adjustment to analyze.

For an explicit *illustrative local* adjustment, set $x(t) = g_a(t) - g_w$ and assume

$$\dot{x} = \chi x, \qquad \chi > 0 \quad \Longrightarrow \quad x(t) = x(0)e^{\chi t}. \tag{63}$$

This formalizes the amplifying response just described. It is not Harrod's original equation or a global account of the business cycle. Expectations, financing, utilization, and investment lags could change the adjustment rule. The identity and warranted-rate condition alone establish neither stability nor instability.

3.3.3 Natural growth and the employment constraint

Let L be the available labor force, growing at n. In a simplified representation, labor productivity at normal operation, A_L, grows at γ. Sustainable output with full employment then grows at

$$Y^N = A_L L, \qquad g_n = n + \gamma. \tag{64}$$

This is a modern shorthand for the full-employment growth constraint, holding labor-supply and productivity paths fixed. It is not a limit on every observed year's GDP growth: an economy can temporarily grow faster by bringing idle workers into employment.

To see the distinction, assume employed workers E produce $Y = A_L E$. The employment rate $e = E/L$ then satisfies

$$e = \frac{Y}{A_L L}, \qquad \frac{\dot{e}}{e} = g_a - g_n. \tag{65}$$

If warranted growth persistently exceeds natural growth, following the warranted path would eventually require $e > 1$; the path cannot continue under these labor and productivity assumptions. If warranted growth is below natural growth, output can follow it while employment fails to keep pace with the labor force. A path satisfying investment plans need not maintain full employment.

Two different questions. The comparison g_a versus g_w concerns adjustment after investment plans are disappointed. The comparison g_w versus g_n concerns whether a warranted path can persist with full employment. Their equality does not follow from accounting, and their difference is not, by itself, a complete instability proof.

3.4 Fixed proportions, compatibility, and the route to Solow

A common benchmark collects part of these concerns into a fixed-proportions production function:

$$Y = \min\left\{\frac{K}{v}, \frac{AL}{a}\right\}, \qquad v, a > 0, \qquad \frac{\dot{A}}{A} = \gamma. \tag{66}$$

Full use of both inputs requires $K = vY$ and $AL = aY$. Restore depreciation $\delta \geq 0$ and now interpret $I = sY$ as *gross* investment. Differentiating $K = vY$ and substituting accumulation gives

$$v\dot{Y} = \dot{K} = sY - \delta K = (s - \delta v)Y, \tag{67}$$

$$\frac{\dot{Y}}{Y} = \frac{s}{v} - \delta. \tag{68}$$

The effective-labor requirement also implies $\dot{Y}/Y = n + \gamma$. Hence

$$\frac{s}{v} - \delta = n + \gamma. \tag{69}$$

There is no mechanism here that makes independently specified parameters satisfy this equality. With zero depreciation and compatible investment coefficients, this combines the warranted and natural rates into $g_w = g_n$. It is a useful consistency benchmark, not a complete reproduction of both original papers.

In Domar's simplest constant-productivity case, identifying capacity as $Q = K/v$ gives $\kappa = 1/v$, and the required rate becomes $s\kappa = s/v$. The matching expression does not erase the difference in questions: Domar compares demand creation with capacity creation; Harrod also asks how firms respond when actual growth fails to validate investment plans.

The same algebra $g = s/v$ can even appear in an AK model that assumes all output is supplied and absorbed. Its presence alone therefore says little about employment or stability. The conclusions depend on which decisions and adjustment mechanisms accompany it.

Solow's response is to let factor proportions change, so the capital-output ratio can adjust rather than remain a fixed recipe. We follow that experiment in Chapter 4. Maintaining full employment there permits a clear study of capital adjustment; it does not resolve demand failures by assumption alone. The present chapter leaves investment behavior and expectations largely specified from outside. Ramsey will later replace the fixed saving rule with an intertemporal household choice.

4 Solow Model

Chapter 3 separated the consistency of a growth path from the behavior that might support or destabilize it. Solow changes one part of the experiment: firms can vary capital intensity. We ask how this permits capital adjustment, while maintaining full employment and a given saving rule.

4.1 From the growth question to the model

4.1.1 What if production can adjust?

The fixed-proportions benchmark in (69) requires $s/v - \delta = n + \gamma$. Its parameters need not satisfy that equality. Before concluding that sustained full employment requires a coincidence, ask what made v fixed. We described production as a single recipe: additional machines could not economize on workers, and additional workers could not use a less capital-intensive technique.

Solow's 1956 paper made this restriction central to its response to the Harrod–Domar framework. His later account also questioned treating saving, labor-force growth, and the capital-output ratio as independently given constants.[8] This addresses the production-side compatibility problem. Harrod's investment responses and Domar's demand problem remain distinct questions, as the previous chapter showed.

4.1.2 Choose one adjustment mechanism

Several responses are possible. Saving might adjust; labor supply or employment might change; firms might use different factor proportions. Each response would produce a different model. To isolate the last one, keep a simple saving rule and an exogenous labor supply, maintain full employment, and allow substitution in production. The sequence here reconstructs the published argument as a modeling exercise, not the author's private thought process.

Start without technical change and replace the fixed recipe by $Y = F(K, L)$. More machines per worker can raise output, while diminishing returns reduce output per unit of capital. In

$$\frac{\dot{K}}{K} = s\frac{Y}{K} - \delta, \tag{70}$$

Y/K can now fall as capital accumulates. Capital growth can slow toward labor-force growth without a change in s or n. This is a candidate adjustment mechanism; the properties of F and the dynamics below must still establish whether it works. Maintaining full employment makes that experiment possible, rather than proving that actual economies always achieve it.

4.1.3 Close the system and choose a useful intensity

Choosing a flexible F is not enough. In $\dot{K} = sF(K, L) - \delta K$, both K and L could vary. To study capital adjustment first, specify $L(t) = L_0 e^{nt}$ and give $K(0)$. This leaves one inherited stock to solve for. An alternative with wage-sensitive labor supply would require another decision rule; it is a possible extension, not an algebraic rearrangement.

[8] Robert M. Solow (1956), "A Contribution to the Theory of Economic Growth," *Quarterly Journal of Economics*, 70(1), 65–94, Sections I–III, `https://doi.org/10.2307/1884513`; and Solow (1987), "Growth Theory and After," Nobel lecture, `https://www.nobelprize.org/prizes/economic-sciences/1987/solow/lecture/`.

Next, a large economy has more capital partly because it has more workers. The relevant object for this question is K/L. Constant returns allows $Y/L = F(K/L, 1)$, reducing the accumulation problem to a single capital intensity. The ratio removes scale while preserving the mechanism we want to study.

Finally, if labor efficiency rises, even constant equipment per worker may fail to keep up with productive opportunities. Introduce labor-augmenting technology A and use $K/(AL)$ instead. We can distinguish accumulating capital relative to effective labor from growth in output per worker. We take A as given here; explaining innovation would add a different question.

The assumptions below collect these choices: a resource constraint, a production experiment, a saving rule, exogenous labor and technology paths, and a normalization. The full-employment assumption removes demand failures from the dynamics being studied; the resulting convergence claim must be read within that experiment.

4.2 Economic environment and assumptions

The conditions are grouped by their economic role. The cards give the full specification; the following paragraphs explain the notation and implications.

01 Economy and sectors

A closed economy contains households and firms, with no government or foreign trade. One good can be consumed or invested. Goods-market clearing requires $Y_t = C_t + I_t$; in this two-sector economy, aggregate saving equals investment.

02 Households and the saving rule

There are many homogeneous households. They own capital and firms, receive factor incomes and any distributed profits, and supply labor inelastically. Each worker supplies one unit of labor. Consumption and saving follow an imposed rule rather than an intertemporal utility-maximization problem:

$$S_t = sY_t, \qquad C_t = (1 - s)Y_t, \qquad I_t = S_t.$$

The rule permits $s \in [0, 1]$. The positive-capital, positive-consumption benchmark below uses $0 < s < 1$; the endpoints are separate limiting cases.

03 Firms and competitive markets

There are many price-taking firms using the same production technology. They hire labor and rent capital in competitive factor markets. A representative firm's problem is

$$\max_{K,L} \; F(K, A_t L) - q_t K - w_t L.$$

Common technology and constant returns allow aggregation into the production function below. Household saving is imposed; firms still maximize profits. Homogeneity, competition, and optimization are distinct assumptions.

04 Endowments, ownership, and depreciation

Initial stocks K_0, A_0, L_0 are given and positive. Households provide labor at wage w_t and rent their capital to firms at rental price q_t. Depreciation removes a fraction δ per discrete period, with $0 \le \delta \le 1$; investment is usable next period:

$$K_{t+1} = (1-\delta)K_t + I_t.$$

In continuous time, $\dot{K} = I - \delta K$ and the net capital return is $r = q - \delta$. The original course notation R for the rental price corresponds to q here. With a common saving share applied to all income and no household-specific constraints, the distribution of ownership does not change aggregate saving; its distributional consequences are outside this model.

05 Technology and exogenous paths

Technology is freely and publicly available to all firms. There are no adoption costs, patents, or research choices in the benchmark. Labor and labor efficiency grow exogenously:

$$A_{t+1} = (1+g)A_t, \qquad L_{t+1} = (1+n)L_t, \qquad g, n \ge 0.$$

The corresponding continuous-time paths satisfy $\dot{A}/A = g$ and $\dot{L}/L = n$; rates in the two time conventions need not be numerically identical. The interior steady-state analysis assumes $\delta + n + g > 0$.

06 Production and its restrictions

Technology is labor-augmenting (Harrod-neutral): $Y_t = F(K_t, A_t L_t)$. F is continuous, twice differentiable for positive inputs, positive there, and has positive, diminishing marginal products of each input. It has constant returns to capital and effective labor:

$$F(\lambda K, \lambda AL) = \lambda F(K, AL), \qquad \lambda > 0.$$

Write $f(k) = F(k, 1)$. The intensive-form restrictions used below are

$$f(0) = 0, \qquad f'(k) > 0, \qquad f''(k) < 0 \quad (k > 0),$$
$$\lim_{k \downarrow 0} f'(k) = \infty, \qquad \lim_{k \to \infty} f'(k) = 0.$$

The last two are Inada restrictions on marginal products. They are sufficient benchmark assumptions, rather than consequences of competition.

07 Market clearing and factor incomes

All supplied labor and capital are employed. With bars denoting supplied endowments and superscripts denoting supply and demand,

$$L_t^d = L_t^s = \bar{L}_t, \qquad K_t^d = K_t^s = \bar{K}_t.$$

Competitive prices equal marginal products, so constant returns implies

$$q_t = F_1(K_t, A_t L_t), \qquad w_t = A_t F_2(K_t, A_t L_t), \qquad Y_t = q_t K_t + w_t L_t.$$

Firms are owned by households. Their economic profits are zero after paying both factors; the owners still receive capital rental income.

08 Normalization and the closed system

Use quantities per effective worker:

$$k_t = \frac{K_t}{A_t L_t}, \qquad y_t = \frac{Y_t}{A_t L_t} = f(k_t), \qquad c_t = \frac{C_t}{A_t L_t} = (1-s) f(k_t).$$

Given initial stocks and the paths of A_t, L_t, the three main equations are

$$Y_t = F(K_t, A_t L_t), \qquad I_t = S_t = s Y_t, \qquad K_{t+1} = (1-\delta) K_t + I_t.$$

The resource constraint, full employment, and competitive factor prices make this a closed aggregate system. They are maintained assumptions of the experiment, not conclusions about demand adjustment in actual economies.

$$\underbrace{K_t, A_t, L_t}_{\text{Given at } t} \longrightarrow \underbrace{Y_t \to C_t, I_t}_{\text{Production and allocation}} \longrightarrow \underbrace{K_{t+1}}_{\text{Available at } t+1}.$$

4.2.1 Resources, timing, and behavior

The economy is closed, has no government, and produces one good. Households own capital and supply labor inelastically. Each worker supplies one unit of labor; changes in hours, participation, and employment rates are omitted. Firms hire labor and rent capital in competitive markets.

In discrete time, K_t is available at the start of period t. Given K_t, A_t, and L_t, firms produce Y_t. Households consume part of output and save the fraction $s \in (0,1)$. Investment becomes available next period:

$$Y_t = C_t + I_t, \qquad I_t = S_t = s Y_t, \tag{71}$$

$$K_{t+1} = (1-\delta) K_t + s Y_t, \qquad 0 \le \delta \le 1. \tag{72}$$

The saving rule is imposed. Firms still maximize profits; the absence of a household optimization problem does not imply that firms are non-optimizing.

Technology and labor follow

$$A_{t+1} = (1+g) A_t, \qquad L_{t+1} = (1+n) L_t, \qquad g, n \ge 0. \tag{73}$$

Initial K_0, A_0, L_0 are positive. We assume $\delta + n + g > 0$ to obtain a finite positive steady state below. Technology is labor-augmenting: an increase in A makes a worker provide more effective labor. Its availability and growth are given, rather than explained by research or adoption decisions.

4.2.2 Production and normalization

Let

$$Y_t = F(K_t, A_t L_t). \tag{74}$$

F has constant returns in capital and effective labor:

$$F(\lambda K, \lambda AL) = \lambda F(K, AL), \qquad \lambda > 0. \tag{75}$$

Replicating all inputs replicates output. This rules out a fixed factor, such as land, that becomes scarcer as the economy expands. It also rules out aggregate increasing returns in these two inputs.

One motivation for constant returns is to treat any gains from organizing production at a larger scale as already exhausted at the scale being modeled. Another is to abstract from scarce inputs other than capital and effective labor. These are ways to motivate the assumption, not proofs that it holds.

Divide by effective labor and define

$$k_t \equiv \frac{K_t}{A_t L_t}, \qquad y_t \equiv \frac{Y_t}{A_t L_t} = F(k_t, 1) \equiv f(k_t), \qquad c_t \equiv \frac{C_t}{A_t L_t} = (1 - s) f(k_t). \tag{76}$$

Here lower-case variables are *per effective worker*. Output per worker is $Y_t/L_t = A_t y_t$, and capital per worker is $K_t/L_t = A_t k_t$. This differs from the per-hour normalization used for level accounting in Chapter 2. With $A_t = 1$, the two capital normalizations coincide.

Labor-augmenting technology also makes a balanced growth path convenient to describe. If k settles at a positive constant, then $K/Y = k/f(k)$ is constant. The functional form alone does not make that ratio constant during a transition. Hicks-neutral technology instead takes the form $Y = A_H F(K, L)$. It is a different placement of productivity; with Cobb–Douglas, $A_H = A^{1-\alpha}$ converts between the two representations. The exogenous growth rate must be converted as well.

Assume $f(0) = 0$ and, for $k > 0$,

$$f'(k) > 0, \qquad f''(k) < 0, \qquad \lim_{k \downarrow 0} f'(k) = \infty, \qquad \lim_{k \to \infty} f'(k) = 0. \tag{77}$$

The last two are the Inada conditions on *marginal products*, not output levels. With the other assumptions, they make capital sufficiently productive near zero and insufficiently productive at very high intensity to guarantee an interior crossing of investment and replacement needs. They are sufficient benchmark restrictions, not empirical laws for every technology.

Our running example is Cobb–Douglas:

$$F(K, AL) = K^{\alpha}(AL)^{1-\alpha}, \qquad f(k) = k^{\alpha}, \qquad 0 < \alpha < 1. \tag{78}$$

Diminishing returns here means adding machines to a fixed amount of effective labor produces progressively smaller additions to output. It does not mean that output falls when capital rises.

Cobb–Douglas is a convenient specification satisfying these restrictions; it is not deduced uniquely from economic behavior. Its constant factor shares are an additional restriction. Likewise, Inada conditions and an exogenous technology path delimit the experiment: the former restrict behavior at extreme capital intensities, while the latter leaves innovation unexplained. Relaxing either calls for a new argument about the resulting dynamics.

4.2.3 Prices and market clearing

At each date, let q_t be the real rental price of capital and w_t the wage per worker. A price-taking firm solves

$$\max_{K,L} \; F(K, A_t L) - q_t K - w_t L. \tag{79}$$

At an interior solution, denoting derivatives with respect to the two arguments of F by F_1, F_2,

$$\frac{\partial \pi}{\partial K} = F_1(K, A_t L) - q_t = 0, \tag{80}$$

$$\frac{\partial \pi}{\partial L} = A_t F_2(K, A_t L) - w_t = 0. \tag{81}$$

$$q_t = F_1(K_t, A_t L_t), \qquad w_t = A_t F_2(K_t, A_t L_t). \tag{82}$$

Constant returns and Euler's theorem for homogeneous functions give the following calculation. For scale $v > 0$,

$$F(vK, vAL) = vF(K, AL), \tag{83}$$

$$F_1(vK, vAL)K + F_2(vK, vAL)AL = F(K, AL), \tag{84}$$

$$F_1(K, AL)K + F_2(K, AL)AL = F(K, AL) \qquad (v = 1). \tag{85}$$

$$Y_t = F_1 K_t + F_2 A_t L_t = q_t K_t + w_t L_t. \tag{86}$$

Hence $\pi_t = Y_t - q_t K_t - w_t L_t = 0$. To express the same prices in intensive units, differentiate $F(K, AL) = ALf(K/(AL))$:

$$F_K = ALf'(k)\frac{1}{AL} = f'(k), \tag{87}$$

$$F_L = Af(k) + ALf'(k)\left(-\frac{K}{AL^2}\right), \tag{88}$$

$$= A[f(k) - kf'(k)]. \tag{89}$$

Zero profits do not mean zero capital income. Economic profits are zero after paying both factors, including capital owned by households. This is a property of this competitive constant-returns benchmark, not a claim that owners receive no capital income.

For Cobb–Douglas, $q_t = \alpha k_t^{\alpha-1}$ and $w_t = (1-\alpha)A_t k_t^{\alpha}$. Capital deepening lowers the rental return and raises the wage per effective worker. Prices support full use of the supplied capital and labor. Goods-market clearing and the saving rule then determine investment. Given the initial stocks and exogenous paths, the model is closed.

The mechanism in one loop. Capital and effective labor determine output; the saving rule turns some output into investment; investment and depreciation determine next period's capital. Factor prices describe the returns along this path. The saving rate does not respond to those returns until we change the household side of the model.

4.3 Solow model in discrete time

4.3.1 Deriving the law of motion

Question: which capital variable makes the problem stationary? Start from $K_{t+1} = (1-\delta)K_t + sY_t$. Dividing by L_{t+1} describes capital per worker, but growing labor efficiency still changes production at each date. Dividing by $A_{t+1}L_{t+1}$ instead gives a stationary map in capital per effective worker when s, n, g, δ are constant. The derivation below keeps the exact growth denominator; the per-worker special case follows in the blue note. Choosing a normalization is part of solving the economic problem, rather than merely shortening its notation.

Define $D \equiv (1+n)(1+g)$, the gross growth factor of effective labor. Divide (72) by $A_{t+1}L_{t+1}$:

$$k_{t+1} = \frac{(1-\delta)K_t + sY_t}{A_{t+1}L_{t+1}}, \tag{90}$$

$$= \frac{(1-\delta)K_t + sY_t}{(1+g)A_t(1+n)L_t}, \tag{91}$$

$$= \frac{(1-\delta)K_t/(A_tL_t) + sY_t/(A_tL_t)}{D}, \tag{92}$$

$$= \frac{(1-\delta)k_t + sf(k_t)}{D} \equiv \Phi(k_t). \tag{93}$$

The numerator adds surviving capital and investment per current effective worker. The denominator spreads it across next period's effective labor. New workers need equipment, and increasing labor efficiency requires more equipment per worker to keep k constant.

Subtract k_t and define $d_D \equiv D - 1 + \delta$:

$$k_{t+1} - k_t = \frac{(1-\delta)k_t + sf(k_t)}{D} - \frac{Dk_t}{D}, \tag{94}$$

$$= \frac{sf(k_t) + [1-\delta-D]k_t}{D}, \tag{95}$$

$$= \frac{sf(k_t) - (D-1+\delta)k_t}{D}, \tag{96}$$

$$D - 1 + \delta = (1+n)(1+g) - 1 + \delta = \delta + n + g + ng. \tag{97}$$

$$k_{t+1} - k_t = \frac{sf(k_t) - d_Dk_t}{D}, \qquad d_D = \delta + n + g + ng. \tag{98}$$

Break-even investment d_Dk_t keeps capital per effective worker unchanged. The ng term is part of the exact discrete-time expression; dropping it is a small-rate approximation. Setting $g = 0$ gives $d_D = \delta + n$; setting $g = n = 0$ gives $d_D = \delta$.

Reading the per-worker version. Let $\bar{k}_t = K_t/L_t$. Dividing the aggregate accumulation equation by L_{t+1} instead gives the exact law

$$\bar{k}_{t+1} = \frac{(1-\delta)\bar{k}_t + sF(\bar{k}_t, A_t)}{1+n}.$$

There is no $1+g$ in this denominator: technology remains inside production. With constant $A_t = \bar{A}$, define $\bar{f}(\bar{k}) = F(\bar{k}, \bar{A})$. The map is then stationary; setting $n = 0$ also removes the population-growth denominator. These are valid restricted cases. When technology grows, a stationary map in $k = K/(AL)$ requires the effective-labor normalization above.

4.3.2 A steady state and the path toward it

A positive normalized steady state satisfies

$$k^* = \frac{(1-\delta)k^* + sf(k^*)}{D}, \tag{99}$$

$$Dk^* - (1-\delta)k^* = sf(k^*), \tag{100}$$

$$(D-1+\delta)k^* = sf(k^*), \tag{101}$$

$$d_D k^* = sf(k^*), \tag{102}$$

$$\frac{k^*}{f(k^*)} = \frac{s}{d_D}. \tag{103}$$

$$k^* = \Phi(k^*), \qquad sf(k^*) = d_D k^*, \qquad \frac{K}{Y} = \frac{k^*}{f(k^*)} = \frac{s}{d_D}. \tag{104}$$

A stable capital-output ratio is a conclusion *on this path*, not something that follows from labor-augmenting notation alone.

For existence and uniqueness, divide the middle equation by $k^* > 0$. Strict concavity and $f(0) = 0$ imply $f'(k) < f(k)/k$, so

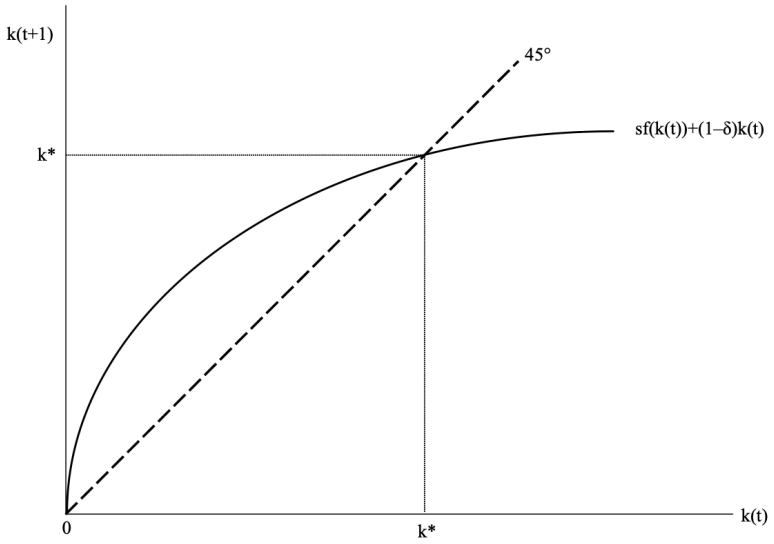
$$\frac{d}{dk}\frac{f(k)}{k} = \frac{kf'(k) - f(k)}{k^2} < 0. \tag{105}$$

The Inada conditions imply $f(k)/k$ falls from infinity to zero. Hence $sf(k)/k = d_D > 0$ has exactly one positive solution. Zero is also an absorbing boundary under $f(0) = 0$; our convergence result starts from $k_0 > 0$.

Below k^*, saving exceeds break-even investment and k rises. Above k^*, saving is insufficient to maintain that intensity and k falls. Moreover,

$$\Phi'(k) = \frac{1 - \delta + sf'(k)}{D} > 0. \tag{106}$$

Since Φ is increasing and $\Phi(k^*) = k^*$, an initial $k < k^*$ maps below k^*, while an initial $k > k^*$ maps above it. The sequence is therefore monotone and bounded toward k^*, and continuity makes its limit a fixed point. The unique positive fixed point is the limit for every $k_0 > 0$.



Reading the diagram. This diagram shows the special case $n = g = 0$, with $\delta > 0$. The curved map is next period's capital; the 45-degree line is unchanged capital. Starting on the horizontal axis, move vertically to the map, then horizontally to the 45-degree line to carry that value into the next iteration. For $n > 0$ or $g > 0$, divide the map by D as in (93).

A steady state is not the definition of equilibrium. A steady state is one particular equilibrium path. A transition from an arbitrary initial stock is also an equilibrium path under the imposed saving rule. Convergence does not make the transition cease to be an equilibrium.

4.3.3 What grows on a balanced growth path?

If $k_t = k^*$, then $y_t = f(k^*)$ and $c_t = (1 - s)f(k^*)$ are constant. Yet

$$\frac{Y_t}{L_t} = A_t f(k^*), \qquad \frac{K_t}{L_t} = A_t k^*, \qquad Y_t = A_t L_t f(k^*). \tag{107}$$

Output and capital per worker grow at gross rate $1 + g$; their aggregates grow at D. With no technical progress, there is no sustained growth per worker on this path, even if the population and total output keep growing.

For Cobb–Douglas, solve $s(k^*)^\alpha = d_D k^*$:

$$s = d_D (k^*)^{1-\alpha} \qquad (k^* > 0), \tag{108}$$

$$(k^*)^{1-\alpha} = s/d_D, \tag{109}$$

$$k^* = (s/d_D)^{1/(1-\alpha)}, \tag{110}$$

$$y^* = (k^*)^\alpha = \left[(s/d_D)^{1/(1-\alpha)}\right]^\alpha = (s/d_D)^{\alpha/(1-\alpha)}. \tag{111}$$

$$k^* = \left(\frac{s}{d_D}\right)^{1/(1-\alpha)}, \qquad y^* = \left(\frac{s}{d_D}\right)^{\alpha/(1-\alpha)}. \tag{112}$$

A higher saving rate raises the long-run level of output per effective worker. Holding the technology path fixed, it does not change the long-run growth rate of output per worker. Its steady-state level elasticity is obtained from

$$\log y^* = \frac{\alpha}{1-\alpha}(\log s - \log d_D).$$
$$\frac{\partial \log y^*}{\partial \log s} = \frac{\alpha}{1-\alpha}. \tag{113}$$

This comparison holds other determinants fixed. It is not a causal estimate from a cross-country correlation between saving and income.

4.3.4 Saving, rebuilding, and conditional convergence

Suppose an economy starts on its old balanced growth path and permanently raises s. At the time of the change, K, A, and L are inherited, so output is unchanged and consumption falls. Higher investment raises future K; output per worker then grows faster during the transition. As diminishing returns reduce the extra investment generated by another unit of capital, k approaches its new, higher steady state. Long-run growth per worker returns to g.

This explains why an investment expansion can have a lasting level effect without a lasting growth-rate effect. It does not establish that any observed investment campaign will be productive: the one-good benchmark abstracts from project selection, idle capacity, financing, and heterogeneous returns.

A one-time loss of physical capital, with technology, labor, and saving behavior unchanged, lowers k. The high marginal product at the lower stock supports rebuilding toward the same path. Actual disasters can also destroy skills, institutions, and technology, so this experiment isolates a capital-loss mechanism rather than predicting every recovery.

The convergence result is conditional on the determinants of k^* and the technology path. A low-income economy with a low saving rate or low productivity need not grow faster than a richer economy already close to a higher path. Income alone does not tell us distance from the relevant steady state.

4.3.5 Local convergence speed

Let $x_t = k_t - k^*$. A first-order Taylor expansion at k^* gives

$$k_{t+1} = \Phi(k^*) + \Phi'(k^*)x_t + O(x_t^2), \tag{114}$$
$$x_{t+1} \approx \phi x_t, \qquad \phi \equiv \frac{1-\delta+sf'(k^*)}{D}. \tag{115}$$

Because $sf'(k^*) < sf(k^*)/k^* = d_D$, we have $0 < \phi < 1$. Iterating the linearized system yields

$$x_1 = \phi x_0, \tag{116}$$
$$x_2 = \phi x_1 = \phi(\phi x_0) = \phi^2 x_0, \tag{117}$$
$$x_3 = \phi x_2 = \phi^3 x_0, \tag{118}$$
$$x_t = \phi^t x_0. \tag{119}$$

These equalities solve the linearized recursion; for the nonlinear model the path below is a local approximation. The course's relative deviation $\hat{k}_t = (k_t - k^*)/k^* = x_t/k^*$ obeys the same recursion:

$$\hat{k}_{t+1} = \frac{x_{t+1}}{k^*} = \frac{\phi x_t}{k^*} = \phi \hat{k}_t.$$

$$x_t \approx \phi^t x_0, \qquad \lambda_D = -\log\phi > 0, \qquad T_{1/2} \approx \frac{\log 2}{\lambda_D}. \tag{120}$$

To find the time to a target fraction $0 < \varepsilon < 1$ of a nonzero initial deviation, take logarithms explicitly:

$$\frac{|x_T|}{|x_0|} = \phi^T = \varepsilon, \tag{121}$$

$$T\log\phi = \log\varepsilon, \tag{122}$$

$$T_\varepsilon = \frac{\log\varepsilon}{\log\phi}, \tag{123}$$

$$T_{1/2} = \frac{\log(1/2)}{\log\phi} = \frac{\log 2}{-\log\phi}. \tag{124}$$

For integer dates, round T_ε up to obtain the first date at or below that fraction in the linearized model. Setting $\varepsilon = 0$ would require an infinite time rather than a logarithm of zero. ϕ is persistence; a larger ϕ means slower adjustment. λ_D is the corresponding decay rate per period, and $T_{1/2}$ is the local half-life of a deviation. The recursion for the linearized system has an exact geometric solution; the approximation is in using that system to describe the original model. The nonlinear path need not have this half-life far from its steady state.

For Cobb–Douglas, the steady-state equation gives

$$s(k^*)^\alpha = d_D k^*, \tag{125}$$

$$s(k^*)^{\alpha-1} = d_D, \tag{126}$$

$$sf'(k^*) = s\alpha(k^*)^{\alpha-1} = \alpha d_D, \tag{127}$$

$$\phi = \frac{1-\delta+\alpha d_D}{D} = \frac{D - d_D + \alpha d_D}{D}. \tag{128}$$

Here $1 - \delta = D - d_D$. Therefore

$$\phi = 1 - \frac{(1-\alpha)d_D}{D}. \tag{129}$$

Adjustment is gradual because productive capacity is a stock. Investment can change immediately, but only accumulated investment changes that stock.

4.4 Solow model in continuous time

4.4.1 The same mechanism with continuous adjustment

Now use continuous rates $\dot{L}/L = n$, $\dot{A}/A = g$, and depreciation rate $\delta \geq 0$, with $b \equiv n + g + \delta > 0$. These are rates per unit of time; they are not numerically identical to finite-period percentage changes. Write

$$\dot{K} = sF(K, AL) - \delta K, \qquad k = \frac{K}{AL}, \qquad y = f(k). \tag{130}$$

Log differentiation of the definition of k, followed by substitution, gives the same result as the quotient rule. First display the latter:

$$\frac{d(AL)}{dt} = \dot{A}L + A\dot{L}, \tag{131}$$

$$\dot{k} = \frac{\dot{K}(AL) - K(\dot{A}L + A\dot{L})}{(AL)^2}, \tag{132}$$

$$= \frac{\dot{K}}{AL} - \frac{K}{AL}\left(\frac{\dot{A}}{A} + \frac{\dot{L}}{L}\right), \tag{133}$$

$$= \frac{sY - \delta K}{AL} - (g+n)k. \tag{134}$$

Dividing by $k = K/(AL) > 0$ also gives the log-differentiation form:

$$\frac{\dot{k}}{k} = \frac{\dot{K}}{K} - \frac{\dot{A}}{A} - \frac{\dot{L}}{L}, \tag{135}$$

$$= \frac{sY}{K} - \delta - g - n, \tag{136}$$

$$\dot{k} = s\frac{Y}{AL} - (n+g+\delta)k = sf(k) - bk \equiv G(k). \tag{137}$$

With $g = 0$ and $A = 1$, this becomes $\dot{k} = sf(k) - (n+\delta)k$. That familiar constant-technology equation is an exact special case.

New investment increases capital intensity; depreciation, population growth, and technical progress reduce it for a given aggregate stock. Technical progress still raises productive possibilities: it reduces k here because each worker now represents more effective labor.

For a short time interval Δt, effective labor grows by approximately $1 + (n+g)\Delta t$ and depreciation is approximately $\delta\Delta t$. Applying the discrete accounting over that interval gives

$$k(t+\Delta t) - k(t) = [sf(k(t)) - bk(t)]\,\Delta t + o(\Delta t). \tag{138}$$

Divide by Δt and take its limit to obtain (137). This explains why the continuous break-even rate is b, whereas the exact discrete expression contains the product ng.

The positive steady state satisfies $sf(k^*) = bk^*$; $G(k) > 0$ below it and $G(k) < 0$ above it. On the balanced growth path, Y/L and K/L grow at g, Y and K grow at $n+g$, and $K/Y = s/b$. Under competitive pricing the net instantaneous real return is $r = f'(k) - \delta$.

4.4.2 Local approximation and an exact Cobb–Douglas solution

Linearizing the law of motion around its steady state gives

$$G(k) = G(k^*) + G'(k^*)(k-k^*) + O((k-k^*)^2), \tag{139}$$

$$G(k^*) = 0, \qquad G'(k^*) = sf'(k^*) - b. \tag{140}$$

$$\dot{k} \approx G'(k^*)(k-k^*), \tag{141}$$

$$G'(k^*) = sf'(k^*) - b < 0, \tag{142}$$

$$k(t) - k^* \approx [k(0) - k^*]e^{-\lambda_C t}, \qquad \lambda_C \equiv b - sf'(k^*) > 0. \tag{143}$$

The exponential expression solves the *linearized* system. Obtain it by setting $x = k - k^*$ and separating variables:

$$\dot{x} = -\lambda_C x, \tag{144}$$

$$\int_{x(0)}^{x(t)} \frac{dv}{v} = -\lambda_C \int_0^t ds, \tag{145}$$

$$\log|x(t)| - \log|x(0)| = -\lambda_C t, \tag{146}$$

$$\left|\frac{x(t)}{x(0)}\right| = e^{-\lambda_C t}, \tag{147}$$

$$x(t) = x(0)e^{-\lambda_C t}. \tag{148}$$

The sign of a nonzero $x(0)$ is preserved in this scalar linear equation; if $x(0) = 0$, its solution is zero. The exponential path is generally only a local approximation to the original nonlinear equation. For Cobb–Douglas, $s(k^*)^{\alpha-1} = b$ implies $sf'(k^*) = \alpha b$, so $\lambda_C = b - \alpha b = (1-\alpha)b$. Its local half-life is $\log 2/[(1-\alpha)b]$.

Cobb–Douglas also permits an exact solution using a different variable. Set $z(t) = k(t)^{1-\alpha}$ and differentiate:

$$\dot{z} = (1-\alpha)k^{-\alpha}\dot{k}, \tag{149}$$

$$= (1-\alpha)k^{-\alpha}(sk^{\alpha} - bk) = (1-\alpha)(s - bz). \tag{150}$$

This is a linear equation in z, with steady state $z^* = s/b$. Its solution can also be obtained without quoting a linear-ODE formula. With $a = (1-\alpha)b > 0$, multiply $\dot{z} + az = (1-\alpha)s$ by e^{at}:

$$e^{at}\dot{z} + ae^{at}z = (1-\alpha)se^{at}, \tag{151}$$

$$\frac{d}{dt}(e^{at}z) = (1-\alpha)se^{at}, \tag{152}$$

$$e^{at}z(t) - z(0) = (1-\alpha)s\int_0^t e^{av}\,dv, \tag{153}$$

$$= \frac{(1-\alpha)s}{a}(e^{at} - 1) = \frac{s}{b}(e^{at} - 1), \tag{154}$$

$$z(t) = z(0)e^{-at} + \frac{s}{b}(1 - e^{-at}). \tag{155}$$

Regrouping the constants and taking the power $1/(1-\alpha)$ gives

$$z(t) = \frac{s}{b} + \left[z(0) - \frac{s}{b}\right]e^{-(1-\alpha)bt}, \tag{156}$$

$$k(t) = \left\{\frac{s}{b} + \left[k(0)^{1-\alpha} - \frac{s}{b}\right]e^{-(1-\alpha)bt}\right\}^{1/(1-\alpha)}. \tag{157}$$

Capital approaches its steady state without arriving there at a finite date unless it starts there. "Time to convergence" must therefore specify a tolerance or a half-life, rather than an exact date of arrival.

4.5 The Golden Rule and the cost of saving

More saving raises steady-state output, but output is not consumption. To maintain a larger capital stock, the economy must devote more resources to investment. The Golden Rule asks which balanced growth path provides the largest consumption per effective worker, holding the technology and labor paths fixed.

In continuous time, a stationary k requires investment bk, so feasible steady-state consumption is

$$c^* = f(k) - bk. \tag{158}$$

Maximize this expression over k:

$$\frac{dc^*}{dk} = f'(k) - b, \tag{159}$$

$$f'(k_{\mathrm{GR}}) = b = n + g + \delta, \qquad \frac{d^2c^*}{dk^2} = f''(k) < 0. \tag{160}$$

One extra unit of capital adds $f'(k)$ to output but requires b units of investment per unit of time to maintain its intensity. At the Golden Rule, these marginal amounts are equal. Equivalently, the net marginal return $f'(k_{\mathrm{GR}}) - \delta$ equals $n + g$.

The saving rate that supports this stock satisfies

$$s_{\mathrm{GR}} = \frac{bk_{\mathrm{GR}}}{f(k_{\mathrm{GR}})}. \tag{161}$$

For Cobb–Douglas, substitute $\alpha k_{\mathrm{GR}}^{\alpha-1} = b$:

$$k_{\mathrm{GR}}^{1-\alpha} = \alpha/b, \tag{162}$$

$$k_{\mathrm{GR}} = (\alpha/b)^{1/(1-\alpha)}, \tag{163}$$

$$s_{\mathrm{GR}} = \frac{bk_{\mathrm{GR}}}{k_{\mathrm{GR}}^{\alpha}} = bk_{\mathrm{GR}}^{1-\alpha} = b(\alpha/b) = \alpha. \tag{164}$$

$$k_{\mathrm{GR}} = \left(\frac{\alpha}{b}\right)^{1/(1-\alpha)}, \qquad s_{\mathrm{GR}} = \alpha. \tag{165}$$

In discrete time the argument is the same with d_D replacing b: $c^* = f(k) - d_D k$ and $f'(k_{\mathrm{GR}}) = d_D$. The Cobb–Douglas saving share is again α under this timing convention.

> **A welfare boundary.** The Golden Rule maximizes consumption *across balanced growth paths*. With constant technology and strictly increasing utility $u(c)$, maximizing stationary consumption also maximizes utility evaluated at that stationary consumption. Solow has not specified discounting or how to value consumption losses during a transition. It therefore does not establish that moving toward the Golden Rule maximizes intertemporal welfare.

If the initial steady state lies below k_{GR}, raising saving toward s_{GR} increases eventual consumption but requires sacrificing consumption while capital is built. If it lies above k_{GR}, reducing saving toward the Golden Rule can raise both current and eventual consumption. Investment is then maintaining more capital than is useful for maximizing steady-state consumption. A high investment share alone is therefore not a welfare criterion. Ramsey will specify preferences and make the transition trade-off part of the saving decision; OLG will examine dynamic efficiency in an economy with different generations.

4.6 Extensions: which assumption needs to change?

An extension should answer a question left open by the benchmark. Adding an input, changing the investment technology, and explaining productivity are different changes; they need not have the same implications for long-run growth. The examples below use continuous time.

4.6.1 Investment spending and productive capital

In the baseline, one unit of investment creates one unit of productive capital. To separate spending from productive capacity, suppose only a fraction $\eta \in (0, 1]$ becomes usable capital:

$$I = sY, \qquad C = (1 - s)Y, \qquad \dot{K} = \eta I - \delta K. \tag{166}$$

The unconverted part is lost in installation or implementation; it is not additional consumption. With the same labor and technology paths,

$$\dot{k} = \eta s f(k) - bk, \qquad k^* = \left(\frac{\eta s}{b}\right)^{1/(1-\alpha)} \quad \text{for } f(k) = k^\alpha. \tag{167}$$

Higher investment spending and more effective conversion into capital both raise the steady-state stock, but only the former directly reduces the consumption share. Poor project selection or implementation can therefore weaken the link between a reported investment share and productive capacity. This scalar η is an illustrative reduced form: it does not yet explain which projects are chosen or how finance allocates resources.

At a stationary k, maintaining it requires $I/(AL) = bk/\eta$, so $c = f(k) - bk/\eta$. The calculation is

$$0 = \eta I/(AL) - bk, \tag{168}$$

$$I/(AL) = bk/\eta, \tag{169}$$

$$c = y - I/(AL) = f(k) - bk/\eta, \tag{170}$$

$$dc/dk = f'(k) - b/\eta = 0. \tag{171}$$

The Golden Rule becomes $f'(k) = b/\eta$. Welfare comparisons must account for the resources spent creating capital, not only the stock that eventually appears in production.

4.6.2 Accumulating human capital

Machines need skills to be useful. The augmented Solow model of Mankiw, Romer, and Weil adds a reproducible human-capital stock to the production function.[9] Write

$$Y = K^\alpha H^\beta (AL)^{1-\alpha-\beta}, \qquad \alpha, \beta > 0, \quad \alpha + \beta < 1. \tag{172}$$

Here H denotes a stock of human capital, rather than the quality-adjusted hours denoted by H in Chapter 2. To keep the distinction visible, use $u \equiv H/(AL)$ for human capital per effective worker. Then

$$y = k^\alpha u^\beta. \tag{173}$$

[9] N. Gregory Mankiw, David Romer, and David N. Weil (1992), "A Contribution to the Empirics of Economic Growth," *Quarterly Journal of Economics*, 107(2), 407–437, Section II. https://doi.org/10.2307/2118477.

Suppose output can be transformed into either stock, with fixed investment shares $s_K, s_H > 0$, $s_K + s_H < 1$, and the same depreciation rate for both:

$$\dot{K} = s_K Y - \delta K, \qquad \dot{H} = s_H Y - \delta H, \tag{174}$$

$$\dot{k} = s_K y - bk, \qquad \dot{u} = s_H y - bu. \tag{175}$$

There are now two state variables. Consumption is $c = (1 - s_K - s_H)y$; investment in skills uses resources rather than appearing costlessly in production.

At a positive steady state, the two accumulation equations imply

$$k^* = \frac{s_K}{b} y^*, \qquad u^* = \frac{s_H}{b} y^*. \tag{176}$$

Substitute both into production and collect the powers of y^*:

$$y^* = \left(\frac{s_K y^*}{b}\right)^{\alpha} \left(\frac{s_H y^*}{b}\right)^{\beta}, \tag{177}$$

$$= \frac{s_K^{\alpha} s_H^{\beta}}{b^{\alpha+\beta}} (y^*)^{\alpha+\beta}, \tag{178}$$

$$(y^*)^{1-\alpha-\beta} = \frac{s_K^{\alpha} s_H^{\beta}}{b^{\alpha+\beta}}, \tag{179}$$

$$y^* = \left(\frac{s_K^{\alpha} s_H^{\beta}}{} b^{\alpha+\beta}\right)^{1/(1-\alpha-\beta)}. \tag{180}$$

The condition $\alpha + \beta < 1$ preserves diminishing returns to the two accumulated inputs together. The extension can change income levels and transition dynamics, while long-run growth per worker on its balanced growth path still comes from A. A higher human-capital investment share is not, by itself, a permanent growth-rate increase in this specification.

Years of schooling are a proxy for skills, not a direct observation of H or s_H. Pritchett (2001) highlights why schooling expansion need not translate uniformly into aggregate growth: learning quality, demand for educated labor, and the activities rewarded by institutions can differ.[10]

What can weak schooling–growth estimates tell us? Weak estimates in a particular specification do not establish that skills have no productive value. They call for checking the proposed accumulation mechanism, measurement, and identification.

4.6.3 Technology adoption and catch-up

Giving every economy the same A path assumes away differences in access to and use of technology. To illustrate an adoption mechanism, let frontier efficiency A_F grow at g_F and posit, for a follower with $A \le A_F$,

$$\frac{\dot{A}}{A} = g_F + \mu \log \frac{A_F}{A}, \qquad \mu > 0. \tag{181}$$

[10] Lant Pritchett (2001), "Where Has All the Education Gone?," *World Bank Economic Review*, 15(3), 367–391, especially Section II. https://doi.org/10.1093/wber/15.3.367.

This is an illustrative adoption rule, not a consequence of the original saving equation. Define the log technology gap $v = \log(A_F/A)$. Then

$$\dot{v} = g_F - \frac{\dot{A}}{A} = -\mu v, \tag{182}$$

$$v(t) = v(0)e^{-\mu t}. \tag{183}$$

A larger gap creates scope for faster adoption; μ governs how quickly that scope is used. Technology catch-up differs from rebuilding a low capital stock. It can change output directly and also induce capital accumulation.

During catch-up, $\dot{A}/A$ varies with the gap. The capital equation is now

$$\dot{k} = sf(k) - [n + \delta + g_F + \mu v(t)]\, k. \tag{184}$$

We cannot use the earlier constant-g transition unchanged. As the technology gap closes, this equation approaches the benchmark with $g = g_F$. Training, organizational changes, and technology access could help determine μ, but this rule does not model their costs or incentives. Explaining those choices is a further step toward endogenous productivity.

4.6.4 Removing diminishing returns: the AK benchmark

What if returns to the reproducible stock do not diminish? Let

$$Y = \mathscr{A}K, \qquad \mathscr{A} > 0 \text{ constant}, \qquad \dot{K} = (s\mathscr{A} - \delta)K. \tag{185}$$

With labor growth n, output per worker satisfies

$$g_{Y/L} = s\mathscr{A} - \delta - n. \tag{186}$$

A permanent increase in saving now raises the growth rate permanently. The reason is the changed production assumption: each additional unit of capital has the same product. The Solow Inada and strict-concavity conditions no longer hold, and there is no positive stationary K/L when this growth rate is positive.

K can be interpreted as a broad reproducible composite only if we explain how its components can be accumulated without diminishing aggregate returns. The AK equation shows the role of that restriction; it does not by itself supply a theory of innovation, externalities, or optimal saving. Those mechanisms belong in the endogenous growth chapter.

4.6.5 Fixed land and the transition to modern growth

The baseline omits a fixed input that becomes scarce. Add land X:

$$Y = BK^a L^\ell X^{1-a-\ell}, \qquad a, \ell > 0, \quad a + \ell < 1. \tag{187}$$

B is Hicks-neutral productivity, as in Chapter 2. Dividing by labor gives

$$\frac{Y}{L} = B\left(\frac{K}{L}\right)^a \left(\frac{X}{L}\right)^{1-a-\ell}. \tag{188}$$

At given B and K/L, population growth lowers land per worker and hence output per worker. If population growth also responds to income, productivity gains may support more people rather than persistent gains in living standards. The land equation alone does not establish this outcome; it needs a demographic mechanism.

Hansen and Prescott (2002) study a transition in which a less land-intensive production process becomes profitable as technology advances. Both production processes are available throughout; which one is used changes in equilibrium. Their analysis retains exogenous technical progress, so it explains a transition in production rather than the ultimate source of innovation.[11]

Where the benchmark ends. Lower labor-force growth reduces dilution in Solow, holding everything else fixed. This is not a complete assessment of ageing: participation, pensions, and age-specific saving are absent. Likewise, the model contains no money, banks, or foreign assets. Monetary, financial, or trade extensions need those objects and their decision rules; their effects cannot be inferred merely from their absence here.

The main unresolved choice is already inside the model: why should households save a fixed share when their income and returns change? Ramsey replaces that rule with an intertemporal problem, making consumption, patience, and the return to saving jointly determine the accumulation path.

[11] Gary D. Hansen and Edward C. Prescott (2002), "Malthus to Solow," *American Economic Review*, 92(4), 1205–1217. https://doi.org/10.1257/00028280260344731.

5 Ramsey Model

Solow determines accumulation from a given saving share. Ramsey asks how much to consume and save when households value consumption at different dates. The change adds a choice, a price for postponing consumption, and a boundary condition that selects an admissible path.

5.1 From a saving rule to a saving decision

Suppose two economies have the same current income. In one, income is expected to rise; in the other, a disruption has temporarily reduced production. A rule $S = sY$ makes saving move proportionally with current output in both cases. It cannot tell us whether households want to smooth a temporary loss, prepare for lower future income, or invest in a newly productive opportunity.

One response would be to make s a function of income or the interest rate. But then we would still need to explain that function and how it changes when expectations or policy change. Instead, specify what consumption is worth, what saving buys, and which paths the household can finance. A saving rate will emerge from those choices rather than being the starting rule.

Start with the experiment of consuming one unit less today. Saving that unit provides a return tomorrow. The household compares the utility lost now with the discounted utility of that future consumption. This suggests an optimality condition across adjacent dates. It cannot yet determine the level of consumption: a household with little wealth cannot finance every path with the right slope. We will need feasibility and a terminal restriction as well.

The planner and household formulations then answer related questions. The planner allocates real resources; a household takes wages and returns as given. Their allocations coincide in the competitive benchmark under the conditions stated below. Taxes can separate the private return from the social product, which is why decentralization matters for the final section.

Ramsey's original contribution was a saving problem; the presentation here is the modern discounted Ramsey–Cass–Koopmans benchmark, rather than a literal reproduction of the 1928 objective.[12]

[12] Frank P. Ramsey (1928), "A Mathematical Theory of Saving," *Economic Journal*, 38(152), 543–559. https://doi.org/10.2307/2224098.

5.2 Assumptions

The conditions below state the technology, ownership, choices, and market structure explicitly. The subsequent notation and solvency sections complete the specification before the optimization problem is solved.

01 Economy and production technology

A closed, one-good economy contains households and firms, without government in the baseline. Output can be consumed or invested:

$$Y = F(K, AL), \qquad Y = C + I, \qquad \dot{K} = I - \delta K.$$

Technology is labor-augmenting (Harrod-neutral), freely and publicly available. F is continuous, twice differentiable for positive inputs, positive there, with positive and diminishing marginal products of capital and effective labor. Constant returns means $F(\lambda K, \lambda AL) = \lambda F(K, AL)$ for $\lambda > 0$. With $k = K/(AL)$ and $f(k) = F(k, 1)$, assume

$$f(0) = 0, \qquad f'(k) > 0, \qquad f''(k) < 0 \quad (k > 0),$$
$$\lim_{k \downarrow 0} f'(k) = \infty, \qquad \lim_{k \to \infty} f'(k) = 0.$$

These are the explicit Solow production restrictions inherited here.

02 Given stocks and exogenous growth

Initial $K(0), A(0), L(0)$ are positive and given. Technology and population follow

$$A(t) = A(0)e^{gt}, \qquad L(t) = L(0)e^{nt}.$$

The population of each dynasty grows at the same rate n, so fixed dynasty shares aggregate to total population growth n. Each member supplies one unit of labor. Growth rates n, g and depreciation δ are exogenous. The course benchmark sets $\delta = 0$; general formulas keep δ visible. The original notation a for technology growth is written consistently as g.

03 Households, ownership, and income

There are many identical, price-taking, infinitely lived dynasties. Households own capital and firms, supply labor inelastically, and choose consumption and saving over time. Household income consists of wages, capital income, and its share of firm profits. In aggregate, before imposing zero profits,

$$\dot{B} = rB + wL + \Pi - C, \qquad r = q - \delta.$$

Here B is household financial wealth, q the rental price before depreciation, w the wage per worker, and Π distributed economic profit. Under the competitive constant-returns benchmark below, $\Pi = 0$ and $B = K$ in equilibrium. Ownership remains an assumption even when profits vanish.

04 Preferences and the household objective

Each dynasty maximizes the discounted sum of its members' utility, subject to its budget and solvency constraints. With consumption per member $x = C/L$,

$$\max_{\{C(t)\}} \int_0^\infty e^{-\rho t} L(t) u(x(t))\, dt, \qquad u'(x) = x^{-\theta}, \quad \theta > 0.$$

Using aggregate L here changes the objective only by a positive constant relative to a dynasty of fixed population share. Patience is measured by ρ; in the separate discrete-time formulation the discount factor is $\beta = 1/(1+\rho)$, with $0 < \beta < 1$. Continuous-time discounting uses $e^{-\rho t}$. The parameter restrictions used below are

$$\rho > n \geq 0, \qquad g \geq 0, \qquad \eta \equiv \rho - n - (1-\theta)g > 0.$$

Saving is an optimal choice rather than Solow's imposed share of income.

05 Firms and competitive factor markets

There are many firms with a common production technology. They take prices as given, hire labor and rent capital in competitive markets, and maximize profits at each date:

$$\max_{K,L} \; F(K, AL) - qK - wL.$$

Common technology and constant returns permit the representative-firm formulation. Homogeneous technology is not itself the definition of perfect competition. Interior factor demands satisfy

$$q = F_1(K, AL) = f'(k), \qquad w = AF_2(K, AL) = A[f(k) - kf'(k)].$$

Factor payments exhaust output, $Y = qK + wL$, and profits are zero. Households receive the factor payments and would receive any profits through ownership.

06 Market clearing and equilibrium

All supplied labor and capital are employed. With bars denoting aggregate endowments,

$$L^d = L^s = \bar{L}, \qquad K^d = K^s = \bar{K}, \qquad Y = C + I, \qquad B = K.$$

Household optimality, firm optimality, and these clearing conditions hold jointly. There are no taxes, externalities, market power, income uncertainty, or borrowing frictions beyond solvency in the baseline. These conditions underlie the planner–competitive allocation equivalence used below.

07 Feasibility and terminal restrictions

Capital is inherited and consumption is chosen. Feasible paths obey the resource constraint with $K \geq 0$ and $C > 0$. Household debt must satisfy a no-Ponzi restriction; optimal accumulation must also satisfy a transversality condition. They serve different purposes and are written explicitly in the solvency subsection below. An Euler equation alone does not select the path, and refinancing debt is not by itself a Ponzi scheme.

5.2.1 Notation and the course simplifications

Use K, C, Y for aggregates and $x = C/L$ for consumption per person. Reserve $k = K/(AL)$ and $c = C/(AL)$ for quantities per effective worker, so $x = Ac$. Technology growth is denoted by g throughout. We retain the course's zero-depreciation benchmark; δ remains visible in the formulas so that setting $\delta = 0$ is explicit. Full depreciation, $\delta = 1$, will be used only in a separately labeled discrete-time example.

Preferences have marginal utility $u'(x) = x^{-\theta}$, $\theta > 0$:

$$u(x) = \begin{cases} x^{1-\theta}/(1-\theta), & \theta \neq 1, \\ \log x, & \theta = 1. \end{cases} \tag{189}$$

In this deterministic model, $1/\theta$ measures willingness to substitute consumption across time. Higher ρ reduces the weight on future utility. Assume $\rho > n \geq 0$, $g \geq 0$, and

$$\eta \equiv \rho - n - (1-\theta)g > 0. \tag{190}$$

These are sufficient restrictions for finite utility on the balanced growth path used below. Weighting by $L(t)$ means the dynasty values the utility of all its members; choosing a different population welfare criterion would change the normalization. The original lists are retained in comments immediately above this section for comparison.

5.2.2 No-Ponzi restrictions and transversality

Solvency and optimality are different restrictions. A household's total wealth B satisfies

$$\dot{B} = rB + wL - C, \tag{191}$$

where r is the net real return and w the wage per person. A no-Ponzi restriction requires

$$\liminf_{T\to\infty} e^{-\int_0^T r(v)\,dv} B(T) \geq 0. \tag{192}$$

Borrowing must ultimately be supported by resources rather than by a negative terminal present value of wealth. Refinancing a loan is not, by itself, a Ponzi scheme: future income can support repayment.

A transversality condition instead concerns optimality. In the planner's normalized problem below it takes the standard form

$$\lim_{t\to\infty} e^{-\eta t} u'(c(t))k(t) = 0. \tag{193}$$

It rules out leaving resources with positive limiting discounted utility value when consuming more would improve the allocation. It does not require the capital stock itself to approach zero. Feasibility and an Euler equation alone need not select the optimal infinite-horizon path.

5.3 Solving the consumption–saving problem

5.3.1 The planner's problem and decentralization

For continuous time, the planner chooses consumption with inherited $K(0)$:

$$\begin{aligned} &\max_{\{C(t)\}} \int_0^\infty e^{-\rho t} L(t) u(C(t)/L(t))\, dt, \\ &\quad \text{s.t. } \dot{K} = F(K, AL) - \delta K - C, \qquad K \geq 0,\ C > 0. \end{aligned} \tag{194}$$

Consumption uses current output; the residual, net of depreciation, changes the capital stock. In a decentralized economy, a household uses its budget constraint instead. Competitive factor pricing gives

$$q = f'(k), \qquad r = q - \delta, \qquad w = A\left[f(k) - k f'(k)\right]. \tag{195}$$

q is the rental payment before depreciation. The net return $r = q - \delta$ and a discrete-time gross return $R = 1 + q - \delta = 1 + r$ are different objects.

With constant returns, factor incomes exhaust output. Firms' dividends are not an additional source of aggregate goods on top of Y. In equilibrium, aggregate household wealth equals capital, $B = K$, so the household budgets consolidate into (194)'s resource constraint. With identical preferences, competitive prices, and no taxes, externalities, or financing frictions, the planner allocation can be supported as a competitive allocation. This equivalence must be checked again when the environment changes.

Question: why do three methods give the same two-period Euler equation? Before solving for a whole path, isolate two dates with $A = L = 1$, known incomes y_0, y_1, and a given gross return $R = 1 + r > 0$. There are no borrowing frictions or terminal bequests. With zero initial and terminal assets, the budgets give

$$\begin{aligned} c_0 + a_1 &= y_0, \qquad c_1 = y_1 + R a_1, \\ a_1 &= y_0 - c_0, \\ c_1 &= y_1 + R(y_0 - c_0), \\ c_0 + c_1/R &= y_0 + y_1/R \equiv W. \end{aligned}$$

For an interior choice of positive consumption, solve

$$\max_{c_0, c_1} u(c_0) + \beta u(c_1), \qquad c_0 + \frac{c_1}{R} = y_0 + \frac{y_1}{R} \equiv W.$$

Elimination. Substitute $c_1 = R(W - c_0)$ into utility and differentiate:

$$\begin{aligned} \frac{d}{dc_0}\left[u(c_0) + \beta u(R(W - c_0))\right] &= u'(c_0) + \beta u'(c_1)(-R) = 0, \\ u'(c_0) &= \beta R u'(c_1). \end{aligned}$$

Geometry. Along an indifference curve, $u'(c_0)dc_0 + \beta u'(c_1)dc_1 = 0$, so its slope is $-u'(c_0)/[\beta u'(c_1)]$. The budget line has slope $-R$. Tangency therefore requires $u'(c_0) = \beta R u'(c_1)$.

Lagrangian. Write

$$\mathscr{L} = u(c_0) + \beta u(c_1) + \lambda\left(W - c_0 - \frac{c_1}{R}\right).$$

Differentiate each choice:

$$\begin{aligned}
\mathscr{L}_{c_0} &= u'(c_0) - \lambda = 0,\\
\mathscr{L}_{c_1} &= \beta u'(c_1) - \lambda/R = 0,\\
\mathscr{L}_{\lambda} &= W - c_0 - c_1/R = 0,\\
u'(c_0) &= \lambda = \beta R u'(c_1).
\end{aligned}$$

All three methods price the same marginal transfer: giving up one unit today buys R units tomorrow. The Euler condition fixes the preferred trade-off; the budget fixes the affordable consumption levels. In infinite time the adjacent-date logic survives, but the full budget and terminal conditions still have to be imposed. A binding borrowing constraint would replace the interior tangency by the appropriate constrained condition.

5.3.2 Lagrangian: the adjacent-date trade-off

From two dates to a finite horizon. First keep the household exercise separate from production. Dates are $t = 1, \ldots, T$, with $T \geq 2$, known incomes y_t, a constant gross return $R > 0$, $0 < \beta < 1$, and no borrowing limit. Let a_t be assets after date t consumption. Set $a_0 = a_T = 0$, assume an interior positive-consumption solution, and write

$$\max \sum_{t=1}^{T} \beta^{t-1} u(c_t), \tag{196}$$

$$c_1 + a_1 = Ra_0 + y_1 = y_1, \tag{197}$$

$$c_2 + a_2 = Ra_1 + y_2, \tag{198}$$

$$\vdots$$

$$c_T + a_T = Ra_{T-1} + y_T. \tag{199}$$

Divide each date-t budget by R^{t-1} and sum. The asset terms telescope:

$$\sum_{t=1}^{T} \frac{c_t}{R^{t-1}} = \sum_{t=1}^{T} \frac{y_t}{R^{t-1}} + \sum_{t=1}^{T} \left(\frac{a_{t-1}}{R^{t-2}} - \frac{a_t}{R^{t-1}}\right), \tag{200}$$

$$= \sum_{t=1}^{T} \frac{y_t}{R^{t-1}} + Ra_0 - \frac{a_T}{R^{T-1}}, \tag{201}$$

$$= \sum_{t=1}^{T} \frac{y_t}{R^{t-1}} \equiv W_T. \tag{202}$$

The single present-value constraint gives the Lagrangian

$$\mathscr{L}_T = \sum_{t=1}^{T} \beta^{t-1} u(c_t) + \Lambda\left(W_T - \sum_{t=1}^{T} \frac{c_t}{R^{t-1}}\right).$$

Differentiate separately at adjacent dates:

$$\frac{\partial \mathscr{L}_T}{\partial c_t} = \beta^{t-1} u'(c_t) - \frac{\Lambda}{R^{t-1}} = 0, \tag{203}$$

$$\frac{\partial \mathscr{L}_T}{\partial c_{t+1}} = \beta^{t} u'(c_{t+1}) - \frac{\Lambda}{R^{t}} = 0, \tag{204}$$

$$\Lambda = (\beta R)^{t-1} u'(c_t) = (\beta R)^{t} u'(c_{t+1}), \tag{205}$$

$$u'(c_t) = \beta R u'(c_{t+1}), \qquad t = 1, \ldots, T-1. \tag{206}$$

The last Euler comparison is between $T-1$ and T; there is no consumption date $T+1$ in this problem.

We can obtain the same condition without first collapsing the budgets:

$$\mathscr{L}_T = \sum_{t=1}^{T} \beta^{t-1} \{ u(c_t) + \lambda_t [R a_{t-1} + y_t - c_t - a_t] \}.$$

For $t < T$, a_t enters two adjacent budgets. Its derivative shows both terms:

$$\frac{\partial \mathscr{L}_T}{\partial c_t} = \beta^{t-1} [u'(c_t) - \lambda_t] = 0, \tag{207}$$

$$\frac{\partial \mathscr{L}_T}{\partial a_t} = -\beta^{t-1} \lambda_t + \beta^{t} R \lambda_{t+1} = 0, \tag{208}$$

$$\lambda_t = u'(c_t), \qquad \lambda_t = \beta R \lambda_{t+1}, \tag{209}$$

$$u'(c_t) = \beta R u'(c_{t+1}). \tag{210}$$

The finite endpoint $a_T = 0$ is given here, not another interior asset choice. In an infinite horizon, adjacent-date conditions remain, but the endpoint must be replaced by the appropriate solvency and transversality restrictions. Taking $T \to \infty$ does not mean requiring the economy's physical capital to approach zero.

The infinite-horizon production economy. First isolate discrete time with $A = L = 1$ and $n = g = 0$. This makes timing visible without carrying growing denominators; it is a restricted version of the model. The planner solves

$$\max_{\{c_t, k_{t+1}\}} \sum_{t=0}^{\infty} \beta^t u(c_t), \qquad c_t + k_{t+1} = f(k_t) + (1-\delta) k_t, \tag{211}$$

where k_0 is given, $0 < \beta < 1$, and $0 \le \delta \le 1$. Writing $\beta = 1/(1+\rho)$ defines a discrete-time rate of time preference; the continuous-time convention uses $e^{-\rho t}$.

Use current-utility multipliers λ_t:

$$\mathscr{L} = \sum_{t=0}^{\infty} \beta^t \left\{ u(c_t) + \lambda_t [f(k_t) + (1-\delta) k_t - c_t - k_{t+1}] \right\}. \tag{212}$$

At an interior solution, differentiate with respect to c_t and k_{t+1}:

$$\frac{\partial \mathscr{L}}{\partial c_t} = \beta^t [u'(c_t) - \lambda_t] = 0, \tag{213}$$

$$\frac{\partial \mathscr{L}}{\partial k_{t+1}} = -\beta^t \lambda_t + \beta^{t+1} \lambda_{t+1} [f'(k_{t+1}) + 1 - \delta] = 0, \tag{214}$$

$$\lambda_t = u'(c_t), \tag{215}$$

$$\lambda_t = \beta \lambda_{t+1} [f'(k_{t+1}) + 1 - \delta]. \tag{216}$$

Substitution gives the Euler equation

$$u'(c_t) = \beta R_{t+1} u'(c_{t+1}), \qquad R_{t+1} = f'(k_{t+1}) + 1 - \delta. \tag{217}$$

One unit saved at t earns its return at $t+1$, when the new capital is used. The utility cost today equals the discounted marginal gain tomorrow. For CRRA,

$$c_t^{-\theta} = \beta R_{t+1} c_{t+1}^{-\theta}, \tag{218}$$

$$\frac{c_t^{-\theta}}{c_{t+1}^{-\theta}} = \beta R_{t+1}, \tag{219}$$

$$\left(\frac{c_{t+1}}{c_t}\right)^{\theta} = \beta R_{t+1}, \tag{220}$$

$$\frac{c_{t+1}}{c_t} = (\beta R_{t+1})^{1/\theta}. \tag{221}$$

This determines consumption growth given the return, not the entire consumption path. With no bequest motive and the usual concavity conditions, the terminal condition is $\lim_{t\to\infty} \beta^t u'(c_t) k_{t+1} = 0$. In a finite-horizon exercise a terminal stock may be set to zero; that is not the same as imposing $k_t \to 0$ in this infinite-horizon economy.

5.3.3 Hamiltonian: saving with continuous adjustment

Return to continuous time with population and technology growth. From $k = K/(AL)$ and $c = C/(AL)$,

$$\dot{k} = f(k) - c - bk, \qquad b \equiv n + g + \delta. \tag{222}$$

For $\theta \neq 1$, substitute $x = Ac$ and $L = L_0 e^{nt}$ into utility:

$$e^{-\rho t} L u(Ac) = e^{-\rho t} L_0 e^{nt} \frac{(A_0 e^{gt} c)^{1-\theta}}{1-\theta}, \tag{223}$$

$$= L_0 A_0^{1-\theta} e^{[-\rho+n+(1-\theta)g]t} \frac{c^{1-\theta}}{1-\theta}, \tag{224}$$

$$= L_0 A_0^{1-\theta} e^{-\eta t} \frac{c^{1-\theta}}{1-\theta}, \tag{225}$$

$$\eta = \rho - n - (1-\theta)g. \tag{226}$$

The positive constant in front does not change choices. For log utility, $\log(Ac) = \log A + \log c$; the $\log A$ term is independent of the allocation. The normalized problem therefore discounts $u(c)$ at η in either case.

Where the costate equation comes from. Write $h(k,c) = f(k) - c - bk$. To derive interior conditions, first use a finite interval $[0,T]$ with both capital endpoints held fixed. Let $p(t)$ be a present-value multiplier. The augmented objective is

$$\mathscr{A} = \int_0^T \{e^{-\eta t} u(c) + p[h(k,c) - \dot{k}]\}\, dt, \tag{227}$$

$$= \int_0^T \mathscr{H}^P\, dt - \int_0^T p\dot{k}\, dt, \tag{228}$$

$$\mathscr{H}^P = e^{-\eta t} u(c) + p h(k,c). \tag{229}$$

Integration by parts keeps the boundary term visible:

$$\int_0^T p\dot{k}\,dt = [pk]_0^T - \int_0^T \dot{p}\,k\,dt, \tag{230}$$

$$\mathscr{A} = \int_0^T [\mathscr{H}^P + \dot{p}\,k]\,dt - [pk]_0^T. \tag{231}$$

Let $v_c(t)$ and $v_k(t)$ be small variations, with $v_k(0) = v_k(T) = 0$. The boundary variation vanishes, leaving

$$\delta\mathscr{A} = \int_0^T [\mathscr{H}_c^P v_c + (\mathscr{H}_k^P + \dot{p})v_k]\,dt.$$

For arbitrary interior variations, the two coefficients must vanish. Variation of p also imposes the original resource equation:

$$\mathscr{H}_c^P = e^{-\eta t}u'(c) - p = 0, \tag{232}$$

$$\mathscr{H}_k^P + \dot{p} = p[f'(k) - b] + \dot{p} = 0, \tag{233}$$

$$\mathscr{H}_p^P = h(k,c) = \dot{k}. \tag{234}$$

Thus $p = e^{-\eta t}u'(c)$ and $\dot{p} = -p[f'(k) - b]$. Holding endpoints fixed was used only to derive the interior conditions. The infinite-horizon optimum still needs its own feasibility and transversality condition; a boundary term must not be discarded by asserting that the capital stock tends to zero.

Now convert units from present utility to date-t utility. Define $\mu = e^{\eta t}p$ and multiply the Hamiltonian by $e^{\eta t}$:

$$e^{\eta t}\mathscr{H}^P = u(c) + e^{\eta t}p\,h(k,c) = u(c) + \mu h(k,c) \equiv \mathscr{H}, \tag{235}$$

$$\dot{\mu} = \eta e^{\eta t}p + e^{\eta t}\dot{p}, \tag{236}$$

$$= \eta\mu - e^{\eta t}p[f'(k) - b], \tag{237}$$

$$= [\eta + b - f'(k)]\mu. \tag{238}$$

With zero discounting on this finite interval, $\eta = 0$ makes $p = \mu$ and $\dot{\mu} = -\mathscr{H}_k$. With discounting, the additional $\eta\mu$ term comes from differentiating $e^{\eta t}$; it is not an optional convention.

The current-value Hamiltonian is therefore

$$\mathscr{H} = u(c) + \mu[f(k) - c - bk]. \tag{239}$$

The control condition and current-value costate equation are

$$\mathscr{H}_c = 0 \quad \Longrightarrow \quad \mu = u'(c), \tag{240}$$

$$\dot{\mu} = \eta\mu - \mathscr{H}_k = [\eta + b - f'(k)]\,\mu. \tag{241}$$

μ is the current utility value of an additional unit of normalized capital. The factor η belongs in the current-value costate equation; a present-value costate would instead absorb the discount factor. These are two representations of the same problem, not an assertion that patience is irrelevant.

Differentiate $\mu = c^{-\theta}$ and use $\eta + b = \rho + \delta + \theta g$:

$$\dot{\mu} = -\theta c^{-\theta-1}\dot{c}, \tag{242}$$

$$\frac{\dot{\mu}}{\mu} = -\theta \frac{c^{-\theta-1}}{c^{-\theta}}\dot{c} = -\theta\frac{\dot{c}}{c}, \tag{243}$$

$$\frac{\dot{\mu}}{\mu} = -\theta\frac{\dot{c}}{c} = \eta + b - f'(k), \tag{244}$$

$$\theta\frac{\dot{c}}{c} = f'(k) - \eta - b, \tag{245}$$

$$\eta + b = \rho - n - (1-\theta)g + n + g + \delta = \rho + \theta g + \delta, \tag{246}$$

$$\frac{\dot{c}}{c} = \frac{f'(k) - \delta - \rho - \theta g}{\theta}. \tag{247}$$

Consumption per person is $x = Ac$, so

$$\begin{aligned} \frac{\dot{x}}{x} &= \frac{\dot{A}}{A} + \frac{\dot{c}}{c} \\ &= g + \frac{f'(k) - \delta - \rho - \theta g}{\theta} \\ &= \frac{\theta g + f'(k) - \delta - \rho - \theta g}{\theta} = \frac{r - \rho}{\theta}. \end{aligned} \tag{248}$$

A high return makes future consumption relatively attractive; impatience favors consumption now. Curvature determines how strongly consumption growth responds. Population growth affects resources and welfare weights, but cancels from this per-person Euler equation under the dynasty objective specified above.

Setting $\delta = 0$ recovers the course benchmark. Setting $g = n = 0$ also removes the growth normalization. At $\rho = 0$ the intertemporal trade-off changes, and an infinite-horizon undiscounted problem needs its own welfare criterion; it is not obtained simply by deleting discounting from every infinite integral.

5.3.4 Bellman and HJB: valuing the remaining opportunities

For the stationary discrete-time version, let $V(k)$ be the value of starting with capital k. Write $k_0 = k$ and $c_t = f(k_t) + (1-\delta)k_t - k_{t+1} > 0$, with feasible $k_{t+1} \geq 0$. Splitting off the first date gives

$$V(k) = \max_{\{k_{t+1}\}} \sum_{t=0}^{\infty} \beta^t u(c_t), \tag{249}$$

$$= \max_{\{k_{t+1}\}} \left\{ u(c_0) + \sum_{t=1}^{\infty} \beta^t u(c_t) \right\}, \tag{250}$$

$$= \max_{k_1} \left\{ u(c_0) + \beta \max_{\{k_{t+1}\}_{t\geq 1}} \sum_{j=0}^{\infty} \beta^j u(c_{j+1}) \right\}, \tag{251}$$

$$= \max_{k_1} \{ u(f(k) + (1-\delta)k - k_1) + \beta V(k_1) \}. \tag{252}$$

The continuation problem has the same technology and preferences, starting from k_1. Write $k' = k_1$. Its first-order condition is

$$0 = u'(f(k) + (1-\delta)k - k')(-1) + \beta V'(k'), \tag{253}$$

$$u'(c) = \beta V'(k'). \tag{254}$$

Show the policy derivative before canceling it. Let the optimal next-capital policy be $k' = \psi(k)$, assumed differentiable at the interior point being considered. On that policy,

$$V(k) = u(f(k) + (1-\delta)k - \psi(k)) + \beta V(\psi(k)).$$

Differentiate the entire expression, including the response of the choice:

$$V'(k) = u'(c)[f'(k) + 1 - \delta - \psi'(k)] + \beta V'(\psi(k))\psi'(k), \tag{255}$$

$$= u'(c)[f'(k) + 1 - \delta] + [\beta V'(\psi(k)) - u'(c)]\psi'(k), \tag{256}$$

$$= u'(c)[f'(k) + 1 - \delta] + 0 \cdot \psi'(k), \tag{257}$$

$$= u'(c)[f'(k) + 1 - \delta]. \tag{258}$$

The bracket is zero by the first-order condition. This cancellation is the envelope theorem in this calculation; it is not an instruction to ignore $\psi'(k)$ before differentiating.

Now write the two conditions at their actual dates:

$$u'(c_t) = \beta V'(k_{t+1}), \tag{259}$$

$$V'(k_{t+1}) = u'(c_{t+1})[f'(k_{t+1}) + 1 - \delta], \tag{260}$$

$$u'(c_t) = \beta u'(c_{t+1})[f'(k_{t+1}) + 1 - \delta], \tag{261}$$

$$\frac{u'(c_t)}{u'(c_{t+1})} = \beta[f'(k_{t+1}) + 1 - \delta] = \beta R_{t+1}. \tag{262}$$

This recovers (217). The Lagrangian and Bellman methods price the same marginal transfer of resources between dates.

A case with an exact saving rule. Take $u(c) = \log c$, $f(k) = k^\alpha$, $0 < \alpha < 1$, and full depreciation $\delta = 1$ only for this example. Guess $V(k) = a + d\log k$. Substitute the guess before differentiating:

$$a + d\log k = \max_{k'}\{\log(k^\alpha - k') + \beta[a + d\log k']\}, \tag{263}$$

$$0 = -\frac{1}{k^\alpha - k'} + \frac{\beta d}{k'}, \tag{264}$$

$$k' = \beta d(k^\alpha - k'), \tag{265}$$

$$(1+\beta d)k' = \beta d k^\alpha, \tag{266}$$

$$k' = sk^\alpha, \qquad s \equiv \frac{\beta d}{1+\beta d}, \qquad c = (1-s)k^\alpha. \tag{267}$$

Insert that policy into the Bellman equation and expand both logarithms:

$$a + d\log k = \log[(1-s)k^\alpha] + \beta[a + d\log(sk^\alpha)], \tag{268}$$

$$= \log(1-s) + \alpha\log k + \beta a + \beta d\log s + \alpha\beta d\log k, \tag{269}$$

$$= \log(1-s) + \beta a + \beta d\log s + \alpha(1+\beta d)\log k. \tag{270}$$

Equality for every $k > 0$ requires equal coefficients of $\log k$:

$$d = \alpha(1 + \beta d), \tag{271}$$

$$d - \alpha\beta d = \alpha, \tag{272}$$

$$d = \frac{\alpha}{1 - \alpha\beta}, \tag{273}$$

$$s = \frac{\beta\alpha/(1 - \alpha\beta)}{} 1 + \beta\alpha/(1 - \alpha\beta) = \alpha\beta. \tag{274}$$

The constant term also has to match:

$$a = \log(1 - s) + \beta a + \beta d \log s, \tag{275}$$

$$a = \frac{\log(1 - \alpha\beta) + \dfrac{\alpha\beta}{1 - \alpha\beta}\log(\alpha\beta)}{1 - \beta}. \tag{276}$$

Thus the complete policy is

$$d = \frac{\alpha}{1 - \alpha\beta}, \qquad k' = \alpha\beta k^{\alpha}, \qquad c = (1 - \alpha\beta)k^{\alpha}. \tag{277}$$

A constant saving share can emerge from optimization in this special case. Its value depends on preferences and technology; the result does not make a constant saving share general once depreciation or utility changes.

Build the continuous-time HJB from a short interval. In effective-worker units let $h(k, c) = f(k) - c - bk$. Over a short interval Δt, hold the chosen control fixed to first order. Then

$$k(t + \Delta t) = k + h(k, c)\Delta t + o(\Delta t), \tag{278}$$

$$e^{-\eta\Delta t} = 1 - \eta\Delta t + o(\Delta t), \tag{279}$$

$$V(k(t + \Delta t)) = V(k) + V'(k)h(k, c)\Delta t + o(\Delta t). \tag{280}$$

The dynamic-programming identity over this interval becomes

$$V(k) = \max_{c}\{u(c)\Delta t + e^{-\eta\Delta t}V(k(t + \Delta t)) + o(\Delta t)\}, \tag{281}$$

$$= \max_{c}\{V(k) + [u(c) + V'(k)h(k, c) - \eta V(k)]\Delta t + o(\Delta t)\}. \tag{282}$$

Subtract $V(k)$, divide by Δt, and take the limit:

$$0 = \max_{c}\{u(c) + V'(k)[f(k) - c - bk] - \eta V(k)\}, \tag{283}$$

$$\eta V(k) = \max_{c}\{u(c) + V'(k)[f(k) - c - bk]\}. \tag{284}$$

The control derivative is $u'(c) - V'(k) = 0$. To obtain the costate equation, write the optimal control as $c = \gamma(k)$ and differentiate the HJB:

$$\eta V'(k) = u'(c)\gamma'(k) + V''(k)h(k, c) + V'(k)[f'(k) - b - \gamma'(k)], \tag{285}$$

$$= V''(k)h(k, c) + V'(k)[f'(k) - b] + [u'(c) - V'(k)]\gamma'(k), \tag{286}$$

$$= V''(k)\dot{k} + V'(k)[f'(k) - b]. \tag{287}$$

Finally, with $\mu = V'(k)$, the chain rule along the path gives

$$\dot{\mu} = V''(k)\dot{k}, \tag{288}$$

$$\dot{\mu} = [\eta + b - f'(k)]\mu, \tag{289}$$

$$\mu = c^{-\theta} \quad \Longrightarrow \quad -\theta\frac{\dot{c}}{c} = \eta + b - f'(k). \tag{290}$$

The value function therefore yields the same costate and consumption conditions as the Hamiltonian. Feasibility and terminal restrictions remain part of the underlying problem.

5.4 Intertemporal equilibrium and economic adjustment

5.4.1 Steady state and the two loci

The continuous-time equilibrium combines

$$\begin{aligned} \dot{k} &= f(k) - c - bk, \\ \dot{c} &= \frac{c}{\theta}[f'(k) - \delta - \rho - \theta g], \end{aligned} \tag{291}$$

with given $k(0)$ and (193). A positive normalized steady state satisfies

$$0 = \frac{c^*}{\theta}[f'(k^*) - \delta - \rho - \theta g], \tag{292}$$

$$0 = f'(k^*) - \delta - \rho - \theta g \qquad (c^* > 0), \tag{293}$$

$$f'(k^*) = \delta + \rho + \theta g, \tag{294}$$

$$0 = f(k^*) - c^* - bk^*, \tag{295}$$

$$c^* = f(k^*) - bk^*. \tag{296}$$

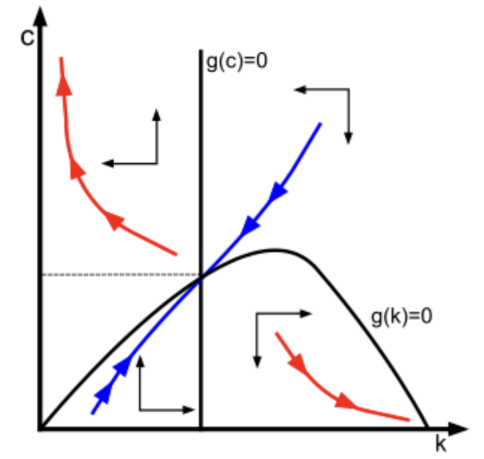
$$f'(k^*) = \delta + \rho + \theta g, \qquad c^* = f(k^*) - bk^*. \tag{297}$$

The production assumptions give a unique positive k^*. Strict concavity gives $f(k^*)/k^* > f'(k^*) > b$, where the second inequality is $\eta > 0$, so $c^* > 0$. On this balanced growth path, consumption per person grows at g, aggregate consumption at $n + g$, and the net return is $r^* = \rho + \theta g$.

For $c > 0$, the $\dot{c} = 0$ locus is vertical at k^*. Consumption grows to its left and falls to its right. The $\dot{k} = 0$ locus is $c = f(k) - bk$; capital grows below it and falls above it. If $b > 0$, its maximum is the Golden Rule stock, where $f'(k_{\mathrm{GR}}) = b$. Neither locus, by itself, is an equilibrium path. The shape of the resource locus follows by differentiation:

$$\frac{dc}{dk} = f'(k) - b, \qquad \frac{d^2c}{dk^2} = f''(k) < 0.$$

Its slope is positive below the Golden Rule stock and negative above it. The $\dot{c} = 0$ locus fixes $k = k^*$ for every $c > 0$, which is why it is vertical; no division by a zero derivative is needed.



Reading the diagram. The vertical line is $\dot{c} = 0$; the curved resource locus is $\dot{k} = 0$. The blue curve depicts the stable path, with arrows toward their intersection. The red paths illustrate departures that do not satisfy the required long-run conditions. The sketch assumes a hump-shaped resource locus and $k^* < k_{\mathrm{GR}}$.

5.4.2 Why consumption can jump while capital cannot

A state is inherited; a control is chosen. Capital is a state inherited from previous investment. Consumption is a control: a household can change spending immediately, but cannot instantly create a new stock of machines. For a given $k(0)$, different initial consumption choices produce different candidate paths under (291). Most fail feasibility or the terminal condition. The saddle path selects initial consumption as well as the later transition; it does not require every variable to start at its steady state.

A one-time destruction of physical capital, with all other fundamentals unchanged, moves the state leftward. Consumption adjusts onto the stable path. The low stock supports a high return and rebuilding, while the loss of resources constrains current spending. Actual disasters may also damage skills or technology, which would change the destination rather than just the initial state.

A credible announcement about future productivity or taxation can change consumption before it changes current output: the household evaluates future income and returns when choosing today's spending. This model isolates that intertemporal mechanism. It omits borrowing constraints, income risk, and different marginal propensities to consume, so its representative consumption response is not an empirical claim about every household.

5.5 Linearization and the saddle path

5.5.1 The local system and its eigenvalues

Let $\Delta k = k - k^*$ and $\Delta c = c - c^*$. In the order (k, c), the first-order system follows by differentiating both right-hand sides. Write $P = f(k) - c - bk$ and $Q = c[f'(k) - \delta - \rho - \theta g]/\theta$:

$$P_k^* = f'(k^*) - b = (\delta + \rho + \theta g) - (n + g + \delta) = \eta, \tag{298}$$

$$P_c^* = -1, \tag{299}$$

$$Q_k^* = \frac{c^*}{\theta} f''(k^*) \equiv z < 0, \tag{300}$$

$$Q_c^* = \frac{f'(k^*) - \delta - \rho - \theta g}{\theta} = 0, \tag{301}$$

$$\dot{\Delta k} \approx P_k^* \Delta k + P_c^* \Delta c = \eta \Delta k - \Delta c, \tag{302}$$

$$\dot{\Delta c} \approx Q_k^* \Delta k + Q_c^* \Delta c = z \Delta k. \tag{303}$$

The constant terms vanish because $P(k^*, c^*) = Q(k^*, c^*) = 0$. In matrix form, the system is

$$\begin{pmatrix} \dot{\Delta k} \\ \dot{\Delta c} \end{pmatrix} \approx J \begin{pmatrix} \Delta k \\ \Delta c \end{pmatrix}, \qquad J = \begin{pmatrix} \eta & -1 \\ z & 0 \end{pmatrix}, \quad z \equiv \frac{c^* f''(k^*)}{\theta} < 0. \tag{304}$$

The top-left entry is $f'(k^*) - b = \eta$; the bottom-right entry vanishes because the consumption growth bracket is zero at the steady state. Thus

$$\det(\lambda I - J) = \det \begin{pmatrix} \lambda - \eta & 1 \\ -z & \lambda \end{pmatrix}, \tag{305}$$

$$= (\lambda - \eta)\lambda - (1)(-z), \tag{306}$$

$$= \lambda^2 - \eta\lambda + z = 0, \tag{307}$$

$$\lambda_\pm = \frac{\eta \pm \sqrt{(-\eta)^2 - 4z}}{2} = \frac{\eta \pm \sqrt{\eta^2 - 4z}}{2}. \tag{308}$$

Since $z < 0$, $\lambda_- < 0 < \lambda_+$. The local system has one stable and one unstable direction. This is the mathematical reason a boundary condition is needed to select consumption for the inherited capital stock.

For a stable eigenvector $(1, m)$, multiply the matrix explicitly:

$$J \begin{pmatrix} 1 \\ m \end{pmatrix} = \begin{pmatrix} \eta - m \\ z \end{pmatrix} = \lambda_- \begin{pmatrix} 1 \\ m \end{pmatrix}, \tag{309}$$

$$\eta - m = \lambda_-, \qquad z = \lambda_- m, \tag{310}$$

$$m = \eta - \lambda_- > 0. \tag{311}$$

In the linearized system, the two modes have $m_\pm = \eta - \lambda_\pm$ and constants A_-, A_+:

$$\Delta k(t) = A_- e^{\lambda_- t} + A_+ e^{\lambda_+ t}, \tag{312}$$

$$\Delta c(t) = m_- A_- e^{\lambda_- t} + m_+ A_+ e^{\lambda_+ t}. \tag{313}$$

Convergence to this steady state requires $A_+ = 0$, because $\lambda_+ > 0$. Then $A_- = \Delta k(0)$ and $\Delta c(0) = m_- \Delta k(0)$. On the stable direction, separate the scalar differential equation:

$$\dot{\Delta k} = \lambda_- \Delta k, \tag{314}$$

$$\int_{\Delta k(0)}^{\Delta k(t)} \frac{dv}{v} = \int_0^t \lambda_- \, ds, \tag{315}$$

$$\log \left| \frac{\Delta k(t)}{\Delta k(0)} \right| = \lambda_- t, \tag{316}$$

$$\Delta k(t) = \Delta k(0) e^{\lambda_- t}. \tag{317}$$

The integration assumes $\Delta k(0) \neq 0$; if it is zero, the stable linearized path remains at zero. Translating back to the nonlinear model, the local stable path is therefore

$$\Delta c(0) \approx m \Delta k(0), \tag{318}$$

$$\Delta k(t) \approx \Delta k(0) e^{\lambda_- t}, \tag{319}$$

$$\Delta c(t) \approx m \Delta k(0) e^{\lambda_- t}. \tag{320}$$

To halve the absolute deviation, solve

$$e^{\lambda_- T_{1/2}} = \tfrac{1}{2}, \tag{321}$$

$$\lambda_- T_{1/2} = -\log 2, \tag{322}$$

$$T_{1/2} = \frac{\log 2}{-\lambda_-}. \tag{323}$$

The positive convergence speed is $-\lambda_-$. The exponential solution is exact for the linearized system, and generally approximate for the original nonlinear transition.

Comparing convergence in Solow and Ramsey. Both local stable paths have exponential decay in effective-worker capital:

$$\text{Solow:} \quad k(t) - k^* \approx [k(0) - k^*] e^{-\lambda_C t},$$
$$\text{Ramsey:} \quad k(t) - k^* \approx [k(0) - k^*] e^{\lambda_- t}.$$

Here $\lambda_C = b - s f'(k^*) > 0$ is defined in (143), while $\lambda_- < 0$ is Ramsey's stable eigenvalue. The positive speeds are therefore λ_C and $-\lambda_-$, and a negative stable eigenvalue is not a negative convergence speed. Each steady state belongs to its own model. In Ramsey, initial consumption must also lie on the stable arm; arbitrary consumption does not produce this convergent path. These are local approximations to the nonlinear models.

5.5.2 The same restriction by undetermined coefficients

Recover the original second-order route. In this calculation all equalities refer to the linearized system. Differentiate its first equation and substitute its second:

$$\dot{\Delta k} = \eta \Delta k - \Delta c, \tag{324}$$

$$\ddot{\Delta k} = \eta \dot{\Delta k} - \dot{\Delta c}, \tag{325}$$

$$= \eta \dot{\Delta k} - z \Delta k, \tag{326}$$

$$\ddot{\Delta k} - \eta \dot{\Delta k} + z \Delta k = 0. \tag{327}$$

Conjecture a nonzero mode $\Delta k(t) = Be^{\lambda t}$. Its derivatives are $\dot{\Delta k} = \lambda B e^{\lambda t}$ and $\ddot{\Delta k} = \lambda^2 B e^{\lambda t}$. Substitution gives

$$\lambda^2 B e^{\lambda t} - \eta \lambda B e^{\lambda t} + z B e^{\lambda t} = 0, \tag{328}$$

$$B e^{\lambda t} (\lambda^2 - \eta \lambda + z) = 0, \tag{329}$$

$$\lambda^2 - \eta \lambda + z = 0. \tag{330}$$

The roots are exactly λ_- and λ_+ above. Keeping only the decaying mode and using $k(0)$ gives $k(t) = k^* + [k(0) - k^*]e^{\lambda_- t}$ for the linearized transition.

The equivalent policy-slope route. Alternatively, conjecture a local policy $\Delta c = m \Delta k$. From the first linearized equation, $\dot{\Delta k} = (\eta - m)\Delta k$. Differentiating the conjecture and using the second equation gives

$$\begin{aligned} \dot{\Delta c} = m \dot{\Delta k} = m(\eta - m)\Delta k = z \Delta k, \\ m(\eta - m) = z, \\ m\eta - m^2 - z = 0, \\ m^2 - \eta m + z = 0. \end{aligned} \tag{331}$$

Choose the positive root $m = \lambda_+$, so $\eta - m = \lambda_- < 0$. This recovers the same stable policy. Choosing the decaying direction is not a free change to initial capital; it is the restriction on the consumption choice implied by admissibility and optimality.

5.6 Maximizing welfare

5.6.1 The object being maximized is a consumption path

A higher future consumption level is attractive, but reaching it may require less consumption today. Welfare makes that trade-off explicit. Compare two feasible paths starting from the same $k(0)$, with the same technology, population, preferences, and information. In effective-worker units, the allocation-dependent objective is

$$W[c] = \int_0^\infty e^{-\eta t} u(c(t))\, dt. \tag{332}$$

For CRRA utility this differs from the original dynasty objective only by a positive constant factor; for log utility there is also an additive term independent of the allocation. Thus the ordering of feasible paths is the same.

The Euler equation balances utility across adjacent dates. Feasibility limits how much consumption can be financed, and the transversality condition rules out leaving valuable resources unused forever.

Together, under the concavity conditions of this chapter, they select the welfare-maximizing path. An Euler equation by itself does not establish that a proposed path is optimal.

5.6.2 Golden Rule versus the optimal steady state

The Golden Rule answers a narrower question: which constant k supports the highest steady consumption? With $b = n + g + \delta > 0$,

$$\bar{c}(k) = f(k) - bk, \qquad \bar{c}'(k) = f'(k) - b, \qquad f'(k_{\mathrm{GR}}) = b. \tag{333}$$

The Ramsey steady state instead satisfies

$$f'(k^*) = \delta + \rho + \theta g = b + \eta > b, \qquad k^* < k_{\mathrm{GR}}. \tag{334}$$

Diminishing marginal product gives the last inequality. If $b = 0$, there is no finite interior Golden Rule stock under $f'(k) > 0$; the Ramsey steady-state condition can still determine a finite k^*.

Higher eventual consumption need not mean higher welfare. Why stop before the consumption-maximizing stock? Building extra capital requires forgone consumption during the transition. The Golden Rule compares steady states without pricing that sacrifice. Ramsey values the sacrifice and the later benefit along a common time path. Starting at k^*, a policy that forces more saving may eventually raise consumption, yet reduce welfare. The later gain alone does not settle the comparison.

This also clarifies the benchmark's policy meaning. With no externalities, market power, borrowing frictions, or tax wedges, the competitive allocation implements the planner's optimum. That statement uses identical dynasties and the specified welfare weights; it does not settle distributional questions when households differ.

5.6.3 Measuring a welfare difference in consumption units

A difference in utility units is hard to interpret. Instead, ask what permanent proportional change to baseline consumption would make a household indifferent between the baseline path c_B and an alternative c_A. Define λ by

$$W[(1+\lambda)c_B] = W[c_A]. \tag{335}$$

A positive λ means the alternative is better. This is a welfare index; the scaled baseline need not itself be feasible under the original resources.

For $\theta \neq 1$, let $J_i = \int_0^\infty e^{-\eta t} c_i(t)^{1-\theta}\, dt$, for $i = A, B$. Substitution into CRRA utility gives

$$W[(1+\lambda)c_B] = \frac{1}{1-\theta}\int_0^\infty e^{-\eta t}[(1+\lambda)c_B(t)]^{1-\theta}\, dt, \tag{336}$$

$$= \frac{(1+\lambda)^{1-\theta}}{1-\theta} J_B = \frac{1}{1-\theta} J_A, \tag{337}$$

$$(1+\lambda)^{1-\theta} = J_A/J_B, \tag{338}$$

$$1 + \lambda = (J_A/J_B)^{1/(1-\theta)}, \tag{339}$$

$$\lambda = (J_A/J_B)^{1/(1-\theta)} - 1. \tag{340}$$

For $\theta = 1$, use $\int_0^\infty e^{-\eta t} dt = 1/\eta$ to obtain

$$W[(1+\lambda)c_B] = \int_0^\infty e^{-\eta t}[\log(1+\lambda) + \log c_B(t)]\, dt, \tag{341}$$

$$= \frac{\log(1+\lambda)}{\eta} + W[c_B] = W[c_A], \tag{342}$$

$$\log(1+\lambda) = \eta\{W[c_A] - W[c_B]\}, \tag{343}$$

$$\lambda = \exp(\eta\{W[c_A] - W[c_B]\}) - 1. \tag{344}$$

For example, if consumption is 1% higher at every date, both formulas give $\lambda = 0.01$. A 1% difference only in eventual steady consumption generally does not: the transition also enters the integral.

5.7 Taxation and capital accumulation

5.7.1 Separate financial transfers from real resource use

A tax can change who receives income, how many goods remain for private use, and the return to saving. These are different mechanisms. To separate them, introduce a balanced-budget government without debt. Let T_L be aggregate lump-sum taxes, $\mathscr{R}$ aggregate transfers, and G government purchases. A constant rate $0 \le \tau < 1$ taxes *net* capital income rK, where $r = f'(k) - \delta$. Depreciation is deductible under this convention.

The household budget and government budget are

$$\dot{B} = (1-\tau)rB + wL - C - T_L + \mathscr{R}, \tag{345}$$

$$\tau rK + T_L = G + \mathscr{R}. \tag{346}$$

In equilibrium $B = K$ and $rK + wL = Y - \delta K$. Substitute the government budget into the household budget:

$$\mathscr{R} = \tau rK + T_L - G, \tag{347}$$

$$\dot{K} = (1-\tau)rK + wL - C - T_L + (\tau rK + T_L - G), \tag{348}$$

$$= rK - \tau rK + wL - C - T_L + \tau rK + T_L - G, \tag{349}$$

$$= rK + wL - C - G = Y - \delta K - C - G, \tag{350}$$

$$\dot{k} = \frac{\dot{K}}{AL} - (n+g)k, \tag{351}$$

$$= f(k) - \delta k - c - \mathscr{G} - (n+g)k. \tag{352}$$

$$\dot{K} = Y - \delta K - C - G, \qquad \dot{k} = f(k) - c - bk - \mathscr{G}, \quad \mathscr{G} \equiv \frac{G}{AL}. \tag{353}$$

Taxes and transfers cancel when consolidating budgets. Government purchases use goods and therefore remain. Here G does not enter production or household utility; public investment or valued public services would require an explicit additional mechanism.

5.7.2 Lump-sum taxation: a resource and wealth effect

Set $\tau = 0$. A lump-sum tax does not depend on an individual household's saving, so it leaves the marginal return unchanged:

$$\frac{\dot{c}}{c} = \frac{f'(k) - \delta - \rho - \theta g}{\theta}. \tag{354}$$

This does not mean that financing G has no effect. Private consumption and investment must share fewer resources, and households adjust spending when they learn about future taxes and purchases.

For constant $\mathscr{G}$, an interior steady state has the same k^* as the untaxed benchmark, but

$$c_G^* = f(k^*) - bk^* - \mathscr{G} > 0. \tag{355}$$

The unchanged long-run stock follows from the unchanged Euler condition; consumption absorbs the steady resource cost. The transition can still differ. If $G = 0$ and lump-sum taxes are returned to the same identical dynasties, the net payment cancels and the benchmark allocation is unchanged.

5.7.3 Capital income taxation: the private saving wedge

Now set $T_L = 0$ and rebate capital tax revenue as lump-sum transfers, with $G = 0$. There are many identical price-taking dynasties. Each takes the transfer as given: saving an extra unit does not give that household the entire increase in aggregate tax rebates. Its marginal return is $(1 - \tau)r$. To obtain the household Euler condition, use consumption per person x and private assets per person a_h. The private budget has $\dot{a}_h = [(1 - \tau)r - n]a_h + w - x + \mathscr{R}/L$. The dynasty discounts per-person utility at $\rho - n$. With marginal value $\nu = u'(x) = x^{-\theta}$, its costate condition gives

$$\frac{\dot{\nu}}{\nu} = (\rho - n) - [(1 - \tau)r - n] = \rho - (1 - \tau)r, \tag{356}$$

$$\dot{\nu} = -\theta x^{-\theta-1}\dot{x}, \tag{357}$$

$$-\theta\frac{\dot{x}}{x} = \rho - (1 - \tau)r, \tag{358}$$

$$\frac{\dot{x}}{x} = \frac{(1 - \tau)r - \rho}{\theta}. \tag{359}$$

Since $x = Ac$, $\dot{c}/c = \dot{x}/x - g$. Substitute $r = f'(k) - \delta$:

$$\frac{\dot{x}}{x} = \frac{(1 - \tau)r - \rho}{\theta}, \tag{360}$$

$$\frac{\dot{c}}{c} = \frac{(1 - \tau)[f'(k) - \delta] - \rho - \theta g}{\theta}. \tag{361}$$

The consolidated resource constraint is still the original one. The tax changes the private incentive even when its revenue is fully rebated. Treating the representative household as if it chose aggregate rebates would erase this wedge and would solve a different problem.

For a permanent constant tax, the interior steady state satisfies

$$0 = (1 - \tau)[f'(k_\tau) - \delta] - \rho - \theta g, \tag{362}$$

$$(1 - \tau)[f'(k_\tau) - \delta] = \rho + \theta g, \tag{363}$$

$$f'(k_\tau) - \delta = \frac{\rho + \theta g}{1 - \tau}. \tag{364}$$

$$f'(k_\tau) = \delta + \frac{\rho + \theta g}{1 - \tau}, \qquad c_\tau = f(k_\tau) - bk_\tau. \tag{365}$$

Differentiate the first equation:

$$f''(k_\tau)\frac{dk_\tau}{d\tau} = \frac{\rho + \theta g}{(1-\tau)^2}, \qquad \frac{dk_\tau}{d\tau} < 0. \tag{366}$$

Explicitly, division by the negative marginal-product derivative gives

$$\frac{dk_\tau}{d\tau} = \frac{\rho + \theta g}{(1-\tau)^2 f''(k_\tau)} < 0.$$

A lower after-tax share requires a higher pre-tax marginal product to support zero growth of c. With diminishing returns, that means less capital. Since $k_\tau < k^* < k_{\mathrm{GR}}$ when $b > 0$, steady private consumption also falls. For $b = 0$, $f(k)$ is increasing and the same consumption ranking follows. Per-person output still grows at g on the balanced growth path: this tax changes its level, while technology growth remains exogenous.

The tax base matters. A tax on gross rental income $f'(k)K$, without depreciation deductibility, would leave the net return $(1-\tau)f'(k) - \delta$ instead. Its steady-state formula would therefore differ when $\delta > 0$.

5.7.4 The transition need not resemble the long-run comparison

A tax can raise consumption initially and lower it eventually. Suppose the economy starts at the untaxed steady state and an unexpected small permanent capital tax is introduced, with full rebates and $G = 0$. The $\dot{c} = 0$ line shifts left, while the $\dot{k} = 0$ curve is unchanged. Capital cannot jump. Consumption can, and locally it jumps *up*: households now want a smaller eventual capital stock and begin consuming part of the inherited wealth. Then capital and consumption decline along the new stable path. The initial consumption increase is consistent with lower long-run consumption.

To see the local logic, write the new stable path as $c - c_\tau \approx m_\tau(k - k_\tau)$, with $m_\tau > 0$. At zero tax its slope is $m = \eta - \lambda_- > \eta$, whereas the resource curve has slope η. A small fall in the target stock therefore places the new stable path above the old steady-state consumption at the inherited k^*. This argument is local and uses the stated rebate rule; government spending, announcements, or other tax bases require their own transition analysis. Write $\Delta k_\tau = k_\tau - k^* < 0$ for a small tax. Expanding the resource curve and the new local policy around zero tax makes the sign calculation explicit:

$$c_\tau - c^* = [f'(k^*) - b]\Delta k_\tau + O(\tau^2) = \eta \Delta k_\tau + O(\tau^2), \tag{367}$$

$$c(0^+) = c_\tau + m_\tau(k^* - k_\tau) + O(\tau^2), \tag{368}$$

$$c(0^+) - c^* = \eta \Delta k_\tau - m\Delta k_\tau + O(\tau^2), \tag{369}$$

$$= (\eta - m)\Delta k_\tau + O(\tau^2) = \lambda_- \Delta k_\tau + O(\tau^2) > 0. \tag{370}$$

Here $m_\tau = m + O(\tau)$, so multiplying its change by $\Delta k_\tau = O(\tau)$ contributes only at second order. Both λ_- and the first-order Δk_τ are negative.

For a numerical level comparison, take $f(k) = k^{1/3}$, $\delta = 0$, $\rho = 0.04$, $\theta = 2$, $n = 0.01$, and $g = 0.02$.

Then $b = 0.03$ and $\eta = 0.05$. The steady-state formulas give

$$f'(k) = \tfrac{1}{3}k^{-2/3} = q, \tag{371}$$

$$k^{-2/3} = 3q, \qquad k = (3q)^{-3/2}, \tag{372}$$

$$q_0 = 0.04 + 2(0.02) = 0.08, \tag{373}$$

$$q_{0.20} = \frac{0.04 + 2(0.02)}{1 - 0.20} = 0.10, \tag{374}$$

$$c = k^{1/3} - 0.03k. \tag{375}$$

Evaluating these expressions gives

Capital tax rate	**Required $f'(k)$**	k	c
0	0.08	8.505	1.786
0.20	0.10	6.086	1.643

These are hypothetical steady states. Their consumption difference is not a consumption-equivalent welfare loss, because that calculation must include the transition from a common initial stock.

In this frictionless benchmark, forcing a capital tax wedge with no valued public use of revenue moves the allocation away from the planner optimum. This is a comparison of given policies, not a solution for optimal fiscal policy. Redistribution, productive public expenditure, or market failures would change the policy problem and must be modeled before drawing a broader conclusion.

6 OLG Model

Ramsey assigns intertemporal choices to a dynasty that continues indefinitely. Now separate a person's lifetime from the economy's lifetime. People work, retire, and leave; new people arrive. Saving remains an optimal choice, but the ownership of assets and the identity of the people paying for transfers must be tracked by generation.

6.1 From retirement needs to overlapping generations

6.1.1 An economy continues after its current savers have died

Imagine looking at households' balance sheets at one date. Some people are building retirement assets from wages; others are spending assets accumulated earlier. A single consumption total conceals these opposite positions. A policy that taxes workers and pays retirees changes the resources of different people, even when it leaves aggregate resources unchanged at the moment of payment.

One response would be to add age labels to the Ramsey household. But if an altruistic dynasty still pools all resources and values every descendant, those labels need not change the underlying allocation problem. To isolate generational turnover, instead let each person value only consumption within his or her own finite life. Give the young labor income and the old a need to finance consumption without working. Then ask what assets can carry resources between these stages, and who supplies the counterpart to those claims.

Two periods are the smallest useful life cycle: one for work and saving, one for retirement and spending. New workers arrive while earlier workers retire. The approximation does not describe every age or predict a literal retirement age. It makes the relation between life-cycle saving, asset ownership, and population replacement visible with only two age groups.

6.1.2 Choose a way to finance retirement, then close the economy

Start with physical capital as the only asset. A young person acquires capital claims; next period those claims pay rental income and the value of surviving capital. Retirees consume the proceeds. A wage and a return determine an individual saving choice, but they are not arbitrary economy-wide inputs: workers' saving determines the next capital stock, which determines the return those same workers receive when old. This closes the market model.

An alternative is a transfer from next period's workers. That alternative cannot simply be assumed to be a private enforceable contract with people who have not yet been born. A pension system or a government issuing debt adds an institution capable of continuing across generations. We first solve the capital-only economy, then introduce those institutions explicitly.

The production economy follows Diamond's OLG framework; the sequence of questions above is a teaching reconstruction, not a claim about the author's private thoughts.[13]

[13] Peter A. Diamond (1965), "National Debt in a Neoclassical Growth Model," *American Economic Review*, 55(5), 1126–1150. Diamond's original model uses no depreciation; we keep general δ and identify the full-depreciation course example separately. `https://www.aeaweb.org/aer/top20/55.5.1126-1150.pdf`.

Meeting only once does not rule out intertemporal trade. A young person can save in capital and consume its payoff when old. That is already an intertemporal allocation. The important limitation is that currently living people cannot make ordinary private contracts with unborn generations. Without bequests or altruistic links, no household represents the interests of the infinite sequence of future entrants. Finite individual lives and an infinite economic horizon can therefore have different welfare implications from a single infinite-lived dynasty.

6.2 Assumptions and the cohort timeline

The cards state the full environment before solving it. Within a cohort people are identical; across ages, their income, remaining lifetime, and asset position differ. There is no single representative person for both age groups.

01 Dates, cohorts, and population

Time is discrete, $t = 0, 1, \ldots$. A cohort of L_t people is born at each date, lives for two periods with certainty, and dies after its second period. At date t, L_t young coexist with L_{t-1} old:

$$L_t = NL_{t-1}, \qquad N \equiv 1 + n > 0, \qquad L_t = N^t L_0.$$

The baseline takes $n \geq 0$; the population comparison later also permits $-1 < n < 0$ when the other conditions remain feasible. Each young person supplies one unit of labor inelastically; the old supply none. Labor is L_t, whereas total population is $L_t + L_{t-1}$. A model period is a life stage, not necessarily a calendar year.

02 Goods, capital, and technology growth

A closed economy produces one good, usable for consumption and capital. There is no government, foreign trade, money, or public debt in the baseline. Technology is fixed and normalized to one: $Y_t = F(K_t, L_t)$. This is an explicit zero-technology-growth benchmark, so $k_t = K_t/L_t$ is capital per worker, not per effective worker. Capital depreciates at $0 \leq \delta \leq 1$ and surviving capital can be sold or consumed:

$$Y_t + (1 - \delta)K_t = C_t + K_{t+1}.$$

General formulas retain δ. The numerical and explicit full-depreciation examples set $\delta = 1$; that restriction is not needed for the general model.

03 Production restrictions

Firms share a freely available technology. F is continuous, twice differentiable for positive inputs, positive there, with positive diminishing marginal products and constant returns:

$$F(\lambda K, \lambda L) = \lambda F(K, L), \qquad f(k) \equiv F(k, 1), \qquad Y_t / L_t = f(k_t).$$

For $k > 0$, assume

$$f(0) = 0, \qquad f'(k) > 0, \qquad f''(k) < 0,$$
$$\lim_{k \downarrow 0} f'(k) = \infty, \qquad \lim_{k \to \infty} f'(k) = 0.$$

Cobb–Douglas, $f(k) = k^{\alpha}$ with $0 < \alpha < 1$, is a specified example, not the only technology consistent with these conditions.

04 Households, ownership, and initial conditions

There are many identical, price-taking people in each cohort. They receive wages when young and asset income when old; firms belong to households and any profits would accrue to their owners. No one values descendants' utility, leaves a bequest, or receives an inheritance. The initially old are the exception to the no-inherited-wealth statement: they enter date zero already owning the given $K_0 > 0$, equally within their cohort. The young start with labor income and no assets. Initially old consumption must be counted even though their earlier saving decision lies outside the model's initial date.

05 Preferences, information, and choices

A person born at t chooses saving a_t and consumption at ages one and two:

$$U_t = u(c_{1,t}) + \beta u(c_{2,t+1}), \qquad 0 < \beta < 1,$$

$$u(c) = \begin{cases} (c^{1-\theta} - 1)/(1 - \theta), & \theta > 0,\ \theta \neq 1, \\ \log c, & \theta = 1. \end{cases}$$

There is no aggregate or individual uncertainty; the next return is correctly anticipated. People take wages and returns as given, despite their joint effect on equilibrium prices. There is no labor choice or liquidity shock. Consumption is positive. The baseline interior optimum has $0 < a_t < w_t$; no borrowing or short sales of physical capital are allowed. Later pension formulas must be checked against this asset constraint.

06 Firms, factor prices, and returns

Competitive firms rent capital and hire labor at each date:

$$\max_{K,L} F(K,L) - q_t K - w_t L.$$

Interior demands imply $q_t = f'(k_t)$ and $w_t = f(k_t) - k_t f'(k_t)$. Economic profits are zero under constant returns. The owner of a unit of capital receives the rental payment and its remaining capital value:

$$R_t = 1 - \delta + q_t, \qquad r_t = R_t - 1 = q_t - \delta.$$

R_t is a gross payoff, r_t a net return, and q_t a rental price. Only with $\delta = 1$ does $R_t = f'(k_t)$.

07 Budgets and age-specific consumption

A person born at t obeys

$$c_{1,t} + a_t = w_t, \qquad c_{2,t+1} = R_{t+1} a_t.$$

The old consume all their wealth and leave no assets after death. At date t, $c_{2,t}$ is the consumption of cohort $t-1$, not the future consumption of cohort t. Aggregate consumption and young people's asset purchases are

$$C_t = L_t c_{1,t} + L_{t-1} c_{2,t}, \qquad \mathscr{A}_t = L_t a_t.$$

We use a_t for saving per young person to distinguish it from Solow's saving share s. $\mathscr{A}_t$ corresponds to the original course notation for aggregate young saving; it is not automatically national saving.

08 Market clearing and the equilibrium path

Capital and labor demands equal their supplied stocks, $K_t^d = K_t$ and $L_t^d = L_t$. Capital is the only asset, so young people's wealth at the end of t must equal next period's capital:

$$K_{t+1} = \mathscr{A}_t = L_t a_t, \qquad N k_{t+1} = a_t.$$

Goods clearing must also hold:

$$c_{1,t} + \frac{c_{2,t}}{N} + N k_{t+1} = f(k_t) + (1-\delta) k_t.$$

An equilibrium is a path of allocations and prices satisfying household and firm optimality, correct return expectations, and these clearing conditions, starting from the specified initial ownership. No infinite-lived household's transversality condition is imposed on the economy as a whole.

6.2.1 Read a cohort's lifetime across dates

Cohort	Date $t-1$	Date t	Date $t+1$
Born at $t-1$	Young, saves	Old, consumes	No longer alive
Born at t	Not yet born	Young, saves	Old, consumes
Born at $t+1$	Not yet born	Not yet born	Young, saves

Reading across a row gives one person's lifetime budget. Reading down a column gives the economy's resources at one date. This distinction explains why $c_{1,t} + c_{2,t+1}/R_{t+1} = w_t$ is an individual present-value budget, whereas $c_{1,t} + c_{2,t}/N$ is current aggregate consumption per worker. The two formulas combine different people and different dates.

6.3 Household optimization and life-cycle saving

6.3.1 Budgets, multipliers, and the Euler condition

Take $w_t > 0$ and $R_{t+1} > 0$ as given. Monotonic utility makes the two budgets bind. With multipliers on resources remaining, write

$$\begin{aligned}\mathscr{L} &= u(c_{1,t}) + \beta u(c_{2,t+1}) \\ &\quad + \lambda_{1,t}(w_t - c_{1,t} - a_t) + \lambda_{2,t+1}(R_{t+1}a_t - c_{2,t+1}).\end{aligned} \tag{376}$$

At an interior optimum,

$$\frac{\partial \mathscr{L}}{\partial c_{1,t}} = u'(c_{1,t}) - \lambda_{1,t} = 0, \tag{377}$$

$$\frac{\partial \mathscr{L}}{\partial c_{2,t+1}} = \beta u'(c_{2,t+1}) - \lambda_{2,t+1} = 0, \tag{378}$$

$$\frac{\partial \mathscr{L}}{\partial a_t} = -\lambda_{1,t} + R_{t+1}\lambda_{2,t+1} = 0, \tag{379}$$

$$u'(c_{1,t}) = \lambda_{1,t}, \tag{380}$$

$$\beta u'(c_{2,t+1}) = \lambda_{2,t+1}, \tag{381}$$

$$\lambda_{1,t} = R_{t+1}\lambda_{2,t+1}. \tag{382}$$

Eliminating both multipliers gives

$$u'(c_{1,t}) = \lambda_{1,t} = R_{t+1}\lambda_{2,t+1} = \beta R_{t+1}u'(c_{2,t+1}).$$

$$u'(c_{1,t}) = \beta R_{t+1}u'(c_{2,t+1}). \tag{383}$$

Giving up one consumption unit while young buys R_{t+1} units when old. The Euler condition prices that marginal transfer. The two budgets still supply the levels. No infinite-horizon terminal condition is needed for this person: after age two, there is neither consumption nor a bequest to choose.

6.3.2 CRRA saving: an explicit solution and its derivatives

Under CRRA, substitute the budgets into (383):

$$(w_t - a_t)^{-\theta} = \beta R_{t+1}(R_{t+1}a_t)^{-\theta}, \tag{384}$$

$$\left(\frac{w_t - a_t}{a_t}\right)^{-\theta} = \beta R_{t+1}^{1-\theta}. \tag{385}$$

Define $h(R) \equiv \beta^{-1/\theta} R^{(\theta-1)/\theta}$. Taking powers and rearranging, step by step,

$$\left(\frac{w_t - a_t}{a_t}\right)^{-\theta} = \beta R_{t+1}^{1-\theta}, \tag{386}$$

$$\frac{w_t - a_t}{a_t} = (\beta R_{t+1}^{1-\theta})^{-1/\theta}, \tag{387}$$

$$= \beta^{-1/\theta} R_{t+1}^{-(1-\theta)/\theta} = h(R_{t+1}), \tag{388}$$

$$\frac{w_t}{a_t} - 1 = h(R_{t+1}), \tag{389}$$

$$w_t = a_t[1 + h(R_{t+1})]. \tag{390}$$

$$\frac{w_t - a_t}{a_t} = h(R_{t+1}), \qquad \frac{w_t}{a_t} = 1 + h(R_{t+1}), \tag{391}$$

so the saving function is

$$a_t = a(w_t, R_{t+1}) = \frac{w_t}{1 + \beta^{-1/\theta} R_{t+1}^{(\theta-1)/\theta}}. \tag{392}$$

The utility constant $-1/(1-\theta)$ has no effect on choices. The log case will be obtained directly rather than substituting $\theta = 1$ into that utility expression.

Holding the return fixed, differentiate only w. Holding the wage fixed, differentiate the denominator through $h(R)$:

$$a_w = \frac{\partial}{\partial w}[w(1 + h(R))^{-1}] = (1 + h(R))^{-1} \in (0, 1), \tag{393}$$

$$h'(R) = \beta^{-1/\theta} \frac{\theta - 1}{\theta} R^{(\theta-1)/\theta - 1}, \tag{394}$$

$$= \frac{\theta - 1}{\theta R}[\beta^{-1/\theta} R^{(\theta-1)/\theta}] = \frac{\theta - 1}{\theta R} h(R), \tag{395}$$

$$a_R = w[-(1 + h)^{-2}] h'(R), \tag{396}$$

$$= -\frac{w}{(1 + h)^2} \frac{\theta - 1}{\theta R} h. \tag{397}$$

$$a_R = -\frac{w h'(R)}{(1 + h)^2} = \frac{1 - \theta}{\theta} \frac{wh}{R(1 + h)^2}. \tag{398}$$

Thus a_R is positive for $\theta < 1$, zero for $\theta = 1$, and negative for $\theta > 1$. These are partial derivatives; an equilibrium price change may also change wages and expected future returns together.

A higher return need not raise retirement saving. A higher return makes postponing consumption more rewarding, but it also allows a given retirement expenditure to be financed with fewer assets. For $\theta < 1$, willingness to substitute consumption across ages is strong and saving rises. For $\theta > 1$, the income effect dominates in this retirement-only-income specification and saving falls. Under log utility, these effects cancel in the saving share out of wages. The result depends on the assumed income profile; it is not a universal empirical saving law.

6.3.3 The log case: a constant share of wages

For $u(c) = \log c$, (383) implies

$$\frac{c_{2,t+1}}{c_{1,t}} = \beta R_{t+1}. \tag{399}$$

Substitute $c_{1,t} = w_t - a_t$ and $c_{2,t+1} = R_{t+1}a_t$:

$$R_{t+1}a_t = \beta R_{t+1}(w_t - a_t), \tag{400}$$

$$a_t = \beta w_t - \beta a_t \qquad (R_{t+1} > 0), \tag{401}$$

$$(1+\beta)a_t = \beta w_t, \tag{402}$$

$$a_t = \frac{\beta}{1+\beta} w_t. \tag{403}$$

$$R_{t+1}a_t = \beta R_{t+1}(w_t - a_t) \quad \Longrightarrow \quad a_t = \kappa w_t, \qquad \kappa \equiv \frac{\beta}{1+\beta}. \tag{404}$$

Saving is endogenous even though this example gives a constant share of *wages*. Solow imposes a share of total output; those are different income bases. Log utility does not make retirement consumption independent of the return: $c_{2,t+1} = R_{t+1}\kappa w_t$ still changes with R_{t+1}.

6.4 Firm optimization and the return to ownership

At date t, firms choose K, L in

$$\max_{K,L} \; F(K,L) - q_t K - w_t L. \tag{405}$$

Differentiate with respect to actual factor inputs before using constant returns to express the prices per worker:

$$F(K,L) = Lf(K/L) = Lf(k), \tag{406}$$

$$\pi_K = Lf'(k)(1/L) - q_t = f'(k) - q_t = 0, \tag{407}$$

$$\pi_L = f(k) + Lf'(k)(-K/L^2) - w_t = 0, \tag{408}$$

$$q_t = f'(k_t), \qquad w_t = f(k_t) - k_t f'(k_t), \tag{409}$$

$$R_t = (1-\delta) + q_t. \tag{410}$$

$$q_t = f'(k_t), \qquad w_t = f(k_t) - k_t f'(k_t), \qquad R_t = 1 - \delta + f'(k_t). \tag{411}$$

Euler's theorem gives $Y_t = q_t K_t + w_t L_t$, so profits are zero after factor payments. Capital owners still receive both rental income and the value of surviving assets. That second item is an asset sale or liquidation, not additional production in GDP.

For Cobb–Douglas,

$$q_t = \alpha k_t^{\alpha-1}, \qquad w_t = (1-\alpha)k_t^{\alpha}, \qquad R_t = 1 - \delta + \alpha k_t^{\alpha-1}. \tag{412}$$

A larger inherited capital intensity raises the current wage and lowers the current return. A young saver cares about the *next* return, determined by k_{t+1}, because that is when the retirement payoff is received.

The course formula $R = f'(k)$ is a full-depreciation case. With $\delta = 1$, no capital survives after production, so the gross payoff is the rental product alone. With $\delta = 0$, the owner also recovers the unit of capital and $R = 1 + f'(k)$. Both conventions can be useful. The household Euler equation uses the gross payoff; replacing it by a rental price without stating depreciation changes the saving problem.

6.5 Capital accumulation and equilibrium dynamics

6.5.1 Young wealth versus aggregate saving

At the start of date t, the old own K_t. Since $K_t = L_{t-1}a_{t-1}$, their aggregate consumption is $L_{t-1}c_{2,t} = R_tK_t$; the same relation holds for the initial owners at date zero. The young consume $L_t(w_t - a_t)$. Add them:

$$C_t = L_tw_t - L_ta_t + R_tK_t, \tag{413}$$

$$= Y_t - L_ta_t + (1-\delta)K_t. \tag{414}$$

In the second line, substitute $R_t = q_t + 1 - \delta$ and $Y_t = w_tL_t + q_tK_t$. Rearranging gives

$$Y_t - C_t = L_ta_t - (1-\delta)K_t, \tag{415}$$

$$K_{t+1} = (1-\delta)K_t + (Y_t - C_t), \tag{416}$$

$$= (1-\delta)K_t + L_ta_t - (1-\delta)K_t = L_ta_t, \tag{417}$$

$$\frac{K_{t+1}}{L_{t+1}}\frac{L_{t+1}}{L_t} = \frac{L_ta_t}{L_t}, \tag{418}$$

$$Nk_{t+1} = a_t. \tag{419}$$

$$K_{t+1} = L_ta_t, \qquad Nk_{t+1} = a_t. \tag{420}$$

Thus the asset-clearing equation also satisfies goods clearing.

Young saving is not national saving. The young acquire assets, while the old liquidate theirs. In this one-good model, gross investment and net national saving are respectively

$$I_t \equiv K_{t+1} - (1-\delta)K_t = L_t a_t - (1-\delta)K_t,$$

$$S_t^{\text{net}} \equiv Y_t - C_t - \delta K_t = K_{t+1} - K_t = L_t a_t - K_t.$$

Writing $K_{t+1} = (1-\delta)K_t + L_t a_t$ would count surviving capital twice: young people's purchases already include acquiring that capital from the old. With full depreciation, gross investment does equal young saving; with general depreciation, the distinction must remain visible.

6.5.2 An implicit law of motion under general preferences

Substitute the saving function and competitive prices into capital clearing:

$$N k_{t+1} = a\big(f(k_t) - k_t f'(k_t),\ 1 - \delta + f'(k_{t+1})\big)\,. \tag{421}$$

This is generally implicit in k_{t+1}: a larger capital stock changes the return that induced people to accumulate it. Solving household saving at an arbitrary fixed return is not yet a general-equilibrium solution.

At a positive stationary equilibrium,

$$N k^* = a\big(f(k^*) - k^* f'(k^*),\ 1 - \delta + f'(k^*)\big)\,. \tag{422}$$

Cobb–Douglas makes this condition

$$N k^* = \frac{(1-\alpha)(k^*)^\alpha}{1 + \beta^{-1/\theta}\big[1 - \delta + \alpha (k^*)^{\alpha-1}\big]^{(\theta-1)/\theta}}. \tag{423}$$

This scalar equation replaces the unfinished general steady-state expression in the course notes. A numerical root may be needed; the explicit log result below is a separate tractable case.

To see why stability needs conditions, differentiate (421). Since $w'(k) = -k f''(k) > 0$,

$$w'(k) = f'(k) - \big[f'(k) + k f''(k)\big] = -k f''(k), \tag{424}$$

$$N \frac{dk_{t+1}}{dk_t} = a_w w'(k_t) + a_R R'(k_{t+1}) \frac{dk_{t+1}}{dk_t}, \tag{425}$$

$$= a_w\big[-k_t f''(k_t)\big] + a_R f''(k_{t+1}) \frac{dk_{t+1}}{dk_t}, \tag{426}$$

$$\big[N - a_R f''(k_{t+1})\big] \frac{dk_{t+1}}{dk_t} = a_w\big[-k_t f''(k_t)\big]. \tag{427}$$

$$\big[N - a_R f''(k_{t+1})\big] \frac{dk_{t+1}}{dk_t} = a_w\big[-k_t f''(k_t)\big]. \tag{428}$$

If the bracket is nonzero, this gives the local slope. A locally defined map is stable at its steady state when the absolute slope is below one. For $\theta \le 1$, $a_R \ge 0$ makes the bracket positive. For $\theta > 1$, the return response can overturn that sign or make the equation locally singular. The log example's uniqueness and convergence cannot be asserted for every OLG specification.

6.5.3 Log utility and Cobb–Douglas: solve the whole path

Using (404), capital clearing reduces to

$$k_{t+1} = \frac{\kappa(1-\alpha)}{N} k_t^{\alpha} \equiv M k_t^{\alpha}, \qquad M \equiv \frac{\kappa(1-\alpha)}{N} > 0. \tag{429}$$

This formula holds for general δ: log saving does not respond to the return, and current wages do not depend on depreciation. Depreciation still changes the old's consumption and welfare.

At a positive steady state, divide $k^* = M(k^*)^{\alpha}$ by $(k^*)^{\alpha}$:

$$\frac{k^*}{(k^*)^{\alpha}} = M, \tag{430}$$

$$(k^*)^{1-\alpha} = M, \tag{431}$$

$$k^* = M^{1/(1-\alpha)}, \tag{432}$$

$$M = \frac{\kappa(1-\alpha)}{N} = \frac{\beta(1-\alpha)}{(1+\beta)N}. \tag{433}$$

$$(k^*)^{1-\alpha} = M, \qquad k^* = \left[\frac{\beta(1-\alpha)}{(1+\beta)N}\right]^{1/(1-\alpha)}. \tag{434}$$

The map is increasing; $k_{t+1}/k_t = M k_t^{\alpha-1}$ exceeds one below k^* and is below one above it. It maps either side toward the steady state without crossing it. Thus every $k_0 > 0$ converges monotonically; zero is an absorbing boundary, not the initial condition of the positive-capital result.

There is also an exact solution in log deviations. Divide the map by its steady-state version and take logarithms:

$$\frac{k_{t+1}}{k^*} = \frac{M k_t^{\alpha}}{M(k^*)^{\alpha}} = \left(\frac{k_t}{k^*}\right)^{\alpha}, \tag{435}$$

$$\log(k_{t+1}/k^*) = \alpha \log(k_t/k^*), \tag{436}$$

$$\log(k_t/k^*) = \alpha^t \log(k_0/k^*), \tag{437}$$

$$k_t/k^* = \exp[\alpha^t \log(k_0/k^*)] = (k_0/k^*)^{\alpha^t}. \tag{438}$$

$$\log(k_{t+1}/k^*) = \alpha \log(k_t/k^*), \qquad k_t = k^* \left(\frac{k_0}{k^*}\right)^{\alpha^t}. \tag{439}$$

The local slope is $\alpha \in (0,1)$. Capital-level deviations satisfy $k_t - k^* \approx \alpha^t (k_0 - k^*)$ only locally; log deviations decay exactly at that factor. This separates convergence of a path from efficiency of the allocation to which it converges.

What the log example does and does not establish. Young people optimally save a fixed share of wages. Wages depend on the inherited stock; their asset purchases become the next stock. This gives a unique positive steady state and a stable path in this example. It does not show that all OLG economies converge, that capital is socially optimal, or that a household's constant saving share equals Solow's imposed output share.

6.6 Comparing Solow, Ramsey, and OLG

Use a common discrete-time benchmark here: fixed technology, population factor N, and depreciation δ. Let σ denote Solow's imposed saving share to keep it distinct from individual saving a_t.

	Solow	**Ramsey dynasty**	**Two-period OLG**
Saving	Given share of output	Chosen for a continuing dynasty	Chosen for each cohort's two ages
People	No age structure in the saving rule	Descendants linked by the objective	New cohorts enter; no altruistic link
Main closure	Accumulation equation	Accumulation, Euler, and terminal conditions	Cohort budgets and asset clearing
Efficiency	Not guaranteed by the saving rule	First-best under its stated assumptions	Overaccumulation is possible

The corresponding aggregate resource and accumulation conditions are

$$\text{Solow:} \quad Nk_{t+1} = (1-\delta)k_t + \sigma f(k_t), \tag{440}$$

$$\text{Ramsey:} \quad Nk_{t+1} = f(k_t) + (1-\delta)k_t - c_t, \tag{441}$$

$$\text{OLG:} \quad Nk_{t+1} = a(w_t, R_{t+1}) \tag{442}$$

$$= f(k_t) + (1-\delta)k_t - c_{1,t} - c_{2,t}/N. \tag{443}$$

In the Ramsey row, c_t is aggregate dynasty consumption per worker. Its population-weighted objective in this discrete convention gives the dynasty Euler equation $u'(c_t) = \beta R_{t+1} u'(c_{t+1})$, together with an appropriate infinite-horizon terminal condition. The OLG Euler equation instead links two ages of one person. The equations look similar but have different coverage of future generations.

> **The common resource constraint does not make the models identical.** In OLG the old spend asset income and the young buy assets, so clearing gives $Nk_{t+1} = a_t$. In Ramsey the continuing household chooses its next wealth while valuing its members' future consumption. Solow specifies saving without that optimization. The distinction is who makes the choice and whose future is included, not whether one model respects accounting and another does not.

6.7 Dynamic efficiency and intergenerational welfare

6.7.1 Count consumption at a date before comparing steady states

Divide goods clearing by L_t:

$$c_{1,t} + \frac{c_{2,t}}{N} + Nk_{t+1} = f(k_t) + (1-\delta)k_t. \tag{444}$$

The factor $1/N = L_{t-1}/L_t$ counts retirees relative to workers. Define $\bar{c}_t \equiv C_t/L_t = c_{1,t} + c_{2,t}/N$. This is consumption per worker, not consumption per resident; the latter is $\bar{c}_t/(1+1/N)$. At a stationary capital intensity,

$$\bar{c}(k) = f(k) - (n+\delta)k. \tag{445}$$

When $n + \delta > 0$, steady consumption per worker is maximized at

$$f'(k_{\mathrm{GR}}) = n + \delta, \qquad R_{\mathrm{GR}} = 1 - \delta + f'(k_{\mathrm{GR}}) = N. \tag{446}$$

For Cobb–Douglas, $k_{\mathrm{GR}} = [\alpha/(n+\delta)]^{1/(1-\alpha)}$. If $n + \delta \leq 0$, there is no finite interior maximum under $f'(k) > 0$; the positive steady-state overaccumulation comparison below then does not apply.

The course condition $f'(k_{\mathrm{GR}}) = 1 + n$ is exactly (446) with $\delta = 1$. The comparable net-return statement is $r_{\mathrm{GR}} = n$, not $R_{\mathrm{GR}} = n$.

6.7.2 Why overaccumulation can make every generation worse off

Suppose a competitive steady state satisfies $k^* > k_{\mathrm{GR}}$. Diminishing marginal product makes this equivalent to

$$f'(k^*) < n + \delta \quad \Longleftrightarrow \quad R^* < N \quad \Longleftrightarrow \quad r^* < n. \tag{447}$$

The economy gives up so many goods to maintain its capital stock that a slightly smaller stock could support more consumption indefinitely. To show why this is more than a comparison of eventual outcomes, construct a feasible change from the same initial capital.

Question: can consuming more today avoid a sacrifice tomorrow? Start at $k_0 = k^* > k_{\mathrm{GR}}$. Choose $k^a = k^* - \varepsilon$ with $k_{\mathrm{GR}} < k^a < k^*$, and set the next and all later capital intensities to k^a. At date zero, inherited output and capital are unchanged, while required next capital falls by $N\varepsilon$ per worker. Hence

$$\bar{c}_0^a - \bar{c}^* = N\varepsilon > 0.$$

At all later dates,

$$\bar{c}_t^a - \bar{c}^* = \left[f(k^a) - (n+\delta)k^a\right] - \left[f(k^*) - (n+\delta)k^*\right] > 0.$$

The inequality follows because $\bar{c}(k)$ decreases to the right of k_{GR}. At each date, a positive gain $\Delta\bar{c}_t$ can be split as $\Delta c_{1,t} = \Delta\bar{c}_t/2$ and $\Delta c_{2,t} = N\Delta\bar{c}_t/2$. Both currently living age groups gain, and every future cohort gains at both ages. This is a feasible Pareto improvement, including for the initially old, rather than a claim that higher future consumption compensates losers today.

The construction changes allocations through a coordinated transfer and investment policy; it is not the action of one price-taking saver. At given prices, each person's saving is optimal. Yet no person chooses on behalf of the infinite sequence of entrants. Competitive optimization within each finite life therefore does not impose the dynasty terminal restriction that rules out this overaccumulation in the Ramsey benchmark. An overlapping-generations structure permits inefficiency; it does not make every stationary equilibrium inefficient.

6.7.3 An explicit log example and the three-model comparison

In the full-depreciation log/Cobb–Douglas example, substitute (434) into the return:

$$(k^*)^{1-\alpha} = \frac{\kappa(1-\alpha)}{N}, \tag{448}$$

$$(k^*)^{\alpha-1} = \frac{N}{\kappa(1-\alpha)}, \tag{449}$$

$$R^* = \frac{\alpha N}{\kappa(1-\alpha)}. \tag{450}$$

All quantities divided below are positive. Thus

$$R^* < N \iff \frac{\alpha}{\kappa(1-\alpha)} < 1, \tag{451}$$

$$\iff \alpha < \kappa(1-\alpha), \tag{452}$$

$$\iff \alpha(1+\beta) < \beta(1-\alpha), \tag{453}$$

$$\iff \alpha + \alpha\beta < \beta - \alpha\beta, \tag{454}$$

$$\iff \alpha(1+2\beta) < \beta, \tag{455}$$

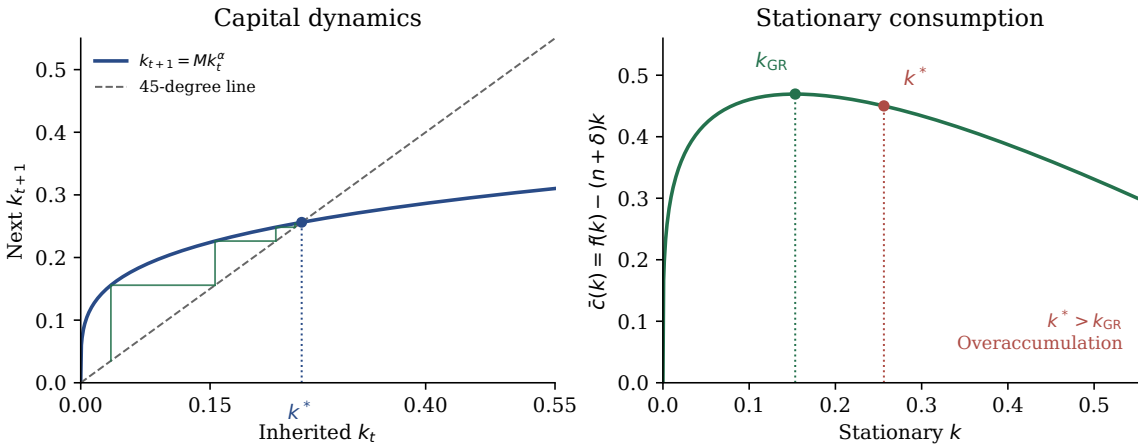
$$\iff \alpha < \frac{\beta}{1+2\beta}. \tag{456}$$

$$R^* = \alpha(k^*)^{\alpha-1} = \frac{\alpha N}{\kappa(1-\alpha)}, \tag{457}$$

$$R^* < N \iff \alpha < \kappa(1-\alpha) \iff \alpha < \frac{\beta}{1+2\beta}. \tag{458}$$

Both efficient and overaccumulating cases are possible by changing the parameters. This is not an automatic consequence of finite lifetimes.

For illustration, take $\alpha = 0.25$, $\beta = 0.96$, $N = 1.02$, and $\delta = 1$. The formulas give $k^* \simeq 0.2562$, $k_{\mathrm{GR}} \simeq 0.1534$, and $R^* \simeq 0.6942 < N$. Steady consumption per worker is about 0.4501 at k^*, versus 0.4694 at the Golden Rule stock. These are hypothetical life-stage parameters, not annual estimates or a calibration to a particular pension system. The Pareto argument above supplies the transition comparison; the two steady consumption numbers alone would not establish it.



A stable path can lead to overaccumulation. The left panel plots (429) against the 45-degree line. The right plots $\bar{c}(k) = k^{\alpha} - Nk$ under full depreciation, with k^* to the right of its consumption-maximizing stock. Stability and welfare answer different questions. Both panels use the illustrative parameters stated above.

For Solow with fixed technology and $n + \delta > 0$, the Cobb–Douglas stocks are

$$k_S^* = \left(\frac{\sigma}{n+\delta}\right)^{1/(1-\alpha)}, \qquad k_{\mathrm{GR}} = \left(\frac{\alpha}{n+\delta}\right)^{1/(1-\alpha)}. \tag{459}$$

Thus overaccumulation occurs if $\sigma > \alpha$, and is not inevitable. The stock comparisons use the same positive power $1 - \alpha$:

$$\sigma(k_S^*)^\alpha = (n+\delta)k_S^*, \tag{460}$$

$$(k_S^*)^{1-\alpha} = \frac{\sigma}{n+\delta}, \tag{461}$$

$$\alpha k_{\mathrm{GR}}^{\alpha-1} = n+\delta, \tag{462}$$

$$k_{\mathrm{GR}}^{1-\alpha} = \frac{\alpha}{n+\delta}, \tag{463}$$

$$k_S^* > k_{\mathrm{GR}} \iff \sigma > \alpha. \tag{464}$$

For the population-weighted Ramsey dynasty in the common discrete benchmark, stationary consumption and its Euler equation instead give

$$1 = \beta R^*, \qquad R^* = 1/\beta. \tag{465}$$

In the course special case $N = 1$, the algebra also reads

$$1 = \beta[1 - \delta + f'(k^*)], \tag{466}$$

$$1 + \rho = 1 - \delta + f'(k^*) \qquad (\beta = 1/(1+\rho)), \tag{467}$$

$$f'(k^*) = \rho + \delta > \delta = f'(k_{\mathrm{GR}}), \tag{468}$$

$$k^* < k_{\mathrm{GR}} \qquad (f'' < 0). \tag{469}$$

With population growth restored, compare $R^* = 1/\beta$ with $R_{\mathrm{GR}} = N$, rather than keeping the $N = 1$ Golden Rule unchanged. Under $\beta N < 1$, the finite-welfare restriction for this stationary benchmark, $R^* > N$ and $k^* < k_{\mathrm{GR}}$. This matches the role of the effective-discount restriction in the previous continuous-time chapter. The protection comes from the dynasty objective and admissibility conditions, not simply from the existence of an Euler equation. OLG has an Euler equation too, but it links two different age consumptions, which need not be equal even at a stationary equilibrium.

6.7.4 A planner needs weights across cohorts, not just within a life

The private discount factor β compares the two ages of one person. A planner also needs a rule for weighing different birth cohorts. To retain that distinction, let cohort weights be $\omega_t = \chi^t$, with $\chi > 0$, and consider

$$\begin{aligned} &\max \omega_{-1}\beta L_{-1}u(c_{2,0}) \\ &\quad + \sum_{t=0}^{\infty} \omega_t L_t \big[u(c_{1,t}) + \beta u(c_{2,t+1})\big], \end{aligned} \tag{470}$$

subject to (444), given k_0, and feasible positive consumption. The first term explicitly includes the initially old; their young-age utility is already sunk. For the stationary comparisons here, $\chi N < 1$ is a sufficient summability restriction. These weights are a normative choice, not a preference inferred from household saving.

Let μ_t be the multiplier on the aggregate date-t resource constraint. The aggregate resource constraint gives the following derivatives:

$$0 = \omega_t L_t u'(c_{1,t}) - \mu_t L_t, \tag{471}$$

$$0 = \beta\omega_{t-1} L_{t-1} u'(c_{2,t}) - \mu_t L_{t-1}, \tag{472}$$

$$0 = -\mu_t + \mu_{t+1}[f'(k_{t+1}) + 1 - \delta]. \tag{473}$$

Cancel L_t from the first equation and L_{t-1} from the second. Then

$$\mu_t = \omega_t u'(c_{1,t}) = \beta\omega_{t-1}u'(c_{2,t}), \tag{474}$$

$$\mu_t = R_{t+1}\mu_{t+1}. \tag{475}$$

Using the age-two condition at $t+1$ recovers the within-cohort Euler equation:

$$\mu_{t+1} = \beta\omega_t u'(c_{2,t+1}), \tag{476}$$

$$\omega_t u'(c_{1,t}) = R_{t+1}\beta\omega_t u'(c_{2,t+1}), \tag{477}$$

$$u'(c_{1,t}) = \beta R_{t+1}u'(c_{2,t+1}). \tag{478}$$

At an interior stationary planner allocation, however, $\mu_{t+1}/\mu_t = \chi$, so its capital condition is $R^* = 1/\chi$. Changing social cohort weights changes the planner problem. The ratio follows directly from constant young consumption:

$$\frac{\mu_{t+1}}{\mu_t} = \frac{\omega_{t+1}u'(c_1^*)}{\omega_t u'(c_1^*)} = \chi, \tag{479}$$

$$1 = R^*\frac{\mu_{t+1}}{\mu_t} = R^*\chi, \tag{480}$$

$$R^* = 1/\chi. \tag{481}$$

Maximizing $\bar{c}(k)$ across stationary paths supplies the Golden Rule; it is not the solution of every weighted lifetime-welfare problem. Moreover, a planner optimum under one set of weights and a Pareto improvement for all cohorts are different claims.

6.8 Population, pensions, and public debt

The benchmark now gives a way to examine how retirement is financed. Introduce each arrangement separately. Taxes and benefits below are lump-sum; there is still no uncertainty, valued government consumption, or endogenous retirement. A more realistic pension system would need those mechanisms before making quantitative policy recommendations.

6.8.1 Population change: a support ratio and a capital effect

With two age groups, the number of retirees per worker is $1/N$. A lower N raises that ratio. In the log/Cobb–Douglas capital-only steady state,

$$\frac{\partial \log k^*}{\partial \log N} = -\frac{1}{1-\alpha}. \tag{482}$$

Lower cohort growth spreads saving over fewer future workers, raising long-run capital per worker, increasing wages and reducing returns. With $A = 1$, this is a comparison of stationary levels; it is not permanently faster productivity growth. An unexpected change in cohort size also changes labor supply and the ratio K/L while inherited aggregate K is fixed; its full transition must track when the affected cohorts enter work.

Real population aging can also reflect longer lives, changing employment, migration, and retirement ages. Those channels are not all captured by a change in N. The support ratio is a useful first accounting step, not a complete empirical explanation of aging.

6.8.2 Pay-as-you-go pensions: current workers finance current retirees

Let each young person pay $T_t \geq 0$ and each old person receive P_t. A balanced pension budget is

$$L_t T_t = L_{t-1} P_t, \qquad P_t = N T_t. \tag{483}$$

A cohort paying at t receives the benefit at $t+1$, so its budgets become

$$c_{1,t} + a_t = w_t - T_t, \qquad c_{2,t+1} = R_{t+1} a_t + P_{t+1}. \tag{484}$$

Private assets still satisfy $N k_{t+1} = a_t$: the tax paid to retirees is not an additional physical asset.[14]

Under log utility, the interior saving choice follows from

$$\frac{1}{w_t - T_t - a_t} = \frac{\beta R_{t+1}}{R_{t+1} a_t + P_{t+1}}, \tag{485}$$

$$R_{t+1} a_t + P_{t+1} = \beta R_{t+1} (w_t - T_t - a_t), \tag{486}$$

$$R_{t+1} a_t + \beta R_{t+1} a_t = \beta R_{t+1} (w_t - T_t) - P_{t+1}, \tag{487}$$

$$(1+\beta) R_{t+1} a_t = \beta R_{t+1} (w_t - T_t) - P_{t+1}, \tag{488}$$

$$a_t = \kappa (w_t - T_t) - \frac{P_{t+1}}{(1+\beta) R_{t+1}}. \tag{489}$$

Current contributions reduce disposable wages, and expected benefits reduce the assets needed for retirement. Both lower private saving at given prices. If this expression is negative, the no-borrowing restriction binds at $a_t = 0$; the interior formula must not be used beyond its feasible range.

A permanent, constant contribution T gives a future benefit NT. At fixed prices its effect on present-value lifetime resources is

$$(w_t - T) + \frac{NT}{R_{t+1}} - w_t = T \left(\frac{N}{R_{t+1}} - 1 \right). \tag{490}$$

This is positive if $N > R_{t+1}$. The initially old can receive a windfall, and suitable transfers can improve an overaccumulating allocation. But this fixed-price comparison is not a proof that every contribution rate benefits every cohort: equilibrium wages, returns, asset constraints, and the entire transition must also be checked. In an efficient benchmark, reducing capital need not create the resources needed to compensate all generations.

A pension's implicit return depends on its funding rule. The gross return N above assumes a constant real contribution per worker. For a constant wage contribution rate τ, $T_t = \tau w_t$ and $P_{t+1}/T_t = N w_{t+1}/w_t$. Wage growth then matters as well as cohort size. Holding benefits fixed instead, (483) requires $T_t = P_t/N$: fewer workers per retiree raise the required contribution. These are accounting relations, not risk-adjusted comparisons with observed financial yields. Benefits, wages, and taxes in actual pension systems are not all fixed or certain.

[14] For the pension budget and the contrast with funded accounts, see MIT 14.472, *Public Economics II* (2004), "Social Security—OLG Settings," lecture notes.

6.8.3 Funded pensions: distinguish compulsory from total saving

Suppose a mandatory contribution D_t is invested in capital and earns the same return as voluntary assets. Benefits are $R_{t+1}D_t$. The household faces

$$c_{1,t} + a_t = w_t - D_t, \qquad c_{2,t+1} = R_{t+1}(a_t + D_t). \tag{491}$$

Define total retirement assets $A_t^h \equiv a_t + D_t$. Substitution gives $c_{1,t} + A_t^h = w_t$ and $c_{2,t+1} = R_{t+1}A_t^h$, exactly the original lifetime budget. If voluntary saving can adjust without a binding constraint, a household offsets a one-unit compulsory contribution by one unit less voluntary saving. Total assets, capital, and consumption are unchanged:

$$Nk_{t+1} = A_t^h, \qquad A_t^h = \kappa w_t \quad \text{under log utility and an interior offset.} \tag{492}$$

If $a_t \geq 0$ binds, the contribution exceeds desired assets, or the account has different returns, fees, insurance, or tax treatment, this neutrality need not hold. A funded account and a transfer to the current old cannot be treated as the same capital-market flow.

Moving from pay-as-you-go to funding also leaves a transition question: who finances benefits already promised to existing retirees while workers build their own accounts? A comparison of eventual fund balances alone cannot determine the welfare of the transition cohorts.

6.8.4 Public debt: a retirement asset with a fiscal counterpart

Now consider government bonds instead of a pension program. Let B_t denote principal issued at $t-1$, due to its old holders at date t. New bonds B_{t+1} are sold to the young for their principal amount at date t. Deterministic bonds and physical capital must offer the same gross payoff R_{t+1} if both are held. With no government purchases, a lump-sum tax T_t^D per young person satisfies

$$R_t B_t = L_t T_t^D + B_{t+1}. \tag{493}$$

A negative tax is a transfer. This budget identifies who pays the old bondholders; a bond is not an extra good available for consumption.

Asset clearing and its per-worker version become

$$L_t a_t = K_{t+1} + B_{t+1}, \qquad a_t = N(k_{t+1} + b_{t+1}), \quad b_t \equiv B_t/L_t. \tag{494}$$

The fiscal budget is $T_t^D = R_t b_t - N b_{t+1}$. Consolidating households and government still gives (444): public liabilities cancel public claims, while physical capital remains a productive resource. Debt changes portfolios and intergenerational payments, not the resource identity itself. Its financing matters because the young and the old are different people without offsetting bequests.

For a constant debt stock per worker, $b_t = b$, the tax is $T_t^D = (R_t - N)b$. If $R_t < N$, new issuance can cover debt service and a transfer at these model prices; if $R_t > N$, maintaining that debt ratio requires a positive tax. This does not establish that an arbitrary debt path is feasible: returns respond to capital, and all future budgets and household choices must hold.

Under log utility, combine $a_t = \kappa(w_t - T_t^D)$ with asset clearing:

$$N(k_{t+1} + b) = \kappa[w(k_t) - (R(k_t) - N)b], \tag{495}$$

$$Nk_{t+1} = \kappa w(k_t) - \kappa R(k_t)b + \kappa N b - Nb, \tag{496}$$

$$= \kappa w(k_t) - [\kappa R(k_t) + (1-\kappa)N]b. \tag{497}$$

$$Nk_{t+1} = \kappa w(k_t) - \big[\kappa R(k_t) + (1-\kappa)N\big]b. \tag{498}$$

Near the no-debt log/Cobb–Douglas steady state, a small permanent increase in b reduces stationary capital. Differentiating the stationary equation at $b = 0$ gives the following chain. Differentiate before setting b to zero:

$$N\frac{dk^*}{db} = \kappa w'(k^*)\frac{dk^*}{db} - \kappa R'(k^*)b\frac{dk^*}{db} - [\kappa R(k^*) + (1-\kappa)N], \tag{499}$$

$$[N - \kappa w'(k^*)]\left.\frac{dk^*}{db}\right|_{b=0} = -[\kappa R^* + (1-\kappa)N]. \tag{500}$$

$$\left.\frac{dk^*}{db}\right|_{b=0} = -\frac{\kappa R^* + (1-\kappa)N}{N - \kappa w'(k^*)} < 0, \qquad N - \kappa w'(k^*) = N(1-\alpha) > 0. \tag{501}$$

Indeed, $w'(k) = \alpha(1-\alpha)k^{\alpha-1}$ and the no-debt steady state gives $\kappa(1-\alpha)(k^*)^{\alpha-1} = N$. Thus $\kappa w'(k^*) = \alpha N$ and the denominator is $N - \alpha N = N(1-\alpha)$. This is a local result in the stated example, including its tax-financing rule, rather than a universal one-for-one displacement of capital by bonds. Whether the reduction helps welfare depends on initial accumulation and on how gains and costs are distributed along the transition.

Who holds the asset, and who supplies its payoff? Physical capital pays from production and surviving assets. A funded pension owns such assets on the worker's behalf. A pay-as-you-go pension pays from current contributors, while government bonds pay from taxes and new issuance. All can help a person finance retirement, but they need not have the same effect on productive capital or on different generations' consumption. Tracing both sides of the balance sheet is the central OLG lesson.

OLG makes retirement finance and population replacement part of the equilibrium, and shows why individually optimal saving need not guarantee intergenerational efficiency. It still takes productivity as given and abstracts from realistic lifetimes, risk, labor choices, and within-cohort inequality. The next growth chapter changes the source of productivity and long-run growth; later fiscal chapters can build richer policy problems on the cohort budgets established here.

7 Endogenous Growth and Resource Allocation

Intermediate macroeconomics introduces AK, education, and innovation as ways to explain sustained growth. This chapter asks what closes those explanations: who accumulates the relevant stock, what it costs, which return guides the choice, and whether that return is the one society receives. We reuse Ramsey's optimization tools rather than treating endogenous growth as an unrelated list of production functions.

7.1 From intermediate macroeconomics and Ramsey to a growth engine

7.1.1 What the familiar diagrams leave to be explained

A machine, a trained worker, and a usable design can all raise output, but they are not accumulated in the same way. Building equipment uses goods. Training uses time and other resources. Developing a design can be costly even when another firm can subsequently use the same instructions without taking them away from the first. These differences affect both the resource constraint and the incentive to invest.

It is tempting to put each item into a production function and declare growth explained. Instead, ask which scarcity remains after accumulation. If machines are added while skills or usable methods remain fixed, returns may fall. If skills and equipment can grow together, the relevant return might not fall. If ideas can be shared, the social gain from discovery may exceed what the inventor can charge. We need to identify that mechanism before choosing the law of motion or discussing policy.

The familiar intermediate-level statements are therefore starting points. "Saving can affect growth" needs a technology that sustains returns and a rule determining saving. "Education raises productivity" needs a skill accumulation technology and a time allocation. "Research creates ideas" needs a resource cost and a reason someone pays it. The models below fill these particular gaps; they do not repeat the Solow derivation.

7.1.2 Ramsey endogenizes saving, not automatically long-run growth

With fixed technology and population, the Ramsey benchmark has

$$\dot{k} = f(k) - c - \delta k, \qquad \frac{\dot{c}}{c} = \frac{f'(k) - \delta - \rho}{\theta}. \tag{502}$$

Saving is chosen optimally. Yet with diminishing marginal product, accumulating k lowers its return. At the stationary optimum, $f'(k^*) = \delta + \rho$ and consumption stops growing. In the earlier version with exogenous labor-efficiency growth, consumption per person can grow, but that long-run technology growth rate is still supplied from outside the model.

The next question is thus about the return to *all reproducible inputs*, not whether households have an Euler equation. Keep the household problem, change the accumulation opportunities, and check what replaces the stationary capital target. OLG changed who lives, saves, and receives transfers. Here we return explicitly to a continuing Ramsey household to isolate a different change: the mechanism that makes productive capacity grow.

Endogenous saving and endogenous growth are different claims. An optimal saving decision can lead to a stationary economy. Conversely, an AK equation with an imposed saving share can generate a growth rate without explaining saving. To obtain both, combine a specified accumulation technology with household optimization, market or planner closure, and the appropriate terminal conditions. Changing notation from an exogenous g to a function called g does not provide those mechanisms.

The following models are separate experiments. Technology coefficients such as η and α are defined within each specification; a training efficiency, a production elasticity, and a research efficiency are not assumed to be the same empirical parameter.

7.2 AK: retain Ramsey's choice, change the return

7.2.1 Complete conditions for the one-stock benchmark

01 Time, population, goods, and states

Time is continuous. A closed economy has one good, no government in the baseline, no uncertainty, and a fixed population normalized to one. A continuing household owns the given stock $k(0) = k_0 > 0$ and firms. Consequently aggregate and per-person quantities coincide in this benchmark. There is no age structure, leisure choice, or exogenous technology growth.

02 Technology and accumulation

Production is linear in a reproducible stock:

$$y = Bk, \qquad B > 0 \text{ constant}, \qquad \dot{k} = (B - \delta)k - c, \quad \delta \geq 0.$$

The stock can be interpreted as a broad productive composite only if its components can be reproduced with the assumed technology. Goods can be converted into the stock at unit cost. Strictly diminishing returns and the Solow Inada limits are deliberately replaced; this is not Cobb–Douglas with the same fixed factor left implicit.

03 Preferences and household choice

The household chooses a positive consumption path and feasible $k(t) \geq 0$:

$$\max \int_0^\infty e^{-\rho t} u(c(t))\, dt, \qquad u'(c) = c^{-\theta}, \quad \rho > 0, \quad \theta > 0.$$

Use $u(c) = c^{1-\theta}/(1-\theta)$ for $\theta \neq 1$ and $u(c) = \log c$ for $\theta = 1$. Saving is the residual of output after consumption, not an independently imposed share.

04 Markets, ownership, and returns

In the competitive interpretation, firms rent the productive stock at $q = B$ and earn zero economic profit after paying it. There is no separate productive labor input in this one-stock technology. The household owns the stock, receives the net return $r = B - \delta$, and can adjust investment without costs. Without externalities or other wedges, its budget coincides with the aggregate resource constraint and implements the planner allocation.

05 Admissibility and the growing solution

The planner's transversality condition is

$$\lim_{t\to\infty} e^{-\rho t} c(t)^{-\theta} k(t) = 0.$$

For the interior positive-growth case studied below, impose

$$g \equiv \frac{B-\delta-\rho}{\theta} > 0, \qquad \xi \equiv \rho - (1-\theta)g > 0.$$

The second restriction ensures finite allocation-dependent utility and the terminal condition along the proposed balanced growth path. For $\theta < 1$ it restricts how productive accumulation can be relative to discounting; a formal Euler solution alone need not be a well-posed optimum.

7.2.2 The same Hamiltonian now gives a constant growth rate

The current-value Hamiltonian and its control and costate conditions are

$$\mathscr{H} = u(c) + \mu[(B-\delta)k - c], \tag{503}$$
$$\mathscr{H}_c = u'(c) - \mu = 0, \tag{504}$$
$$\mathscr{H}_k = \mu(B-\delta), \tag{505}$$
$$\dot{\mu} = \rho\mu - \mathscr{H}_k = [\rho - (B-\delta)]\mu, \tag{506}$$
$$\mu = c^{-\theta}, \qquad \dot{\mu} = [\rho - (B-\delta)]\mu. \tag{507}$$

Since $\dot{\mu}/\mu = -\theta\dot{c}/c$, the growth equation follows by

$$\dot{\mu} = -\theta c^{-\theta-1}\dot{c}, \tag{508}$$
$$-\theta\frac{\dot{c}}{c} = \rho - (B-\delta), \tag{509}$$
$$\theta\frac{\dot{c}}{c} = B - \delta - \rho. \tag{510}$$
$$\frac{\dot{c}}{c} = \frac{B-\delta-\rho}{\theta} = g. \tag{511}$$

The marginal product no longer falls as the stock rises. Waiting remains rewarded at the same rate, so the optimal consumption growth rate need not fade toward zero. This is the change in technology doing the work; the intertemporal preference condition is still Ramsey's.

The Euler equation gives $c(t) = c_0 e^{gt}$ but not c_0. Set $a \equiv B - \delta$ and integrate $\dot{k} = ak - c_0 e^{gt}$, using $\xi = a - g = \rho - (1-\theta)g > 0$:

$$\frac{\dot{c}}{c} = g, \qquad \log(c(t)/c_0) = gt, \qquad c(t) = c_0 e^{gt}, \tag{512}$$

$$\dot{k} - ak = -c_0 e^{gt}, \tag{513}$$

$$e^{-at}\dot{k} - ae^{-at}k = -c_0 e^{-(a-g)t}, \tag{514}$$

$$\frac{d}{dt}(e^{-at}k) = -c_0 e^{-\xi t}, \tag{515}$$

$$e^{-at}k(t) - k_0 = -c_0 \int_0^t e^{-\xi v}\, dv, \tag{516}$$

$$\int_0^t e^{-\xi v}\, dv = \left[-\frac{e^{-\xi v}}{\xi}\right]_0^t = \frac{1 - e^{-\xi t}}{\xi}. \tag{517}$$

Also $\theta g = a - \rho$ gives $a - g = \rho + \theta g - g = \rho - (1-\theta)g$. Multiplying the integrated capital equation by e^{at} yields

$$k(t) = e^{at}\left[k_0 - c_0 \int_0^t e^{-(a-g)v}\, dv\right], \tag{518}$$

$$= \left(k_0 - \frac{c_0}{\xi}\right)e^{at} + \frac{c_0}{\xi}e^{gt}. \tag{519}$$

If $c_0 > \xi k_0$, the first term eventually makes capital negative. If $c_0 < \xi k_0$, capital grows at the faster rate a and its limiting discounted utility value is positive: the household leaves valuable resources unused. The terminal expression makes that last statement explicit:

$$e^{-\rho t}u'(c(t))k(t) = c_0^{-\theta}\left[\left(k_0 - \frac{c_0}{\xi}\right)e^{(a-\rho-\theta g)t} + \frac{c_0}{\xi}e^{[(1-\theta)g-\rho]t}\right], \tag{520}$$

$$= c_0^{-\theta}\left[k_0 - \frac{c_0}{\xi} + \frac{c_0}{\xi}e^{-\xi t}\right], \tag{521}$$

$$\lim_{t\to\infty} e^{-\rho t}u'(c(t))k(t) = c_0^{-\theta}\left(k_0 - \frac{c_0}{\xi}\right). \tag{522}$$

Here $a - \rho - \theta g = 0$ and $(1-\theta)g - \rho = -\xi < 0$. Feasibility and transversality therefore select

$$c_0 = \xi k_0, \qquad k(t) = k_0 e^{gt}, \qquad c(t) = \xi k(t). \tag{523}$$

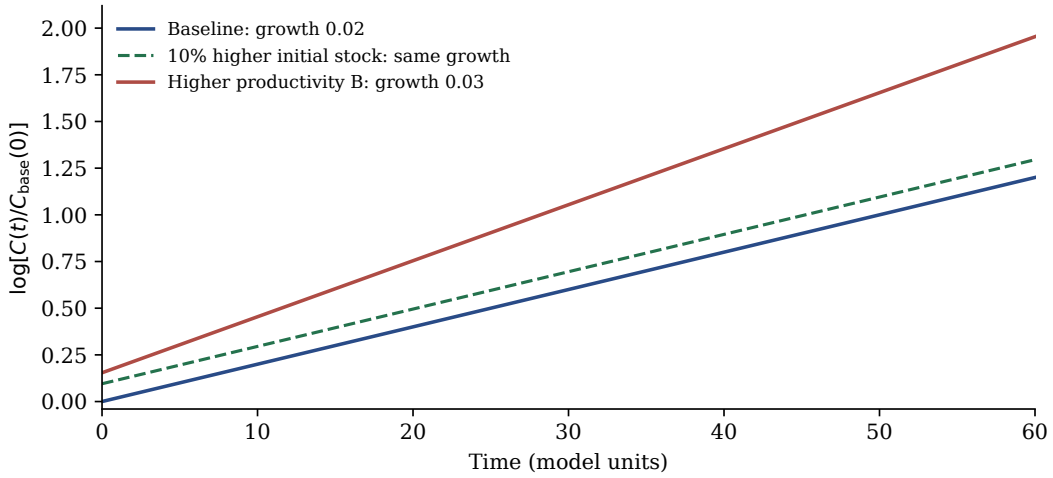
Indeed, along this path the terminal expression is proportional to $e^{-[\rho-(1-\theta)g]t} = e^{-\xi t}$ and vanishes.

A balanced growth path replaces the stationary capital target. There is no finite positive stationary k on this growing solution. Consumption jumps to the feasible optimal ratio $c/k = \xi$ for the inherited stock, then both grow at g. The saving share is an outcome, $s^* = (\delta + g)/B = 1 - \xi/B$, rather than Solow's input. Under the stated positive-growth conditions it lies between zero and one.

7.2.3 A level change, a return change, and a policy wedge

A one-time increase in initial capital, holding B and preferences fixed, raises the levels of capital and consumption but leaves g unchanged. A permanent change in B changes the return and hence g; comparing the new paths requires the new initial consumption ratio as well as the slope. The absence of a transition in the ratio is a special feature of linear technology and these preferences, not a general result for growth models.

For example, $B = 0.10$, $\delta = 0.02$, $\rho = 0.04$, and $\theta = 2$ imply $g = 0.02$, $c/k = 0.06$, and $s^* = 0.40$. Raising k_0 by 10% raises the whole path by 10%. Raising B to 0.12 instead gives $g = 0.03$ and $c/k = 0.07$. These are illustrative model numbers, not empirical return estimates.



Levels versus growth rates. Parallel log-consumption paths have the same growth rate. A larger inherited stock shifts the path, whereas a higher B changes its slope and its optimal initial consumption ratio. Each path starts at date zero with its stated parameters; this is not a simulation of an unanticipated policy change partway through a transition. The numbers are the hypothetical AK example above.

A tax rate τ on net capital income, rebated as a lump-sum transfer, leaves aggregate resources unchanged but changes the household's marginal return. Each price-taking household treats its rebate as given, as in Section 5.7. For an interior admissible growing equilibrium,

$$g_\tau = \frac{(1-\tau)(B-\delta)-\rho}{\theta}, \qquad \frac{c}{k} = B - \delta - g_\tau. \tag{524}$$

The finite-welfare restriction $\rho - (1-\theta)g_\tau > 0$ must still hold. This is a permanent growth effect under the AK assumption. The corresponding Ramsey tax experiment with diminishing returns changed stationary levels; that difference comes from the accumulation technology. An empirical claim that a tax changes trend growth would require evidence for the relevant mechanism, not this formula alone.

7.2.4 Private diminishing returns can coexist with aggregate AK

The one-stock competitive benchmark need not contain a market failure. An alternative construction does: suppose firm i has

$$y_i = Bk_i^{\alpha}(\bar{k}\ell_i)^{1-\alpha}, \qquad 0 < \alpha < 1, \tag{525}$$

where $\bar{k} = K/L$ is an economy-wide knowledge spillover associated with capital, taken as given by each small firm. Labor satisfies $\sum_i \ell_i = 1$. Firms have constant returns in their private inputs k_i, ℓ_i, and take factor prices and $\bar{k}$ as given. In a symmetric allocation with the same capital per worker across firms, aggregate output is $Y = BK$.

Differentiate the firm's own technology before imposing symmetry. Then

$$\frac{\partial y_i}{\partial k_i} = \alpha B k_i^{\alpha-1}(\bar{k}\ell_i)^{1-\alpha}, \tag{526}$$

$$\frac{\partial y_i}{\partial \ell_i} = (1-\alpha)Bk_i^{\alpha}\bar{k}^{1-\alpha}\ell_i^{-\alpha}. \tag{527}$$

At a common capital intensity $k_i = \bar{k}\ell_i$, these become

$$q_D = \alpha B(\bar{k}\ell_i)^{\alpha-1}(\bar{k}\ell_i)^{1-\alpha} = \alpha B, \tag{528}$$

$$w = (1-\alpha)B(\bar{k}\ell_i)^{\alpha}\bar{k}^{1-\alpha}\ell_i^{-\alpha} = (1-\alpha)B\bar{k}. \tag{529}$$

Since $L = 1$, $\bar{k} = K$. Also $y_i = B\bar{k}\ell_i$, so $\sum_i y_i = B\bar{k}\sum_i \ell_i = BK$.

$$q_D = \alpha B, \qquad w = (1-\alpha)BK, \qquad r_D = \alpha B - \delta. \tag{530}$$

Factor payments still exhaust output. A planner who internalizes how K raises the common spillover instead obtains marginal product B. With the same household preferences and appropriate admissibility restrictions,

$$g_D = \frac{\alpha B - \delta - \rho}{\theta}, \qquad g_P = \frac{B - \delta - \rho}{\theta}. \tag{531}$$

An aggregate AK equation does not identify the private return. Where both growing solutions are feasible, $g_D < g_P$. These are two different interpretations of the same aggregate $Y = BK$: either B is privately earned, or part of the social return is external. Aggregate linearity alone cannot tell us which policy problem is present. Learning during investment or shared production experience can motivate a spillover, but its size has to be specified and measured rather than inferred from the AK notation.

7.3 Human capital: explain the reproducible composite

7.3.1 Why adding a second stock is not enough

The augmented Solow model added physical and human capital with diminishing returns to their joint accumulation. Its balanced growth still depended on exogenous technology. To change that result, give skill formation its own technology: time spent learning today affects the ability to produce and learn tomorrow. The question is now how much time to divert from current output. This motivates a two-sector Uzawa–Lucas-style benchmark. We omit average-skill externalities to isolate the private accumulation mechanism.[15]

[15] Robert E. Lucas Jr. (1988), "On the Mechanics of Economic Development," *Journal of Monetary Economics*, 22(1), 3–42, Section 4. The simplified model here is not the full externality specification in that paper.

01 Household, stocks, and the scope of the experiment

Keep the continuous-time, fixed-population household and CRRA preferences above, with $K_0, H_0 > 0$. H is a stock of skill in the continuing household, not hours worked or the number of school years. The household chooses consumption $C > 0$ and a work fraction $v \in [0, 1]$. The remaining fraction $1 - v$ is used for learning; leisure is omitted. There is no uncertainty, knowledge spillover to other households, or public subsidy in this benchmark.

02 Production and the goods constraint

Goods production is

$$Y = K^{\alpha}(vH)^{1-\alpha}, \qquad 0 < \alpha < 1, \qquad \dot{K} = Y - C - \delta K.$$

Goods can be consumed or accumulated as physical capital. Current production uses only the skill-weighted time assigned to work. The technology is constant returns in K, H at fixed v, despite a diminishing marginal product of K when H is fixed. The household/planner controls the two uses of its resources.

03 Learning technology and its cost

Skills accumulate according to

$$\dot{H} = [\eta(1 - v) - \delta_H]H, \qquad \eta > 0, \quad \delta_H \geq 0.$$

Learning uses time, so its resource cost is forgone production, even though there is no separate goods expenditure in this stripped-down specification. A given learning fraction adds skill in proportion to the existing stock. Skill depreciation is explicit; finite careers, cohort replacement, and education quality differences are not separately modeled. Indefinite aggregate skill accumulation is an assumption about this continuing household, not an assertion that one person can study forever.

04 Optimality, boundaries, and balanced growth

This is a planner/household production problem with no externality. The Hamiltonian conditions below apply to an interior allocation $0 < v < 1$, $K, H, C > 0$, with terminal conditions

$$\lim_{t \to \infty} e^{-\rho t}\lambda(t)K(t) = 0, \qquad \lim_{t \to \infty} e^{-\rho t}\mu(t)H(t) = 0.$$

For a positive-growth balanced path require $g = (\eta - \delta_H - \rho)/\theta > 0$ and $\rho - (1 - \theta)g > 0$. They imply the interior work fraction derived below; if the implied fraction lies outside $[0, 1]$, the interior solution cannot be used. Capital and skill are both inherited; an arbitrary initial ratio need not already be on the balanced growth path.

https://doi.org/10.1016/0304-3932(88)90168-7.

7.3.2 Time allocation and the two shadow values

Write the current-value Hamiltonian as

$$\mathscr{H} = u(C) + \lambda[Y - C - \delta K] + \mu[\eta(1 - v) - \delta_H]H. \tag{532}$$

The consumption condition is $\lambda = C^{-\theta}$. The work-time condition and the physical-capital costate equation are obtained by differentiating each use of the two stocks and of work time:

$$Y_v = (1 - \alpha)Y/v, \quad Y_K = \alpha Y/K, \quad Y_H = (1 - \alpha)Y/H, \tag{533}$$

$$\mathscr{H}_C = u'(C) - \lambda = C^{-\theta} - \lambda = 0, \tag{534}$$

$$\mathscr{H}_v = \lambda(1 - \alpha)Y/v - \mu\eta H = 0, \tag{535}$$

$$\dot{\lambda} = \rho\lambda - \lambda[\alpha Y/K - \delta], \tag{536}$$

$$\dot{\lambda}/\lambda = \rho + \delta - \alpha Y/K. \tag{537}$$

$$\lambda(1 - \alpha)Y/v = \mu\eta H, \tag{538}$$

$$\dot{\lambda}/\lambda = \rho + \delta - \alpha Y/K. \tag{539}$$

An extra unit of work time produces goods now; an extra unit of learning time produces skills valued at μ. The first equation compares those two uses of time in the same utility units.

For the skill stock,

$$\dot{\mu} = \rho\mu - \lambda(1 - \alpha)Y/H - \mu[\eta(1 - v) - \delta_H]. \tag{540}$$

Use (538) to replace $\lambda(1 - \alpha)Y/H$ by $\mu\eta v$. Then

$$\lambda(1 - \alpha)Y/v = \mu\eta H, \tag{541}$$

$$\lambda(1 - \alpha)Y/H = \mu\eta v, \tag{542}$$

$$\dot{\mu} = \rho\mu - \mu\eta v - \mu[\eta(1 - v) - \delta_H], \tag{543}$$

$$\frac{\dot{\mu}}{\mu} = \rho - \eta v - \eta + \eta v + \delta_H = \rho - \eta + \delta_H. \tag{544}$$

$$\frac{\dot{\mu}}{\mu} = \rho - \eta + \delta_H. \tag{545}$$

The return to skill includes its contribution to further skill formation. Looking only at its current wage contribution would miss that part of the accumulation opportunity.

7.3.3 Balanced growth and the work–learning split

On a balanced path, K, H, C grow at the same rate g and v is constant. Equation (538) then makes μ/λ constant, because Y/H is constant. Therefore $\dot{\mu}/\mu = \dot{\lambda}/\lambda = -\theta g$. Using (545) and the skill law gives

the following sequence:

$$\frac{\mu}{\lambda} = \frac{(1-\alpha)Y}{\eta v H} \quad \text{constant on the balanced path,} \tag{546}$$

$$\frac{\dot{\mu}}{\mu} - \frac{\dot{\lambda}}{\lambda} = 0, \tag{547}$$

$$\frac{\dot{\lambda}}{\lambda} = -\theta\frac{\dot{C}}{C} = -\theta g, \tag{548}$$

$$-\theta g = \rho - \eta + \delta_H, \tag{549}$$

$$g = (\eta - \delta_H - \rho)/\theta, \tag{550}$$

$$g = \dot{H}/H = \eta(1 - v^*) - \delta_H, \tag{551}$$

$$\eta v^* = \eta - \delta_H - g = \rho + \theta g - g, \tag{552}$$

$$v^* = 1 - (g + \delta_H)/\eta = [\rho - (1-\theta)g]/\eta. \tag{553}$$

$$g = \frac{\eta - \delta_H - \rho}{\theta}, \qquad v^* = 1 - \frac{g + \delta_H}{\eta} = \frac{\rho - (1-\theta)g}{\eta}. \tag{554}$$

The stated restrictions ensure $0 < v^* < 1$. The physical Euler condition and resource constraint also require

$$\alpha Y/K = \rho + \delta + \theta g, \qquad C/K = Y/K - \delta - g > 0. \tag{555}$$

Indeed, $\dot{K} = Y - C - \delta K$ and $\dot{K}/K = g$ imply $g = Y/K - C/K - \delta$; moving C/K to the other side gives the second relation. Since $Y/K = (vH/K)^{1-\alpha}$, the balanced stock ratio is obtained by isolating that ratio before taking powers:

$$\alpha(vH/K)^{1-\alpha} = \rho + \delta + \theta g, \tag{556}$$

$$(vH/K)^{1-\alpha} = (\rho + \delta + \theta g)/\alpha, \tag{557}$$

$$vH/K = [(\rho + \delta + \theta g)/\alpha]^{1/(1-\alpha)}, \tag{558}$$

$$K/H = v[\alpha/(\rho + \delta + \theta g)]^{1/(1-\alpha)}. \tag{559}$$

$$(K/H)^* = v^* \left(\frac{\alpha}{\rho + \delta + \theta g}\right)^{1/(1-\alpha)}. \tag{560}$$

These equations characterize a balanced path, not the complete transition from arbitrary K_0/H_0. Consumption and time allocation can adjust on impact; the two stocks cannot be chosen afresh to match this ratio.

Where did diminishing returns go? Adding machines while holding skill fixed still lowers their marginal product. But scaling K and H together at a fixed work fraction scales output by the same factor. The learning technology also reproduces H linearly. Persistent growth uses those joint reproduction opportunities, rather than assuming that machines alone have escaped diminishing returns. If learning became progressively harder as H rose, this conclusion would need to be rederived.

More schooling time at a given H raises immediate skill accumulation while reducing current goods production. A one-time higher H_0 is different from a permanent improvement in training productivity η. School attendance, actual learning, skill depreciation, and the use of skills at work are not interchangeable measures. The model specifies one way those margins interact; it does not imply that every extra school year permanently raises a country's growth rate. Shared ideas introduce another issue: unlike a person's time, the same design can be used in many places at once.

7.4 Ideas: from a research share to innovation incentives

7.4.1 Nonrival use and costly invention are different margins

A trained engineer cannot supply the same hour to two laboratories at once. A design can guide production in both laboratories without being depleted by its first use. That is nonrivalry. Whether others are allowed to use it without paying is a separate question of excludability. The distinction motivates Romer's innovation framework.[16]

If every additional use can be supplied at negligible cost, competitive pricing of use alone may not recover the original discovery cost. Patent rents, research contracts, public funding, or other institutions could supply the reward. We specify one arrangement instead of assuming that a positive research share appears without someone bearing its cost.

The worked model adds varieties of a differentiated input. It uses flow inputs and labor-based research to keep the accounting transparent; it is a Romer-style teaching model, not a literal reproduction of all sectors in the 1990 paper.[17]

01 Household, population, states, and ownership

Time is continuous and the population $L > 0$ is constant. A continuing household supplies labor inelastically, owns firms and patents, and chooses consumption and financial saving under the earlier CRRA objective. Total consumption is C; since L is constant, using C rather than C/L does not change optimal growth. The inherited state is the number of designs $A(0) > 0$. There is no aggregate uncertainty, physical storage stock, population growth, or leisure choice in this worked model.

[16] Paul M. Romer (1990), "Endogenous Technological Change," *Journal of Political Economy*, 98(5), Part 2, S71–S102. https://doi.org/10.1086/261725.

[17] For expanding-variety constructions and alternative research technologies, see Daron Acemoglu, MIT, *Advanced Economic Growth*, Lecture 2 (September 10, 2007), lecture notes.

02 Final production and intermediate resource use

Competitive final-good firms use production labor L_Y and a continuum of input flows indexed by their designs:

$$Y = L_Y^{1-\alpha} \int_0^A x_j^{\alpha}\, dj, \qquad 0 < \alpha < 1.$$

Each unit of x_j costs one unit of the final good to produce and is fully used within the period. It is not a durable capital stock. Final goods left for consumption satisfy $C = Y - \int_0^A x_j\, dj$. There are no government purchases or other goods uses. Gross output and net consumption must not be confused when calculating the cost of innovation.

03 Exclusive production rights and pricing

Each design has a perpetual patent with one producer of its input, no expiry, and no creative destruction. The owner chooses the input price taking final firms' demand as given; marginal production cost is one. Designs are symmetric but their producers have monopoly power over their own inputs. Final-good firms and factor markets are competitive. Patents protect production rights; existing knowledge is available to researchers without a separate royalty.

04 Research, free entry, and labor clearing

Research employs L_R workers and produces designs at

$$\dot{A} = \eta A L_R, \qquad \eta > 0, \qquad L_Y + L_R = L.$$

Each research firm takes the common stock A, wage w, and value V per new patent as given. Positive research with free entry requires $\eta A V = w$: the value of new designs per worker equals its wage cost. Knowledge does not depreciate. The linear dependence on A and the absence of congestion are substantive accumulation assumptions, not definitions.

05 Asset pricing, budgets, and the interior path

A patent pays flow profit π. With no risk or displacement, $rV = \pi + \dot{V}$, where r is the instantaneous net financial return. Household financial wealth is $W = AV$ and obeys

$$\dot{W} = rW + wL - C, \qquad \frac{\dot{C}}{C} = \frac{r - \rho}{\theta}.$$

Impose solvency and $\lim_{t\to\infty} e^{-\rho t} C(t)^{-\theta} W(t) = 0$. The interior balanced path below requires $\alpha\eta L > \rho$ and $\rho - (1-\theta) g_D > 0$. Planner comparisons also require the stated planner interiority and finite-welfare restrictions. If profitable research does not cover its cost, a zero-research corner replaces the interior formula.

7.4.2 Demand, the markup, wages, and patent profits

Final firms maximize $Y - wL_Y - \int_0^A p_j x_j\,dj$. Demand for input j satisfies

$$0 = \frac{\partial Y}{\partial x_j} - p_j = \alpha L_Y^{1-\alpha} x_j^{\alpha-1} - p_j, \tag{561}$$

$$p_j x_j^{1-\alpha} = \alpha L_Y^{1-\alpha}, \tag{562}$$

$$x_j^{1-\alpha} = (\alpha/p_j) L_Y^{1-\alpha}, \tag{563}$$

$$x_j = L_Y(\alpha/p_j)^{1/(1-\alpha)}. \tag{564}$$

$$p_j = \alpha L_Y^{1-\alpha} x_j^{\alpha-1}, \qquad x_j = L_Y(\alpha/p_j)^{1/(1-\alpha)}. \tag{565}$$

The input owner maximizes $(p_j - 1)x_j(p_j)$. Since demand has price elasticity $1/(1-\alpha) > 1$, its marginal-revenue condition gives the following explicit calculation in quantities:

$$\pi(x) = [p(x) - 1]x = \alpha L_Y^{1-\alpha} x^{\alpha} - x, \tag{566}$$

$$\pi'(x) = \alpha^2 L_Y^{1-\alpha} x^{\alpha-1} - 1 = 0, \tag{567}$$

$$\alpha^2 L_Y^{1-\alpha} x^{\alpha-1} = 1, \tag{568}$$

$$x^{1-\alpha} = \alpha^2 L_Y^{1-\alpha}, \tag{569}$$

$$x_D = \alpha^{2/(1-\alpha)} L_Y = \zeta_D L_Y, \tag{570}$$

$$p_D = \alpha L_Y^{1-\alpha} x_D^{\alpha-1} = \alpha/\alpha^2 = 1/\alpha. \tag{571}$$

$$p_D = 1/\alpha, \qquad x_D = \zeta_D L_Y, \qquad \zeta_D \equiv \alpha^{2/(1-\alpha)}. \tag{572}$$

Equivalently, differentiating profit in x gives $\alpha^2 L_Y^{1-\alpha} x^{\alpha-1} = 1$. Labor's marginal product and profit per patent are therefore computed by

$$w = (1-\alpha) L_Y^{-\alpha} \int_0^A x_j^{\alpha}\,dj, \tag{573}$$

$$= (1-\alpha) L_Y^{-\alpha} A(\zeta_D L_Y)^{\alpha} = (1-\alpha) A \zeta_D^{\alpha}, \tag{574}$$

$$\pi = (p_D - 1)x_D = (1/\alpha - 1)\zeta_D L_Y = \frac{1-\alpha}{\alpha}\zeta_D L_Y, \tag{575}$$

$$Y = L_Y^{1-\alpha} A(\zeta_D L_Y)^{\alpha} = A L_Y \zeta_D^{\alpha}, \tag{576}$$

$$C = Y - A x_D = A L_Y \zeta_D^{\alpha} - A L_Y \zeta_D = A L_Y(\zeta_D^{\alpha} - \zeta_D). \tag{577}$$

$$w = (1-\alpha) A \zeta_D^{\alpha}, \qquad \pi = \frac{1-\alpha}{\alpha}\zeta_D L_Y. \tag{578}$$

Every design can be used alongside the same production labor. Consequently $Y = A L_Y \zeta_D^{\alpha}$ and total intermediate cost is $A L_Y \zeta_D$. Net consumption is

$$C = A L_Y(\zeta_D^{\alpha} - \zeta_D) > 0. \tag{579}$$

The production firm's wage payment and the input owners' profits sum to this net product. A larger set of designs expands productive possibilities; the markup limits how intensively each is used.

7.4.3 Free entry and Ramsey pricing determine research effort

On a balanced path with constant L_Y, L_R, wages grow with A, while profit *per patent* is constant. Free entry then makes patent value constant:

$$V = \frac{w}{\eta A} = \frac{(1-\alpha)\zeta_D^{\alpha}}{\eta}. \tag{580}$$

Asset pricing reduces to $r = \pi/V$. Since $\zeta_D^{1-\alpha} = \alpha^2$, division of profit by patent value gives

$$r = \pi/V = \frac{[(1-\alpha)/\alpha]\zeta_D L_Y}{}(1-\alpha)\zeta_D^{\alpha}/\eta, \tag{581}$$

$$= \frac{\eta L_Y}{\alpha}\zeta_D^{1-\alpha} = \frac{\eta L_Y}{\alpha}\alpha^2 = \alpha\eta L_Y. \tag{582}$$

$$r = \eta L_Y \frac{\zeta_D^{1-\alpha}}{\alpha} = \alpha\eta L_Y. \tag{583}$$

Consumption grows at the rate of designs on this path. Combine the household Euler equation, research technology, and labor clearing:

$$g_D = \eta L_R = \frac{\alpha\eta(L - L_R) - \rho}{\theta}, \tag{584}$$

$$\theta\eta L_R = \alpha\eta L - \alpha\eta L_R - \rho, \tag{585}$$

$$\theta\eta L_R + \alpha\eta L_R = \alpha\eta L - \rho, \tag{586}$$

$$\eta(\theta + \alpha)L_R = \alpha\eta L - \rho, \tag{587}$$

$$L_R^D = \frac{\alpha\eta L - \rho}{\eta(\theta + \alpha)}, \qquad g_D = \frac{\alpha\eta L - \rho}{\theta + \alpha}. \tag{588}$$

The research share has now been derived from incentives, not inserted into $\dot{A}/A$ as a given number. Under the stated restriction it lies strictly between zero and L. If $\alpha\eta L \le \rho$, the interior expression would be nonpositive; the no-research corner has $L_R = 0$, rather than negative research employment.

> **Research uses labor even when no research goods appear in the budget.** New patents cost $V\dot{A} = wL_R$. That expenditure pays research workers, whose time is no longer available to final production. Household saving buys new equity: with constant V, $\dot{W} = V\dot{A}$. Indeed, $rW + wL - C = A\pi + w(L_Y + L_R) - (A\pi + wL_Y) = wL_R$. The financial transaction and the labor opportunity cost describe the same allocation. Treating all L as production labor while also employing L_R in research would manufacture resources that the economy does not have.

This mechanism connects to laboratories, software development, and design work: people must be paid before there is a product to sell, and the expected claim on future profits helps finance that cost. Patent duration, imitation, entry barriers, or the ability to sell a design would change that claim. Those margins are deliberately fixed here; the formula is not a measured return to every kind of research.

7.4.4 The planner's research problem is another AK problem

A planner does not need the monopoly markup to finance discovery. At given A, L_Y, it first maximizes net output, $L_Y^{1-\alpha}\int x_j^\alpha dj - \int x_j dj$. The optimal input use satisfies

$$0 = \frac{\partial}{\partial x_j}[L_Y^{1-\alpha}x_j^\alpha - x_j] = \alpha L_Y^{1-\alpha}x_j^{\alpha-1} - 1, \tag{589}$$

$$x_P^{1-\alpha} = \alpha L_Y^{1-\alpha}, \tag{590}$$

$$x_P = \alpha^{1/(1-\alpha)}L_Y = \zeta_P L_Y. \tag{591}$$

$$\alpha L_Y^{1-\alpha}x_P^{\alpha-1} = 1, \qquad x_P = \zeta_P L_Y, \quad \zeta_P \equiv \alpha^{1/(1-\alpha)} > \zeta_D. \tag{592}$$

Let $D_P \equiv \zeta_P^\alpha - \zeta_P > 0$. Its remaining resource constraint is $C = D_P A(L - L_R)$. Solve it for L_R and substitute into research:

$$C/(D_P A) = L - L_R, \tag{593}$$

$$L_R = L - C/(D_P A), \tag{594}$$

$$\dot{A} = \eta A[L - C/(D_P A)] = \eta LA - \frac{\eta}{D_P}C. \tag{595}$$

$$\dot{A} = \eta LA - \frac{\eta}{D_P}C. \tag{596}$$

Define a transformed state $Z \equiv D_P A/\eta$. Then

$$\dot{Z} = \frac{D_P}{\eta}\dot{A}, \tag{597}$$

$$= \frac{D_P}{\eta}\left(\eta LA - \frac{\eta}{D_P}C\right), \tag{598}$$

$$= D_P LA - C = \eta LZ - C. \tag{599}$$

$$\dot{Z} = \eta LZ - C. \tag{600}$$

This is exactly the one-stock AK resource equation, with net return ηL. Z is a planner's transformed state, not the market patent wealth AV. Reusing the earlier optimization, rather than maximizing growth as a separate objective, gives the Hamiltonian conditions

$$\mathscr{H}_P = u(C) + \nu(\eta LZ - C), \tag{601}$$

$$\mathscr{H}_{P,C} = C^{-\theta} - \nu = 0, \tag{602}$$

$$\dot{\nu} = \rho\nu - \eta L\nu = (\rho - \eta L)\nu, \tag{603}$$

$$-\theta\dot{C}/C = \dot{\nu}/\nu = \rho - \eta L, \tag{604}$$

$$g_P = \dot{C}/C = (\eta L - \rho)/\theta. \tag{605}$$

$$g_P = \frac{\eta L - \rho}{\theta}, \qquad L_R^P = g_P/\eta, \tag{606}$$

provided $0 < g_P < \eta L$ and $\rho - (1-\theta)g_P > 0$. With $\alpha\eta L > \rho$ and both interior admissible paths, $g_P > g_D$. The sign can be checked without comparing two complicated fractions by eye:

$$g_P - g_D = \frac{(\eta L - \rho)(\theta + \alpha) - \theta(\alpha\eta L - \rho)}{\theta(\theta + \alpha)}, \tag{607}$$

$$= \frac{\theta\eta L(1-\alpha) + \alpha(\eta L - \rho)}{\theta(\theta + \alpha)} > 0. \tag{608}$$

Here $\alpha\eta L > \rho$ implies $\eta L > \rho$. Private innovation rewards omit part of the value of future knowledge, and monopoly pricing also restricts input use. The growth comparison belongs to this specified welfare problem; it does not establish that any intervention raising measured research must improve welfare.

A subsidy to invention and a policy changing input markups address different margins. Their financing, targeting, and effect on entry would have to be modeled before proposing an optimal instrument. In settings with duplication, displacement of existing innovators, or other external effects, the welfare comparison would need to be redone. Nonrivality alone does not identify one universally correct patent or subsidy policy.

7.5 What changes when ideas become harder to find?

7.5.1 The scale prediction is an assumption worth testing

The worked model has $\dot{A}/A = \eta L_R$: a permanently larger research workforce supports a permanently faster growth rate. In particular, (588) makes the scale L matter. This follows from linear research productivity in existing knowledge and no congestion. It is not implied by the definition of a nonrival idea.

Jones (1995) questioned the scale predictions of early R&D models using the historical evidence available in that paper.[18] One way to investigate the restriction is to change the research technology:

$$\dot{A} = \eta A^{\phi} L_R^{\lambda}, \qquad \phi < 1, \quad 0 < \lambda \leq 1. \tag{609}$$

These are new assumptions. The preceding optimal research-share and patent formulas must not be carried over unchanged.

7.5.2 A conditional balanced-growth calculation

For this calculation, allow labor to grow at exogenous rate $n \geq 0$ and hold the research share $s_R = L_R/L$ constant and positive. This is a technology experiment; it does not yet solve the new household and entry problem. Since

$$g_A \equiv \dot{A}/A = \eta A^{\phi-1}(s_R L)^{\lambda}, \tag{610}$$

a positive constant g_A on a balanced path requires the logarithmic differentiation

$$\log g_A = \log\eta + (\phi - 1)\log A + \lambda \log s_R + \lambda \log L, \tag{611}$$

$$\frac{\dot{g}_A}{g_A} = (\phi - 1)\frac{\dot{A}}{A} + \lambda\frac{\dot{L}}{L}, \tag{612}$$

$$0 = (\phi - 1)g_A + \lambda n, \tag{613}$$

$$(1 - \phi)g_A = \lambda n. \tag{614}$$

[18] Charles I. Jones (1995), "R&D-Based Models of Economic Growth," *Journal of Political Economy*, 103(4), 759–784. The comparison concerns that paper's models and evidence, not a claim that current data settle every innovation mechanism. https://doi.org/10.1086/262002.

$$0 = \frac{d \log g_A}{dt} = (\phi - 1) g_A + \lambda n, \qquad g_A = \frac{\lambda n}{1 - \phi}. \tag{615}$$

For $n > 0$, the corresponding balanced-path relation is

$$A^{1-\phi} = \frac{\eta s_R^{\lambda}}{g_A} L^{\lambda}. \tag{616}$$

A permanent increase in the research share changes this level relation and the transition, while the balanced growth rate in this specification is unchanged. Research remains a costly productive activity; a level effect is not evidence that research is useless.

When $n = 0$ and L_R is constant, integrate (609) directly:

$$A^{-\phi} \dot{A} = \eta L_R^{\lambda}, \tag{617}$$

$$\frac{d}{dt} A^{1-\phi} = (1 - \phi) A^{-\phi} \dot{A} = (1 - \phi) \eta L_R^{\lambda}, \tag{618}$$

$$\int_0^t \frac{d}{dv} A(v)^{1-\phi} \, dv = (1 - \phi) \eta L_R^{\lambda} \int_0^t dv, \tag{619}$$

$$A(t)^{1-\phi} - A(0)^{1-\phi} = (1 - \phi) \eta L_R^{\lambda} t. \tag{620}$$

$$A(t)^{1-\phi} = A(0)^{1-\phi} + (1 - \phi) \eta L_R^{\lambda} t. \tag{621}$$

New ideas keep arriving, but their percentage addition to a growing stock shrinks and g_A tends to zero. This illustrates why a model can endogenize research decisions or technological change without making every permanent research policy change alter the asymptotic growth rate.

7.6 Allocation: more output from the same stocks

7.6.1 A distinct question from inventing or accumulating inputs

Two economies with the same total capital and available production methods need not use them equally well. Capital might be tied to a less productive firm while a more productive firm cannot finance expansion. This changes aggregate productivity without creating another machine or design. Separate that allocation margin from the growth engines already studied.

The example below is a static resource-allocation problem, not a new balanced growth model. It isolates a mechanism used in research on misallocation. Hsieh and Klenow (2009) apply a richer framework to manufacturing establishments; the simple numbers below are not their empirical estimates.[19]

[19] Chang-Tai Hsieh and Peter J. Klenow (2009), "Misallocation and Manufacturing TFP in China and India," *Quarterly Journal of Economics*, 124(4), 1403–1448. https://doi.org/10.1162/qjec.2009.124.4.1403.

01 Firms, fixed resources, and technology

At one date there are two firms with fixed labor and other complementary inputs. They produce the same good at price one:

$$y_i = z_i k_i^{\alpha}, \qquad z_i > 0, \quad 0 < \alpha < 1, \quad i = 1, 2.$$

The total available stock is $k_1 + k_2 = K > 0$, with $k_i \geq 0$. There is no investment, research, uncertainty, or adjustment cost in this one-date experiment. The z_i and the complementary inputs are held fixed.

02 Efficient allocation and the private problem

A planner maximizes $y_1 + y_2$ subject to the fixed stock. In an undistorted competitive rental market, firms maximize $z_i k_i^{\alpha} - q k_i$ at the common rental price q. Households own the stock and firms. Labor costs and incomes can be suppressed because their quantities are fixed here; they cannot be inferred to be absent from a full growth economy.

03 A precisely specified allocation wedge

For the distorted comparison, firm i faces user cost $(1 + \tau_i)q$, with $1 + \tau_i > 0$. It chooses capital to maximize $z_i k_i^{\alpha} - (1 + \tau_i) q k_i$. Think of a rental tax rebated lump-sum to owners, so the payment itself is not a resource loss. Different effective financing costs could motivate a wedge, but this simple tax experiment does not derive a credit constraint. Market clearing still requires $k_1 + k_2 = K$.

7.6.2 Equalize marginal products, not capital quantities

With multiplier ν on the capital stock, the planner's interior conditions come from

$$\mathscr{L} = z_1 k_1^{\alpha} + z_2 k_2^{\alpha} + \nu(K - k_1 - k_2), \tag{622}$$

$$\mathscr{L}_{k_1} = \alpha z_1 k_1^{\alpha - 1} - \nu = 0, \tag{623}$$

$$\mathscr{L}_{k_2} = \alpha z_2 k_2^{\alpha - 1} - \nu = 0. \tag{624}$$

Thus

$$\alpha z_1 k_1^{\alpha - 1} = \nu = \alpha z_2 k_2^{\alpha - 1}. \tag{625}$$

Rearrange the ratio, then use resource clearing:

$$z_1 k_1^{\alpha - 1} = z_2 k_2^{\alpha - 1}, \tag{626}$$

$$(k_1/k_2)^{\alpha - 1} = z_2/z_1, \tag{627}$$

$$(k_1/k_2)^{1 - \alpha} = z_1/z_2, \tag{628}$$

$$k_1/k_2 = (z_1/z_2)^{1/(1-\alpha)}. \tag{629}$$

Set $b_i = z_i^{1/(1-\alpha)}$ and $S = b_1 + b_2$. Since $k_1/k_2 = b_1/b_2$ and $k_1 + k_2 = K$,

$$k_1 = (b_1/b_2) k_2, \tag{630}$$

$$K = (1 + b_1/b_2) k_2, \tag{631}$$

$$k_2 = K b_2/S, \qquad k_1 = K b_1/S. \tag{632}$$

$$\frac{k_1}{k_2} = \left(\frac{z_1}{z_2}\right)^{1/(1-\alpha)}, \qquad k_i = K\frac{z_i^{1/(1-\alpha)}}{z_1^{1/(1-\alpha)} + z_2^{1/(1-\alpha)}}. \tag{633}$$

Substituting into output and collecting terms gives the full sum

$$Y^* = \sum_{i=1}^{2} z_i (K b_i / S)^{\alpha}, \tag{634}$$

$$= K^{\alpha} S^{-\alpha} \sum_{i=1}^{2} z_i b_i^{\alpha}, \tag{635}$$

$$z_i b_i^{\alpha} = z_i^{1+\alpha/(1-\alpha)} = z_i^{1/(1-\alpha)} = b_i, \tag{636}$$

$$Y^* = K^{\alpha} S^{-\alpha} (b_1 + b_2) = K^{\alpha} S^{1-\alpha}. \tag{637}$$

$$Y^* = K^{\alpha} \left[z_1^{1/(1-\alpha)} + z_2^{1/(1-\alpha)}\right]^{1-\alpha}. \tag{638}$$

Capital goes to the firm with the higher marginal product until the marginal products match. Equal capital would be efficient only if the technologies and complementary inputs make that allocation satisfy the same condition.

Distorted private choices instead satisfy

$$0 = \frac{d}{dk_i}\left[z_i k_i^{\alpha} - (1+\tau_i) q k_i\right] = \alpha z_i k_i^{\alpha-1} - (1+\tau_i) q, \tag{639}$$

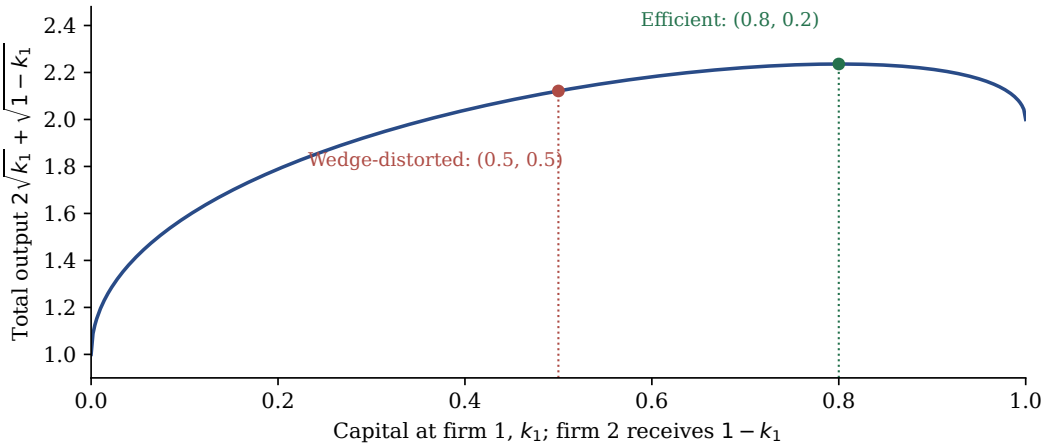
$$q = \alpha \frac{z_i}{1+\tau_i} k_i^{\alpha-1}. \tag{640}$$

Applying the same ratio calculation to $z_i/(1+\tau_i)$ gives

$$\frac{\alpha z_i k_i^{\alpha-1}}{1+\tau_i} = q, \qquad \frac{k_1}{k_2} = \left[\frac{z_1/(1+\tau_1)}{z_2/(1+\tau_2)}\right]^{1/(1-\alpha)}. \tag{641}$$

Both firms optimize, but their physical marginal products differ when their wedges differ. Removing the wedge changes where capital works; it does not change the given total stock.

For $\alpha = 1/2$, $z_1 = 2$, $z_2 = 1$, and $K = 1$, the efficient allocation is $(k_1, k_2) = (0.8, 0.2)$ and $Y^* = \sqrt{5} \simeq 2.2361$. If $\tau_1 = 1, \tau_2 = 0$, both adjusted productivities are one, so market clearing gives $(0.5, 0.5)$ and $Y = 3/\sqrt{2} \simeq 2.1213$. The output loss is about 5.13% of the efficient level, despite identical total capital and unchanged firm technologies. This is a hypothetical accounting comparison, not an estimate of the loss from a specific financial institution.



Allocation changes output without changing total capital. The stock is one throughout. Moving capital toward the more productive firm raises total output until marginal products match at $k_1 = 0.8$. The peak is an allocation at one date, not a growth rate or a new stock of knowledge.

A productivity level gain is not automatically a permanent growth gain. Reallocating the existing stock can raise output now. Repeating that comparison with an unchanged efficient allocation does not create ever faster growth. To affect trend growth, the allocation change must also alter a continuing mechanism such as research entry, learning, or the return to reproducible inputs. A measured productivity residual can reflect allocation as well as available technology; identifying its source requires more evidence.

Observed differences in marginal revenue products are not by themselves a complete diagnosis of financing distortions. Market power, adjustment costs, risk, prices, product quality, and measurement can matter. A wedge model provides a disciplined counterfactual once those choices are specified; it does not turn every cross-firm gap into a causal estimate of misallocation.

7.7 How to read an endogenous-growth claim

The same Ramsey household has appeared with different accumulation opportunities. The AK problem made the marginal return constant; the skill model explained one way reproducible inputs can grow together; the idea model made invention and its rewards explicit. The allocation experiment changed the use of existing resources rather than supplying another growth engine.

Mechanism	What is determined	What remains assumed
Ramsey benchmark	Optimal consumption and saving	Technology growth, if present
AK	Growth from the chosen saving path	A non-diminishing reproducible return
Skill accumulation	Work–learning split and balanced growth	A linear skill-formation opportunity
Innovation	Research effort, patent value, and growth	Research productivity, rights, and market structure
Semi-endogenous experiment	Conditional technology growth	Research share and labor growth in the calculation
Static allocation	Output from given stocks	Technologies, complementary inputs, and wedges

Endogenous does not mean parameter-free. Preferences, learning productivity, research technology, and legal rules remain specified primitives. The claim is that the accumulation or innovation choice, and the resulting growth, have been determined inside the stated environment.

Ask which stock grows, what resources it requires, whose marginal return supports the choice, and which restriction keeps that return from fading. Then distinguish a higher level, a faster transition, and a permanently higher growth rate. None is unimportant, but they are different conclusions.

These models isolate growth mechanisms rather than explaining every stage of development. Adoption, institutions, openness, demography, and distribution can change the relevant constraints and returns. The open-economy and policy parts will add those links. The next chapter compares the growth mechanisms through concrete economic situations and brings them together before we turn to fluctuations.

8 Growth Models: Mechanisms, Comparisons, and Reality

A region builds factories, trains workers, starts research laboratories, and eventually faces slower growth and an older population. Which growth model describes it? Several may help, because each isolates a different question. The useful comparison is between mechanisms, rather than between labels for countries or stages of development.

This chapter follows that region through a sequence of questions. The region is a teaching construction, not a disguised account of a particular country. We add mechanisms as the questions require them, keep their assumptions visible, and use historical evidence at specific points to connect the construction to actual experience. The aim is to learn how to move from an observation to a model and then back to an observation that could test it.

8.1 A map of the questions, not a ranking of the models

Growth accounting asks where measured output growth appears: capital, labor, or a residual. A growth model asks why those quantities change. Harrod and Domar ask whether spending and capacity expansion remain compatible. Solow asks how capital intensity adjusts when inputs can substitute. Ramsey asks who gives up consumption to finance accumulation. OLG changes who is alive, who owns assets, and which generations can trade. Endogenous-growth models ask what keeps the return to reproducible inputs or new knowledge from fading.

Production, saving, and ownership.

Framework	Question it closes	Situation that makes the question concrete
Growth accounting	Which measured inputs and residual accompany growth?	Output rises, but employment and capital also rise. How much is left to explain?
Harrod–Domar	Which investment path keeps demand, capacity, and plans compatible?	Building factories creates spending today and capacity that needs customers later.
Solow	How does capital per effective worker adjust under a given saving rule?	More machines initially help a capital-scarce region; their marginal contribution then falls.
Ramsey	How do households choose consumption and saving over time?	Higher expected returns make postponing consumption more attractive, but the sacrifice has a cost.
OLG	How do life-cycle saving and trade between living cohorts determine assets?	Workers buy retirement assets from older owners; future workers cannot sign contracts today.

Skills, ideas, and the use of resources.

Mechanism	What must be explained	Situation that makes the question concrete
AK	Why can a reproducible stock retain its marginal return?	Machines, skills, or other complementary inputs may expand together; the aggregate linearity still needs a justification.
Human capital	How much time goes to work rather than learning?	A technician leaves production for training, reducing output now to improve future capability.
Innovation	Who pays to create a reusable idea, and who receives its return?	A firm funds a design team; rivals may benefit from knowledge that the firm cannot fully charge for.
Semi-endogenous growth	How does research effort interact with the difficulty of discovery?	More researchers can be needed just to maintain a previous pace of technical improvement.
Allocation	Which producers use the existing inputs?	A productive workshop cannot expand while a less productive firm retains equipment or financing.

These are distinct dimensions. OLG does not follow Ramsey because it has more technology; it changes the household structure. An OLG economy can also innovate. A Ramsey economy can face demand constraints once the required market mechanisms are added. An allocation problem can matter at low or high income. Combining them requires a new, consistent equilibrium, not the addition of conclusions obtained under incompatible assumptions.

Add the mechanism that answers the next question. Start with what is scarce, who chooses its use, and what prevents adjustment. Then ask whether the unresolved issue concerns demand, saving, generations, skills, discovery, or allocation. A later model in these notes is not a claim that every economy must pass through the same development sequence.

8.2 A region builds factories: demand, capacity, and diminishing returns

8.2.1 First thought: more investment should mean more growth

Suppose the region has workers, basic skills, and access to an existing production method, but few machines. A factory project employs construction workers and buys equipment. Once completed, it can produce goods for years. It is natural to start with investment as the driver of growth. Two questions are already hidden in that statement: will the new capacity be used, and how much output will another unit of equipment add?

Harrod–Domar makes the first question visible. In Domar's experiment,

$$\dot{Q} = \kappa I, \qquad Y^d = I/s, \qquad \dot{Y}^d = \dot{I}/s,$$

where Q is capacity, Y^d is demand-supported output, s is a fixed saving propensity, and κ is a constant capacity increment per unit of investment. With no depreciation in this experiment, constant positive investment keeps adding capacity but leaves demand-supported output unchanged. Maintaining an initially matched path requires $\dot{I}/I = s\kappa$. Section 3 derives the conditions; the relation is a compatibility requirement, not a forecast of investment.

Harrod adds a related planning question: does realized growth justify the capacity that firms intended to install? With fixed saving propensity s and desired capital–output ratio $v_r > 0$, his no-depreciation benchmark gives warranted growth $g_w = s/v_r$. Labor and labor productivity instead set natural growth $g_n = n + g_A$. Matching these rates is an additional condition, not an identity. If sales disappoint and firms cut investment, the resulting loss of spending can further undermine sales. That feedback depends on the stated investment response; fixed proportions alone do not prove instability.

The real counterpart is a new industrial park whose construction creates business for local suppliers, but whose finished factories still need buyers. A factory is a productive asset; construction spending is a flow of demand. Confusing them makes it easy to count the spending effect and silently assume that all future output will be sold. Whether customers are domestic or foreign, and whether prices or credit adjust, requires additional mechanisms.

8.2.2 Second thought: can every new factory be as productive as the first?

Now suppose demand is sufficient and markets clear. The region can alter its mix of workers and equipment. Solow isolates this different question: with $k = K/(AL)$ and $y = Y/(AL)$,

$$y = f(k), \qquad \dot{k} = sf(k) - (n + g_A + \delta)k.$$

Here A is labor efficiency, L grows at rate n, A grows at an exogenous rate g_A, and δ is depreciation. Diminishing $f'(k)$ means that another machine adds less output when each effective worker already has more equipment. Capital intensity adjusts rather than requiring one fixed capital–output ratio to remain compatible with every growth path.

For a numerical check, hold A and L fixed at a date and let $f(k) = k^{1/3}$. A 20% increase in k raises y by

$$\frac{y_1}{y_0} = 1.2^{1/3} \simeq 1.0627, \qquad \frac{f'(k_1)}{f'(k_0)} = 1.2^{-2/3} \simeq 0.8855.$$

Output rises by about 6.27%, while the marginal product falls by about 11.45%. There is no contradiction: total output and the return to the next unit of capital are different objects. These are illustrative quantities, not estimates of an industrialization program.

Solow does not prove that newly built capacity always finds buyers. Harrod–Domar does not prove that substitution and price adjustment never occur. The region can have a short-run utilization problem and a long-run capital-intensity problem at the same time. The models differ in the adjustment mechanism they choose to expose.

8.3 Someone must save: from an investment target to household choices

8.3.1 A higher eventual income is not a free transition

Suppose officials propose raising the saving rate from 0.20 to 0.30. In the continuous-time Cobb–Douglas Solow benchmark,

$$k^* = \left(\frac{s}{n + g_A + \delta}\right)^{1/(1-\alpha)}, \qquad y^* = \left(\frac{s}{n + g_A + \delta}\right)^{\alpha/(1-\alpha)}.$$

With $\alpha = 1/3$, unchanged technology growth, and $n + g_A + \delta > 0$, the eventual ratio of output per effective worker is

$$\frac{y^*_{\text{new}}}{y^*_{\text{old}}} = \left(\frac{0.30}{0.20}\right)^{1/2} \simeq 1.2247.$$

But capital is inherited at the policy date. If the saving change is imposed immediately, output initially stays fixed and consumption changes by

$$\frac{c(0^+)_{\text{new}}}{c(0^-)_{\text{old}}} = \frac{1-0.30}{1-0.20} = 0.875.$$

An eventual output gain of about 22.47% comes with an initial consumption cut of 12.5%. This calculation compares two imposed saving rules. It does not establish the welfare ranking of their entire consumption paths.

The next question is unavoidable: whose consumption falls, and why would they agree? Ramsey replaces the saving instruction with preferences, resources, prices, and a choice. In the fixed-population, fixed-technology, competitive CRRA benchmark, with c and k per person,

$$\frac{\dot{c}}{c} = \frac{f'(k)-\delta-\rho}{\theta}, \qquad \rho > 0, \quad \theta > 0.$$

A unit not consumed today can earn a net return. Consumption grows when that return exceeds impatience, with θ governing the response. The Euler condition must still be combined with the resource constraint, initial capital, feasibility, and the terminal condition. It is not a saving rate by itself.

In both the Solow and standard Ramsey benchmarks, continued growth of output per worker on a balanced path still comes from the specified growth of labor efficiency. Optimizing saving changes the accumulation path and its welfare interpretation; it does not remove that technological assumption. The OLG benchmark changes generations rather than supplying a new technology engine. This is the gap addressed by endogenous growth.

For the region, a credible improvement in future production opportunities may change desired saving. A temporary income loss may instead encourage households to use assets to protect consumption. Taxes, borrowing limits, and uncertainty can alter both responses. The basic Ramsey model supplies the intertemporal trade-off; it does not establish every household's response when its omitted constraints become important.

8.3.2 The same residents are also different generations

Now let the region have young workers, retirees, and future entrants. The question is no longer only how one continuing household spreads consumption over time. Workers must acquire claims that will pay in retirement, while current retirees sell assets or receive transfers.

In the two-period OLG benchmark, cohort t chooses $c_{1,t}$ and $c_{2,t+1}$ with lifetime utility $u(c_{1,t}) + \beta u(c_{2,t+1})$, where $\beta > 0$. Without taxes, transfers, or bequests, its budgets are

$$c_{1,t} + a_{t+1} = w_t, \qquad c_{2,t+1} = R_{t+1}a_{t+1}.$$

Here a_{t+1} is saving per young worker, w_t is labor income, and R_{t+1} is the gross payoff on saving. A worker's saving can become the next period's physical capital, but the economy's net saving also reflects retirees' decumulation. The number and resources of each cohort matter.

A pension reform therefore has several margins: contributions reduce workers' disposable income, expected benefits may replace private retirement saving, and the first recipients and transition cohorts face different budgets. Fewer future workers can also lower a pay-as-you-go system's support ratio. None of these observations alone fixes the effect of aging on aggregate saving or capital: longevity, retirement ages, wages, transfers, and behavior must be specified.

An aging economy is not automatically dynamically inefficient. In the deterministic, stationary OLG benchmark with no technology growth, overaccumulation means $r < n$, equivalently $R < 1 + n$, under the chapter's depreciation convention. Age structure or a low observed safe interest rate alone does not verify that condition. Risky capital returns, government claims, and a changing population are not interchangeable objects.

Ramsey and OLG thus offer different ownership and welfare comparisons for the same investment program. Ramsey tracks a continuing household's consumption path. OLG asks which living and future cohorts gain or lose, including the initially old. Greater eventual output need not compensate those who finance the transition.

8.4 The next machine needs a skill; the next skill needs an idea

8.4.1 A machine shortage becomes a capability question

The region now has equipment, but technicians cannot maintain it, managers cannot coordinate production, and workers need time to master new methods. Buying the same machine again may not solve these problems. What looked like a capital shortage can include missing complementary capabilities.

Human capital makes one part explicit. In the two-sector benchmark,

$$Y = K^{\alpha}(vH)^{1-\alpha}, \qquad \dot{H} = [\eta(1-v) - \delta_H]H,$$

where H is the skill stock, $v \in [0,1]$ is the fraction of time spent working, $1 - v$ is learning time, $0 < \alpha < 1$, $\eta > 0$ is learning productivity, and $\delta_H \geq 0$ is skill depreciation. Moving time into learning reduces current production but raises future productive capability. The model's two shadow values put that opportunity cost next to its benefit.

This helps distinguish three interventions. More training hours changes $1 - v$; more effective instruction changes η; a better inherited skill base changes $H(0)$. They are different experiments. School enrollment or a training budget does not directly measure acquired skills or learning productivity. Credit constraints, teaching quality, and employer incentives would need to be added to explain those gaps.

AK provides a useful warning at this point. Writing $Y = BK$ assumes that the reproducible aggregate retains its marginal product. It does not explain why installing more identical machines is sufficient. Joint skill and capital accumulation can provide one mechanism, under the conditions derived in Section 7. An aggregate externality provides another, but then private and social returns differ. The equation alone cannot choose between them.

8.4.2 Learning an existing method differs from creating a new one

Suppose local engineers learn a production method already used abroad. This improves local productivity without expanding the world's stock of ideas. Adoption requires capability and often costly adaptation; the distance from existing best practice matters. At the frontier, the engineers must instead discover a method that is not yet available.

Korea provides a historical connection. A 2007 World Bank–Korea Development Institute study describes learning through imported equipment, production arrangements, training, and reverse engineering, followed by a greater role for domestic innovation.[20] This illustrates why equipment,

[20] Joonghae Suh and Derek H. C. Chen, eds. (2007), *Korea as a Knowledge Economy: Evolutionary Process and Lessons Learned*, World Bank, especially Chapters 6–8 and pp. 168–169 on technological adoption and innovation. World Bank report.

absorptive skills, and research can be complements. It does not identify one model as the complete explanation of Korean growth, or turn the sequence into a causal estimate. Trade, finance, institutions, and policy also belong in that account.

Return to our region. A design team uses workers who could have produced goods today. Once created, an idea may be usable by many producers without being depleted. A patent or another means of earning revenue can fund the team, but its private return need not capture all future knowledge benefits. This is the additional question in the innovation model: research effort must be supported by expected rewards and market entry, not just entered as a positive term in a production function.

Three objects that a productivity residual can hide. Adoption brings an existing method into local use. Human capital enables people to use or develop methods. Innovation adds new ideas. A measured productivity improvement may contain all three, as well as allocation and measurement effects. Growth accounting locates a residual; it does not separate these mechanisms without further information.

8.4.3 More researchers need not mean permanently faster growth

In the earlier linear research technology, $\dot{A}/A = \eta L_R$, a permanently larger research workforce L_R permanently raises idea growth. Changing the technology to

$$\dot{A} = \eta A^{\phi} L_R^{\lambda}, \qquad \phi < 1, \quad 0 < \lambda \leq 1,$$

changes that conclusion. With a fixed positive research share and labor growing at $n > 0$, a balanced path instead has $g_A = \lambda n/(1-\phi)$. A larger research share then raises the level path and changes the transition, while this balanced growth rate is unchanged. The calculation is conditional on the research share; it is not a new solution for optimal research entry.

Bloom, Jones, Van Reenen, and Webb (2020) examine research effort and research productivity in historical aggregate and sectoral evidence. Their finding that rising effort can accompany declining research productivity motivates checking the research technology rather than assuming a constant return per researcher.[21] It does not prove that one value of ϕ explains every industry or period. The practical question is how much effective research is needed to sustain the next improvement, not simply how many laboratories are built.

8.5 The same resources can produce more, but who gains?

Even after new factories, training, and research, the region can waste productive opportunities. Imagine two firms with different technologies. One has idle equipment; the other could use additional equipment productively but faces an explicit tax or financing obstacle. More aggregate investment is only one possible response. Moving the existing stock can also raise output, provided the gain exceeds movement, adjustment, and other costs.

The static allocation experiment in Section 7 holds total capital and complementary inputs fixed. With $Y = 2\sqrt{k_1} + \sqrt{k_2}$ and $k_1 + k_2 = 1$, equal allocation gives $Y = 3/\sqrt{2}$; the efficient allocation $(0.8, 0.2)$ gives $Y = \sqrt{5}$. No new equipment or idea is required. The mechanism is a change in where capital operates. Repeating an already completed reallocation does not generate another permanent increase in the growth rate.

[21] Nicholas Bloom, Charles I. Jones, John Van Reenen, and Michael Webb (2020), "Are Ideas Getting Harder to Find?" *American Economic Review*, 110(4), 1104–1144. https://doi.org/10.1257/aer.20180338.

Hsieh and Klenow's study of manufacturing in China and India makes the allocation question empirical by constructing counterfactuals under a specified firm-level model.[22] The connection is the distinction between available technologies and the allocation of inputs across producers. The counterfactual remains conditional on how productivity, prices, markups, and wedges are measured. A difference in observed revenue per unit of capital is not sufficient to diagnose a credit distortion.

For residents, a productive reallocation can still create losers. Workers may need to move, existing asset values can fall, and a firm owner may lose a protected position. Our static calculation has neither relocation costs nor a distributional account. Those costs must be introduced before claiming that every observed obstacle is removable at no sacrifice.

> **Three comparisons that must stay separate.** More output from given inputs is a production comparison. More consumption after replacement and accumulation is a resource comparison. A preferred consumption path, or an improvement for every cohort, is a welfare comparison. Solow's Golden Rule, Ramsey welfare, and OLG Pareto improvements answer different questions. None follows from an investment or GDP target alone.

At the research frontier, allocation and growth can interact. Moving skilled workers toward research may alter future knowledge creation, while raising the rewards to incumbent firms may weaken entry. To sign a growth effect, specify those margins and solve the resulting decisions. The static output gain itself does not settle the dynamic incentive question.

8.6 A growth slowdown: work backward from observations

Suppose the region's GDP growth falls after years of expansion. One account says it has exhausted capital accumulation; another blames aging; another calls for more research. All are hypotheses. Start by asking what actually slowed and which observations would distinguish the mechanisms.

8.6.1 Separate production per person from the number of people working

For total hours L, employment E, and population N,

$$\frac{Y}{N} = \frac{Y}{L}\frac{L}{E}\frac{E}{N}, \qquad g_{Y/N} = g_{Y/L} + g_{L/E} + g_{E/N}.$$

The identity does not require a growth model. Falling employment per resident can reduce output per person even if output per hour continues improving. Aging can affect that ratio, but participation and retirement behavior can also change. A decline in total GDP growth could additionally reflect slower population growth. Specify the denominator before choosing the explanation.

[22] Chang-Tai Hsieh and Peter J. Klenow (2009), "Misallocation and Manufacturing TFP in China and India," *Quarterly Journal of Economics*, 124(4), 1403–1448. https://doi.org/10.1162/qjec.2009.124.4.1403.

Different diagnoses of the same headline.

Observation to investigate	Candidate mechanism	What would help distinguish it?
Capital per worker keeps rising; its additional output contribution falls	Solow transition toward a conditional balanced path	Capital quality, effective labor, returns, and the distance from the conditional target
Output per person slows more than output per hour	Employment, hours, and demographic composition	Participation, age-specific employment, hours, and population measures
Equipment is installed, but cannot be used effectively	Skills, complementary inputs, or adoption costs	Training outcomes, vacancies, maintenance, and technology use
Research effort rises without a matching pace of improvement	Research productivity, duplication, or measurement	Effective research inputs and comparable measures of useful innovation
Productive firms remain small while others retain inputs	Allocation wedges, adjustment costs, or market power	Firm prices, quantities, costs, risk, and a specified counterfactual
Orders and utilization fall before installed capacity changes much	Spending, expectations, or short-run adjustment	Orders, inventories, hours, credit conditions, and the timing of capacity changes

The last column contains useful evidence, not automatic causal tests. For example, low measured returns can reflect expectations or risk, and unfilled vacancies can reflect wages as well as skills. A diagnosis becomes stronger when a proposed mechanism explains several observations and survives a counterfactual comparison.

8.6.2 Separate using capacity from expanding it

For a simple diagnostic illustration, define Q as output capacity under a stated operating convention and $u = Y/Q > 0$ as utilization. Then

$$Y = uQ, \qquad \Delta \log Y = \Delta \log u + \Delta \log Q.$$

This is an identity once Q is defined, not an independent measure of capacity or a behavioral theory. If capacity grows by 3% over a year but utilization falls from 0.90 to 0.81, actual output changes by

$$\frac{Y_1}{Y_0} = \frac{0.81}{0.90} \times 1.03 = 0.927.$$

Output falls by 7.3% although capacity rises. If utilization subsequently recovers, fast observed growth need not mean that the growth rate of capacity has permanently increased. The chosen numbers are an accounting example.

Figure 1 compares three distinct changes in output paths. They are deliberately constructed illustrations, not estimated responses. In an actual economy, capacity and utilization are difficult to separate and can affect one another: prolonged weak sales can discourage investment, training, or entry. The short run can leave a long-run trace.

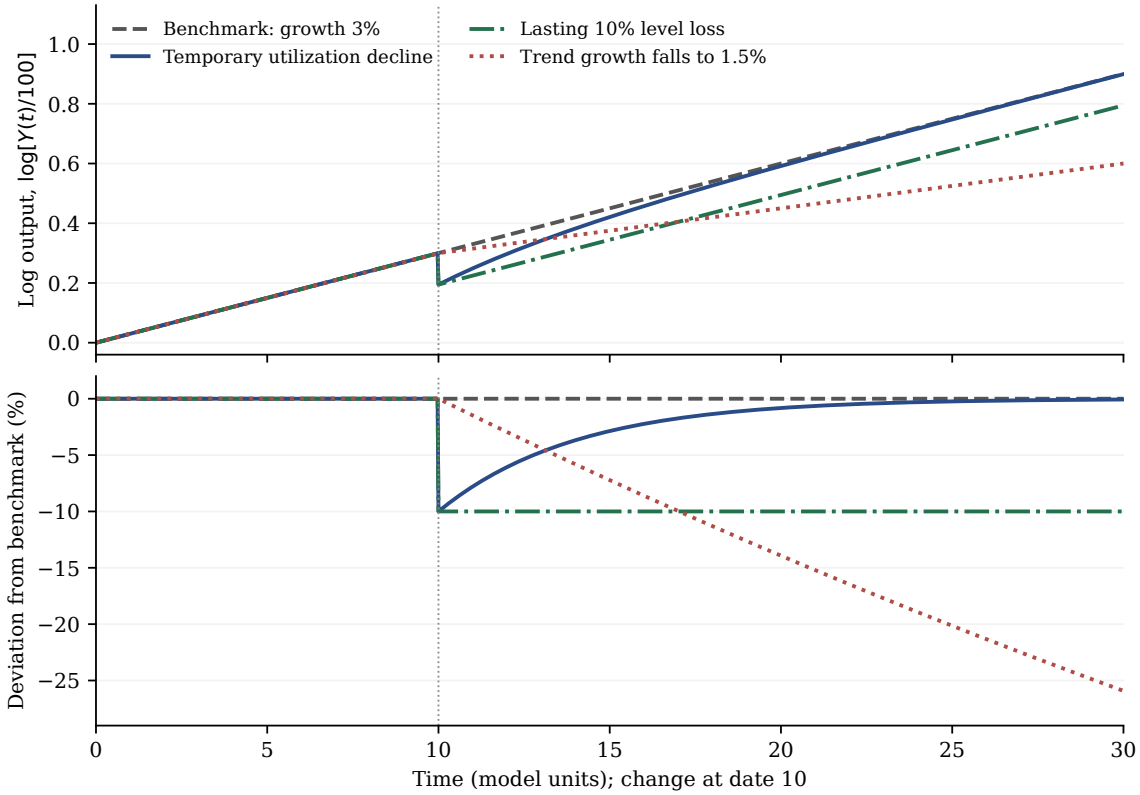


Figure 1: A temporary utilization decline, a lasting level loss, and slower trend growth produce different paths relative to the same benchmark. The shock occurs at model date 10. Both panels show constructed output series; they are not a decomposition estimated from data.

Question: what would your explanation predict next? If the slowdown is mainly a closing catch-up gap, restoring yesterday's investment share need not restore yesterday's growth. If it is mainly weak utilization, installing more capacity may not solve the immediate problem. If it reflects a weaker innovation mechanism, a temporary spending recovery does not by itself repair that mechanism. State a counterfactual and follow the quantities, decisions, and returns that it changes.

8.7 A longer view: learning to exercise economic judgment

Knowing how to solve each model is the beginning of economic judgment. The harder task is to decide which relationship deserves attention in an economy where many things change together. The preceding models help us practice that judgment: they reveal different costs of moving resources through time, across people, and toward new uses. Read them together and a few habits become especially useful.

8.7.1 Look for the constraint whose relaxation would change behavior

Suppose the region has both idle factories and small firms that want to expand. Calling it short of capital or burdened with excess capital is too quick. Equipment may be in the wrong firms, in the wrong locations, or suited to yesterday's products. Some producers may lack customers; others may have orders but lack working funds or skilled workers. An aggregate capital stock can conceal all of these situations.

Ask what would happen if one particular obstacle were relaxed. Would cheaper finance lead a firm to produce more, or would it merely refinance existing debt? Would an additional technician allow equipment to operate, or would output still be limited by sales? The value of relaxing a constraint is the intuition behind its shadow price. Observed interest rates, wages, and profits can help, but taxes, risk, market power, and contractual restrictions can separate them from social marginal values.

This is why the models should sharpen the next question asked of a firm or household. "Investment is low" names an outcome. A diagnosis identifies the decision that would change, the obstacle preventing it, and the evidence that would distinguish that obstacle from another one.

8.7.2 Recognize that success changes the next problem

An economy that has built basic equipment may need a different next investment from one that has scarcely begun. Training can make existing machines useful; capable engineers can make adaptation feasible; adaptation can bring production close enough to existing best practice that further progress requires discovery. These possibilities explain why a successful policy need not remain the best use of resources indefinitely.

The lesson is not that capital ceases to matter when ideas appear. Research needs laboratories and people, and new methods may require new equipment. The relevant complement and the return at the margin can change. Solow's diminishing returns, the skill model's joint accumulation, and the innovation model's costly discovery each expose a different part of that process.

A long horizon also changes how we read disappointment. A plant installed under reasonable expectations can become unprofitable after an unforeseen change in technology or demand. Its later loss does not establish that its original decision was irrational. Conversely, repeated losses followed by new investment require an explanation of incentives, financing, and exit. Distinguish a bad outcome from a bad decision, then ask whether the decision rule learns from the outcome.

8.7.3 Follow resources, financial claims, and expectations together

Finance gives someone command over resources and specifies who receives future payments. It does not by itself create the engineers, machines, or future goods needed to honor those payments. A financing arrangement can nevertheless be productive if it lets resources reach a use that was previously blocked. The real allocation and the financial contract must be examined together.

For example, a guarantee can make funding cheaper for a factory. Ask who bears the loss if sales disappoint and whether the guarantee changes project selection. If losses become public obligations, their cost reappears in someone's future budget. If cheaper funding enables a genuinely productive project that could not otherwise proceed, that benefit belongs in the comparison too. Neither the funding rate nor the spending total settles it.

Expectations connect the contract to current choices. A firm pays for training when it expects to use the resulting skills. A researcher pursues an idea when future rewards justify its present cost. A household saves against the consumption opportunities it expects assets to provide. Institutions matter here through concrete features: the reliability of contracts, the rules for entry and exit, the treatment of losses, and the

credibility of future policy. Naming "institutions" without specifying such a channel leaves the mechanism unexplained.

8.7.4 Judge a transition by the people who must live through it

A better eventual allocation does not tell us how to reach it. Capital is installed in particular places; skills belong to particular people; retirement claims were acquired under particular expectations. Moving toward a new production structure can strand some of those investments. The aggregate gain and the individual loss can both be real.

Ramsey teaches us to count consumption sacrificed before later benefits arrive. OLG teaches us that the people who sacrifice and the people who benefit may be different. An allocation model shows a production gain; it does not automatically provide the taxes, transfers, mobility, or information needed to compensate the losers. Even when compensation is feasible on paper, its financing and delivery require their own account.

The distinction matters for feasibility as well as fairness. A reform can be reversed if those bearing its early costs have no credible prospect of sharing its benefits. Yet preserving every existing claim can prevent resources from moving to better uses. Evaluate the sequence, the temporary support, and the commitment behind it. Neither immediate liquidation nor indefinite protection follows from the growth models alone.

8.7.5 Ask whether the opportunity can be repeated

Repairing a bridge, moving equipment to a better producer, and inventing a new method can all raise output. Their contribution to future growth depends on what remains to be done after the first gain. Once the bridge works or the equipment is well allocated, repeating the same action need not yield the same benefit. Research can continue to find new opportunities, but its resource requirements and difficulty may change.

For every growth proposal, ask what the economy will accumulate or discover next, what supports the return to that next step, and who will pay for it. AK makes continued returns explicit as an assumption. Human-capital and innovation models give those returns a more specific mechanism. The semi-endogenous experiment shows how a change in research difficulty can alter the long-run conclusion even while research remains valuable.

This also prevents an excessive preference for growth-rate effects. A lasting improvement in health, consumption, or productive efficiency can be worthwhile without making growth permanently faster. The welfare question concerns the opportunities people gain over their lives, including their costs. A growth rate is one description of those opportunities.

8.7.6 Give a conditional judgment and say what would change it

An experienced economist should be able to state a view clearly while remaining explicit about what could overturn it. Begin with the leading mechanism, explain why it fits the available observations, and identify the strongest competing explanation. Then ask which decision or observation would separate them. More equations cannot substitute for that comparison.

If orders are weak but productive capacity remains usable, the immediate question differs from one in which orders are strong and capable producers cannot expand. If both problems coexist, the policy sequence and its effects on future capacity must be considered together. An intervention that improves spending today may also change debt, entry, training, and the willingness to invest tomorrow.

Some decisions can be revised cheaply; others commit resources for years. Where learning is possible and delay is not too costly, a limited experiment with stated evaluation criteria may be preferable to an irreversible expansion. Where delay destroys productive relationships or skills, waiting also has a cost. The comparison requires the size of the potential gain, the consequences of being wrong, and the value of information obtained along the way.

A useful judgment has a mechanism and a revision rule. State which constraint appears to matter, whose behavior would change if it were relaxed, how the change reaches production and consumption, and who bears the transition cost. Then state what evidence would make you revise the diagnosis. The purpose of experience is to make these connections clearer, including when the available evidence cannot yet choose between plausible explanations.

8.8 What we carry into fluctuations

Growth models leave us with inherited stocks, resource constraints, decision rules, and a benchmark path. Fluctuation models ask how production and choices respond when opportunities or spending conditions change over time. The same physical accounting continues to hold. In a closed, one-good economy without government, in discrete time,

$$Y_t = A_t F(K_t, \ell_t), \qquad K_{t+1} = (1-\delta)K_t + Y_t - C_t,$$

where A_t is productivity, ℓ_t is labor input, and C_t is aggregate consumption. Here A_t denotes a multiplicative productivity shifter, rather than necessarily the labor-efficiency index in Solow. An inherited K_t cannot jump because households revise their expectations; consumption, hours, utilization where modeled, and investment can respond sooner.

The RBC benchmark adds changing real opportunities and intertemporal responses, with flexible prices and market clearing. New Keynesian models then add nominal adjustment restrictions and policy transmission. In neither case is a statistical trend automatically productive capacity, an efficient allocation, or the model's natural-output path. Those are different objects whose relationship must be established.

The region may also import equipment, export its products, or borrow abroad. Our closed-economy resource examples deliberately abstracted from those transactions. Adding them changes the goods and asset accounts and introduces relative prices and external financing. The open-economy part will develop those links; the domestic saving rate then need not equal the domestic investment rate.

The region's factories, skills, ideas, and age structure remain relevant when demand falls or productivity changes. Conversely, a recession can change those stocks and future opportunities. Moving to fluctuations changes the question and the adjustment mechanisms under examination; it does not require assuming that all shocks wash out or that long-run growth happens outside the economy's decisions.

When reading a new episode, first measure the outcome, then identify the margin of adjustment, the decision-maker, and the missing constraint. Choose a model that closes those relationships. Its value is the mechanism it makes explicit and the observations it helps us interpret.

Part III
Fluctuations

How do disturbances change production, spending, employment, and prices? We first build intuition about the description, sources, propagation, horizons, and consequences of fluctuations. We then develop a real benchmark, introduce nominal rigidities, and examine labor-market adjustment. Part V adds financial and distributional mechanisms and develops the policy comparison.

9 Understanding Economic Fluctuations

A factory receives fewer orders. At first it reduces overtime and lets inventories rise. If weak sales continue, it postpones equipment purchases and stops recruiting. Its suppliers lose business; some workers reduce spending. The factory's machines and its workers' skills have not suddenly vanished, yet production, income, and employment can change together.

This is a useful starting puzzle. What changed first? Why did other people respond? Why did adjustment take time? And when orders recover, will the economy simply return to its former path? The factory is a teaching example, not an explanation of every recession. It lets us construct the questions before choosing between competing models of the answers.

9.1 Why study growth before fluctuations?

The order in these notes is an analytical convenience, not a rule that all courses follow or that economies obey. Growth theory first gives us an account of productive resources, accumulation, consumption choices, and intertemporal consistency. We can then ask how those choices respond when conditions change unexpectedly or adjustment is incomplete.

A household facing a temporary income loss still compares consumption today with future opportunities. A firm revising investment still inherits its existing capital. A government trying to stabilize activity still uses resources or reallocates claims on resources. The growth models establish these relationships before uncertainty, labor adjustment, nominal contracts, and financial constraints complicate their interaction.

This sequence also provides benchmarks. To say that output is unusually weak, we need to specify weak relative to what: its previous level, an estimated trend, a forecast made before the disturbance, or a model's feasible or efficient allocation. Growth theory helps us think about the benchmark; it does not make that benchmark observable or immune to shocks.

Historically and conceptually, growth and fluctuations also pose joint questions. A period of weak activity can change investment, skills, entry, and innovation. A change in expected future productivity can change spending today. Studying a deterministic growth model first isolates some mechanisms; it does not imply that growth happens first in calendar time or independently of the disturbances that arrive along the way.

> **Growth gives us a path and a structure; fluctuations ask how they move.** Carry forward the inherited stocks, budgets, and choices. Then ask what changed, what can adjust now, and what must adjust later. A useful benchmark need not be a fixed trend, and a fluctuation need not leave that benchmark unchanged.

9.2 What exactly are we describing as a fluctuation?

We use economic fluctuations to describe uneven movements in aggregate production, spending, employment, and related variables over time. The object is a joint economic episode, not necessarily a deviation from one fixed smooth line. Begin with the observed movements before assigning them to a trend, a cycle, or a particular cause.

9.2.1 Levels, growth rates, and deviations answer different questions

Let $Y_t > 0$ denote real output measured consistently across dates. The log growth rate is $\Delta \log Y_t = \log Y_t - \log Y_{t-1}$. If output rises from 100 to 105 and then to 107, growth slows but output does not fall. If output instead falls from 100 to 95 and then rises to 98, the final period has positive growth but output remains below its initial level. Returning from 95 to 100 requires an increase of about 5.26%, not 5%. These are arithmetic examples, not recession-dating rules.

For a chosen reference path $\bar{Y}_t > 0$, define a log deviation

$$d_t = \log Y_t - \log \bar{Y}_t.$$

It is approximately the proportional deviation for small gaps. It can be negative while actual output is growing: a growing economy can lag behind an even faster-growing reference path. Conversely, output can fall and still lie above a chosen reference. The sign of growth and the sign of the deviation are different descriptions.

There are several possible references, with different interpretations.

Reference	What the comparison asks	What it does not establish
Previous peak	Has the previous level of activity been regained?	Whether the economy has caught up with a growing counterfactual path
Statistical trend	How far is output from a fitted smooth component?	That the smooth component is productive capacity or an efficient allocation
Earlier forecast	How far is activity from the path previously expected?	That every forecast error is an inefficient output loss
Model benchmark	How does output compare with the stated model allocation?	That the model's technology, frictions, or welfare criterion fit the episode

Potential output is an estimate of sustainable production under specified conditions. Natural output often denotes a model's flexible-price allocation under the other stated frictions; efficient output uses a welfare criterion. These need not coincide. A statistical filter cannot establish their equality. The reference must be named before a deviation is interpreted as slack, overheating, or a welfare loss.

Detrending is therefore part of the measurement problem. The estimated trend can change when new data arrive, especially near the end of the sample; an actual change in trend can also be assigned to the cycle by an unsuitable method. Hamilton's critique of the HP filter illustrates how detrending can introduce misleading dynamics.[23] For a first description, report the actual path alongside the deviation and explain how the reference was constructed.

9.2.2 An aggregate episode has breadth, timing, and composition

A decline at one firm is not automatically an aggregate recession. It might reflect customers moving to a competitor, a seasonal lull, or a sector shrinking while another expands. The business-cycle question is why activity moves across many firms, sectors, or households, and how those movements relate to one another.

[23] James D. Hamilton (2018), "Why You Should Never Use the Hodrick–Prescott Filter," *Review of Economics and Statistics*, 100(5), 831–843; working-paper version: `https://www.nber.org/papers/w23429`. The methodological lesson here is to examine the decomposition, not to presume that another filter reveals the true economic benchmark.

Describe the amplitude of a movement, its duration, and its persistence. Then examine comovement: do employment, hours, consumption, investment, and output move together or in opposite directions? Timing matters too. Orders, production, hiring, unemployment, and wages need not turn at the same date. A variable is called procyclical or countercyclical relative to a stated activity measure; leading and lagging describe timing, not causation.

Stock and Watson's study of postwar US macroeconomic time series organizes the evidence around these kinds of relationships.[24] Consumption categories should also be separated: a household may postpone a car purchase while maintaining food consumption. Investment and durable purchases can move sharply because their timing is adjustable. A story about aggregate consumption that ignores this composition may miss the margin on which households respond.

Before interpreting a series, specify whether it is real or nominal, aggregate or per person, and seasonally adjusted where appropriate. Predictable holiday spending is different from an unexpected demand change. A rise in nominal sales can reflect higher prices rather than more goods. A revision to GDP estimates can change our description without changing what firms actually produced at the time.

9.2.3 A business cycle is not a regular clock

Expansion, peak, contraction, and trough describe phases of aggregate activity. They do not require a repeating sine wave or a known period. An economy can be hit again before it has adjusted to an earlier disturbance. Different sectors and countries can also move at different times.

For US recession dating, the NBER considers the size, breadth, and duration of declines across several indicators. Its chronology is not mechanically the rule of two consecutive quarters of falling real GDP.[25] A trough marks a turning point into expansion; it does not certify that employment, household incomes, or output relative to a previous forecast have recovered. Other countries and institutions may use different dating procedures. Keep the descriptive convention separate from the explanation.

9.3 Where can fluctuations come from?

9.3.1 Begin with the first change in opportunities, spending, or financing

Return to the factory. Fewer orders could begin with a buyer losing income, a rise in financing costs, or a decision to postpone purchases. Output could also fall because an essential input becomes unavailable, even when buyers still want the product. Similar observed production declines can begin from different changes in the economic environment.

[24] James H. Stock and Mark W. Watson (1999), "Business Cycle Fluctuations in US Macroeconomic Time Series," *Handbook of Macroeconomics*, 1A, 3–64; working-paper version: `https://www.nber.org/papers/w6528`. The data coverage is historical US evidence, not a claim of identical regularities in every country or period.

[25] NBER, "Business Cycle Dating Procedure: Frequently Asked Questions," explains its US chronology, indicators, and treatment of depth, diffusion, and duration. NBER dating FAQ.

Starting disturbance	Concrete situation	First question for the mechanism
Production or input availability	A component shortage prevents assembly; energy becomes more costly	Which feasible production opportunities or marginal costs changed?
Private spending	Households postpone durable purchases; firms revise expected sales	Why did desired spending change at the existing prices and incomes?
Financing and balance sheets	Lenders cut credit limits; collateral loses value	Who cannot fund a planned transaction, and which losses or constraints caused it?
Policy	The policy rate, taxes, purchases, or a regulatory rule changes	Which part was anticipated or a response to economic conditions?
News and uncertainty	A future technology improvement is announced; income risk becomes harder to assess	Which expectations or risks changed before current production did?
Reallocation	Demand shifts between sectors needing different skills or equipment	How costly is moving people and resources to the new uses?

These categories overlap. An input shortage can reduce household income and spending. Lower asset prices can reflect worse expected profits, then reduce collateral and restrict credit. A rise in uncertainty can encourage some households to build buffers or some firms to delay irreversible projects, with the direction and size depending on preferences, constraints, and available options. A label identifies a candidate channel, not a complete causal account.

9.3.2 A shock is defined relative to information and a model

In a model, a shock commonly means a new realization or innovation that was not predicted from the information available at the previous decision date. The level of a tax, productivity, or an interest rate is not itself necessarily a shock. A policy action can partly be a predictable response to inflation or activity; interpreting the whole action as an independent policy disturbance would confuse response with cause.

An anticipated future change can affect activity before its implementation. If a buyer expects new equipment to be much better next year, today's orders may change now. The new information arrives today, even though the physical improvement occurs later. Rational expectations, where assumed, do not mean that agents foresee every realization or agree that nothing can go wrong.

Exogeneity is also a modeling boundary. A foreign demand change can be external to a small region while being the outcome of households and firms in the rest of the world. A productivity residual may reflect utilization or measurement, as well as technology. Calling a residual a technology shock does not identify its economic source.

Some models generate fluctuations through feedback, delayed adjustment, financial fragility, or changes in expectations without a sequence of large external disturbances. Self-fulfilling expectations require conditions such as strategic complementarities and an equilibrium-selection account; they cannot be inferred merely because confidence and activity fall together. A stable deterministic model without such mechanisms will not start cycling simply because its agents make choices.

An observed surprise is not yet an identified cause. A forecast error says that the outcome differed from an expectation. A structural explanation specifies the change in opportunities, preferences, constraints, information, or policy that produced it. Identification asks what evidence separates that explanation from competing changes occurring at the same time.

9.4 How does a disturbance become an aggregate movement?

9.4.1 Separate the impulse from propagation and amplification

The impulse is the initiating change. Propagation is how its effects move across time, sectors, and people. Amplification means that interactions make the response larger than a specified direct-effect comparison. Persistence means that the response remains after the initiating innovation; it need not imply periodic oscillation.

A missing component interrupts a producer. Its customers then face delays, its suppliers lose orders, and its workers may lose income. Networks make a local disturbance relevant elsewhere. A customer who postpones a purchase reduces a seller's revenue; if that seller cuts wages or jobs and the affected households cannot smooth consumption, another round of spending can fall. The strength of this feedback depends on price adjustment, alternative buyers, insurance, financial access, and policy responses.

Falling collateral can create another loop. A borrower loses funding, reduces purchases or sells assets, and those sales can further reduce asset prices under limited market absorption. Conversely, buffers, insurance, flexible substitution, and stabilizing policy can damp a disturbance. The initiating shock and the strength of these loops are separate objects.

A small disturbance can therefore produce a large movement in one setting and a modest movement in another. But a large observed decline does not prove a multiplier or a financial accelerator: the initiating disturbance might itself have been large. The direct effect, the comparison economy, and the feedback must be specified.

9.4.2 A temporary innovation can have a persistent response

A deliberately simple recursion makes the timing distinction visible:

$$x_t = \rho x_{t-1} + \varepsilon_t, \qquad 0 \le \rho < 1.$$

Here x_t is a dimensionless activity deviation, ρ is an assumed persistence coefficient, and ε_t is an innovation. Start from $x_{-1} = 0$, set $\varepsilon_0 = -1$, and set every later innovation to zero. Substitution gives

$$x_0 = -1, \qquad x_1 = -\rho, \qquad x_2 = -\rho^2, \qquad x_t = -\rho^t \quad (t \ge 1).$$

The innovation lasts one period. Unless $\rho = 0$, its response lasts longer. Higher ρ slows its decay; these paths do not repeatedly change sign. This is a timing illustration, not a household problem, an estimated response, or proof of a particular source of persistence. In a structural model, capital, job matching, contracts, or balance sheets would have to generate the relevant dynamics.

The accumulated deviation in this stable example is $\sum_{t=0}^{\infty} x_t = -1/(1-\rho)$. That is an area in units of the chosen deviation, not a welfare measure or an identified amount of GDP lost. It shows why impact size alone is insufficient to describe a disturbance. Figure 2 holds the impact fixed to isolate persistence; a full model can change impact and duration together.

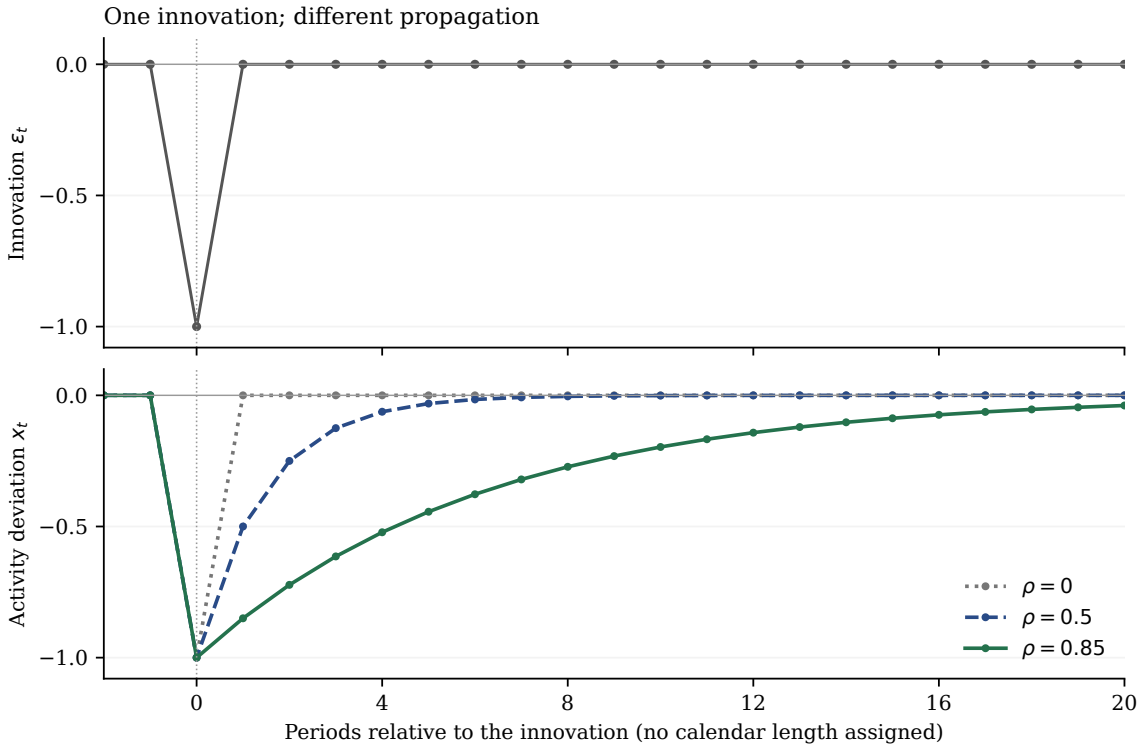


Figure 2: The same one-period negative innovation with different assumed persistence coefficients. All paths have the same impact. The lower panel shows subsequent activity, not additional shocks. Periods have no assigned calendar length, and the curves are not estimates.

9.4.3 Individual plans meet through income, prices, and constraints

A household can try to improve its finances by spending less. If many households do so together, their suppliers' revenues change. It is then necessary to explain how saving reaches investment, how prices and interest rates respond, and whether output or employment adjusts. Planned saving is not an extra stock of goods available independently of those responses.

In a flexible-price economy with effective intermediation, some adjustment may occur through relative prices and intertemporal substitution. With sticky prices, pessimistic expected sales, or binding financing constraints, the same initial spending intention can produce a different output response. The paradox of thrift is a conditional general-equilibrium mechanism, not a statement that saving is always harmful. It is consistent with the growth chapters' finding that accumulation can improve future opportunities.

The same principle applies to a firm cutting wages or costs: an individually sensible response changes other people's income and choices. Individual optimization does not by itself guarantee an efficient aggregate outcome when externalities or other distortions are present. Conversely, adjustment to a genuinely reduced production opportunity can lower output even in an efficient equilibrium. Falling output alone does not choose between these accounts.

9.5 Do fluctuations belong to the short, medium, or long run?

9.5.1 The relevant clock is the adjustment mechanism

Business-cycle analysis often focuses on changes over quarters and years. That is a useful scale for production, spending, and employment decisions; it is not a definition that all effects expire within a fixed number of years. Short, medium, and long run describe which adjustments have had time to occur. Those adjustments overlap rather than occupying sealed calendar boxes.

Perspective	Adjustments to examine	Why an effect may remain
Short run	Orders, inventories, hours, spending, prices, and contracts	Existing commitments or constraints can prevent immediate reallocation
Medium run	Hiring, matching, capital installation, sectoral movement, and debt repair	Stocks, skills, and financial positions take time to rebuild or reorganize
Long run	Productive capacity, institutions, research, and paths of technology and population	Earlier decisions can alter the stocks or accumulation mechanisms that support later activity

A price can change quickly while a borrower remains heavily indebted. Construction can stop immediately although an unfinished plant takes years to repurpose. Employment can recover in aggregate while displaced workers still need different skills. The appropriate horizon depends on the question and the constraints that remain, not simply on how much time has passed.

Financial movements can also have a different timescale from measured business-cycle movements. Drehmann, Borio, and Tsatsaronis (2012) document longer movements in credit and property prices in their historical financial- cycle analysis.[26] A short recovery in spending need not complete balance-sheet repair. Nor does a longer financial movement establish that every business cycle began in finance.

9.5.2 Shock duration, recovery time, and lasting damage are different

A shipping interruption might end quickly while inventories, production schedules, and customers' orders take longer to normalize. A persistent rise in energy costs may require a new production mix instead of restoration of the old one. A brief period without credit might close a firm whose customer relationships cannot be recreated immediately.

The growth chapters make these links explicit. Less investment changes future capital. Less training can change skills. Research cuts or reduced entry can change later innovation. Not every temporary reduction produces a permanent effect: replacement investment, rehiring, catch-up, or new entry can restore the previous path under suitable conditions. The adjustment mechanism must determine whether and how that happens.

In the strict sense, hysteresis means that history changes the economy's long-run outcome, rather than merely producing slow convergence to the same one. Blanchard and Summers' analysis of European unemployment is a classic application of this idea.[27] The term should not substitute for a mechanism. Lost skills, disrupted job matching, and a temporary hiring delay have different implications for whether an effect eventually disappears.

[26] Mathias Drehmann, Claudio Borio, and Kostas Tsatsaronis (2012), "Characterising the Financial Cycle: Don't Lose Sight of the Medium Term!" *BIS Working Papers*, no. 380. `https://www.bis.org/publ/work380.htm`. This is evidence under that paper's definition and sample, not a universal period for every credit cycle.

[27] Olivier J. Blanchard and Lawrence H. Summers (1986), "Hysteresis and the European Unemployment Problem," *NBER Macroeconomics Annual*, 1, 15–78; working-paper version: `https://www.nber.org/papers/w1950`.

Cerra and Saxena (2008) provide historical cross-country evidence of highly persistent output losses after financial crises and some political crises. [28] This is a reason to investigate lasting losses, not permission to label every estimated trend break as the causal effect of a recession. The earlier forecast may have been too optimistic, and the disturbance may itself have changed productive opportunities.

Long-run adjustment does not mean returning to the old forecast. A model can have flexible prices in the long run and still inherit different capital, skills, debt, or technology after a downturn. A permanent level loss, a permanently lower growth rate, and slow recovery to an unchanged path are different outcomes. The time horizon alone does not tell us which one occurs.

9.6 Why do fluctuations matter?

9.6.1 Lost activity and displaced people are not just a line on a graph

An unsold service today may never be produced later. A later increase in output need not replace the consumption or work experience that was missed. Returning to an earlier output level also falls short of a counterfactual path that would have continued growing. The benchmark must still be justified before a cumulative shortfall is interpreted as avoidable lost production.

Households experience the same aggregate movement differently. A worker losing a job, a borrower facing refinancing, a retiree with fixed nominal claims, and a firm with cash reserves face different risks. Aggregate consumption can appear stable while losses concentrate among people who cannot insure them. Inflation and asset-price changes can redistribute purchasing power as well as change aggregate activity.

The costs include consumption risk, involuntary loss of work, disruption of productive relationships, and possible changes in future opportunities. An expansion can also involve excessive risk-taking or distorted allocation under particular incentives. The welfare object is therefore richer than minimizing the variance of GDP around a fitted trend.

9.6.2 Even fluctuations around the same mean can be costly

A small example isolates consumption risk. Suppose a person has log utility and faces two equally likely consumption outcomes, 0.9 and 1.1, with no way to insure or transfer resources between them. Expected consumption is one, but expected utility is

$$\tfrac{1}{2}\log(0.9) + \tfrac{1}{2}\log(1.1) = \tfrac{1}{2}\log(0.99) < \log(1) = 0.$$

The certain consumption giving the same utility is $\sqrt{0.99} \simeq 0.99499$. Eliminating this particular risk at unchanged mean consumption would be desirable to this person. It does not tell us the aggregate welfare cost of business cycles: insurance, leisure, heterogeneity, expected growth, and the cost of changing the fluctuations are absent.

The example illustrates a reason to care about risk. It does not imply that every observed output movement should be prevented. If an input becomes scarce, an efficient economy may produce less and alter its consumption mix. If a price or credit distortion causes additional unemployment and lost trade, reducing that distortion poses a different welfare problem.

[28] Valerie Cerra and Sweta Chaman Saxena (2008), "Growth Dynamics: The Myth of Economic Recovery," *American Economic Review*, 98(1), 439–457. `https://doi.org/10.1257/aer.98.1.439`. The statement concerns the crises and methods studied there, not an inevitability for every downturn.

Question: the same fall in output, three different economies. In one economy, a missing input makes production infeasible. In another, customers postpone purchases while goods prices adjust slowly. In a third, firms have orders but lose access to working funds. Ask what limits production in each, what happens to prices and employment, and which intervention changes the relevant constraint. The observed fall in GDP is insufficient to prescribe the same response in all three.

Policy can cushion income risk, support spending, repair intermediation, or help resources adjust. Each intervention has a channel, financing, incentive effects, and implementation limits. Supporting demand does not manufacture a missing component; increasing productive capacity does not ensure it will be purchased now. Some episodes require several responses because the initial disturbance creates new constraints as it propagates.

9.7 What must a model of fluctuations explain?

A convincing account should explain a joint path: production and spending, employment and hours, prices and wages, investment and finance. It should say what initiated the change, what was already anticipated, why other agents responded, and why the response took the observed time to unfold. Matching the movement of GDP alone can leave very different explanations observationally similar.

A useful empirical distinction is between an unconditional regularity and a response to a specified shock. Output and employment moving together in the data does not establish how either responds to a technology shock, for example: the data contain several disturbances and policy responses. An impulse response asks what follows a particular innovation relative to a specified comparison path. Its interpretation depends on how that innovation was identified.

The next chapter starts with RBC because it carries the Ramsey logic into changing real opportunities and endogenous labor choices. It gives a benchmark for asking how much individual adjustment and inherited capital can explain under flexible prices. The order of presentation does not establish that technology causes all fluctuations, that money is irrelevant, or that all unemployment is an efficient choice. Those are claims requiring specific assumptions and evidence.

New Keynesian and sticky-wage models then make nominal adjustment part of the explanation. Labor-market models examine hiring and matching, while later chapters develop financial and distributional transmission. The models will earn their place by explaining a mechanism and its limits. Keep the factory's questions in view: what changed first, what carried the change, who bore it, and which opportunities remained different afterward?

10 Real Business Cycle Model

10.1 From a growth path to changing opportunities

10.1.1 What would make a Ramsey household change its working week?

Imagine a group of factories. Their machines have not changed overnight, but an interruption in power supply now makes each working shift less productive. Orders, hours, investment, and consumption may all change. Start with questions rather than labeling the event a "technology shock." Can the factories produce less with the same inputs? Are households poorer? Should they postpone work or investment until conditions improve? Would the answer differ if the interruption lasted one week rather than several years?

Ramsey already gives a way to compare consumption today and tomorrow. But with fixed labor and a known production path, it leaves two margins closed. Households cannot move working time toward more rewarding dates, and they never have to choose before learning tomorrow's production conditions. Let labor be chosen; let future productivity be uncertain. Keep preferences, ownership, production, and market clearing explicit. This produces a tractable real business cycle (RBC) benchmark.

This is a teaching reconstruction of the modeling questions, not a record of any author's private thought process. Kydland and Prescott (1982) and Long and Plosser (1983) are foundational contributions. Their actual environments differ; the one-sector model below is a simpler teaching economy, without the former's multi-period construction technology or the latter's sectoral structure. [29]

10.1.2 Choose the disturbance, then explain its transmission

A scalar A_t summarizes how much output given capital and hours can produce. An unexpected change in it is an innovation. The persistence of A_t describes how long production opportunities change. Household choices and accumulated capital determine how that change travels through the economy. These are three different objects: innovation, persistence, and propagation.

A fall in A_t need not mean that engineers forgot a blueprint. A power or logistics disruption can reduce effective production opportunities. But treating an event as a scalar productivity change is a modeling decision: energy inputs, relative prices, network bottlenecks, and distributional losses have been suppressed. A credit refusal or an absence of orders cannot automatically be reclassified as technology. Their mechanisms need their own constraints.

The growth models explain accumulation and, in some cases, the source of trend growth. Here we hold trend productivity constant to isolate fluctuations around a stationary reference. This does not assert that actual output has no trend. Adding a deterministic trend requires a consistent normalization of production, preferences, and Euler conditions; it cannot be hidden inside the shock term.

[29] Finn E. Kydland and Edward C. Prescott (1982), "Time to Build and Aggregate Fluctuations," *Econometrica*, 50(6), 1345–1370, https://www.minneapolisfed.org/~/media/files/research/prescott/papers/timetobuild.pdf; John B. Long, Jr., and Charles I. Plosser (1983), "Real Business Cycles," *Journal of Political Economy*, 91(1), 39–69, https://doi.org/10.1086/261128.

A real benchmark is a conditional experiment. The original course principles become two questions. How much joint movement can real disturbances produce when agents respond optimally? Which observed movements require additional mechanisms? Money and nominal frictions are excluded from the real allocation block below. Their exclusion does not establish their empirical irrelevance. Government purchases, relative import prices, and preference disturbances can be studied in extensions, but are not silently included in this one-shock economy.

10.2 Complete conditions and the order of decisions

The main model has endogenous capital. The course's fixed-capital benchmark will be recovered as a separate restriction in Section 10.5. Population is one and does not grow; hours are allowed to vary at every date. In particular, zero population growth is not constant labor input.

01 Dates, people, and the time endowment

Time is discrete, $t = 0, 1, \ldots$. A closed economy contains many identical, infinite-lived, price-taking households, aggregated to one representative household of constant mass one. Each household has one unit of time per date: $N_t \in [0, 1]$ is work and $L_t = 1 - N_t$ is leisure. Work denotes an intensive hours margin; there is no separate unemployment, participation, matching, or involuntary rationing margin. Consumption satisfies $C_t > 0$.

02 Goods, states, ownership, and timing

One homogeneous good can be consumed or installed as capital at unit resource cost. Households own the capital and firm shares. The given initial stock is $K_0 > 0$; K_t was chosen at $t - 1$ and is usable at t. The current productivity A_t is observed before choosing C_t, N_t, I_t, K_{t+1} and financial positions. Next period's innovation is not yet observed. Capital is the only accumulated productive stock; there are no inventories, skills, adjustment costs, installation delays, or variable utilization in this benchmark.

03 Production and competitive firms

Firms rent capital and hire labor in competitive spot markets. Their technology is $Y_t = A_t K_t^{\alpha} N_t^{1-\alpha}$, with $A_t > 0$ and $0 < \alpha < 1$. It has constant returns in (K_t, N_t), positive marginal products, and diminishing marginal products holding the other input fixed. Firms take prices and A_t as given; there is no markup, externality, or nominal price or wage rigidity.

04 The stochastic process and information

Write $a_t = \log(A_t/\bar{A})$, where $\bar{A} > 0$, and let

$$a_t = \rho_a a_{t-1} + \varepsilon_t^a, \qquad |\rho_a| < 1, \qquad E_{t-1}\varepsilon_t^a = 0, \quad E[(\varepsilon_t^a)^2] = s_\varepsilon^2 < \infty.$$

Innovations are independent and identically distributed, with a distribution known to households; Gaussian innovations are an optional specification. E_t conditions on the full observed history through t and the equilibrium rules, including current K_t, A_t. Rational expectations means using this conditional law, not knowing the future realization. $a_t = 0$ is the deterministic reference, not a claim that $EA_t = \bar{A}$ under uncertainty.

05 Preferences and the household objective

The household maximizes $E_0 \sum_{t=0}^{\infty} \beta^t u(C_t, N_t)$, with $0 < \beta < 1$, $\sigma > 0$, $\chi > 0$, and $\varphi > 0$, where

$$u(C,N) = \begin{cases} (C^{1-\sigma} - 1)/(1-\sigma) - \chi N^{1+\varphi}/(1+\varphi), & \sigma \neq 1, \\ \log C - \chi N^{1+\varphi}/(1+\varphi), & \sigma = 1. \end{cases}$$

The original course normalization is $\chi = 1$. σ governs consumption curvature; φ governs the curvature of work disutility. We derive local interior conditions with $0 < N_t < 1$. Parameter and shock choices must support finite expected welfare and feasible paths; large shocks may activate corners that a local solution does not describe.

06 Accumulation, prices, and household resources

Capital depreciates at $0 < \delta < 1$ and satisfies $K_{t+1} = (1-\delta)K_t + I_t \geq 0$. The baseline permits reversible investment; there is no additional $I_t \geq 0$ constraint. Let w_t be the real wage, q_t the real capital rental payment before depreciation, and d_t real profits distributed to households. With no financial trade the household budget is

$$C_t + K_{t+1} = w_t N_t + (q_t + 1 - \delta)K_t + d_t.$$

Households face no tax, transfer, borrowing wedge, or idiosyncratic uninsured risk. Prices and returns are taken as given in individual optimization; market clearing is imposed afterward.

07 Financial claims and nominal units

For the course's nominal bond calculation, $P_t > 0$ is the goods price, $W_t = P_t w_t$, and $D_t = P_t d_t$. B_t pays one currency unit per unit at $t+1$ and costs $Q_t > 0$ at t, so $R_t^n = 1/Q_t = 1 + i_t$ is the gross nominal return. The full budget is

$$P_t[C_t + K_{t+1} - (1-\delta)K_t] + Q_t B_t = W_t N_t + P_t q_t K_t + B_{t-1} + D_t.$$

Bonds are in zero aggregate net supply and $B_{-1} = 0$. A household's small financial trade is allowed when deriving its Euler condition; identical-agent market clearing then sets aggregate $B_t = 0$. A risk-free *real* claim is a separate hypothetical payoff, not the nominal bond deflated by a known future price. There is no monetary asset in the core household budget.

08 Market clearing, solvency, and terminal behavior

Household labor and capital supplies equal firms' demands. Goods clear as $Y_t = C_t + I_t$, and constant returns with competitive pricing imply $d_t = 0$. Admissible household plans rule out debt rollovers that finance consumption without resources. For the capital-only aggregate optimum the transversality restriction is

$$\lim_{T \to \infty} E_0[\beta^T C_T^{-\sigma} K_{T+1}] = 0.$$

Solvency restricts admissible borrowing; transversality selects optimal terminal behavior. Neither follows merely from writing the interior Euler equations. No-Ponzi restrictions also apply to any individual financial trades.

09 Equilibrium, reference state, and scope

An equilibrium is a collection of history-contingent allocations and prices satisfying household optimality and admissibility, firm optimality, the productivity law, and all market-clearing conditions for the given initial state. We select the locally stable interior solution near its deterministic steady state. Money, government, foreign trade, monopoly, financial frictions, and household heterogeneity are absent. A nominal policy or quantity closure is an extra specification; the real equations alone do not select P_t.

The within-date sequence is

$$\underbrace{K_t,\ A_t \text{ observed}}_{\text{inherited stock and current information}} \longrightarrow \underbrace{N_t,\ Y_t,\ C_t,\ I_t}_{\text{work, production, and allocation}} \longrightarrow \underbrace{K_{t+1}}_{\text{productive next date}} .$$

An announcement about tomorrow can change today's choices; tomorrow's machine stock cannot jump retroactively. In the experiment below only unanticipated productivity innovations occur.

10.3 Household choice: working now and transferring consumption

10.3.1 Retain the course's nominal bond route

First suppress capital trade to isolate the original asset calculation. Write nominal nonlabor income as $\mathscr{D}_t$; this includes the fixed factor's income in the course benchmark. The budget is

$$P_tC_t + Q_tB_t = B_{t-1} + W_tN_t + \mathscr{D}_t. \tag{642}$$

Use resources minus expenditures in the multiplier term:

$$\mathscr{L} = E_0 \sum_{t=0}^{\infty} \beta^t \{u(C_t, N_t) + \lambda_t [B_{t-1} + W_tN_t + \mathscr{D}_t - P_tC_t - Q_tB_t]\}. \tag{643}$$

For an interior choice at a history observed at t, the first two derivatives are

$$u_{C,t} - \lambda_t P_t = 0, \qquad \lambda_t = u_{C,t}/P_t, \tag{644}$$

$$u_{N,t} + \lambda_t W_t = 0, \qquad -u_{N,t} = \lambda_t W_t. \tag{645}$$

The common factor β^t and the probability of reaching this history cancel from the two conditions. Divide the second condition by the first:

$$\frac{-u_{N,t}}{u_{C,t}} = \frac{\lambda_t W_t}{\lambda_t P_t}, \tag{646}$$

$$-\frac{u_{N,t}}{u_{C,t}} = \frac{W_t}{P_t} = w_t. \tag{647}$$

The marginal disutility of another working hour equals its consumption-valued wage. This is an intratemporal condition: all its objects are known at t.

One unit of B_t costs Q_t now and pays one currency unit in every next-date state. Its two multiplier contributions give

$$0 = -\beta^t \lambda_t Q_t + \beta^{t+1} E_t \lambda_{t+1}, \tag{648}$$

$$\lambda_t Q_t = \beta E_t \lambda_{t+1}, \tag{649}$$

$$Q_t \frac{u_{C,t}}{P_t} = \beta E_t \left[\frac{u_{C,t+1}}{P_{t+1}} \right], \tag{650}$$

$$Q_t = \beta E_t \left[\frac{u_{C,t+1}}{u_{C,t}} \frac{P_t}{P_{t+1}} \right]. \tag{651}$$

Current P_t and $u_{C,t}$ can leave the expectation; future marginal utility and the future price generally cannot. The last line retains the course's nominal Euler equation with a consistent sign and information convention.

Differentiate the stated utility explicitly:

$$\frac{d}{dC} \frac{C^{1-\sigma} - 1}{1-\sigma} = \frac{(1-\sigma)C^{-\sigma}}{1-\sigma} = C^{-\sigma}, \tag{652}$$

$$\frac{d}{dC} \log C = 1/C = C^{-1}, \tag{653}$$

$$\frac{\partial u}{\partial N} = -\frac{\chi(1+\varphi)N^{\varphi}}{1+\varphi} = -\chi N^{\varphi}. \tag{654}$$

Substitution into the two optimality conditions yields

$$-\frac{-\chi N_t^{\varphi}}{C_t^{-\sigma}} = \chi C_t^{\sigma} N_t^{\varphi} = w_t, \tag{655}$$

$$\frac{u_{C,t+1}}{u_{C,t}} = \frac{C_{t+1}^{-\sigma}}{C_t^{-\sigma}} = \left(\frac{C_{t+1}}{C_t} \right)^{-\sigma}, \tag{656}$$

$$Q_t = \beta E_t \left[\left(\frac{C_{t+1}}{C_t} \right)^{-\sigma} \frac{P_t}{P_{t+1}} \right]. \tag{657}$$

When C_t and thus the marginal value of resources are held fixed,

$$\log N_t = \frac{\log w_t - \log \chi - \sigma \log C_t}{\varphi}, \tag{658}$$

$$\left. \frac{\partial \log N_t}{\partial \log w_t} \right|_{u_{C,t}} = 1/\varphi. \tag{659}$$

This is the Frisch hours elasticity. Along an equilibrium, consumption also changes: a higher wage creates both a reward to working and additional wealth. The Frisch calculation alone does not determine the total hours response.

Keep random prices inside the expectation. In general $E_t(X/Z) \neq E_tX/E_tZ$, and $E_t(XZ) \neq E_tX\,E_tZ$. For two equally likely future states with $(X,Z) = (1,1),(3,2)$,

$$E(X/Z) = \tfrac{1}{2}(1 + 3/2) = 5/4, \qquad EX/EZ = 2/(3/2) = 4/3.$$

The nominal bond pays a certain currency amount, but its purchasing power $1/P_{t+1}$ is uncertain. A risk-free real bond instead has a certain goods payoff. Confusing the two loses the connection between prices and consumption risk.

10.3.2 Capital brings back the Ramsey margin, with uncertain returns

For the full budget in card 07, consumption and hours have the same derivatives. The cost of one extra unit of K_{t+1} is P_t at t. Next date it delivers rental income $P_{t+1}q_{t+1}$ plus surviving capital $P_{t+1}(1-\delta)$. Taking prices as given, the two-date derivative is

$$0 = -\beta^t \lambda_t P_t + \beta^{t+1} E_t[\lambda_{t+1} P_{t+1}(q_{t+1} + 1 - \delta)], \tag{660}$$

$$\lambda_t P_t = \beta E_t[\lambda_{t+1} P_{t+1}(q_{t+1} + 1 - \delta)], \tag{661}$$

$$C_t^{-\sigma} = \beta E_t[C_{t+1}^{-\sigma}(q_{t+1} + 1 - \delta)]. \tag{662}$$

Let $R^K_{t+1} = q_{t+1} + 1 - \delta$ and $M_{t,t+1} = \beta(C_{t+1}/C_t)^{-\sigma}$. Division by current marginal utility gives

$$1 = E_t[M_{t,t+1} R^K_{t+1}], \tag{663}$$

$$Q_t = E_t[M_{t,t+1} P_t/P_{t+1}], \tag{664}$$

$$R^f_t = 1/E_t M_{t,t+1}. \tag{665}$$

The last line prices a hypothetical risk-free real payoff. The bond can help price intertemporal transfers even though identical households hold zero bonds in aggregate. It cannot make aggregate consumption exceed aggregate resources.

Expected capital returns need not equal the risk-free return. Expanding the conditional covariance gives

$$1 = (E_t M_{t,t+1})(E_t R^K_{t+1}) + \mathrm{Cov}_t(M_{t,t+1}, R^K_{t+1}), \tag{666}$$

$$E_t R^K_{t+1} = R^f_t - \frac{\mathrm{Cov}_t(M_{t,t+1}, R^K_{t+1})}{E_t M_{t,t+1}}. \tag{667}$$

A payoff arriving when consumption is scarce is especially valuable. This risk distinction disappears to first order around the deterministic reference; it is not absent from the exact model.

10.4 Firms, resource clearing, and the real equilibrium

10.4.1 Differentiate before normalizing the capital stock

In real goods the firm solves

$$\max_{K^d_t, N^d_t} A_t (K^d_t)^\alpha (N^d_t)^{1-\alpha} - w_t N^d_t - q_t K^d_t.$$

The household owns the stock; the firm rents its current services. Thus the firm's capital demand is a within-date choice, even though aggregate capital supply K_t is predetermined. The input derivatives at an interior optimum are

$$0 = (1-\alpha)A_t(K_t^d)^{\alpha}(N_t^d)^{-\alpha} - w_t, \tag{668}$$

$$0 = \alpha A_t(K_t^d)^{\alpha-1}(N_t^d)^{1-\alpha} - q_t. \tag{669}$$

Impose $K_t^d = K_t$ and $N_t^d = N_t$ afterward:

$$w_t = (1-\alpha)A_t K_t^{\alpha} N_t^{-\alpha} = (1-\alpha)Y_t/N_t, \tag{670}$$

$$q_t = \alpha A_t K_t^{\alpha-1} N_t^{1-\alpha} = \alpha Y_t/K_t. \tag{671}$$

For a prescribed output Y_t, cost minimization over labor alone, with fixed K_t, delivers the same conditional labor requirement. It is not an unconstrained minimization of wage payments with no output requirement. The stock cannot be replaced by the wage in the production function.

Factor income exhausts output:

$$w_t N_t = (1-\alpha)Y_t, \qquad q_t K_t = \alpha Y_t, \tag{672}$$

$$d_t = Y_t - w_t N_t - q_t K_t = Y_t - (1-\alpha)Y_t - \alpha Y_t = 0. \tag{673}$$

Zero profit does not remove households' capital income. Use the household budget, zero bond supply, and these payments to obtain

$$C_t + K_{t+1} = w_t N_t + q_t K_t + (1-\delta)K_t + d_t, \tag{674}$$

$$= Y_t + (1-\delta)K_t, \tag{675}$$

$$Y_t - C_t = K_{t+1} - (1-\delta)K_t = I_t, \tag{676}$$

$$Y_t = C_t + I_t. \tag{677}$$

Saving allocates current goods to capital; bond trade only rearranges claims between households. The labor condition, resource constraint, and capital Euler equation play different roles and must all hold.

10.4.2 The planner route checks the same allocation

With the stated competition, ownership, and absence of wedges, a planner can choose the aggregate allocation under the same technology and information:

$$\mathscr{L}^P = E_0 \sum_{t=0}^{\infty} \beta^t \{u(C_t, N_t) + \mu_t[A_t K_t^{\alpha} N_t^{1-\alpha} + (1-\delta)K_t - C_t - K_{t+1}]\}. \tag{678}$$

Its derivatives are

$$u_{C,t} - \mu_t = 0, \tag{679}$$

$$u_{N,t} + \mu_t(1-\alpha)A_t K_t^{\alpha} N_t^{-\alpha} = 0, \tag{680}$$

$$-\mu_t + \beta E_t[\mu_{t+1}(\alpha A_{t+1} K_{t+1}^{\alpha-1} N_{t+1}^{1-\alpha} + 1 - \delta)] = 0. \tag{681}$$

Substituting $\mu_t = C_t^{-\sigma}$ recovers (655) and (662), with the competitive factor prices. Concavity, feasibility, and the terminal restriction accompany these interior conditions. The equivalence must be reconsidered when a wedge, externality, markup, or market incompleteness changes the environment.

The equilibrium is a loop, not two independent curves. Observed productivity and inherited capital determine productive opportunities. Households choose hours and consumption jointly, using both current wages and expected future returns. Hours change output and factor prices; the consumption choice leaves investment; investment changes tomorrow's capital. Prices must support all of these choices at once. An AS/AD grouping can organize the original equations, but cannot close the economy without its resource, accumulation, expectations, and terminal conditions.

A compact list of the *exact* real conditions is therefore

$$Y_t = A_t K_t^{\alpha} N_t^{1-\alpha}, \tag{682}$$
$$\chi C_t^{\sigma} N_t^{\varphi} = (1-\alpha) Y_t / N_t, \tag{683}$$
$$C_t^{-\sigma} = \beta E_t[C_{t+1}^{-\sigma}(\alpha Y_{t+1}/K_{t+1} + 1 - \delta)], \tag{684}$$
$$Y_t = C_t + I_t, \tag{685}$$
$$K_{t+1} = (1-\delta) K_t + I_t, \tag{686}$$
$$a_t = \rho_a a_{t-1} + \varepsilon_t^a. \tag{687}$$

The initial state and transversality condition remain part of the model even when they are not repeated in this list. Wages, rentals, and bond prices are then recovered from their pricing equations. No nominal variable is required to solve this real block.

10.5 Recover the course's fixed-capital benchmark

Freeze productive capital at $\bar{K} = 1$, remove its investment choice, and assume no depreciation, government purchases, or foreign trade. There is no replacement expenditure, so $I_t = 0$ and $C_t = Y_t$. This is a separate classroom restriction, not the full model with its capital Euler equation still imposed. Owners receive fixed-factor income $\mathscr{D}_t/P_t = \alpha Y_t$; the labor-only firm's residual pays that factor. The course's $\chi = 1$ may also be imposed. The limit $\alpha = 0$ recovers labor-only production, as allowed in the course; the main capital model uses $0 < \alpha < 1$.

The real equations become

$$Y_t = A_t N_t^{1-\alpha}, \qquad C_t = Y_t, \tag{688}$$
$$w_t = (1-\alpha) A_t N_t^{-\alpha}, \tag{689}$$
$$w_t = \chi C_t^{\sigma} N_t^{\varphi}. \tag{690}$$

Solve this restricted allocation without skipping the wealth effect. Equate wages and replace consumption by production:

$$\chi C_t^{\sigma} N_t^{\varphi} = (1-\alpha) A_t N_t^{-\alpha}, \tag{691}$$
$$\chi (A_t N_t^{1-\alpha})^{\sigma} N_t^{\varphi} = (1-\alpha) A_t N_t^{-\alpha}, \tag{692}$$
$$N_t^{\sigma(1-\alpha)+\varphi+\alpha} = \frac{1-\alpha}{\chi} A_t^{1-\sigma}. \tag{693}$$

Let $d_N = \sigma(1-\alpha) + \varphi + \alpha > 0$. At an interior solution,

$$N_t = \left[\frac{1-\alpha}{\chi} A_t^{1-\sigma}\right]^{1/d_N}, \tag{694}$$

$$\log N_t = \frac{\log(1-\alpha) - \log\chi}{d_N} + \frac{1-\sigma}{d_N}\log A_t, \tag{695}$$

$$\frac{d\log N_t}{d\log A_t} = \frac{1-\sigma}{d_N}, \tag{696}$$

$$\frac{d\log Y_t}{d\log A_t} = 1 + (1-\alpha)\frac{1-\sigma}{d_N} = \frac{1+\varphi}{d_N}. \tag{697}$$

The last equality follows by collecting $d_N + (1-\alpha)(1-\sigma) = \varphi + \alpha + 1 - \alpha = 1 + \varphi$. The formulas apply while $0 < N_t < 1$; they do not bypass the time endowment.

Question: does better productivity necessarily raise work? In this fixed-factor economy, $\sigma = 1$ gives $N_t = [(1-\alpha)/\chi]^{1/(1+\varphi)}$, independent of A_t. Higher productivity raises the wage and income, but the work-reward and wealth margins exactly offset. For $\sigma < 1$ hours rise; for $\sigma > 1$ they fall, provided the solution is interior. Endogenous labor is not by itself a prediction that hours always move with output. The investment margin in the full model changes this experiment.

There is no endogenous capital memory here. With serially independent A_t, the real allocation depends only on the current shock. A persistent A_t produces persistence mechanically; zero-net-supply bond prices do not create another aggregate resource state. This is why adding the capital choice changes the model.

10.6 Capital accumulation and the deterministic reference

10.6.1 Find the steady state in the order the conditions determine it

Switch innovations off and set $A_t = \bar{A}$. A deterministic steady state has constant K^*, C^*, N^*, Y^*, I^*. It is a reference for local expansion, not the unconditional mean of the nonlinear stochastic economy. The Euler equation gives

$$(C^*)^{-\sigma} = \beta(C^*)^{-\sigma}(q^* + 1 - \delta), \tag{698}$$

$$1 = \beta(q^* + 1 - \delta), \tag{699}$$

$$q^* = \beta^{-1} - 1 + \delta. \tag{700}$$

Let $v = Y^*/K^* = q^*/\alpha > 0$. Capital stationarity and goods clearing imply

$$K^* = (1-\delta)K^* + I^*, \qquad I^* = \delta K^*, \tag{701}$$

$$C^* = Y^* - \delta K^*, \qquad s_I \equiv I^*/Y^* = \delta/v, \tag{702}$$

$$s_C \equiv C^*/Y^* = 1 - \delta/v, \qquad s_C + s_I = 1. \tag{703}$$

Here $v > \delta$ since $q^* > \delta$ and $\alpha < 1$, so $s_C > 0$. The steady capital–hours ratio $x = K^*/N^*$ follows from the rental equation:

$$q^* = \alpha \bar{A}(K^*)^{\alpha-1}(N^*)^{1-\alpha} = \alpha \bar{A} x^{\alpha-1}, \tag{704}$$

$$x^{1-\alpha} = \alpha \bar{A}/q^*, \tag{705}$$

$$x = (\alpha \bar{A}/q^*)^{1/(1-\alpha)}. \tag{706}$$

Define $z = Y^*/N^* = \bar{A}x^{\alpha}$. Then $C^* = s_C z N^*$ and $w^* = (1-\alpha)z$. The labor condition pins down the scale:

$$\chi(s_C z N^*)^{\sigma}(N^*)^{\varphi} = (1-\alpha)z, \tag{707}$$

$$(N^*)^{\sigma+\varphi} = \frac{(1-\alpha)z}{\chi(s_C z)^{\sigma}}, \tag{708}$$

$$N^* = \left[\frac{(1-\alpha)z}{\chi(s_C z)^{\sigma}}\right]^{1/(\sigma+\varphi)}, \tag{709}$$

$$K^* = xN^*, \quad Y^* = zN^*, \quad C^* = s_C z N^*, \quad I^* = \delta x N^*. \tag{710}$$

If this N^* is outside $(0,1)$, the interior formula is inapplicable. One can instead choose an intended interior N^* and solve the same equation for χ; that is a stated calibration, not an extra equilibrium condition.

10.6.2 Why current investment and current capital must not be confused

A factory can change operating hours after learning A_t, but the stock K_t is already installed. Investment uses goods today and changes K_{t+1}. Even after A_t returns to its reference, the economy can inherit a different capital stock. A single innovation can therefore have effects lasting beyond the disturbance itself. The equations must determine whether the stock builds up or runs down; the accounting identity alone does not choose investment.

10.7 Log-linearization: show the approximation in each equation

10.7.1 The course's two routes describe the same local deviation

For any positive X_t with positive deterministic reference X^*, define $\hat{x}_t = \log X_t - \log X^*$. Lowercase hats always denote log deviations; K_t is a level, $\hat{k}_t$ its deviation. Let $d_t = (X_t - X^*)/X^*$. The logarithm route gives

$$\hat{x}_t = \log(X_t/X^*) = \log(1+d_t), \tag{711}$$

$$\log(1+d_t) = \log 1 + \left.\frac{1}{1+d}\right|_{d=0} d_t + O(d_t^2), \tag{712}$$

$$\hat{x}_t = d_t + O(d_t^2) \approx (X_t - X^*)/X^*. \tag{713}$$

The exponential route starts from the exact inverse:

$$\log X_t = \log X^* + \hat{x}_t, \tag{714}$$

$$X_t = e^{\log X^* + \hat{x}_t} = X^* e^{\hat{x}_t}, \tag{715}$$

$$e^{\hat{x}_t} = 1 + \hat{x}_t + O(\hat{x}_t^2), \tag{716}$$

$$X_t = X^*(1 + \hat{x}_t) + O(\hat{x}_t^2), \tag{717}$$

$$(X_t - X^*)/X^* = \hat{x}_t + O(\hat{x}_t^2). \tag{718}$$

The two derivations retain the original teaching routes. Positivity makes an absolute-value sign unnecessary. An equality between a log deviation and a percentage deviation is only first order. In the figures, $100\hat{x}_t$ is an approximate percentage response, not an exact level change.

10.7.2 Products become exact log differences

Take logs of production and of its steady-state counterpart, then subtract:

$$\log Y_t = \log A_t + \alpha \log K_t + (1 - \alpha) \log N_t, \tag{719}$$

$$\log Y^* = \log \bar{A} + \alpha \log K^* + (1 - \alpha) \log N^*, \tag{720}$$

$$\hat{y}_t = a_t + \alpha \hat{k}_t + (1 - \alpha)\hat{n}_t. \tag{721}$$

This log identity is exact for Cobb–Douglas, before any solution approximation. Do the same for the real wage and rental price:

$$\log w_t = \log(1 - \alpha) + \log Y_t - \log N_t, \tag{722}$$

$$\hat{w}_t = \hat{y}_t - \hat{n}_t = a_t + \alpha \hat{k}_t - \alpha \hat{n}_t, \tag{723}$$

$$\log q_t = \log \alpha + \log Y_t - \log K_t, \tag{724}$$

$$\hat{q}_t = \hat{y}_t - \hat{k}_t = a_t + (\alpha - 1)\hat{k}_t + (1 - \alpha)\hat{n}_t. \tag{725}$$

The labor condition gives

$$\log w_t = \log \chi + \sigma \log C_t + \varphi \log N_t, \tag{726}$$

$$\hat{w}_t = \sigma \hat{c}_t + \varphi \hat{n}_t, \tag{727}$$

$$a_t + \alpha \hat{k}_t - \alpha \hat{n}_t = \sigma \hat{c}_t + \varphi \hat{n}_t, \tag{728}$$

$$(\varphi + \alpha)\hat{n}_t = a_t + \alpha \hat{k}_t - \sigma \hat{c}_t. \tag{729}$$

If writing nominal wages instead, $w_t = W_t/P_t$ implies $\hat{w}_t = \hat{W}_t - \hat{p}_t$. Keeping this distinction avoids mixing a real wage with a nominal wage divided by the price level again.

10.7.3 Sums retain their steady-state weights

Goods clearing is a sum, so its logarithm does not distribute across terms. Instead insert the exponential representations:

$$Y^* e^{\hat{y}_t} = C^* e^{\hat{c}_t} + I^* e^{\hat{\imath}_t}, \tag{730}$$

$$Y^*(1+\hat{y}_t) \approx C^*(1+\hat{c}_t) + I^*(1+\hat{\imath}_t), \tag{731}$$

$$Y^* + Y^*\hat{y}_t \approx C^* + I^* + C^*\hat{c}_t + I^*\hat{\imath}_t, \tag{732}$$

$$Y^*\hat{y}_t \approx C^*\hat{c}_t + I^*\hat{\imath}_t, \tag{733}$$

$$\hat{y}_t \approx s_C\hat{c}_t + s_I\hat{\imath}_t. \tag{734}$$

Here $\hat{\imath}_t = \log(I_t/I^*)$ denotes *investment*, not an interest rate. The local neighborhood has $I_t > 0$ since $I^* > 0$, even though the general budget permits reversible investment. In the zero-investment course benchmark, this log is undefined; $Y = C$ instead gives $\hat{y} = \hat{c}$ exactly.

Apply the same steps to accumulation:

$$K^* e^{\hat{k}_{t+1}} = (1-\delta)K^* e^{\hat{k}_t} + I^* e^{\hat{\imath}_t}, \tag{735}$$

$$K^*(1+\hat{k}_{t+1}) \approx (1-\delta)K^*(1+\hat{k}_t) + I^*(1+\hat{\imath}_t), \tag{736}$$

$$K^*\hat{k}_{t+1} \approx (1-\delta)K^*\hat{k}_t + I^*\hat{\imath}_t, \tag{737}$$

$$\hat{k}_{t+1} \approx (1-\delta)\hat{k}_t + \delta\hat{\imath}_t. \tag{738}$$

The weights sum to one because $I^*/K^* = \delta$. Substituting the resource condition is often more useful for solving:

$$\hat{\imath}_t \approx (\hat{y}_t - s_C\hat{c}_t)/s_I, \tag{739}$$

$$\hat{k}_{t+1} \approx (1-\delta)\hat{k}_t + \frac{\delta}{s_I}(\hat{y}_t - s_C\hat{c}_t), \tag{740}$$

$$= (1-\delta)\hat{k}_t + v(\hat{y}_t - s_C\hat{c}_t). \tag{741}$$

The final coefficient uses $s_I = \delta/v$. It is not permissible to replace investment by consumption when capital is being chosen.

10.7.4 Expand inside the conditional expectation

The capital Euler condition contains both a random marginal utility and a random gross return. Since $R^{K*} = q^* + 1 - \delta = 1/\beta$,

$$R^K_{t+1} = q^* e^{\hat{q}_{t+1}} + 1 - \delta, \tag{742}$$

$$\approx R^{K*} + q^*\hat{q}_{t+1} = R^{K*}(1+\omega\hat{q}_{t+1}), \tag{743}$$

$$\omega \equiv q^*/R^{K*} = \beta q^*. \tag{744}$$

Divide the Euler equation by $(C^*)^{-\sigma}$ and expand:

$$e^{-\sigma \hat{c}_t} = \beta E_t[e^{-\sigma \hat{c}_{t+1}} R^K_{t+1}], \tag{745}$$

$$1 - \sigma \hat{c}_t \approx \beta R^{K*} E_t[(1 - \sigma \hat{c}_{t+1})(1 + \omega \hat{q}_{t+1})], \tag{746}$$

$$= E_t[1 - \sigma \hat{c}_{t+1} + \omega \hat{q}_{t+1} - \sigma\omega \hat{c}_{t+1}\hat{q}_{t+1}], \tag{747}$$

$$-\sigma \hat{c}_t \approx -\sigma E_t \hat{c}_{t+1} + \omega E_t \hat{q}_{t+1}, \tag{748}$$

$$\sigma(E_t \hat{c}_{t+1} - \hat{c}_t) \approx \omega E_t \hat{q}_{t+1}. \tag{749}$$

The product of deviations is second order and has been discarded, not replaced by a product of expectations. The equality signs below refer to the resulting linear approximation. Expected future rental income raises the reward to saving; K_{t+1} is chosen with date-t information, but A_{t+1} and hours next date remain uncertain.

Expectations can be linearized; logarithms cannot simply pass through them. For an integrable local expansion $e^z = 1 + z + O(z^2)$, take E_t after expanding. In general $\log E_t e^z \neq E_t z$. If $z = +h$ or $-h$ with equal probability,

$$Ez = 0, \qquad \log Ee^z = \log[(e^h + e^{-h})/2] > 0 \quad (h \neq 0).$$

The difference is of second order for small h. A first-order solution near a deterministic state omits risk adjustments of that order; it does not justify removing expectations from the exact Euler equation. This refines the original course warning rather than dropping the expectation calculation.

Finally, the shock law is already linear in logs:

$$a_{t+1} = \rho_a a_t + \varepsilon^a_{t+1}, \tag{750}$$

$$E_t a_{t+1} = \rho_a a_t + E_t \varepsilon^a_{t+1} = \rho_a a_t, \tag{751}$$

$$E_t a_{t+2} = \rho_a E_t a_{t+1} = \rho_a^2 a_t, \tag{752}$$

$$E_t a_{t+h} = \rho_a^h a_t. \tag{753}$$

A higher ρ_a changes what households expect before they choose, not just the shape of productivity after they have chosen.

10.8 Solve the model before drawing its responses

10.8.1 An exact special case checks the consumption–investment logic

As a separate special case, set $\delta = 1$ and $\sigma = 1$, while retaining $0 < \alpha < 1$ and the hours disutility. The partial-depreciation baseline will be restored afterward. Guess $K_{t+1} = sY_t$ and $C_t = (1-s)Y_t$, where $0 < s < 1$. The capital Euler equation becomes

$$\frac{1}{C_t} = \alpha\beta E_t\left[\frac{1}{C_{t+1}}\frac{Y_{t+1}}{K_{t+1}}\right], \tag{754}$$

$$\frac{1}{(1-s)Y_t} = \alpha\beta E_t\left[\frac{Y_{t+1}}{(1-s)Y_{t+1}K_{t+1}}\right], \tag{755}$$

$$= \frac{\alpha\beta}{(1-s)K_{t+1}} = \frac{\alpha\beta}{(1-s)sY_t}, \tag{756}$$

$$1 = \alpha\beta/s, \qquad s = \alpha\beta. \tag{757}$$

Tomorrow's output cancels *state by state*, before taking expectations. It has not been replaced by its mean. The labor condition then gives

$$\chi C_t N_t^{\varphi} = (1-\alpha)Y_t/N_t, \tag{758}$$

$$\chi(1-\alpha\beta)Y_t N_t^{1+\varphi} = (1-\alpha)Y_t, \tag{759}$$

$$N_t = \left[\frac{1-\alpha}{\chi(1-\alpha\beta)}\right]^{1/(1+\varphi)} \equiv N^e. \tag{760}$$

Choose parameters with $0 < N^e < 1$. Feasibility and the terminal restriction are also explicit:

$$C_t + K_{t+1} = (1-\alpha\beta)Y_t + \alpha\beta Y_t = Y_t, \tag{761}$$

$$E_0[\beta^T C_T^{-1} K_{T+1}] = \beta^T \frac{\alpha\beta}{1-\alpha\beta} \longrightarrow 0. \tag{762}$$

Under admissible shock moments these policies satisfy the concave planner's conditions. The constant share resembles the exact Ramsey log example, now with uncertainty and hours. It does not extend to arbitrary depreciation or curvature. Indeed, it makes consumption and investment proportional to output, so it cannot explain every relative-volatility fact merely by being an RBC model.

10.8.2 Reduce the partial-depreciation model without hiding its coefficients

Return to $0 < \delta < 1$ and the general curvature parameters. Solve the linear labor condition first, then insert it into production:

$$\hat{n}_t = \frac{\alpha}{\varphi+\alpha}\hat{k}_t + \frac{1}{\varphi+\alpha}a_t - \frac{\sigma}{\varphi+\alpha}\hat{c}_t, \tag{763}$$

$$\hat{y}_t = a_t + \alpha\hat{k}_t + (1-\alpha)\left[\frac{\alpha\hat{k}_t + a_t - \sigma\hat{c}_t}{\varphi+\alpha}\right], \tag{764}$$

$$= b_k\hat{k}_t + b_a a_t + b_c\hat{c}_t, \tag{765}$$

where collection of each coefficient gives

$$b_k = \alpha + \frac{(1-\alpha)\alpha}{\varphi+\alpha} = \frac{\alpha(1+\varphi)}{\varphi+\alpha}, \tag{766}$$

$$b_a = 1 + \frac{1-\alpha}{\varphi+\alpha} = \frac{1+\varphi}{\varphi+\alpha}, \tag{767}$$

$$b_c = -\frac{(1-\alpha)\sigma}{\varphi+\alpha}. \tag{768}$$

Substitute into capital accumulation:

$$\hat{k}_{t+1} = (1-\delta)\hat{k}_t + v[b_k\hat{k}_t + b_a a_t + b_c\hat{c}_t - s_C\hat{c}_t], \tag{769}$$

$$= \gamma_k\hat{k}_t + \gamma_a a_t - \gamma_c\hat{c}_t, \tag{770}$$

$$\gamma_k = 1-\delta+vb_k, \qquad \gamma_a = vb_a, \qquad \gamma_c = v(s_C - b_c) > 0. \tag{771}$$

The Euler equation uses $\hat{q}_{t+1} = \hat{y}_{t+1} - \hat{k}_{t+1}$. Since K_{t+1} is known when chosen,

$$E_t\hat{q}_{t+1} = (b_k-1)\hat{k}_{t+1} + b_a\rho_a a_t + b_c E_t\hat{c}_{t+1}, \tag{772}$$

$$\sigma E_t\hat{c}_{t+1} - \sigma\hat{c}_t = \omega[(b_k-1)\hat{k}_{t+1} + b_a\rho_a a_t + b_c E_t\hat{c}_{t+1}], \tag{773}$$

$$(\sigma - \omega b_c)E_t\hat{c}_{t+1} = \sigma\hat{c}_t + \omega(b_k-1)\hat{k}_{t+1} + \omega b_a\rho_a a_t. \tag{774}$$

Define $\tau = \sigma - \omega b_c$, $\upsilon = \omega(b_k-1)$, and $\zeta = \omega b_a\rho_a$. Then

$$\tau E_t\hat{c}_{t+1} = \sigma\hat{c}_t + \upsilon\hat{k}_{t+1} + \zeta a_t. \tag{775}$$

These coefficients come from stated structural equations; none is an independently fitted persistence parameter.

10.8.3 Match coefficients and select the stable rule

Conjecture a linear consumption rule $\hat{c}_t = p_k\hat{k}_t + p_a a_t$. Equations (770) and the productivity forecast imply

$$\hat{k}_{t+1} = (\gamma_k - \gamma_c p_k)\hat{k}_t + (\gamma_a - \gamma_c p_a)a_t, \tag{776}$$

$$= H_k\hat{k}_t + H_a a_t, \tag{777}$$

$$E_t\hat{c}_{t+1} = p_k\hat{k}_{t+1} + p_a\rho_a a_t. \tag{778}$$

Insert the last line into (775), then collect:

$$\tau(p_k\hat{k}_{t+1} + p_a\rho_a a_t) = \sigma(p_k\hat{k}_t + p_a a_t) + \upsilon\hat{k}_{t+1} + \zeta a_t, \tag{779}$$

$$(\tau p_k - \upsilon)\hat{k}_{t+1} = \sigma p_k\hat{k}_t + (\sigma p_a + \zeta - \tau\rho_a p_a)a_t. \tag{780}$$

Equating the coefficient of the predetermined stock yields

$$(\tau p_k - \upsilon)(\gamma_k - \gamma_c p_k) = \sigma p_k, \tag{781}$$

$$\tau\gamma_k p_k - \tau\gamma_c p_k^2 - \upsilon\gamma_k + \upsilon\gamma_c p_k - \sigma p_k = 0, \tag{782}$$

$$\tau\gamma_c p_k^2 - (\tau\gamma_k + \upsilon\gamma_c - \sigma)p_k + \upsilon\gamma_k = 0. \tag{783}$$

For $d_0 = \tau\gamma_k + \upsilon\gamma_c - \sigma$, the candidate slopes are

$$p_k^{\pm} = \frac{d_0 \pm \sqrt{d_0^2 - 4\tau\gamma_c\upsilon\gamma_k}}{2\tau\gamma_c}. \tag{784}$$

For each candidate compute $H_k = \gamma_k - \gamma_c p_k$. Keep the locally stable root with $|H_k| < 1$; the numerical example below has exactly one. As in Ramsey, an arbitrary consumption slope also admits the unstable mode and does not satisfy the desired terminal behavior near the reference.

For the productivity coefficient, let $J = \tau p_k - \upsilon$. Matching a_t gives

$$J(\gamma_a - \gamma_c p_a) = \sigma p_a + \zeta - \tau\rho_a p_a, \tag{785}$$

$$J\gamma_a - \zeta = (\sigma + J\gamma_c - \tau\rho_a)p_a, \tag{786}$$

$$p_a = \frac{J\gamma_a - \zeta}{\sigma + J\gamma_c - \tau\rho_a}, \qquad H_a = \gamma_a - \gamma_c p_a. \tag{787}$$

The denominator must be nonzero. Changing ρ_a changes p_a and H_a, not just the future path of a_t. Resolve the rule when changing beliefs about persistence.

The solved local state system is

$$\begin{pmatrix} \hat{k}_{t+1} \\ a_{t+1} \end{pmatrix} = \begin{pmatrix} H_k & H_a \\ 0 & \rho_a \end{pmatrix} \begin{pmatrix} \hat{k}_t \\ a_t \end{pmatrix} + \begin{pmatrix} 0 \\ \varepsilon^a_{t+1} \end{pmatrix}. \tag{788}$$

Its two persistence roots are H_k and ρ_a. Given the states, $\hat{c}_t$, $\hat{n}_t$, $\hat{y}_t$, and $\hat{\imath}_t$ follow from the rules already derived. Only the first-order approximation has been solved here; large shocks, corners, and higher-order risk effects require other methods.

10.9 A worked shock experiment and its economic meaning

10.9.1 State the calibration and reproduce the initial response

Use quarters as dates and the illustrative parameters

$$\alpha = 0.33, \quad \beta = 0.99, \quad \delta = 0.025, \quad \sigma = 1, \quad \varphi = 1, \quad \bar{A} = 1.$$

Set $N^* = 1/3$ and choose the preference weight from the steady-state labor condition. These are teaching choices, not estimates or a claimed fit to any country. The steady calculations give

$$q^* = 1/0.99 - 1 + 0.025 \simeq 0.035101, \tag{789}$$

$$\upsilon = q^*/0.33 \simeq 0.106367, \tag{790}$$

$$s_I = 0.025/\upsilon \simeq 0.235036, \qquad s_C \simeq 0.764964, \tag{791}$$

$$K^* \simeq 9.449473, \quad Y^* \simeq 1.005109, \quad C^* \simeq 0.768872, \quad I^* \simeq 0.236237, \tag{792}$$

$$\chi = \frac{(1-\alpha)Y^*/N^*}{(C^*)^\sigma (N^*)^\varphi} \simeq 7.882724. \tag{793}$$

For $\rho_a = 0.95$, coefficient matching selects

$$p_k \simeq 0.553075, \quad p_a \simeq 0.386565, \quad H_k \simeq 0.953146, \quad H_a \simeq 0.107783.$$

The other consumption-stock slope is about -0.236912, implying $H_k \simeq 1.059755 > 1$, and is rejected. Start at the reference stock $\hat{k}_0 = 0$, apply a one-time innovation $\varepsilon_0^a = 0.01$, and set all later innovations to zero. On impact,

$$a_0 = 0.01, \qquad \hat{c}_0 = p_a a_0 \simeq 0.003866, \tag{794}$$

$$\hat{n}_0 = \frac{0.01 - \hat{c}_0}{1.33} \simeq 0.004612, \tag{795}$$

$$\hat{y}_0 = 0.01 + 0.67\hat{n}_0 \simeq 0.013090, \tag{796}$$

$$\hat{\imath}_0 = \frac{\hat{y}_0 - s_C \hat{c}_0}{s_I} \simeq 0.043113, \tag{797}$$

$$\hat{k}_1 = (1-\delta)\hat{k}_0 + \delta\hat{\imath}_0 \simeq 0.001078. \tag{798}$$

For the same 1% log innovation with $\rho_a = 0$, solve again: $p_a \simeq 0.086073$, $H_a \simeq 0.148334$, while p_k and H_k are unchanged in this model. The immediate responses are

Variable	$\boldsymbol{\rho_a = 0.95}$	$\boldsymbol{\rho_a = 0}$
Productivity	1.000	1.000
Output	1.309	1.460
Consumption	0.387	0.086
Investment	4.311	5.933
Hours	0.461	0.687
Capital available at date 0	0.000	0.000

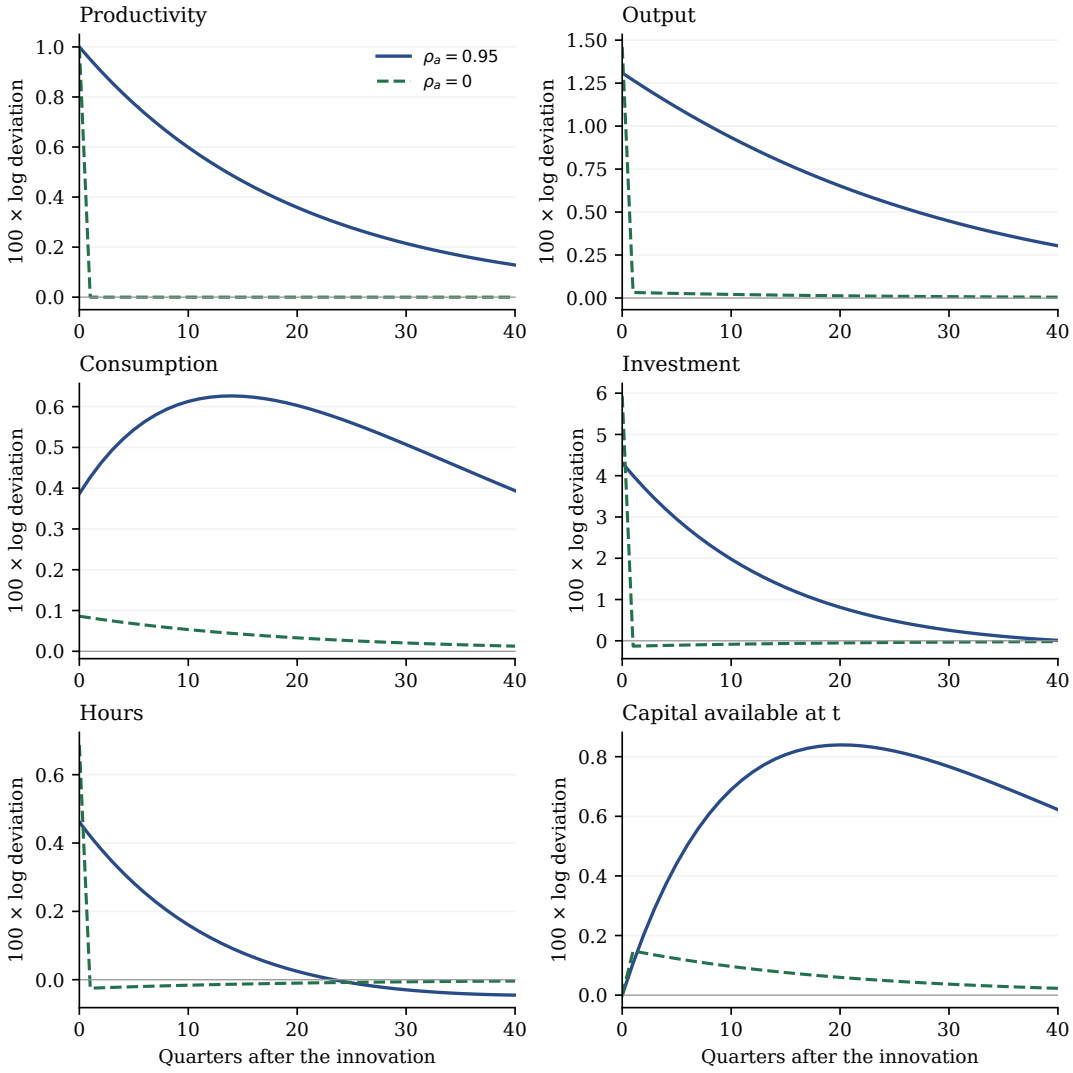
Entries are 100 times log deviations, approximately percentage changes. With $\rho_a = 0$ the opportunity is brief: the household increases current work more and consumes less of the additional output. A persistent improvement raises the value of lifetime resources more, supporting a larger initial consumption increase. This comparison includes both the changed opportunity path and the reoptimized household response.

Output rises immediately because the existing machines and chosen work are more productive. Consumption rises less in this example: households spread resources across dates. Investment absorbs more of the extra output and changes next quarter's machines. Its percentage response is especially large partly because investment is a smaller steady-state share of output. From resources,

$$\hat{\imath}_t = (\hat{y}_t - s_C \hat{c}_t)/s_I;$$

a modest goods reallocation can be a large percentage of I^*. This arithmetic helps interpret relative volatility; an impact ratio is not itself a simulated standard-deviation ratio.

10.9.2 Follow the goods and the inherited stock



Model-generated, first-order responses to one positive productivity innovation. Both paths start with the same capital and $a_0 = 0.01$. All later innovations are zero; the consumption rule is solved separately for each persistence parameter. The plot uses only the equations and hypothetical parameters above, not estimated responses or observed data.

In the dashed path, productivity is back at its reference at date 1, but capital is then above its reference. Output and consumption can remain above it. Hours can be below it: the inherited stock makes households wealthier and changes the wage they need to work. A real shock does not require every variable to move in the same direction at every horizon. In the solid path, both productivity persistence and capital memory operate. Stable roots imply convergence back to this reference after the temporary innovation, rather than a permanent change in trend growth.

The factory interpretation is concrete. More productive shifts free goods for both consumption and replacement or expansion of equipment. Once equipment is installed, its services remain after the unusually productive shift ends. An automation investment, however, can also change tasks and workers' skills; a credit shock can prevent a profitable installation. Those are additional mechanisms, not effects already represented by this single A_t.

Capital propagates; expectations shape the initial allocation. A given current productivity surprise does not determine how much is consumed, worked, or invested. The expected duration and future returns help determine those choices. Capital then carries part of today's reallocation into later production. Compare $\rho_a = 0$ with $\rho_a > 0$ only after resolving the policy; holding the old consumption rule fixed would mix two different experiments.

10.10 Retain nominal pricing without confusing it with real closure

10.10.1 Derive the nominal Euler approximation and the Fisher comparison

For this calculation take the deterministic nominal reference to have a constant price level, hence zero inflation and $R^{n*} = 1/\beta$. Define

$$\hat{\jmath}_t = \log(R_t^n/R^{n*}), \qquad \pi_{t+1} = \log(P_{t+1}/P_t).$$

$\hat{\jmath}_t$ is the log deviation of the *gross nominal interest factor*; it is not the investment deviation $\hat{\imath}_t$, and not generally the percentage deviation of the net rate i_t. Invert Q_t in (657) and divide by its reference:

$$1 = \beta R_t^n E_t[(C_{t+1}/C_t)^{-\sigma} P_t/P_{t+1}], \tag{799}$$

$$= e^{\hat{\jmath}_t} E_t \exp\{-\sigma(\hat{c}_{t+1} - \hat{c}_t) - \pi_{t+1}\}, \tag{800}$$

$$1 \approx (1 + \hat{\jmath}_t)[1 - \sigma(E_t\hat{c}_{t+1} - \hat{c}_t) - E_t\pi_{t+1}], \tag{801}$$

$$0 \approx \hat{\jmath}_t - \sigma(E_t\hat{c}_{t+1} - \hat{c}_t) - E_t\pi_{t+1}, \tag{802}$$

$$\hat{c}_t \approx E_t\hat{c}_{t+1} - \frac{1}{\sigma}(\hat{\jmath}_t - E_t\pi_{t+1}). \tag{803}$$

In the multiplication, products of deviations are discarded. The form resembles the intertemporal demand relation used in the next chapter; here flexible-price production and labor choices determine the supporting allocation jointly.

A real risk-free payoff instead satisfies $1 = \beta R_t^f E_t(C_{t+1}/C_t)^{-\sigma}$. Let $\hat{r}_t^f = \log(R_t^f/(1/\beta))$. Expanding the same way gives

$$0 \approx \hat{r}_t^f - \sigma(E_t\hat{c}_{t+1} - \hat{c}_t), \tag{804}$$

$$\hat{r}_t^f \approx \hat{\jmath}_t - E_t\pi_{t+1}. \tag{805}$$

This is the local Fisher comparison. Under uncertain inflation it is not an exact statement equating the nominal bond's random realized real return to a risk-free real return. Exact pricing retains the covariance terms and the random purchasing power in (665).

10.10.2 What does the original money-demand equation add?

One can append the course's nominal relation in the clearly stated form

$$\frac{M_t}{P_t} = \kappa Y_t (R_t^n)^{-\eta}, \qquad \kappa > 0, \quad \eta > 0. \tag{806}$$

It is a reduced-form attachment, not a household FOC derived from the money-free preferences and budget above. With $R_t^n = 1/Q_t$, the same expression is $M_t/P_t = \kappa Y_t Q_t^{\eta}$. Its log transformation is

$$\log M_t - \log P_t = \log \kappa + \log Y_t - \eta \log R_t^n, \tag{807}$$

$$\hat{m}_t - \hat{p}_t = \hat{y}_t - \eta \hat{j}_t. \tag{808}$$

A genuine monetary model would specify why households hold money, its budget and supply arrangements, and a compatible policy rule and boundary conditions. Neither the real block nor this attachment alone selects a unique nominal price path. The original warning about money neutrality is thus conditional.

For example, start with a nominal price path and claims that support the real allocation and satisfy the attachment. Rescale $M_t, P_t, W_t, B_t, \mathscr{D}_t$ by the same positive constant at all dates, keeping Q_t fixed. Real wages, real balances, real payoffs, and every real allocation are unchanged. No real resource or relative reward has changed. This unit-rescaling result does not establish that an unanticipated monetary intervention has no real effects in an economy with sticky contracts, redistribution, or financial frictions.

> **Money neutrality is a property of a specified environment.** The real core excludes monetary wedges; its equations determine allocation without a price level. A nominal attachment must still be closed consistently. A particular level rescaling, a change in money growth, and a change in the policy rule are different experiments. The core's separation of real and nominal variables is useful for comparison with the next chapter, not evidence that money never matters in actual fluctuations.

10.11 What should an economist learn from the benchmark?

10.11.1 Use joint paths to test mechanisms

A plausible output response is only a starting point. Compare consumption, investment, hours, real wages, and capital jointly, under a common measurement and detrending convention. At a minimum, ask whether the same parameter values produce credible relative volatilities, correlations, persistence, and response timing. A model calibrated to selected moments has not independently explained all those moments. A convenient choice of shock persistence cannot substitute for evidence about its source.

For the production function used here, the measured residual would be

$$\log A_t = \log Y_t - \alpha \log K_t - (1-\alpha) \log N_t, \tag{809}$$

$$\Delta \log A_t = \Delta \log Y_t - \alpha \Delta \log K_t - (1-\alpha) \Delta \log N_t. \tag{810}$$

This is accounting under the assumed technology. An omitted utilization rate, input quality, changing markup, or measurement error can enter that residual. Calling it an exogenous technology innovation requires an identification argument beyond subtracting measured inputs.

Estimated technology responses can challenge the labor mechanism. Galí (1999), using his long-run identification strategy, reports a decline in hours after a positive technology innovation.[30] The response of hours is an empirical question; the fixed-capital calculation already showed why it is not a mathematical consequence of the word "productivity." Different trend, identification, and propagation assumptions must be examined rather than hidden behind an unconditional correlation.

Rebelo's review places the basic framework alongside subsequent extensions and unresolved questions.[31] The teaching exercise here demonstrates an internally consistent mechanism; it does not assign a numerical share of actual recessions to technology.

10.11.2 Efficient adjustment can still be painful

In the undistorted representative-household economy, the competitive allocation matches the planner allocation under the same changing resources and information. A negative productivity shock can reduce attainable consumption even when the response is efficient. Restoring yesterday's quantities is not automatically a feasible improvement. Efficiency of a response does not make the disturbance beneficial.

In actual communities a closure can leave particular workers without income, credit, matching opportunities, or insurance. This model has an intensive hours choice and common ownership, so it cannot turn such losses into a theory of involuntary unemployment or distribution. Removing hours from a work schedule and failing to find a job are economically different events.

The first-order paths also do not suffice for a quantitatively valid welfare comparison under uncertainty. Utility curvature, risk effects, and mean shifts can contribute at second order. Plugging a first-order allocation into nonlinear utility does not automatically supply all terms of a second-order welfare calculation. Welfare claims require an approximation and a feasible policy comparison suited to the question.

10.11.3 Ask which missing constraint would change the diagnosis

If a solvent factory has a profitable order but cannot finance working capital, add the financing constraint. If workers want jobs at the current wage but cannot find matches, add a labor-market mechanism. If firms cut production because sales decline while prices adjust slowly, study demand and nominal rigidity. If imported energy becomes scarce, distinguish its relative price and resource constraint from a neutral productivity scalar. Government spending similarly requires its purchases, taxes or debt, and household wealth effects to be specified before a fiscal prediction can be derived.

The purpose of the benchmark is to make those additions meaningful. We now know which outcomes can arise from uncertain real opportunities, work choice, and accumulation alone, and which assumptions support the welfare interpretation. The next chapter changes price setting. Its flexible-price comparison allocation must not be confused with a fitted statistical trend, an economy with literally no shocks, or a policy target that can be reached without cost.

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[30] Jordi Galí (1999), "Technology, Employment, and the Business Cycle: Do Technology Shocks Explain Aggregate Fluctuations?" *American Economic Review*, 89(1), 249–271, https://doi.org/10.1257/aer.89.1.249. The result is conditional on that paper's identification and measurement, not a definition of a technology shock.

[31] Sergio Rebelo (2005), "Real Business Cycle Models: Past, Present, and Future," *Scandinavian Journal of Economics*, 107(2), 217–238, https://doi.org/10.1111/j.1467-9442.2005.00405.x.

11 New Keynesian Model

DNK is short for Dynamic New Keynesian[32]
This New keynesian Model contains elements being

- dispersive economy, a continuum of agents
- perfect information
- rational expectations, i.e. $E_t(X_{t+1}) = E(X_{t+1}|I_t)$
- perfect competitive labor market
- monopolisitc competitive product market, heterogenous products, nominal rigidities

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11.1 UMP of household

Two stages of optimization

Goal of UMP is to acquire DIS, which represents demand. Household face two stages of optimization. Since goods are heterogenous, UMP begins by choosing items $i \in I$.

Stage I: choosing goods

One way to choose products is by method of maximization.

$$\begin{aligned} &\max_{C_{it}}(\int_0^1 C_{it}^{\frac{\epsilon-1}{\epsilon}} di)^{\frac{\epsilon}{\epsilon-1}} \\ &s.t. \int_0^1 P_{it}C_{it}di \leqslant Z_t \end{aligned} \tag{811}$$

, in which the ϵ stands for current substitute elasticity and $\frac{\epsilon-1}{\epsilon}$ is contemporaneous elasticity of substitution. And Z_t stands for endowment.

The other way to choose products is by method of minimization. In this case Lagrangian multiplier could be shadow price.

$$\begin{aligned} &\min \int_0^1 P_{it}C_{it}di \\ &s.t.(\int_0^1 C_{it}^{\frac{\epsilon-1}{\epsilon}} di)^{\frac{\epsilon}{\epsilon-1}} \geqslant C_t \end{aligned} \tag{812}$$

Form a Lagrangian, we would have

$$\underset{\{C_{it}\}}{\mathscr{L}} = \int_0^1 P_{it}C_{it}di + P_t[(\int_0^1 C_{it}^{\frac{\epsilon-1}{\epsilon}} di)^{\frac{\epsilon}{\epsilon-1}} - C_t] \tag{813}$$

$$\Rightarrow P_{it} = P_t \frac{\epsilon}{\epsilon-1}((\int_0^1 C_{it}^{\frac{\epsilon-1}{\epsilon}} di)^{\frac{\epsilon}{\epsilon-1}-1}) \frac{\epsilon-1}{\epsilon} C_{it}^{\frac{\epsilon-1}{\epsilon}-1} \tag{814}$$

$$\Rightarrow P_{it} = P_t[(\int_0^1 C_{it}^{\frac{\epsilon-1}{\epsilon}} di)^{\frac{\epsilon}{\epsilon-1}})^{\frac{1}{\epsilon}}]C_{it}^{\frac{-1}{\epsilon}} \tag{815}$$

$$\Rightarrow \frac{P_{it}}{P_t} = (\frac{C_{it}}{C_t})^{\frac{-1}{\epsilon}} \tag{816}$$

$$\Rightarrow C_{it} = (\frac{P_{it}}{P_t})^{-\epsilon} C_t \text{ (the demand curve)} \tag{817}$$

[32] This chapter is mainly based on the Gali, 2015.

From $\int_0^1 P_{it}C_{it}di = P_tC_t$ we have

$$\Rightarrow \int_0^1 P_{it}(\frac{P_{it}}{P_t})^{-\epsilon}C_t di = P_tC_t \tag{818}$$

$$\Rightarrow \int_0^1 P_{it}(\frac{P_{it}}{P_t})^{-\epsilon} di = P_t \tag{819}$$

$$\Rightarrow P_t = (\int_0^1 P_{it}^{1-\epsilon}di)^{\frac{1}{1-\epsilon}} \text{ (the expression of total price level)} \tag{820}$$

Apparently $\begin{cases} \max\limits_{C_{it}} (\frac{1}{I}\sum\limits_{i=1}^{I} C_{it}^{\frac{1}{1+n_t}})^{1+n_t} = C_t \\ s.t. \sum\limits_{i=1}^{I} P_{it}C_{it} \leqslant Z_t \end{cases}$ and $\begin{cases} \min\limits_{C_{it}} \sum\limits_{i=1}^{I} P_{it}C_{it} \\ s.t. (\frac{1}{I}\sum\limits_{i=1}^{I} C_{it}^{\frac{1}{1+n_t}})^{1+n_t}) \geqslant C_t \end{cases}$ are the same. But as mentioned before, the minimization approach is commonly prefered.

Using either approach, we can conduct the result that demand curve and expression of total price level being

$$C_{it} = (I \cdot \frac{P_{it}}{P_t})^{\frac{1+n_t}{-\wedge_t}} C_t \tag{821}$$

$$P_t = I(\frac{1}{I}\sum_{i=1}^{I} P_{it}^{\frac{1}{-\wedge_t}})^{-\wedge_t} \tag{822}$$

After picking products, we enter stage II, UMP at period t

$$\begin{aligned} &\max_{C_t, N_t, \frac{M_t}{P_t}} E_0 \sum_{t=0}^{\infty} \beta^t u\left(C_t, N_t, \frac{M_t}{P_t}\right) \leftarrow \text{(MIU)} \\ &s.t. \int_0^1 P_{it}C_{it}di + M_t + B_t \leqslant M_{t-1} + (1+i_{t-1})B_{t-1} + \int_o^1 W_{it}N_{it}di + T_t \end{aligned} \tag{823}$$

constraint of this UMP can be rewritten as

$$\begin{aligned} &\max_{C_t, N_t, \frac{M_t}{P_t}} E_0 \sum_{t=0}^{\infty} \beta^t u\left(C_t, N_t, \frac{M_t}{P_t}\right) \\ &s.t.\ P_tC_t + M_t + B_t \leqslant M_{t-1} + (1+i_{t-1})B_{t-1} + W_tN_t + T_t \end{aligned} \tag{824}$$

Note 1 (Subscript it and t)**.** $C_{it} \neq C_t \Leftrightarrow$ *Monopolistic competitive goods market;* $N_{it} = N_t \Leftrightarrow$ *Perfect competitve labor market*

Form the Lagrangian

$$\mathscr{L} = E_0 \sum_{t=0}^{\infty} \beta^t \{u(C_t, N_t, \frac{M_t}{P_t}) + \lambda_t[P_tC_t + M_t + B_t - M_{t-1} - (1+i_{t-1})B_{t-1} - W_tN_t + T_t]\} \tag{825}$$

Solving this will deliver the F.O.C.s., resulting

$$\Rightarrow \begin{cases} Q_t = \beta E_t \cdot (\frac{u_{C,t+1}}{u_{C,t}} \frac{P_t}{P_{t+1}}) \\ \frac{W_t}{P_t} = -\frac{u_{N_t}}{U_{C_t}} \\ 1 - Q_t = u_{\frac{M_t}{P_t}} \end{cases} \tag{826}$$

This result is basically same as in RBC model with one more equation since money is added to the utility function.

Linearization

Linearization specificly focusing on the third equation of the second equation, as the other two are identical in RBC model's linearization.

$$1 - \frac{1}{1+i_t} = C_t^{\sigma}(\frac{M_t}{P_t})^{-\nu} \text{ (using } Q_t\text{'s defintion)} \tag{827}$$

$$\Rightarrow \frac{i_t}{1+i_t} = C_t^{\sigma}(\frac{M_t}{P_t})^{-\nu} \tag{828}$$

Taking logs

$$\log \frac{i_t}{1+i_t} = \sigma c_t - \nu(m_t - p_t) \text{ (hereafter a lowercase letter } x \text{ denotes } \log x) \tag{829}$$

$$\log \frac{i}{1+i} = \sigma c - \nu(m - p) \tag{830}$$

$$\Rightarrow \log(\frac{i_t}{1+i_t}) - \log(\frac{i}{1+i}) = \sigma \hat{c}_t - \nu(m_t - \hat{p}_t) \tag{831}$$

$$\Rightarrow \nu(m_t - \hat{p}_t) = \sigma \hat{c}_t + \log(\frac{1+i_t}{i_t}) - \log(\frac{1+i}{i}) \tag{832}$$

$$\Rightarrow \nu(m_t - \hat{p}_t) = \sigma \hat{c}_t + [\log \frac{(\frac{1+i}{i}) + 1}{\frac{1+i}{i}} \frac{i - (1+i)}{i^2}(i_t - i) - \log(\frac{1+i}{i})] \tag{833}$$

$$\Rightarrow \nu(m_t - \hat{p}_t) = \sigma \hat{c}_t + \frac{i}{i+1}\frac{i-(1+i)}{i}\frac{i_t - i}{i} \tag{834}$$

$$\Rightarrow \nu(m_t - \hat{p}_t) = \sigma \hat{c}_t - \frac{1}{1+i}\hat{i}_t \tag{835}$$

$$\Rightarrow m_t - \hat{p}_t = \frac{\sigma}{\nu}\hat{c}_t - \frac{1}{(1+i)\nu}\hat{i}_t \tag{836}$$

$$\Rightarrow m_t - \hat{p}_t = \hat{c}_t - \eta\hat{i}_t \tag{837}$$

Note 2 (Must-be in exam: monopolistic competition vs perfect competition)**.** *Zero market power to finite market power, meaning infinite elasticiy to non-zero positive elasticity.*

11.2 PMP of firm

Note 3 (Two features DNK firm has)**.** *Monopolistic competitive goods market and stikcy price.*

11.2.1 Price-setting behavioral equation

We first use PMP to acquire price-setting equation then linearize it. Next we use price-setting equation to derive NKPC.

PMP and price-setting behavioral equation

Set base period on the intial period $t = 0$ for simplicity

$$\max_{P_{i0}} \pi_0^n = E_0 \sum_{t=0}^{\infty} Q_{0,t} \cdot (1-\theta)^t \{P_{i0} Y_{it|0} - [TC_{it|0}^n(Y_{it|0})]\}$$
$$s.t.\ Y_{it|0} = (\frac{P_{i0}}{P_t})^{-\epsilon} Y_t \text{ (since } C_t = Y_t \text{ at equilbrium, the constraint is essentially demand curve)} \tag{838}$$

$|0$ means evaluated at prices from period 0
Y_{it} is total sales of firm i at period t
TC is a function of nominal total cost
$Q_{0,t}$ means discount rate from period 0 to period t
θ means the probability of price adjusting, therefore $1-\theta$ means the probablity to stay put. It's sticky price. [33]

Solving the PMP by putting demand curve into profit function, it would be

$$\pi_0^n = E_0 \sum_{t=0}^{\infty} Q_{0,t} \cdot (1-\theta)^t \{P_{i0} (\frac{P_{i0}}{P_t})^{-\epsilon} Y_t - [TC_{it|0}^n((\frac{P_{i0}}{P_t})^{-\epsilon} Y_t)]\} \tag{839}$$

then let the F.O.C. be 0 (fig 1,2)

$$\frac{\partial \pi_0^n}{\partial P_{i0}} = 0 \tag{840}$$

$$\Rightarrow E_0 \sum_{t=0}^{\infty} Q_{0,t} \cdot (1-\theta)^t \{(1-\theta)(\frac{P_{i0}}{P_t})^{-\epsilon} Y_t - [\frac{\partial TC_{it|0}^n}{\partial P_{it|0}} \frac{\partial P_{it|0}}{\partial P_{i0}}] = 0 \tag{841}$$

$$\Rightarrow E_0 \sum_{t=0}^{\infty} (1-\theta)^t Q_{0,t} [(1-\epsilon)(\frac{P_{i0}}{P_t})^{-\epsilon} Y_t - MC_{it|0}^n(-\epsilon)(\frac{P_{i0}}{P_t})^{-\epsilon-1} Y_t \frac{1}{P_t}] = 0 \tag{842}$$

$$\Rightarrow E_0 \sum_{t=0}^{\infty} (1-\theta)^t Q_{0,t} [(1-\epsilon) Y_t - MC_{it|0}^n(-\epsilon)(\frac{P_{i0}}{P_t})^{-\epsilon-1} Y_t \frac{1}{P_t}] = 0 \tag{843}$$

$$\Rightarrow E_0 \sum_{t=0}^{\infty} (1-\theta)^t Q_{0,t} [(1-\epsilon) Y_{it|0} + \frac{\epsilon}{P_t} MC_{it|0}^n (\frac{P_{i0}}{P_t})^{-1}] = 0 \tag{844}$$

$$\Rightarrow E_0 \sum_{t=0}^{\infty} (1-\theta)^t Q_{0,t} Y_{it|0} [(1-\epsilon) + \epsilon MC_{it|0}^n \cdot (P_{i0})^{-1}] = 0 \tag{845}$$

$$\Rightarrow E_0 \sum_{t=0}^{\infty} (1-\theta)^t \cdot Q_{0,t} \cdot Y_{it|0} \cdot (1-\epsilon) = -E_0 \sum_{t=0}^{\infty} (1-\theta)^t \cdot Q_{0,t} \cdot Y_{it|0} \cdot \epsilon \cdot MC_{it|0}^n \cdot (P_{i0})^{-1} \tag{846}$$

$$\Rightarrow (1-\epsilon) E_0 \sum_{t=0}^{\infty} (1-\theta)^t \cdot Q_{0,t} \cdot Y_{it|0} = -\epsilon (P_{i0})^{-1} E_0 \sum_{t=0}^{\infty} (1-\theta)^t \cdot Q_{0,t} \cdot Y_{it|0} \cdot MC_{it|0}^n \tag{847}$$

restore Q_{0t} and $Y_{it|0}$ back to demand curve

$$\Rightarrow (\epsilon-1) E_0 \sum_{t=0}^{\infty} (1-\theta)^t \beta^t [(\frac{Y_0}{Y_t})^{\rho} \frac{P_0}{P_t}](\frac{P_{i0}}{P_t})^{-\epsilon} Y_t = (P_{i0})^{-1} E_0 \sum_{t=0}^{\infty} (1-\theta)^t \beta^t [(\frac{Y_0}{Y_t})^{\rho} \frac{P_0}{P_t}](\frac{P_{i0}}{P_t})^{-\epsilon} Y_t \cdot MC_{it|0}^n \tag{848}$$

[33] For more of price adjusting, see in Calvo, 1983 for uncertain version; Taylor, 1980 for certain version.

take P_{i0} and P_0 out since it can be treated as constants

$$\Rightarrow (\epsilon-1)\underline{P_{i0}^{-\epsilon}P_0Y_o^{\sigma}}E_0\sum_{t=0}^{\infty}(1-\theta)^t\beta^tP_t^{\epsilon-1}Y_t^{1-\rho}=\epsilon\underline{P_{i0}^{-1}P_{i0}^{-\epsilon}P_0Y_0^{\rho}}E_0\sum_{t=0}^{\infty}(1-\theta)^t\beta^tP_t^{\epsilon-1}Y_t^{1-\sigma}\cdot MC_{it|0}^n \tag{849}$$

eliminates the underlined common factors on RH and LH

$$\Rightarrow (\epsilon-1)E_0\sum_{t=0}^{\infty}(1-\theta)^t\beta^tP_t^{\epsilon-1}Y_t^{1-\rho}=\epsilon E_0\sum_{t=0}^{\infty}(1-\theta)^t\beta^tP_t^{\epsilon-1}Y_t^{1-\sigma}\cdot MC_{it|0}^n \tag{850}$$

$$\Rightarrow P_{i0}^*=\frac{\epsilon}{\epsilon-1}\frac{E_0\sum\limits_{t=0}^{\infty}(1-\theta)^t\beta^tP_t^{\epsilon-1}Y_t^{1-\sigma}\cdot MC_{it|0}^n}{E_0\sum\limits_{t=0}^{\infty}(1-\theta)^t\beta^tP_t^{\epsilon-1}Y_t^{1-\sigma}}\text{ (sticky price form price-setting equation)} \tag{851}$$

Also acquiring optimal price p^*, and $\frac{\epsilon}{\epsilon-1}$ is cost mutltiplier

$$\Rightarrow P_{i0}^*=P_0^*=\frac{\epsilon}{\epsilon-1}\frac{[(1-\epsilon)\beta]^0P_o^{\epsilon-1}Y_0^{1-\sigma}MC_{0|0}^n}{(1-\theta)^0\beta^0P_0^{\epsilon-1}Y_0^{1-\sigma}}=\frac{\epsilon}{\epsilon-1}MC_{0|0}^n\text{ (elastic form price-setting equation)} \tag{852}$$

For elastic form, the intuition is that firms are adjusting price at any time so there is no need to observe profit based on any specific period. So no summation symbol.

Question 1 (what if price is elastic?)**.**

elasticity means adjusting price at all time. it means price sticky price is $0 \Leftrightarrow 1-\theta=0$.

Note 4. *monopolistic competition and elasticity are not the same thing*

next try the period-t-based version in textbook (pic 3)

$$P_t^*=\frac{\epsilon}{\epsilon-1}\frac{E_t\sum\limits_{k=0}^{\infty}(1-\theta)^k\beta^{t+k}P_{t+k}^{\epsilon-1}Y_{t+k}^{1-\sigma}\cdot MC_{t+k|t}^n}{E_t\sum\limits_{k=0}^{\infty}(1-\theta)^k\beta^tP_{t+k}^{\epsilon-1}Y_{t+k}^{1-\sigma}} \tag{853}$$

(here the subscript i is gone because all firms are making unanimous decisions)

$$P_t^*=\frac{\epsilon}{\epsilon-1}MC_{t|t}^n \tag{854}$$

Note 5 (CRTS in DNK)**.** *if production function is CRTS, then all firms would also be making unanimous decisions regardless of time period position*

log-linearization of sticky price form price-setting behavioral equation
the method is still first order Taylor expansion, though bit trickier (pic 3, 4, 5)

dividing P_{t-1} on both side, and try seperately

$$\underbrace{\frac{P_t^*}{P_{t-1}} E_t \sum_{k=0}^{\infty} (1-\theta)^k \beta^t P_{t+k}^{\epsilon-1} Y_{t+k}^{1-\sigma}}_{\text{LHS}} = \underbrace{\frac{1}{P_{t-1}} \frac{\epsilon}{\epsilon-1} E_t \sum_{k=0}^{\infty} (1-\theta)^k \beta^{t+k} P_{t+k}^{\epsilon-1} Y_{t+k}^{1-\sigma} \cdot MC_{t+k|t}^n}_{\text{RHS}} \tag{855}$$

LHS mutil-variate first order Taylor expansion, the constant term is LHS at steady state where all P are equal

$$\begin{aligned}
\text{LHS's 1st order Taylor expansion} = & \sum_{k=0}^{\infty} [(1-\theta)\beta]^k P^{\epsilon-1} Y^{1-\sigma} \\
& + \frac{1}{p} E_t \sum_{k=0}^{\infty} [(1-\theta)\beta]^k P^{\epsilon} Y^{1-\sigma} (\frac{P_t^* - P}{P}) \\
& - \frac{P}{P^2} E_t \sum_{k=0}^{\infty} [(1-\theta)\beta]^k P^{\epsilon-1} Y^{1-\sigma} (\frac{P_{t-1} - P}{P}) \\
& + (\epsilon-1) E_t \sum_{k=0}^{\infty} [(1-\theta)\beta]^k P^{\epsilon-1} Y^{1-\sigma} (\frac{P_{t+k} - P}{P}) \\
& + (1-\sigma) E_t \sum_{k=0}^{\infty} [(1-\theta)\beta]^k P^{\epsilon-1} Y^{1-\sigma} (\frac{Y_{t+k} - Y}{Y})
\end{aligned} \tag{856}$$

Like before, we use lowercase letters denotes log version

$$\Rightarrow \text{LHS} = P^{\epsilon-1} Y^{1-\sigma} E_t \sum_{k=0}^{\infty} [(1-\theta)\beta]^k P^{\epsilon-1} Y^{1-\sigma} [1 + \hat{p}_t^* - \hat{p}_{t-1} + (\epsilon-1)\hat{p}_{t+k} + (1-\sigma)\hat{y}_{t+k}] \tag{857}$$

For RHS,

$$\text{RHS} = \cdots = \frac{\epsilon}{\epsilon-1} \frac{1}{P} P^{\epsilon-1} Y^{1-\sigma} E_t \sum_{k=0}^{\infty} [(1-\theta)\beta]^k P^{\epsilon-1} Y^{1-\sigma} [1 - \hat{p}_{t-1} + (\epsilon-1)\hat{p}_{t+k} + (1-\sigma)\hat{y}_{t+k} + \hat{mc}_{t+k|t}^n] \tag{858}$$

Note 6 (First order Taylor expansion and steady state)**.** *First term (constant term) of 1st Taylor expansion is steady state*

Question 2 (why do we do log-linearization)**.**

to simulate a curve with a line, so there would be a foundable or writable solution.

Now the two sides are log-linearized, let them be in the same equation

$$\underbrace{P^{\epsilon-1} Y^{1-\sigma} E_t \sum_{k=0}^{\infty} [(1-\theta)\beta]^k P^{\epsilon-1} Y^{1-\sigma} [1 + \hat{p}_t^* - \hat{p}_{t-1} + (\epsilon-1)\hat{p}_{t+k} + (1-\sigma)\hat{y}_{t+k}]}_{\text{log-linear LHS}} = \tag{859}$$

$$\underbrace{\frac{\epsilon}{\epsilon-1} \frac{1}{P} P^{\epsilon-1} Y^{1-\sigma} E_t \sum_{k=0}^{\infty} [(1-\theta)\beta]^k P^{\epsilon-1} Y^{1-\sigma} [1 - \hat{p}_{t-1} + (\epsilon-1)\hat{p}_{t+k} + (1-\sigma)\hat{y}_{t+k} + \hat{mc}_{t+k|t}^n]}_{\text{log-linear RHS}} \tag{860}$$

Use sticky form price-setting equation at steady state (which because of steadiness looks like elastic form), we have

$$\frac{MC^n}{P} = \frac{\epsilon - 1}{\epsilon} \tag{861}$$

Since E_t is followed by an infinite series, it is meaningless. Eliminate it.

$$\Rightarrow \hat{p}_t^* \cancel{E_t} \sum_{k=0}^{\infty} [(1-\theta)\beta]^k = E_t \sum_{k=0}^{\infty} [(1-\theta)\beta]^k \hat{mc}_{t+k|t}^n \tag{862}$$

$$\Rightarrow \frac{\hat{p}_t^*}{1-(1-\theta)\beta} = E_t \sum_{k=0}^{\infty} [(1-\theta)\beta]^k MC_{t+k|t}^n \tag{863}$$

$$\Rightarrow \hat{p}_t^* = [1-(1-\theta)\beta] E_t \sum_{k=0}^{\infty} [(1-\theta)\beta]^k \hat{mc}_{t+k|t}^n \text{ (log-linear sticky price setting equation)} \tag{864}$$

Therefore we have the log-linear optimal-price-setting equation for monopolistic competition and sticky price features firms.

11.2.2 Philips curve

First way to acquire Philips curve
next we use this log-linearized price-setting equation to acquire NK Philips curve
originally PC is about inflation and employment
step I, inflation (p7,8,9)

$$P_t = I(\frac{1}{I}\sum_{i=1}^{I} P_{it}^{\frac{1}{-\wedge_t}})^{-\wedge_t} \text{ (expression of total price)} \tag{865}$$

$$= [\int_0^{\theta} (P_{it}^*)^{1-\epsilon} di + \int_{\theta}^{1} P_{t-1}^{1-\epsilon} di]^{\frac{1}{1-\epsilon}} \text{ (θ portion of firms don't adjust price)} \tag{866}$$

$$= [\theta (P_{it}^*)^{1-\epsilon} + (1-\theta) P_{t-1}^{1-\epsilon}]^{\frac{1}{1-\epsilon}} \text{ (expression of total price with θ firms adjusting price)} \tag{867}$$

$$\Rightarrow P_t^{1-\epsilon} = \theta (P_{it}^*)^{1-\epsilon} + (1-\theta) P_{t-1}^{1-\epsilon} \tag{868}$$

$$\Rightarrow (\frac{P_t}{P_{t-1}})^{1-\epsilon} = \theta (\frac{P_t^*}{P_{t-1}})^{1-\epsilon} + (1-\theta) \tag{869}$$

$$\Rightarrow \Pi^{1-\epsilon} = \theta (\frac{P_t^*}{P_{t-1}})^{1-\epsilon} + (1-\theta) \text{ (the capital letter } \Pi \text{ is for total inflation } \frac{P_t}{P_{t-1}}) \tag{870}$$

Then log-linearization by bi-variate first order Taylor expansion

$$\Rightarrow \Pi^{1-\epsilon} + (1-\epsilon)\Pi^{1-\epsilon}(\frac{\Pi_t - \Pi}{\Pi}) = \{\theta (\frac{P}{P})^{1-\epsilon} + \theta(1-\epsilon)(\frac{P}{P})^{1-\epsilon}[\frac{1}{P}(P_t^* - P) - \frac{P}{P^2}\frac{(P_{t-1} - P)}{P}]\}^{1-\theta} \tag{871}$$

Since first term of tay exp is steady state, making $\Pi^{1-\epsilon} = 1$

$$\Rightarrow 1 + (1-\epsilon)\hat{\Pi}_t = \theta + \theta(1-\epsilon)(\hat{P}_t^* - \hat{P}_{t-1}) + (1-\theta) \tag{872}$$

$$\Rightarrow \hat{\Pi}_t = \theta(\hat{P}_t^* - \hat{P}_{t-1}) \tag{873}$$

$$\Rightarrow \frac{1}{\theta}\hat{\Pi}_t = \hat{P}_t^* - \hat{P}_{t-1} \tag{874}$$

Using price-setting equation, which is related to marginal cost, then production. (making nominal price combination of real price and $\hat{p}_{t-1}$)

$$\Rightarrow \frac{1}{\theta}\hat{\Pi}_t = [1-(1-\theta)\beta]E_t \sum_{k=0}^{\infty}[(1-\theta)\beta]^k(\hat{mc}^r_{t+k|t} + \hat{p}_{t+k} - \hat{p}_{t-1}) \tag{875}$$

$$= \underbrace{[1-(1-\theta)\beta]E_t \sum_{k=0}^{\infty}[(1-\theta)\beta]^k \hat{mc}^r_{t+k|t}}_{A} + [1-(1-\theta)\beta]E_t \sum_{k=0}^{\infty}[(1-\theta)\beta]^k(\hat{p}_{t+k} - \hat{p}_{t-1}) \tag{876}$$

Let the untouched part be A for simplicity, open the sum so there will be expression of inflation in log-linearized form

$$= A + [1-(1-\theta)\beta]E_t \sum_{k=0}^{\infty}[(1-\theta)\beta]^k(\hat{p}_{t+k} - \hat{p}_{t-1}) \tag{877}$$

$$\begin{aligned} &= A + [1-(1-\theta)\beta]E_t\{(\hat{p}_t - \hat{p}_{t-1}) + [(1-\theta)\beta](\hat{p}_{t+1} - \hat{p}_t + \hat{p}_t - \hat{p}_{t-1}) \\ &\quad + [(1-\theta)\beta]^2(\hat{p}_{t+2} - \hat{p}_{t+1} + \hat{p}_{t+1} - \hat{p}_t + \hat{p}_t - \hat{p}_{t-1}) + \dots\} \end{aligned} \tag{878}$$

$$= A + [1-(1-\theta)\beta]E_t\{[(\hat{\pi}_t + [(1-\theta)\beta](\hat{\pi}_{t+1} + \hat{\pi}_t) + [(1-\theta)\beta]^2(\hat{\pi}_{t+2} + \hat{\pi}_{t+1} + \hat{\pi}_t) + \dots\} \tag{879}$$

$$\begin{aligned} &= A + E_t\{[(\hat{\pi}_t + [(1-\theta)\beta](\hat{\pi}_{t+1} + \hat{\pi}_t) + [(1-\theta)\beta]^2(\hat{\pi}_{t+2} + \hat{\pi}_{t+1} + \hat{\pi}_t) + \dots\} \\ &\quad - (1-\theta)\beta E_t\{[(\hat{\pi}_t + [(1-\theta)\beta](\hat{\pi}_{t+1} + \hat{\pi}_t) + [(1-\theta)\beta]^2(\hat{\pi}_{t+2} + \hat{\pi}_{t+1} + \hat{\pi}_t) + \dots\} \\ &\quad \text{(making elimination possible)} \end{aligned} \tag{880}$$

$$\begin{aligned} &= A + E_t\{\hat{\pi}_t + [(1-\theta)\beta](\hat{\pi}_{t+1} + \hat{\pi}_t) + [(1-\theta)\beta]^2(\hat{\pi}_{t+2}\hat{\pi}_{t+1} + \hat{\pi}_t) + \dots\} \\ &\quad - [(1-\theta)\beta]\{\hat{\pi}_t - [(1-\theta)\beta]^2 E_t(\hat{\pi}_{t+1} + \hat{\pi}_t) - \dots\} \end{aligned} \tag{881}$$

Make format simple by doing iteration form, the first term is when $k = 0$, but the second still starts from k=0 so there will be some parameters changed

$$= [1-(1-\theta)\beta]E_t \sum_{k=0}^{\infty} \hat{mc}^r_{t+k|t} + E_t \sum_{k=0}^{\infty} \hat{\pi}_{t+k+1} \tag{882}$$

$$= \underbrace{[1-(1-\theta)\beta]mc^r_{t+k|t} + \pi_t}_{k=0} + [1-(1-\theta)\beta]E_t \sum_{k=0}^{\infty}[(1-\theta)\beta]^{k+1}\hat{mc}^r_{t+k+1|t} + E_t \sum_{k=0}^{\infty}[(1-\theta)\beta]^{k+1}\hat{\pi}_{t+k} \tag{883}$$

$$= [1-[(1-\theta)\beta]]\hat{mc}^r_{t|t} + \hat{\pi}_t + [(1-\theta)\beta]\{\qquad\} \tag{884}$$

$$= B \text{ (back to recursive form and let it be B)} \tag{885}$$

$$\Rightarrow \frac{1}{\theta}\hat{\pi}_t = B \tag{886}$$

$$\Rightarrow \hat{\pi}_t = \underbrace{\frac{\theta[1-(1-\theta)\beta]}{1-\theta}\hat{mc}^r_t}_{\text{effect of current marginal cost}} + \underbrace{\beta E_t\hat{\pi}_{t+1}}_{\text{effect of expectation of future inflation}} \quad \text{(New Keynesian Philips Curve marginal cost version)} \tag{887}$$

Question 3 (at which variable are we making derivative of Lagrangian function ?)**.**

Theoratically we can make derivative of any variable if possible but not proper.

NKPC has another version where marginal cost is substitued with real output, thus measuring the effect of output gap.

$$\begin{aligned} &\min \frac{W_t}{P_t} N_t \\ &s.t. Y_t = A_t N_t^{1-\alpha} \end{aligned} \tag{888}$$

Form a Lagrangian

$$\Rightarrow \mathscr{L} = \frac{W_t}{P_t} N_t + MC_t^r \cdot (Y_t - A_t N_t^{1-\alpha}) \text{ (shadow price can also be marginal cost)} \tag{889}$$

Take partial derivative of N_t here since labor is decided by market, the firm only can control demand of labor

$$\Rightarrow \frac{\partial \mathscr{L}}{\partial N_t} = 0 \tag{890}$$

$$\Rightarrow \frac{W_t}{P_t} N_t = MC_t^r \cdot MPN_t \text{ , where } MPN_t = \frac{\partial Y_t}{\partial N_t} = (1-\alpha) A_t N_t^{-\alpha} \tag{891}$$

$$\Rightarrow MC_t^r = \frac{\frac{W_t}{P_t}}{MPN_t} \tag{892}$$

$$\hat{mc}_t^\sigma = \hat{w}_t - \hat{p}_t - (a_t - \alpha \hat{n}_t) \text{ (log-linearized, and using labor supply equation)} \tag{893}$$

$$= \sigma \hat{c}_t + \varphi \hat{n}_t - (a_t - \alpha \hat{n}_t) \tag{894}$$

$$= \sigma \hat{y}_t - a_t - (\varphi + \alpha) \frac{\hat{y}_t - a_t}{1-\alpha} \tag{895}$$

$$= \frac{\sigma(1-\alpha) + (\varphi + \alpha)}{1-\alpha} \hat{y}_t - \frac{1+\varphi}{1-\alpha} a_t \tag{896}$$

To acquire natural ratio of output, we use the fact that elastic marginal cost is zero.

$$\Rightarrow 0 = \hat{mc}_t^r = \frac{\sigma(1-\alpha) + (\varphi + \alpha)}{1-\alpha} \hat{y}_t^f - \frac{1+\varphi}{1-\alpha} a_t \text{ (upscript } f \text{ stands for flexible meaning elastic)} \tag{897}$$

$$\Rightarrow \frac{1+\varphi}{1-\alpha} a_t = \frac{\sigma(1-\alpha) + (\varphi + \alpha)}{1-\alpha} \hat{y}_t^f \tag{898}$$

$$\Rightarrow \hat{mc}_t^r = \frac{\sigma(1-\alpha) + (\varphi + \alpha)}{1-\alpha} (\underbrace{\hat{y}_t - \hat{y}_t^f}_{\equiv \tilde{y}_t, \text{ output gap}}) \text{ (put it back, having the gap of output } \hat{y}_t - y_t^f = \tilde{t}) \tag{899}$$

If we put the last equation back to NKPC marginal cost version, we connect natural output to inflation which is NKPC real output version.

New Keynesian Philips Curve marginal cost version:

$$\hat{\pi}_t = \underbrace{\frac{\theta[1-(1-\theta)\beta]}{1-\theta} \frac{\sigma(1-\alpha) + (\varphi + \alpha)}{1-\alpha} (\hat{y}_t - \hat{y}_t^f)}_{\text{effect of current output gap}} + \underbrace{\beta E_t \hat{\pi}_{t+1}}_{\text{effect of expectation of future inflation}} \tag{900}$$

another way to acquire the NKPC is to concern monopolistic competition, then sticky price.

$$\max_{P_{it}} \pi^r = \frac{P_{it}}{P_t} Y_{it} - \frac{W_t}{P_t} N_{it} \tag{901}$$

$$\Rightarrow \frac{P_{it}}{P_t} = \frac{\epsilon}{\epsilon - 1} MC_t^r \tag{902}$$

Note 7 (indicator of monopolistic competition)**.**

$\frac{\epsilon}{\epsilon-1}$ means monopolistic competition
integral means perfect competitive labor supply for households
ϵ is subtitute elasticity

This is real price, where ϵ is subtitute elasticity. Here MC_i comes without subscript t is because of CRTS production function.

$$\Rightarrow \frac{P_{it}}{P_t} = \frac{\epsilon}{\epsilon - 1} \frac{W_t / P_t}{A_t} \tag{903}$$

$$= \frac{\epsilon}{\epsilon - 1} \frac{Y_t^\sigma N_t^\varphi}{A_t} \tag{904}$$

$$= \frac{\epsilon}{\epsilon - 1} \frac{Y_t^\sigma}{A_t} (\int_0^1 N_{it} di)^\varphi \text{ (because of goods market p.c.)} \tag{905}$$

$$= \frac{\epsilon}{\epsilon - 1} \frac{Y_t^\sigma}{A_t} [\int_0^1 (\frac{Y_{it}}{A_t}) di]^\varphi \text{ (based on CRTS production function)} \tag{906}$$

$$= \frac{\epsilon}{\epsilon - 1} \frac{Y_t^\sigma}{A_t^{1+\varphi}} [\int_0^1 (\frac{P_{it}}{P_t})^{-\epsilon} Y_t di]^\varphi \text{ (using demand curve and move } A_t \text{ out)} \tag{907}$$

$$= \frac{\epsilon}{\epsilon - 1} \frac{Y_t^\sigma}{A_t^{1+\varphi}} (\frac{P_{it}^*}{P_t})^{-\epsilon\varphi} \text{ (star means symmetric and at equil)} \tag{908}$$

Since at equilibrium all prices are same

$$P_{it}^* = P_t^* \Rightarrow \frac{P_{it}^*}{P_t} = \frac{\epsilon}{\epsilon - 1} \frac{Y_t^{\sigma+\varphi}}{A_t^{1+\varphi}} (\frac{P_t^*}{P_t})^{-\epsilon\varphi} \tag{909}$$

Let's consider a situation that

$$\text{If } P_t^* = P_t \text{ (meaning when price elastic)} \tag{910}$$

$$1 = \frac{\epsilon}{\epsilon - 1} \tag{911}$$

$$\Rightarrow Y_t^f \text{ or } \hat{y}_t^f \text{ (to be log-linearized)} \tag{912}$$

Taking logs

$$(1 + \epsilon\varphi)(P_t^* - P_t) = \ln \frac{\epsilon}{\epsilon - 1} - (1 + \varphi) a_t + (\sigma + \varphi) y_t \tag{913}$$

$$= (\sigma + \phi) y_t - (1 + \phi) a_t + \ln(\frac{\epsilon}{\epsilon - 1}) \text{ (change order)} \tag{914}$$

Change order for furthur preparation

$$\Rightarrow P_t^* = P_t + \frac{\sigma+\varphi}{1+\epsilon\varphi} y_t - \frac{1+\varphi}{1+\epsilon\varphi} a_t + \frac{1}{1+\epsilon\varphi} \ln(\frac{\epsilon}{\epsilon-1}) \tag{915}$$

$$\Rightarrow P^* = P + \frac{\sigma+\varphi}{1+\epsilon\varphi} y_t - 0 + \frac{1}{1+\epsilon\varphi} \ln(\frac{\epsilon}{\epsilon-1}) \text{ (steady state by definition)} \tag{916}$$

Two equations above deducted, having the log-linearized result; elastic price means $p_t^* = \hat{p}_t$

$$\hat{p}_t^* = \hat{p}_t + \frac{\sigma+\varphi}{1+\epsilon\varphi} \hat{y}_t - \frac{1+\varphi}{1+\epsilon\varphi} a_t \tag{917}$$

$$P_t^* = \hat{p}_t \text{ (elastic price gives } y \text{ a } f \text{ upscript)} \tag{918}$$

$$\Rightarrow 0 = \frac{\sigma+\varphi}{1+\epsilon\varphi} \hat{y}_t - \frac{1+\varphi}{1+\epsilon\varphi} a_t \tag{919}$$

$$\Rightarrow \hat{y}_t^f = \frac{1+\varphi}{\sigma+\varphi} a_t \tag{920}$$

here notice natural output is a function of technology

$$\Rightarrow \hat{p}_t^* = \hat{p}_t + \frac{\sigma+\varphi}{1+\epsilon\varphi} \hat{y}_t - \frac{\sigma+\varphi}{1+\epsilon\varphi} \hat{y}_t^f \tag{921}$$

$$\Rightarrow \hat{p}_t^* = \hat{p}_t + \gamma(\hat{y}_t - \hat{y}_t^f) \text{ (γ denotes the bundle of coefficients)} \tag{922}$$

$$\Rightarrow \hat{p}_t^* = \hat{p}_t + \gamma\tilde{y}_t \text{ (acceptable price-setting euqation with only assumption of m.c.)} \tag{923}$$

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Note 8 (DNK model is perfect information rational expectation)**.** *Here E_t is not $\bar{E}_t$ because of perfect information rational expectation*

Second way to acquire NKPC

We need to know two methods of price-setting:

- Price adjusting in certainty, Taylor, 1978[35]
- price adjusting in randomness, Calvo, 1983[36]

Note 9 (Price adjusting in certainty vs in randomness)**.**

Taylor

a, b and c denotes three firms in a monopolistic competitive market. they adjust price by turns and one person per period.

Q1: how does one choose price

$P_A^{\triangle} = \bar{3}$, the upscript $\triangle$ stands for staying put in the next three periods.

[34] See Mankiw and Reis 2002 for more details of acceptable price-setting equation

[35] Taylor, John B. "Staggered Wage Setting in a Macro Model." The American Economic Review 69, no. 2 (1979): 108–13.

[36] Calvo, Guillermo A., 1983. "Staggered prices in a utility-maximizing framework," Journal of Monetary Economics, Elsevier, vol. 12(3), pages 383-398, September.

Q2: what is the total price
P_t =
Calvo
each firm adjust price at probablity θ^t at period t.

Now sticky price price-adjusting equation

$$\hat{p}_t^{\triangle} = \underbrace{(1-\theta\beta)\sum_{k=0}^{\infty}(\theta\beta)^k E_t P_{t+k}^*}_{\text{weighted average}} \tag{924}$$

$$\Rightarrow \hat{p}_t^{\triangle} = (1-\theta\beta)\hat{p}_t^* + \theta\beta E_t \hat{p}_{t+1}^{\triangle} \text{ (sticky PAE)} \tag{925}$$

The θ is the opposite of the one in previous PAE, here rather is to stay put. price adjusting based on acceptable price setting equation in future periods, β mean subjective discount.

With the expression of total price being

$$\Rightarrow \hat{p}_t = (1-\theta)\sum_{k=0}^{\infty}\theta^k \hat{p}_{t-k}^{\triangle} \tag{926}$$

$$\Rightarrow \hat{p}_t = (1-\theta)\hat{p}_t^{\triangle} + \theta\hat{p}_{t-1} \text{ (expression of total price based on APSE and PAE)} \tag{927}$$

Based on Acceptable Price-setting Equation (from monopolistic competition) and Total Price (from sticky price), one can quickly conduct NKPC.

Note 10 (prices in DNK)**.**

starred price means acceptable price, triangled price means adjusting price

From the first equation $\hat{p}_t = (1-\theta)\sum_{k=0}^{\infty}\theta^k \hat{p}_{t-k}^{\triangle}$ we have

$$\hat{p}_t = (1-\theta)P_t^{\triangle} + \theta P_{t-1} \tag{928}$$

$$\Rightarrow (1-\theta)P_t^{\triangle} = \hat{p}_t - \theta\hat{p}_{t-1} \tag{929}$$

$$\Rightarrow (1-\theta)E_t P_{t+1}^{\triangle} = E_t\hat{p}_{t+1} - \theta\hat{p}_t \tag{930}$$

$$\Rightarrow (1-\theta)\beta E_t \hat{P}_{t+1}^{\triangle} = \beta E_t \hat{p}_{t+1} - \theta\beta\hat{p}_t \tag{931}$$

From the second equation $\hat{p}_t = (1-\theta)\hat{p}_t^{\triangle} + \theta\hat{p}_{t-1}$, first use sticky price price-adjusting equation $\hat{p}_t^{\triangle} = (1-\theta\beta)\hat{p}_t^* + \theta\beta E_t\hat{p}_{t+1}^{\triangle}$ to substitute

$$\Rightarrow \hat{p}_t - \theta\hat{p}_{t-1} = (1-\theta)\hat{p}_t^{\triangle} = (1-\theta)(1-\theta\beta)\hat{p}_t^* + (1-\theta)\theta\beta E_t P_{t+1}^{\triangle} \tag{932}$$

then use APSE $\hat{p}_t^* = \hat{p}_t + \gamma\tilde{y}_t$ to substitute

$$\Rightarrow \hat{p}_t - \theta\hat{p}_{t-1} = (1-\theta)(1-\theta\beta)(\hat{p}_t + \gamma\tilde{y}_t) + (1-\theta)\theta\beta E_t P_{t+1}^{\triangle} \tag{933}$$

use the conclusion $(1-\theta)E_t P_{t+1}^{\triangle} = E_t\hat{p}_{t+1} - \theta\hat{p}_t$ above

$$\Rightarrow \hat{p}_t - \theta\hat{p}_{t-1} = (1-\theta)(1-\theta\beta)(\hat{p}_t + \gamma\tilde{y}_t) + \theta\beta E_t\hat{p}_{t+1} - \theta^2\beta\hat{p}_t \tag{934}$$

$$\Rightarrow [1+\theta^2\beta - (1-\theta)(1-\theta\beta)]\hat{p}_t - \theta P_{t-1} = (1-\theta)(1-\theta\beta)\gamma\tilde{y}_t + \theta\beta E_t \text{ (break APSE)} \tag{935}$$

$$\Rightarrow [1+\theta^2\beta - (1-\theta)(1-\theta\beta)]\hat{p}_t - \theta P_{t-1} - \theta\beta\hat{p}_t = (1-\theta)(1-\theta\beta)\gamma\tilde{y}_t + \theta\beta(E_t\hat{p}_{t+1} - \hat{p}_t) \tag{936}$$

$$\Rightarrow \theta(\hat{p}_t - \hat{p}_{t-1}) = (1-\theta)(1-\theta\beta)\gamma\tilde{y}_t + \theta\beta(E_t\hat{P}_{t+1} - \hat{p}_t) \text{ (match terms so there is inflation)} \tag{937}$$

$$\Rightarrow \theta\hat{\pi}_t = (1-\theta)(1-\theta\beta)\gamma\tilde{y}_t + \theta\beta E_t\hat{\pi}_{t+1} \text{ (here comes inflation)} \tag{938}$$

$$\Rightarrow \hat{\pi}_t = k\hat{y}_t + \beta E_t\hat{\pi}_{t+1} \text{ (NKPC)} \tag{939}$$

11.2.3 Dynamic IS curve

Recalling that household

$$\hat{y}_t = E_t\hat{y}_{t+1} - \frac{1}{\sigma}(\hat{i}_t - E_t\hat{\pi}_{t+1}) \tag{940}$$

$$\Rightarrow \hat{y}_t^f = E_t\hat{y}_{t+1} - \frac{1}{\sigma}(\hat{i}_t - 0) \text{ (inflation elastic form is assumed as 0)} \tag{941}$$

$$\Rightarrow \tilde{y}_t = E_t\tilde{y}_{t+1} - \frac{1}{\sigma}(\hat{i}_t - E_t\hat{\pi}_{t+1}) \text{ (dynamic IS curve, DIS)} \tag{942}$$

11.3 Equilibrium of model

Now NKPC and DIS each represents supply and demand. The two equations have three unkown variables $\hat{\pi}_t, \tilde{y}_t, \hat{i}_t$, applying the model does not have a solution. It is so that we need to add another variable into the model. There are two ways making DNK model solvable, if not monetary supply rule then Taylor rule.

11.3.1 DNK with money supply rule

$$\text{DNK} \begin{cases} \hat{\pi}_t = \beta E_t\hat{\pi}_{t+1} + k\tilde{y}_t \text{ (NKPC)} \\ \tilde{y}_t = E_t\tilde{y}_{t+1} - \frac{1}{\sigma}(\hat{i}_t - E_t\hat{\pi}_{t+1}) \text{ (DIS)} \\ \hat{m}_t = \hat{p}_t + \hat{y}_t - \eta\hat{i}_t \text{ (Money supply rule, used to be ad hoc in RBC)} \end{cases} \tag{943}$$

Seems now three equations have even more unkown variables $\hat{\pi}_t, \tilde{y}_t, \hat{i}_t, \hat{p}_t, m_t$.

But hat $\pi_t, \hat{p}_t$ are the same thing because

$$\hat{\pi}_t = \hat{p}_t - \hat{p}_{t-1} \tag{944}$$

$$\Rightarrow \hat{p}_t = \hat{\pi}_t + \hat{p}_{t-1} \tag{945}$$

$$\Rightarrow \hat{m}_t = \hat{\epsilon}_t \tag{946}$$

this transfer works either way.

Let the m_t money supply be exogenous for simplicity.

Note 11 (Exogenous variables in DNK model)**.** *Only technology and money supply are considered real exogenous variables here.*

So we are left with three equations and three unkowns, the model is now solutionable. The final result will be a function of m_t.

11.3.2 DNK with interest rate rule

$$\text{DNK} \begin{cases} \hat{\pi}_t = \beta E_t \hat{\pi}_{t+1} + k\tilde{y}_t \text{ (NKPC)} \\ \tilde{y}_t = E_t \tilde{y}_{t+1} - \frac{1}{\sigma}(\hat{i}_t - E_t \hat{\pi}_{t+1}) \text{ (DIS)} \\ \hat{i}_t = \phi_\pi \hat{\pi}_t + \phi_y \hat{y}_t + \nu_t \text{ (Interest rate rule)} \end{cases} \tag{947}$$

$$\text{where } \nu_t = \rho_v \nu_{t-1} + \epsilon_t^\nu \text{ is the exogenous shock of interest rate, with the feature of AR1} \tag{948}$$

Note 12 (Interest rate rule)**.**

$$\hat{i}_t = \phi_\pi \cdot (\hat{\pi}_t - \hat{\pi}^*) + \phi_y \cdot (\hat{y}_t - y_t^f) + \nu_t \tag{949}$$

If the target parameters $\hat{\pi}^*$ and y_t^f are given, it is Taylor Rule.
Operation parameters ϕ_π and ϕ_y are decided based on past experience.

Acquiring equilibrium and prove it is the only equilibrium
Rewrite the DIS

$$\Rightarrow \tilde{y}_t = E_t \tilde{y}_{t+1} - \frac{1}{\sigma}[(\phi_\pi \hat{\pi}_t + \phi_y \hat{y}_t + \nu_t) - E_t \hat{\pi}_{t+1}] \text{ (apply interest rate rule to the second equation)} \tag{950}$$

$$= E_t \tilde{y}_{t+1} - \frac{1}{\sigma}\{[\phi_\pi \cdot (\beta E_t \hat{\pi}_{t+1} + k\tilde{y}_t) + \phi_y \hat{y}_t + \nu_t] - E_t \hat{\pi}_{t+1}\} \text{ (put the first in)} \tag{951}$$

$$= E_t \tilde{y}_{t+1} + \frac{1 - \beta\phi_\pi}{\sigma} E_t \pi_{t+1} - \frac{\phi_\pi k + \phi_y}{\sigma} \tilde{y}_t - \frac{\nu_t}{\sigma} \tag{952}$$

$$= \frac{1}{\sigma + \phi_\pi k + \phi_y}[\sigma E_t \tilde{y}_{t+1} + (1 - \beta\phi_\pi) E_t \hat{\pi}_{t+1} - \nu_t] \text{ (the relationship of } y_t \text{ and } y_{t+1} \text{ and } \pi_{t+1}) \tag{953}$$

Similarly we can change the NKPC

$$\hat{\pi}_t = \frac{1}{\sigma + \phi_\pi k + \phi_y}\{k\sigma E_t \tilde{y}_{t+1} + [k + \beta(\sigma + \phi_y)] E_t \hat{\pi}_{t+1} - \nu_t\} \tag{954}$$

Therefore three equations are two now.

Iteration in vector form

$$\begin{bmatrix}\tilde{y}_t\\ \hat{\pi}_t\end{bmatrix} = \frac{1}{\sigma+\phi_\pi k+\phi_y}\begin{bmatrix}\sigma & 1-\beta\phi_\pi\\ k\sigma & k+\beta(\sigma+\phi_y)\end{bmatrix}\begin{bmatrix}E_t\hat{y}_{t+1}\\ E_t\hat{\pi}_{t+1}\end{bmatrix} - \frac{1}{\sigma+\phi_\pi k+\phi_y}\begin{bmatrix}1\\ k\end{bmatrix}\nu_t \tag{955}$$

$$\Rightarrow \begin{bmatrix}\tilde{y}_t\\ \hat{\pi}_t\end{bmatrix} = A\begin{bmatrix}C_{t-1}\\ K_{t-1}\end{bmatrix} - B\nu_t \text{ (at RHS one period ahead meaning the whole equation is foresight-seeing)} \tag{956}$$

$$\Rightarrow \lambda^2 - \frac{\sigma+k+\beta(\sigma+\phi_y)}{\sigma+\phi_\pi k+\phi_y}\lambda + \frac{\sigma\beta}{\sigma+\phi_\pi k+\phi_y} = 0 \tag{957}$$

Note 13 (Technique in Ramsey and DNK)**.**

From the perspective of techniques in persuiing solution, it is greatly similar to the way we did with Ramsey model, where the model is one period backward at RHS.

Note 14 (HNKPC)**.** *Hybrid New Keynesian Philips Curve*

Since NKPC lacks persistence in practical use (shocks are too slow to effects), we might as well add a 1-period-lagged inflation to the RHS as adpative expectation. This is somehow a nominal shock since it gets into the DNK model through m_t.

$$\hat{\pi}_t = \beta E_t\hat{\pi}_{t+1} + k\tilde{y}_t \text{ (NKPC)} \tag{958}$$

$$\hat{\pi}_t = \hat{\pi}_{t-1} + \beta E_t\hat{\pi}_{t+1} + k\tilde{y}_t \text{ (HNKPC)} \tag{959}$$

a_t is described as real shocks of DNK model. But it shows up too fast.
Later the concept of sticky information is added to the model to solve the problem.

Note 15 (BK conditions)**.**

First brought up by Blanchard and Kahn, 1980

[37]

Note 16.

Lasalle, 1986

$$x^2 + bx + c = 0 \tag{960}$$

$$x_1, x_2 < 1 \iff \begin{cases}|c| < 1\\ |b| < 1 + c\end{cases} \tag{961}$$

[37] Also see Nakajima 2007 for similar techniques.

If the equilibrium exists and it is the only one, then

$$\begin{cases} |\frac{\sigma\beta}{\sigma+\phi_\pi k+\phi_y}| < 1 \Rightarrow \sigma(\beta-1) < 0 < \phi_\pi k + \phi_y \\ |\frac{\sigma+k+\beta(\sigma+\phi_y)}{\sigma+\phi_\pi k+\phi_y}| < 1 + \frac{\sigma\beta}{\sigma+\phi_\pi k+\phi_y} \Rightarrow k(\phi_\pi - 1) + (1-\beta)\phi_y > 0 \end{cases} \tag{962}$$

Method of Undetermined Coefficients:
Let's assume

$$\begin{cases} \tilde{y}_t = \psi_{y\nu}\nu_t \\ \hat{\pi}_t = \psi_{\pi\nu}\nu_t \end{cases} \tag{963}$$

recall that

$$\hat{\pi}_t = \beta E_t\hat{\pi}_{t+1} + k\hat{y}_t \text{ (expected inflation)} \tag{964}$$

$$\psi_{\pi\nu}\nu_t = \beta E_t(\psi_{\pi\nu}\nu_{t+1}) + k\hat{y}_t \text{ (iterating)} \tag{965}$$
$$= \beta E_t[\psi_{\pi\nu}(\rho_\nu\nu_t + \epsilon^\nu_{t+1})] + k\hat{y}_t \text{ (expand } \nu_{t+1} \text{ since it's AR1 shock of } i_t) \tag{966}$$
$$= \beta\psi_{\pi\nu}\rho_\nu\nu_t + k\psi_{y\nu}\nu_t \text{ (preassuming expectation of next period noise is } 0) \tag{967}$$
$$\Rightarrow (1-\beta\rho_\nu)\psi_{\pi\nu} = k\psi_{y\nu} \tag{968}$$
$$\Rightarrow k\psi_{y\nu} = \frac{1-\beta\rho_\nu}{k}\psi_{\pi\nu} \text{ (relationship of two undetermined coefficients)} \tag{969}$$

Using DIS

$$\tilde{y}_t = E_t\tilde{y}_{t+1} - \frac{1}{\sigma}[\phi_\pi\hat{\pi}_t + \phi_y\hat{y}_t + \nu_t) - E_t\hat{\pi}_{t+1}] \tag{970}$$

using interest rate rule

$$\Rightarrow \psi_{y\nu}\nu_t = \psi_{y\nu}E_t\nu_{t+1} - \frac{1}{\sigma}(\phi_\pi\psi_{\pi\nu}\nu_t + \phi_y\psi_{y\nu}\nu_t + \nu_t - \psi_{\pi\nu}E_t\nu_{t+1}) \tag{971}$$
$$= \psi_{y\nu}\rho_\nu\nu_t - \frac{1}{\sigma}(\phi_\pi\psi_{\pi\nu}\nu_t + \phi_y\psi_{y\nu}\nu_t + \nu_t - \psi_{\pi\nu}\rho_\nu\nu_t) \tag{972}$$
$$= \frac{\sigma\rho_\nu - \phi_y}{\sigma}\phi_{y\nu}\nu_t - \frac{\phi_\pi - \rho_\nu}{\sigma}\psi_{\pi\nu}\nu_t - \frac{1}{\sigma}\nu_t \tag{973}$$
$$\Rightarrow \sigma\psi_{y\nu} = (\sigma\rho_\nu - \phi_y)\psi_{y\nu} - (\phi_\pi - \rho_\nu)\psi_{\pi\nu} - 1 \text{ (elimination of } \nu_t) \tag{974}$$
$$\Rightarrow (\sigma - (\sigma\rho_\nu - \phi_y))\psi_{y\nu} + (\phi_\pi - \rho_\nu)\psi_{\pi\nu} = -1 \tag{975}$$
$$\Rightarrow \psi_{\pi\nu} = \frac{-k}{\cdots} = -k\wedge_\nu \tag{976}$$
$$\Rightarrow \psi_{y\nu} = -(1-\beta\rho_\nu)\wedge_\nu \tag{977}$$

11.4 Analyze exogenous shocks

Now policy change will reflect in the parameter ϵ, making policy analysis possible.
we use ν to denote the exogenous shock
last section we have expression of ν
$\Rightarrow$ to analyze the shock would mean fully using it.

Note 17 (what is a shock?)**.**

In a model with uncertainty, variable has the feature of causing change in variation by change itself can cause a shock.

But it was later broadened as change of exogenous variable in a model without uncertainty.

recall output gap at equilibrium

$$\begin{aligned}
\hat{y}_t - \hat{y}_t^f = \hat{y}_t^* &= \psi_{y\nu}\nu_t && (978)\\
&= -(1-\beta\rho_\nu) \wedge_\nu \nu_t && (979)\\
&= -(1-\beta\rho_\nu) \wedge_\nu (\rho_\nu \nu_{t-1} + \epsilon_t^\nu) && (980)\\
&= -(1-\beta\rho_\nu) \wedge_\nu \rho_\nu\nu_{t-1} - (1-\beta\rho_\nu) \wedge_\nu \epsilon_t^\nu && (981)\\
&= \rho_\nu \tilde{y}_{t-1} - (1-\beta\rho_\nu) \wedge_\nu \epsilon_t^\nu \ (\tilde{y}_{t-1} \text{ backward one period}) && (982)
\end{aligned}$$

Note 18. *we can only do analysis at equil*

similiar with inflation at equilibrium

$$\begin{aligned}
\hat{\pi}_t^* &= \psi_{\pi\nu}\nu_t && (983)\\
&= -k \wedge_\nu \nu_t && (984)\\
&= -k \wedge_\nu (\rho_\nu \nu_{t-1} + \epsilon_t^\nu) && (985)\\
&= \rho_\nu \pi_{t-1} - k \wedge_\nu \epsilon_t^\nu && (986)
\end{aligned}$$

interest rate at equilibrium

$$\begin{aligned}
\hat{r}_t^* &= \hat{i}_t - E_t\hat{\pi}_{t+1} \text{ (Real interest rate at equilibrium)} && (987)\\
&= \sigma(E_t\tilde{y}_{t+1} - \hat{y}_t) \leftarrow \text{DIS} && (988)\\
&= \sigma(1-\beta\rho_\nu)(1-\rho_\nu) \wedge_\nu \nu_t && (989)\\
\Rightarrow \hat{i}_t^* &= \hat{r}_t + E_t\hat{\pi}_{t+1} \text{Nominal interest rate at equilibrium} && (990)\\
&= [\sigma(1-\beta\rho_\nu)(1-\rho_\nu) - k\rho_\nu] \wedge_\nu \nu_t && (991)
\end{aligned}$$

moreever we can do analysis with money supply using quantity equation of money

$$\begin{aligned}
\hat{m}_t^* &= \hat{p}_t^* + \hat{y}_t^* - \eta\hat{i}_t^* \text{ (LHS is solvable since all variables at RHS has been solved)} && (992)\\
&= \text{ (assuming } \hat{y}_t^{f*} = 0) && (993)\\
&= && (994)\\
&= \text{ use solutions from above} && (995)\\
\Rightarrow & && (996)\\
= & && (997)\\
\frac{d\hat{m}_t^*}{d\nu_t} &= && (998)\\
&= \text{showing the result is decided by mutiple joint forces} && (999)
\end{aligned}$$

Note 19 (quantity equation of money)**.** $MV = Py$

While this theory was originally formulated by Polish mathematician Nicolaus Copernicus in 1517, it was popularized later by economists Milton Friedman and Anna Schwartz after the publication of their book, "A Monetary History of the United States, 1867-1960," in 1963.

11.5 Natural rates, output gaps, and model limits

Planned section. Distinguish the natural real interest rate, flexible-price output, and efficient output. Check which shocks and distortions enter each object before interpreting policy responses or welfare losses.

12 Sticky Wage Model

Before good market is price-sticky and monopolistic competitive, now the setting goes to labor market.

$$Y_{it} = (\frac{P_{it}}{P_t})^{-\epsilon_P} Y_t \text{ (good demand curve, } \epsilon \text{ is substitute elasticity)} \tag{1000}$$

$$P_t = (\int_0^1 P_{it}^{1-\epsilon} di)^{\frac{1}{1-\epsilon_p}} \text{ (total price)} \tag{1001}$$

$$W_t = W_{it} = \int_0^1 W_{it} di \tag{1002}$$

$$P_{it}^* = c + \alpha \tilde{y}_t \text{ (desired price)} \tag{1003}$$

labor market m.c. (as in classical model) and sticky wage (partial rigidity, in contrary to total rigidity where all wages are rigid).

1) labor demand curve and total wage price level

$$\begin{aligned} &\min \int_0^1 W_{jt} N_{ijt} dj \\ &s.t. A_t [(\int_0^1 N_{ijt}^{\frac{\epsilon_W - 1}{\epsilon_W}} dj)^{\frac{\epsilon_W}{\epsilon_W - 1}}]^{1-\alpha} \geqslant A_t N_{it}^{1-\alpha} \end{aligned} \tag{1004}$$

where i indexing firm, j indexing labor, t is time

Form a Lagrangian

$$\mathscr{L} = \int_0^1 W_{jt} N_{ijt} dj + W_t \{A_t [(\int_0^1 N_{ijt}^{\frac{\epsilon_w - 1}{\epsilon_w}} dj)^{\frac{\epsilon_W}{\epsilon_W - 1}}]^{1-\alpha} - A_t N_{it}^{1-\alpha}\} \tag{1005}$$

where Lagrangian multiplier is total wage level

Solving the optimization problem results

$$\Rightarrow \begin{cases} N_{ijt} = (\frac{W_{jt}}{W_t})^{-\epsilon_W} N_{it} \text{ (labor supply equation)} \\ W_t = (\int_0^1 W_{jt}^{1-\epsilon_W} dj)^{\frac{1}{1-\epsilon_W}} \text{ (expression of total wage level)} \end{cases} \tag{1006}$$

2) UMP of household

in utility function N is labor demand instead of labor supply as before. this is because of m.c. labor market.

$$\begin{aligned} u &= u(C_t, N_t) \\ &= u(C_t, \{N_{jt}\}) \\ &= \frac{C_t^{1-\sigma} - 1}{1-\sigma} - \int_0^1 \frac{N_j^{1+\varphi}}{1+\varphi} dj \text{ (CES style)} \end{aligned} \tag{1007}$$

the UMP can be rewritten as

$$\begin{aligned} \max_{\{C_t, W_t^*\}} \sum_{t-0}^{\infty} \beta^t u(C_t, \int_0^1 N_{jt} dj) \iff \max \sum_{t=0}^{\infty} \beta^t u(C_t, \int_0^1 (\frac{W_t^*}{W_t})^{-\epsilon_W} N_{it} dj) \\ s.t. P_t C_t + Q_t B_t \leqslant \int_0^1 W_t^* (\frac{W_t^*}{W_t})^{-\epsilon_W} N_{it} dj + B_{t-1} \end{aligned} \tag{1008}$$

here W_t^* instead of original because all wages can be adjusted, eventually they will be the same

a simplified version would be

$$\begin{aligned} &\max \sum_{t=0}^{\infty} \beta^t u(C_t, (\frac{W_t^*}{W_t})^{-\epsilon_W} N_{it}) \\ &s.t. P_t C_t + Q_t B_t \leqslant W_t^* (\frac{W_t^*}{W_t})^{-\epsilon_W} + B_{t-1} - T_t \end{aligned} \tag{1009}$$

$$\begin{aligned} &\max_{W_0^*} \sum_{t=0}^{\infty} \beta^t \theta_W^t u(C_{t|0}, N_{t|0}) \\ &s.t. \begin{cases} C_{t|0} = \frac{1}{P_t}[W_0^* (\frac{W_0^*}{W_t})^{-\epsilon_W} N_{it} + B_{t-1|0} - T_{t|0} - Q_{t-1|t} B_{t|0}] \\ N_{t|0} = (\frac{W_0^*}{W_t})^{-\epsilon_W} N_{it} \text{ (with labor demand curve, left with only } W_0^*) \end{cases} \end{aligned} \tag{1010}$$

sticky wage, where $\theta \equiv Prob(\text{stay put})$. θ is before utility function because the whole equation is based on period 0, entailing a prob for the whole equation to stay put.

put two constraints in the max part, now the optimization problem is a unconstrainted one

$$\max_{W_0^*} \sum_{t=0}^{\infty} (\beta\theta_W)^t u(\frac{1}{P_t}[W_0^* (\frac{W_0^*}{W_t})^{-\epsilon_W} N_{it} + B_{t-1|0} - T_{t|0} - Q_{t-1|t} B_{t|0}], (\frac{W_0^*}{W_t})^{-\epsilon_W} N_{it}) \tag{1011}$$

$$FOC \Rightarrow E_0 \sum_{t=0}^{\infty} (\beta\theta_W)^t [u_C(C_{t|0}, N_{t|0}) \frac{1}{P_t} (1-\epsilon_W)(\frac{W_0^*}{W_t})^{-\epsilon_W} N_{it} - u_N(C_{t|0}, N_{t|0}) \epsilon_W (\frac{W_0^*}{W_t})^{-(\epsilon_W)^{-1}} \frac{1}{W_t} N_{it})] \tag{1012}$$

$$= E_0 \sum_{t=0}^{\infty} (\beta\theta_W)^t [u_C(C_{t|0}, N_{t|0}) \frac{1}{P_t} (1-\epsilon_W)(\frac{W_0^*}{W_t})^{-\epsilon_W} W_t^{\epsilon_W} N_{it} - u_N(C_{t|0}, N_{t|0}) \epsilon_W (\frac{W_0^*}{W_t})^{-(\epsilon_W)} W_t^{\epsilon_W} N_{it} \frac{1}{W_0^*})] \tag{1013}$$

move some exogenous variables out since they basically can be treated as constants

$$\Rightarrow (1-\epsilon_W)(W_0^*)^{-\epsilon_W} \ldots \tag{1014}$$

$$= \epsilon_W (W_0^*)^{-\epsilon_W^{-1}} \tag{1015}$$

$$\Rightarrow W_0^* = \frac{\epsilon_W}{1-\epsilon_W} \frac{E_0 \sum_{t=0}^{\infty} (\beta\theta_W)^t u_N(C_{t|0}, N_{t|0}) W_t^{\epsilon_W} N_{it}}{E_0 \sum_{t=0}^{\infty} (\beta\theta_W)^t u_C(C_{t|0}, N_{t|0}) \frac{1}{P_t} W_t^{\epsilon_W} N_{it}} \tag{1016}$$

use marginal substitute ratio, where $MRS_{t|0} \equiv -\frac{U_N(C_{t|0})}{U_C(C_{t|0})}$, the equation is essentially an inflation equation

$$W_0^* = \frac{\epsilon_W}{\epsilon_W - 1} \frac{E_0 \sum_{t=0}^{\infty} (\beta\theta_W)^t u_N(C_{t|0}, N_{t|0}) MRS_{t|0} W_t^{\epsilon_W} N_{it}}{E_0 \sum_{t=0}^{\infty} (\beta\theta_W)^t u_C(C_{t|0}, N_{t|0}) \frac{1}{P_t} W_t^{\epsilon_W} N_{it}} \tag{1017}$$

$$\Rightarrow \frac{W_0^*}{P_0} = \frac{\epsilon_W}{\epsilon_W - 1} MRS_{t|0} \text{ (if elastic wage and m.c., there will be no equation above)} \tag{1018}$$

now based on the perspective at period t

$$W_t^* = \frac{\epsilon_W}{\epsilon_W - 1} \frac{E_t \sum_{k=0}^{\infty} (\beta\theta_W)^k u_C(C_{t+k|t}, N_{t+k|t}) MRS_{t+k|t} W_{t+k|t}^{\epsilon_W}}{E_t \sum_{k=0}^{\infty} (\beta\theta_W)^k u_C(C_{t+k|t}, N_{t+k|t}) \frac{1}{P_{t+k}} W_{t+k|t}^{\epsilon_W}} \text{ (at t period instead 0 period)} \quad (1019)$$

$$\Rightarrow \frac{W_t^*}{P_t} = \frac{\epsilon_W}{\epsilon_W - 1} MRS \text{ (if elasticity wage and monopolistic competitive at t period)} \quad (1020)$$

Note 20 (M.C. firm choice variable for optimization problem)**.** *MC firm can choose price max or quantity max*

13 Labor Markets and Unemployment

Planned chapter.

13.1 Employment, unemployment, and participation

Define the margins hidden by a single aggregate labor-supply variable. Distinguish changes in hours from movements into and out of employment.

13.2 Search, matching, and wage determination

Introduce a basic matching framework to explain vacancies and unemployment. State how wage setting and hiring incentives determine job creation.

13.3 Adjustment and persistence

Compare search frictions with nominal wage rigidity. Examine which mechanisms can make employment recover slowly and connect a recession to longer-run output.

Part IV
Open Economy

An open economy can trade goods and claims on future income. This changes the connection between domestic saving and investment, introduces relative prices between countries, and allows shocks to cross borders. We extend the earlier models before comparing policy choices in Part V.

14 International Accounts and Intertemporal Trade

Planned chapter.

14.1 Trade balances, current accounts, and external wealth

Derive the links between national saving, investment, and the current account. Distinguish transaction flows from valuation changes in the international investment position.

14.2 Consumption smoothing and international borrowing

Start with a small open endowment economy. Derive its intertemporal budget constraint and explain how expected income affects the current account.

14.3 Investment and external adjustment

Add capital accumulation and examine how borrowing finances investment. State borrowing and solvency conditions before discussing external adjustment.

15 Trade and Relative Prices

Planned chapter.

15.1 Comparative advantage, trade gains, and trade costs

Use a simple trade model to explain specialization and gains from exchange. Introduce tariffs and other trade costs to connect trade policy to allocation.

15.2 Terms of trade and the real exchange rate

Define both relative prices with explicit conventions. Trace how their changes affect purchasing power, production, and external balances.

15.3 Tradable goods, nontradable goods, and adjustment

Introduce the distinction between tradable and nontradable production. Explain how expenditure and resources move between sectors during adjustment.

16 Open Economy Fluctuations and Exchange Rates

Planned chapter.

16.1 The small open economy RBC model

Extend the real benchmark to international borrowing and investment. Compare productivity, world-interest-rate, and terms-of-trade shocks.

16.2 Capital flows, interest parity, and risk

Derive the relevant asset-pricing conditions and distinguish covered parity from an uncovered expected-return condition. Introduce risk premia and foreign-currency balance-sheet exposure.

16.3 Nominal rigidities and exchange-rate regimes

Introduce an open economy New Keynesian benchmark. Compare the adjustment mechanisms under fixed and flexible exchange rates, and state the constraints created by capital mobility.

Part V
Money, Finance, and Macroeconomic Policy

The earlier parts establish the real and nominal benchmarks. We now add financial constraints and differences between households, then compare the transmission and welfare effects of monetary and fiscal policy. Open economy mechanisms remain part of that comparison.

17 Money, Banking, and Central Bank Operations

Planned chapter.

17.1 Money, payments, and financial claims

Distinguish money from other assets and identify the services that give it a role in household and firm decisions.

17.2 Banks and balance sheets

Set out the balance sheets of households, commercial banks, and the central bank. Explain lending, deposits, reserves, and the limits to credit creation.

17.3 Implementing monetary policy

Connect the model's policy rate to the central bank's operating framework. Distinguish reserve provision, asset purchases, and changes in borrowing costs.

18 Financial Frictions and Aggregate Adjustment

Planned chapter.

18.1 Borrowing limits and collateral

Derive a simple borrowing constraint and show how it changes optimal choices. Connect collateral values to spending and investment.

18.2 Intermediation and financial amplification

Introduce a constraint on intermediary balance sheets. Explain how losses, leverage, and credit spreads can amplify a real or financial disturbance.

18.3 Financial distress and policy intervention

Distinguish liquidity problems from insolvency. State the mechanisms and trade-offs behind liquidity support, restructuring, and macroprudential policy.

19 Household Heterogeneity and Policy Transmission

Planned chapter.

19.1 Income risk, insurance, and precautionary saving

Introduce uninsurable income risk and borrowing limits. Explain how the distribution of liquid wealth affects saving and consumption.

19.2 Marginal propensities to consume and redistribution

Compare households' spending responses to income and wealth changes. Trace how taxes, transfers, and interest payments redistribute resources.

19.3 From representative to heterogeneous households

Use a two-agent benchmark before motivating a richer distribution of households. Identify which policy questions require the additional heterogeneity.

20 Monetary Policy

This chapter extends the interest-rate rules used to close the New Keynesian model into a systematic policy comparison. The existing welfare derivation is retained below; the preceding sections are outlines.

20.1 Transmission and policy shocks

Planned section. Separate a systematic policy response from an unexpected policy disturbance. Compare intertemporal, credit, redistribution, and exchange-rate channels.

20.2 Rules, commitment, and discretion

Planned section. Compare equilibrium determination with welfare-based policy design. Explain how credibility and promises about future policy affect current choices.

20.3 The interest-rate lower bound and balance-sheet policies

Planned section. State when the policy-rate constraint binds and which transmission channels remain. Examine forward guidance and asset purchases under explicit assumptions.

20.4 Optimal monetary policy

A significant paper Mankiw and Reis 2002 used a new keynesian macroeconomic model to acquire the optimal monetary policy. [38]

Note 21 (why is there welfare loss?)**.**

monopoly and sticky price cause deviation from Pareto optimality.

Note 22 (Optimal monetary policy)**.**

No concept of optimality can be found in policy analysis in the perspective of household or firm. But the government can use welfare loss fucntion in its optimality to maximize social welfare. In the process of achieving maximized social welfare, the monetary policy is the so-called optimal monetary policy.

To acquire the optimal monetary policy, the welfare loss function must be previously introduced.
To acquire WLF, we use second order taylor expansion
first order Taylor expansion

$$\frac{X_t - X}{X} = \ln X_t - \ln X = x_t - x = \hat{x}_t \tag{1021}$$

[38] N. Gregory Mankiw, Ricardo Reis, Sticky Information versus Sticky Prices: A Proposal to Replace the New Keynesian Phillips Curve, The Quarterly Journal of Economics, Volume 117, Issue 4, November 2002, Pages 1295–1328, https://doi.org/10.1162/003355302320935034

second order Taylor expansion

$$\begin{aligned}\frac{X_t - X}{X} &= \frac{1}{X}e^{x_t} - 1 \\ &= (\frac{1}{X}e^{x} - 1) + \frac{1}{X}e^{x}(x_t - x) + \frac{1}{2}\frac{1}{X}e^{x}(x_t - x)^2 \\ &= \hat{x}_t + \frac{1}{2}\hat{x}_t^2\end{aligned} \tag{1022}$$

Similarly

$$\Rightarrow \begin{cases} \frac{C_t - C}{C} \approx \hat{c}_t + \frac{1}{2}\hat{c}_t^2 \\ \frac{N_t - N}{N} \approx \hat{n}_t + \frac{1}{2}\hat{n}_t^2 \end{cases} \tag{1023}$$

Bi-variate 2o taylor expansion for a utility function where people gain utilities from consuming goods and enjoying leisure.

$$u(C_t, N_t) \approx u(C, N) + u_C^C(\frac{C_t - C}{C}) + u_N^N(\frac{N_t - N}{N}) + \frac{1}{2}u_{CC}^{C^2}\frac{(C_t - C)^2}{C} + \frac{1}{2}u_{NN}^{N^2}\frac{(N_t - N)^2}{N} \tag{1024}$$

$$= u(C, N) + u_C C(\hat{c}_t + \frac{1}{2}\hat{c}_t^2) + u_N N(\hat{n}_t + \frac{1}{2}\hat{n}_t^2) + \frac{1}{2}u_{CC}C^2(\hat{c}_t + \frac{1}{2}\hat{c}_t^2) + \frac{1}{2}u_{NN}N^2(\hat{n}_t + \frac{1}{2}\hat{n}_t^2) \tag{1025}$$

$$= u(C, N) + u_C C(\hat{c}_t + \frac{1}{2}\hat{c}_t^2) + u_N N(\hat{n}_t + \frac{1}{2}\hat{n}_t^2) + \frac{1}{2}u_{CC}C^2\hat{c}_t + \frac{1}{2}u_{NN}N^2\hat{n}_t \text{ (reservate only to 2o)} \tag{1026}$$

$$= u(C, N) + u_C C(\hat{c}_t + \frac{1 + \frac{u_{CC}}{u_C}C}{2}\hat{c}_t) + u_N N(\hat{n}_t + \frac{1 + \frac{u_{NN}}{u_N}N}{2}\hat{n}_t) \tag{1027}$$

$$\Rightarrow \frac{u(C_t, N_t) - u(C, N)}{u_C C} = (\hat{c}_t + \frac{1 + \frac{u_{CC}}{u_C}C}{2}\hat{c}_t^2) + \frac{u_N N}{u_C C}(\hat{n}_t + \frac{1 + \frac{u_{NN}}{u_N}}{2}\hat{n}_t^2) \tag{1028}$$

$$\because \hat{c}_t = \hat{y}_t \text{ (market clearing at equilibrium)} \tag{1029}$$

$$u(C_t, N_t) = \frac{C_t^{1-\sigma} - 1}{1 - \sigma} - \frac{N_t^{1+\varphi} - 1}{1 + \varphi} \iff \begin{cases} -\frac{u_{CC}}{u_C}C = \sigma \\ -\frac{u_{NN}}{u_N}N = \varphi \end{cases} \tag{1030}$$

Note 23 (Coefficients of CES function)**.**

σ is intertemporal substitute elasticy, φ is frisch elasticity

continue to the last taylor expanded equation, introducing var_i as the deviation of each firm's price

$$\ldots = (\hat{y}_t + \frac{1 - \sigma}{2}\hat{y}_t^2) + \frac{u_N N}{u_C C}\{\frac{1}{\alpha}(\hat{y}_t - a_t) + A \cdot var_i P_{it}\} + \frac{1 + \varphi}{2}[\frac{1}{\alpha}(\hat{y}_t - a_t) + A \cdot var_i P_{it}]^2 \tag{1031}$$

$$= (\hat{y}_t + \frac{1 - \sigma}{2}\hat{y}_t^2) + \frac{u_N N}{u_C C}\frac{1}{\alpha}\{[(\hat{y}_t - a) + B \cdot var_i P_{it}] + \frac{1 + \varphi}{2\alpha}(\hat{y}_t - a_t)^2\}, B \equiv \alpha A \tag{1032}$$

, here let's ignore many fluctuation terms from taylor expansion as they are extremely small and irrelevant to monetary policy.

$$-\frac{U_N}{U_C} = MP \cdot N = \alpha A N^{\alpha-1} \tag{1033}$$

$$= \alpha \frac{C}{N} \Rightarrow \frac{u_N N}{u_C C} = -\alpha \text{ (can use this to simplify equation before)} \tag{1034}$$

go back to before

$$\Rightarrow \ldots = (\hat{y}_t - \frac{1-\sigma}{2}\hat{y}_t^2) - \{(\hat{y}_t - a_t) + B \cdot var_i P_{it} + \frac{1+\varphi}{2\alpha}(\hat{y}_t - a_t)^2\} \tag{1035}$$

$$= \frac{1-\sigma}{2}\hat{y}_t^2 - B \cdot var_i P_{it} - \frac{1+\varphi}{2\alpha}(\hat{y}_t - a_t)^2 \tag{1036}$$

$$= \frac{1-\sigma}{2}\hat{y}_t^2 - B \cdot var_i P_{it} - \frac{1+\varphi}{2\alpha}(\hat{y}_t^2 - 2\hat{y}_t a_t + a_t^2) \tag{1037}$$

$$= (\frac{1-\sigma}{2} + \frac{1+\varphi}{2\alpha})\hat{y}_t^2 + \frac{\sigma+\varphi}{\alpha}\hat{y}_t a_t - B \cdot var_i P_{it} \tag{1038}$$

$$= \frac{\alpha(1-\alpha)+1+\varphi}{2\alpha}\hat{y}_t^2 + \frac{\sigma+\varphi}{\alpha}\hat{y}_t \hat{y}_t^f - B \cdot var_i P_{it} \text{ (remember when we work out the } \hat{y}_t^f) \tag{1039}$$

$$= \underbrace{-\frac{1}{2}[\underbrace{C(\hat{y}_t - \hat{y}_t^f)}_{\text{deviation from output}} + \underbrace{D(\hat{\pi}_t - \pi^*)^2}_{\text{inflation}}]}_{\text{welfare loss}} \text{ (welfare loss function)} \tag{1040}$$

the goal is to minimize the whole term

Note 24 (paradoxy in welfare loss function)**.**

the two goals are contradicated according to philips curve

21 Fiscal Policy

Planned chapter.

21.1 Government budgets, debt, and Ricardian equivalence

Derive the government flow and intertemporal budget constraints. Explain the conditions for Ricardian equivalence and why they can fail.

21.2 Taxes, purchases, transfers, and public investment

Separate income, substitution, and resource effects. Compare the objectives and transmission mechanisms of different fiscal instruments.

21.3 Fiscal multipliers and financing

Define the multiplier and its horizon. Compare spending responses across financing arrangements, monetary policy rules, household structures, and open economy environments.

22 Policy Interaction and Constraints

Planned chapter.

22.1 Fiscal and monetary regimes

Study how the fiscal backing of public liabilities interacts with interest-rate policy, inflation, and equilibrium determination.

22.2 Debt sustainability and policy space

Derive debt-to-output dynamics and distinguish arithmetic from an assessment of sustainability. Include growth, interest rates, primary balances, and risk.

22.3 Policy combinations in an open economy

Bring together exchange-rate regimes, capital flows, domestic balance sheets, and international spillovers. Compare policy combinations under stated institutional and financial constraints.

Part VI
Topics

The topics combine mechanisms from the earlier parts. Each begins with dated facts and measurement choices, compares competing explanations, and asks what evidence would distinguish them. Policy conclusions remain conditional on that evidence.

23 The Dollar in the International Monetary System

Planned chapter.

23.1 History and measures of international currency use

Separate reserve holdings, invoicing, payments, funding, and exchange-rate anchors. Define what each measure can establish about dollar dominance.

23.2 Network effects, invoicing, and safe assets

Explain the mechanisms that can sustain an international currency's use. Connect dollar pricing and demand for safe assets to the open economy models.

23.3 Benefits, obligations, and the international balance sheet

Distinguish seigniorage, financing advantages, and valuation effects. Examine the obligations and constraints associated with supplying safe and liquid international liabilities.

23.4 Global funding and policy spillovers

Trace dollar borrowing, funding stress, and the international transmission of US policy. Examine liquidity facilities and the role of financial sanctions.

23.5 Persistence, alternatives, and testable changes

Compare explanations of persistence with possible sources of change. Assess claims about de-dollarization separately across the measures defined at the start of the chapter.

24 China's Growth and Adjustment after the Pandemic

Planned chapter.

24.1 The baseline, timeline, and measurement

Separate the pre-pandemic baseline, pandemic years, and reopening period. Distinguish output levels from growth rates and real activity from nominal income. An observation after the pandemic does not identify a pandemic effect.

24.2 Demand, consumption, and employment

Compare cyclical explanations using household income, consumption, prices, and employment. State what each explanation predicts before assessing it.

24.3 Property, balance sheets, and local public finance

Map the links between housing, household and firm balance sheets, banks, and local government finances. Compare expenditure adjustment and debt resolution mechanisms.

24.4 Population, productivity, and resource allocation

Use growth accounting and the growth models to distinguish demographic, investment, and productivity explanations. Examine the allocation of resources across sectors and firms.

24.5 External demand and policy alternatives

Connect domestic adjustment to trade and international conditions. Compare policy alternatives through their mechanisms, constraints, distributional effects, and testable implications.